

COMPASS_Synoptic_SoilFluxes_2025_CH4

Stephanie J. Wilson

2026-01-22

Contents

0.1	Goals of this code:	1
0.2	Set Up & Load Packages	2
0.3	Pull in Smart Chamber Files from the Raw Data Folder	2
0.4	Check if there is a meta-data file, if not, create a meta-data file with a unique Plot_ID	2
0.5	Mark Drop column in Metadata for Retaken Fluxes or other reasons to Drop	2
0.6	Match up the metadata to the raw licor data	3
0.7	Change the gas units from ppb to nmol	3
0.8	Calculate fluxes	3
0.9	Visualize fluxes and sort by r2	3
0.10	check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)	4
0.11	Spot check a few fluxes with R2 > 0.9	6
0.12	Look at the really high or really low ones to see if there is a timing issue or something	147
0.13	Check all fluxes	196
0.14	Do some visualization of QAQC plots to assess fluxes	200
0.15	Read out final dataframe and export for use elsewhere	201

0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label, the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right Start time – would be initially populated by the start time in the OR blank Observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data
 - a. Change drop column if fluxes need to be dropped
 - b. This is where the adjusted times will also be applied
4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data Folder

0.4 Check if there is a meta-data file, if not, create a meta-data file with a unique Plot_ID

```
## [1] "CSV file exists and has been read into the code."
```

0.5 Mark Drop column in Metadata for Retaken Fluxes or other reasons to Drop

```
##### Check For Plot ID

##Sometimes there are other fluxes in the .json files that we don't want to calculate
#change drop column to TRUE if the plot doesn't include a site code in the Plot_ID
metadata <- metadata %>%
  rename(Plot = Plot_ID) %>%
  mutate(
    drop = ifelse(
      !grepl("SWH|GCW|MSM|GWI|GCRew", Plot),
      # Check if Plot_ID does not contain the specified strings
      TRUE, # Set drop to TRUE if it doesn't contain any of the strings
      drop # Keep the original value of drop if it contains one of the strings
    )
  )

##### Check for retakes

##Sometimes we have to retake a flux in the field due to issues or ebullition
##We need to change the drop column to reflect using only the latest flux
#In order to we expect an "a" after the label, but sometimes this is not the case so we will fix that
metadata <- metadata %>%
  mutate(Plot = case_when(
    str_detect(Plot, "-retake") ~ str_replace(Plot, "-retake", "a"),
    str_detect(Plot, "-Retake") ~ str_replace(Plot, "-Retake", "a"),
    str_detect(Plot, "-b") ~ str_replace(Plot, "-b", "b"),
    TRUE ~ Plot # Keeps the original value if none of the conditions match
  ))

#If the flux was taken more than once, flag the others so that we use the latest measurement
# Extract the base plot (e.g., "GCW-TR-1" from "GCW-TR-1", "GCW-TR-1a", etc.)
metadata$base_plot <- str_extract(metadata$Plot, "^[^-]+-[A-Za-z]+-[A-Za-z]+-[A-Za-z]+")

# Extract the replicate number (the number after the last hyphen)
metadata$replicate <- as.numeric(str_extract(metadata$Plot, "(?<=-)[0-9]+(?=[a-zA-Z]?$)"))

# Extract the suffix (if any, the letter after the replicate number)
metadata$suffix <- str_extract(metadata$Plot, "(?<=\\d)[a-zA-Z]$")

# If there's no suffix, set it to an empty string
metadata$suffix[is.na(metadata$suffix)] <- ""
```

```

# Sort by base plot, replicate number, and suffix (to keep highest suffix) - #####INSTEAD WE WANT TO CH
metadata <- metadata %>%
  group_by(base_plot, replicate) %>%
  arrange(base_plot, replicate, desc(suffix), .by_group = TRUE) %>%
  mutate(drop = row_number() != 1) %>% #the entry with the latest suffix will keep false all others wil
  #slice_head(n = 1) %>% #if you wanted to remove a the line rather than flag it
  ungroup()

##### Check Observation lengths

#if the observation length is over 5 mins then change drop column to true
metadata <- metadata %>%
  mutate(drop = if_else(Total_Time > 300, TRUE, drop))

##### Remove columns with drop = TRUE
metadata <- metadata %>%
  filter(drop == "FALSE")

```

0.6 Match up the metadata to the raw licor data

0.7 Change the gas units from ppb to nmol

0.8 Calculate fluxes

0.9 Visualize fluxes and sort by r2

```

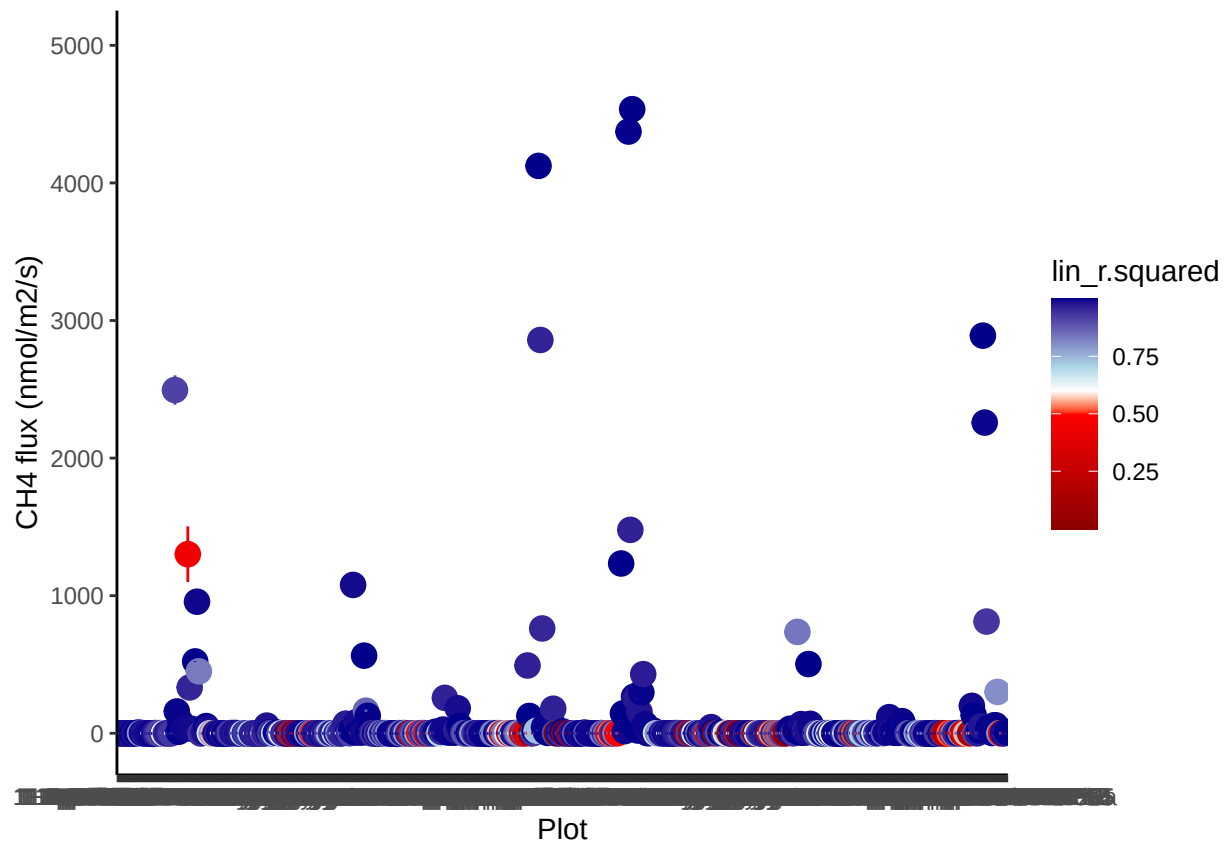
## Warning: Removed 5 rows containing missing values or values outside the scale range
## ('geom_point()').

```

```

## Warning: Removed 5 rows containing missing values or values outside the scale range
## ('geom_segment()').

```



0.10 check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

```
tobechecked <- c(
)

colnames <- c("Plot")

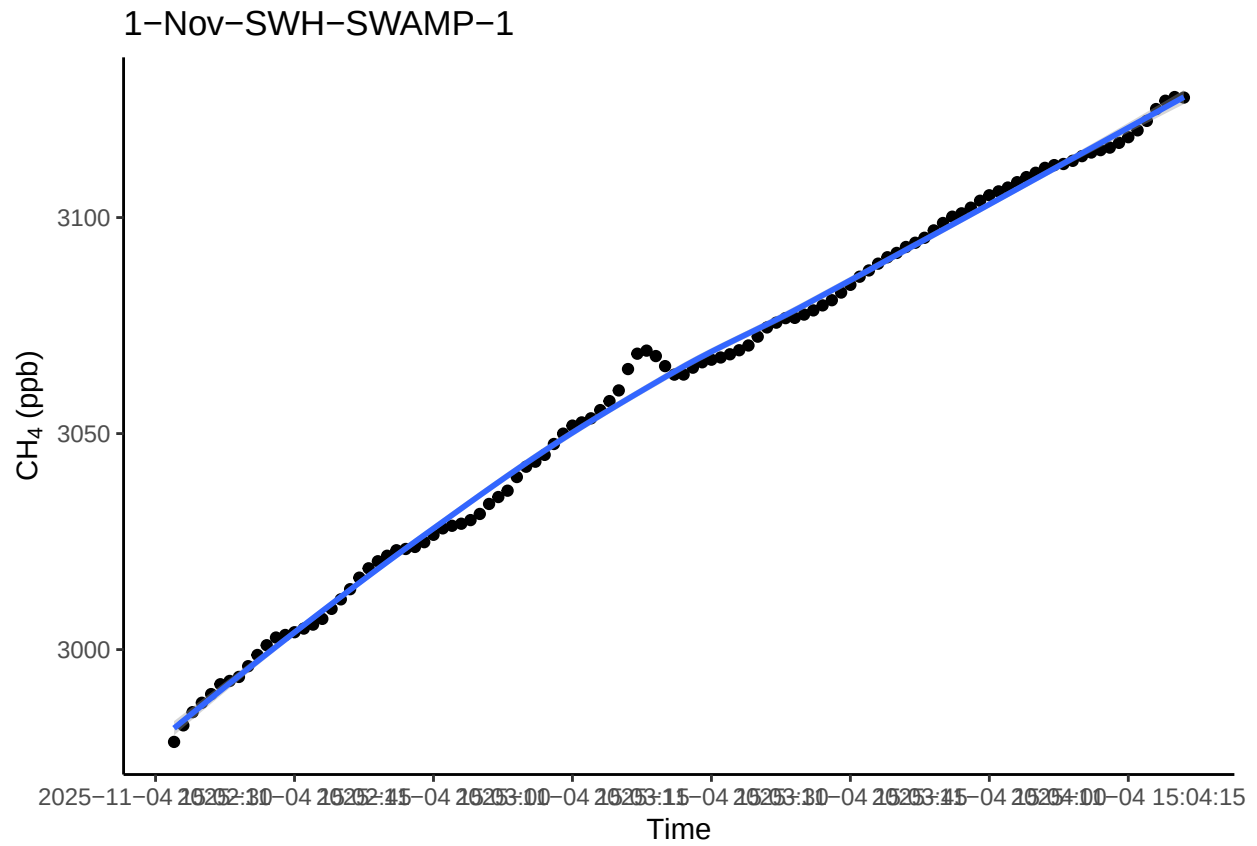
#Loop to run five random plots subset above
for(P in tobechecked){
  r1 <- subset(alldat, Plot == P)
  p1 <- ggplot(r1, aes(as.POSIXct(TIMESTAMP), ch4)) + geom_point() +
    scale_x_datetime(breaks = "15 sec") +
    theme_classic() +
    ylab(expression(paste( CH [4], " (ppb)"))) +
    xlab("Time") + labs(title = P) +
    geom_smooth()
  print(p1)
}

#list of the ones that I changed and checked and looked good?
fixedplots <- c("1-Sep-MSM-TR-1", "1-Sep-MSM-TR-2", "1-Sep-SWH-SWAMP-3",
  "2-May-SWH-SWAMP-3a", "2-May-SWH-UP-3", "2-Sep-SWH-TR-6",
```

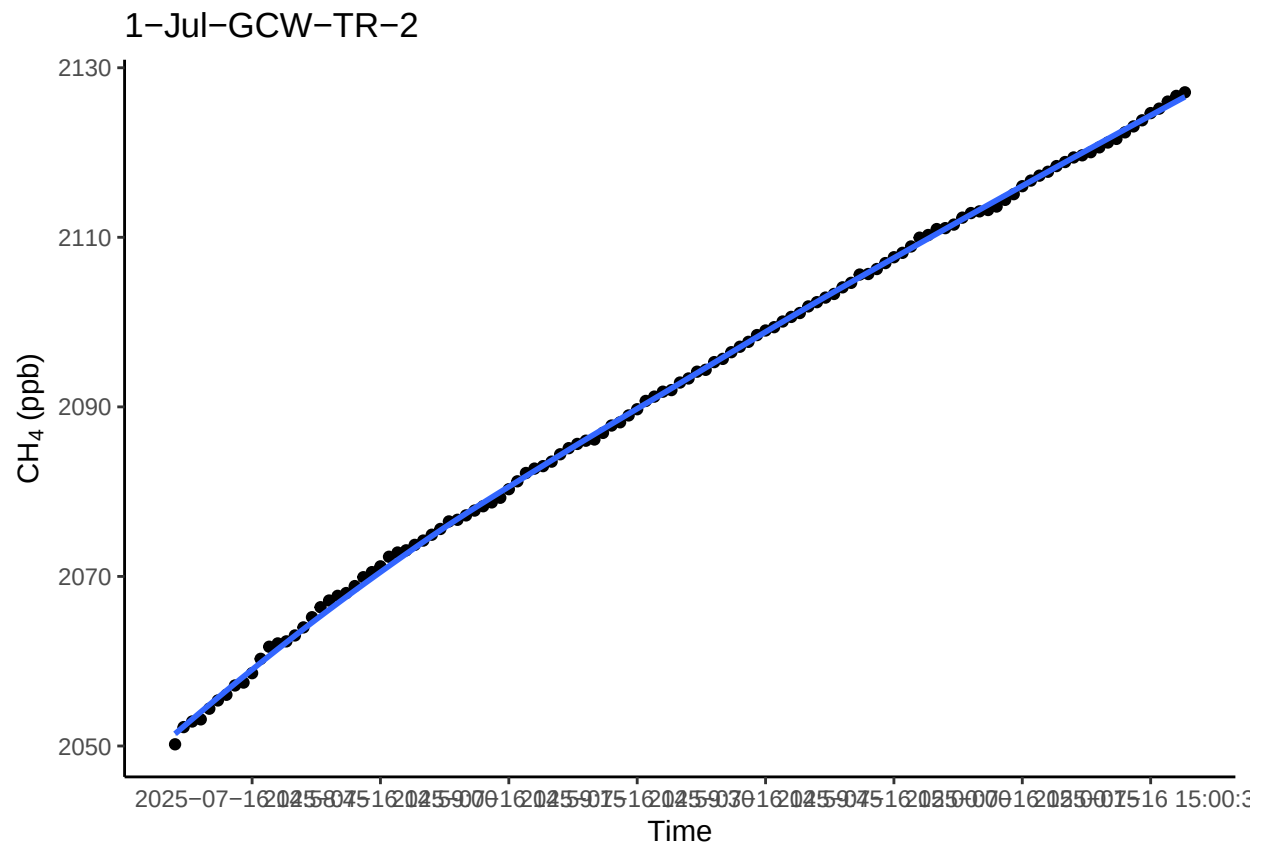
"1-Sep-GCW-TR-3", "1-Nov-SWH-SWAMP-1", "1-May-SWH-TR-4",
"1-May-SWH-TR-3", "1-Sep-SWH-TR-8", "1-May-SWH-TR-9",
"2-May-SWH-SWAMP-1a", "1-Mar-SWH-TR-3", "1-Mar-SWH-TR-5",
"1-May-SWH-SWAMP-4a", "1-Nov-SWH-SWAMP-3b", "1-Jul-SWH-SWAMP-4",
"1-Jul-SWH-TR-10", "1-Jul-SWH-TR-8", "2-Jul-SWH-TR-6",
"2-May-SWH-TR-10", "2-May-SWH-TR-9", "2-Sep-SWH-TR-10a",
"2-Sep-SWH-TR-7", "2-Jul-SWH-TR-8",
"1-May-SWH-TR-5", "1-May-SWH-TR-6a", "1-Sep-GCW-TR-2",
"1-Sep-SWH-TR-5", "1-Sep-SWH-TR-6", "1-Sep-SWH-TR-7",
"1-Sep-SWH-TR-9", "2-Jul-SWH-TR-9", "2-Jul-SWH-UP-1",
"2-Jul-SWH-UP-3", "2-Jul-SWH-UP-4", "2-Jul-SWH-UP-5",
"2-Mar-SWH-TR-7a", "2-May-SWH-SWAMP-2", "2-May-SWH-TR-3",
"2-May-SWH-TR-4", "2-May-SWH-TR-7a", "2-May-SWH-TR-8a",
"2-Nov-SWH-SWAMP-1", "2-Nov-SWH-SWAMP-2", "2-Nov-SWH-SWAMP-3b",
"2-Nov-SWH-SWAMP-4a", "2-Nov-SWH-TR-9a", "2-Sep-GCW-TR-4",
"2-Sep-GCW-TR-5", "2-Sep-SWH-SWAMP-4", "2-Sep-SWH-TR-5", "2-Sep-SWH-TR-7",
"2-Mar-SWH-TR-5", "2-Sep-SWH-UP-4", "2-Sep-SWH-UP-1", "2-Sep-SWH-SWAMP-3",
"2-Sep-MSM-TR-1", "2-Sep-GCW-TR-2", "2-Nov-MSM-TR-5", "2-Nov-GWI-TR-1a",
"2-Nov-SWH-UP-4", "1-Jul-SWH-TR-5", "1-Jul-SWH-UP-2", "1-Jun-GCW-TR-3",
"1-Jun-GCW-TR-4", "1-Mar-SWH-TR-7a", "1-Mar-SWH-UP-1", "1-May-SWH-SWAMP-3a",
"1-May-SWH-SWAMP-5", "1-May-SWH-TR-10", "1-May-SWH-TR-2", "1-May-SWH-UP-2",
"1-Nov-GCW-TR-1", "1-Nov-SWH-TR-5", "1-Nov-SWH-TR-6", "1-Nov-SWH-TR-9a",
"1-Nov-SWH-UP-2a", "1-Sep-GCW-TR-1", "1-Sep-GCW-TR-5", "1-Sep-MSM-TR-2",
"1-Sep-SWH-SWAMP-4", "1-Sep-SWH-SWAMP-5", "2-Jul-SWH-SWAMP-2", "2-Jul-SWH-TR-5",
"2-Jul-SWH-TR-8", "2-Jun-GCW-UP-1", "2-Mar-SWH-TR-1", "2-Mar-SWH-TR-2",
"2-Mar-SWH-TR-4", "2-Mar-SWH-UP-1", "2-May-SWH-SWAMP-3a", "2-May-SWH-UP-2")

0.11 Spot check a few fluxes with $R^2 > 0.9$

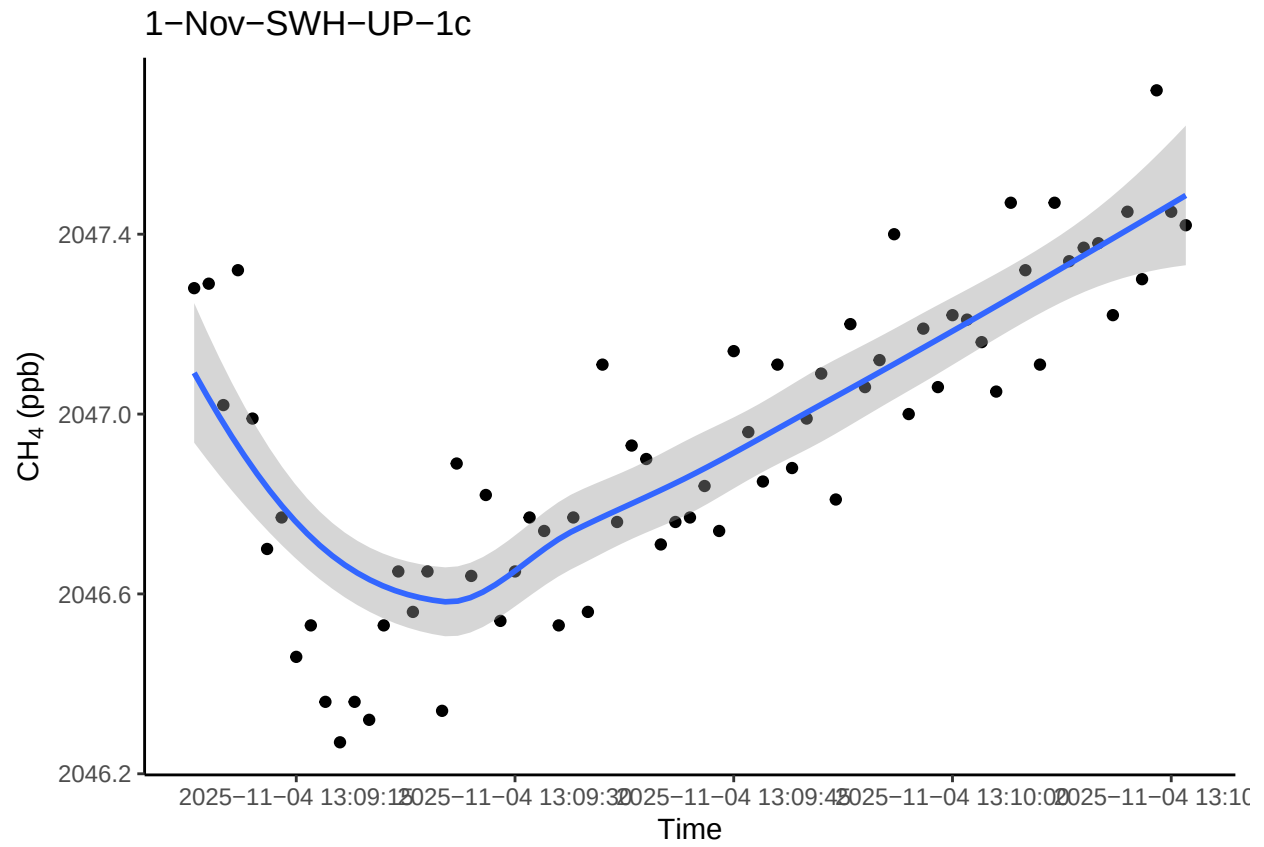
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



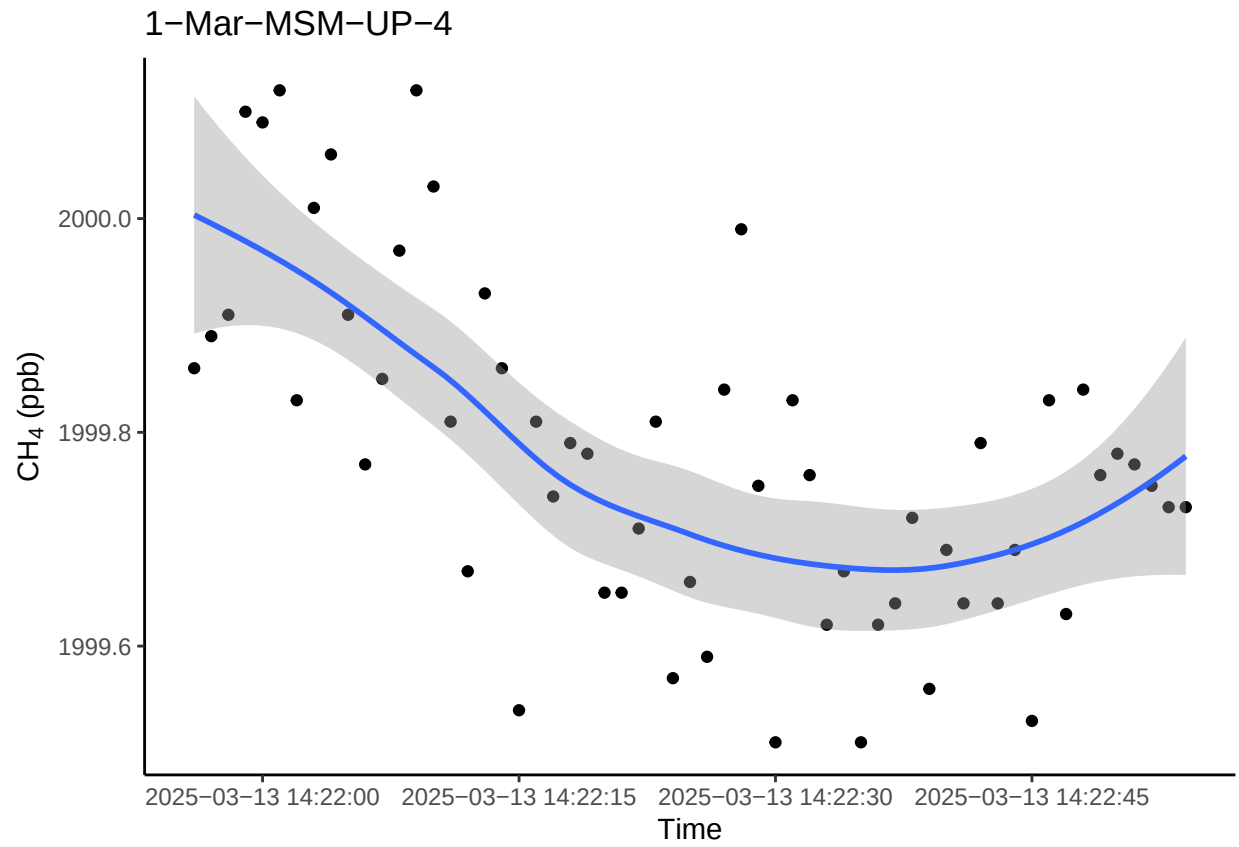
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



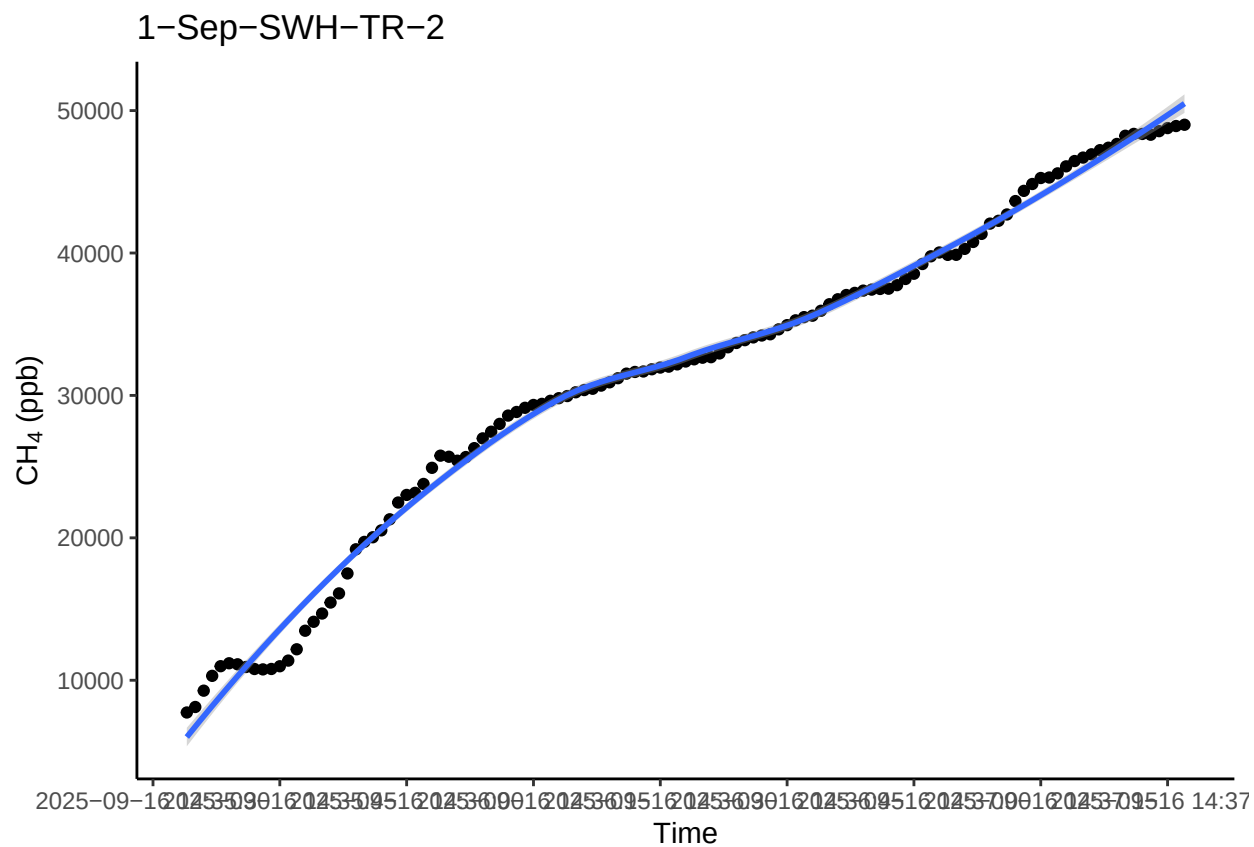
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

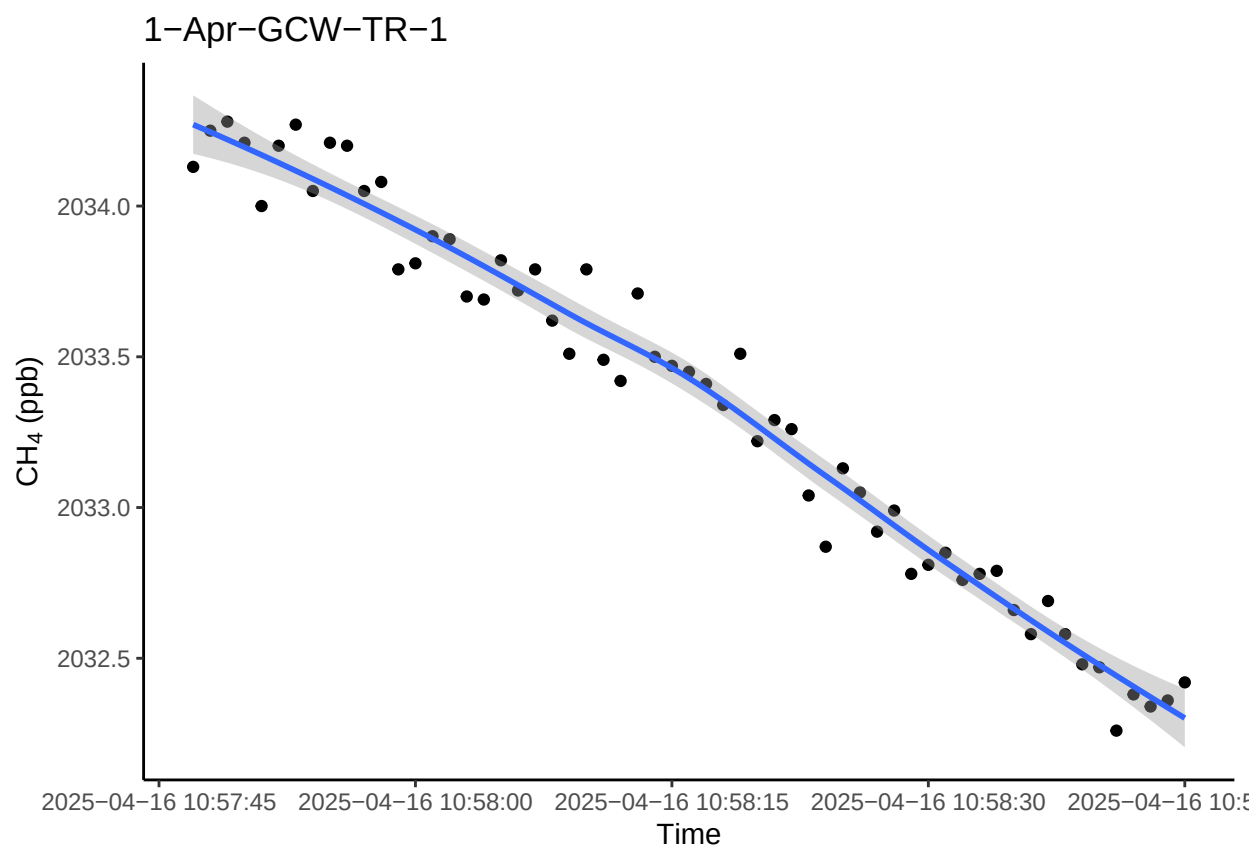



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

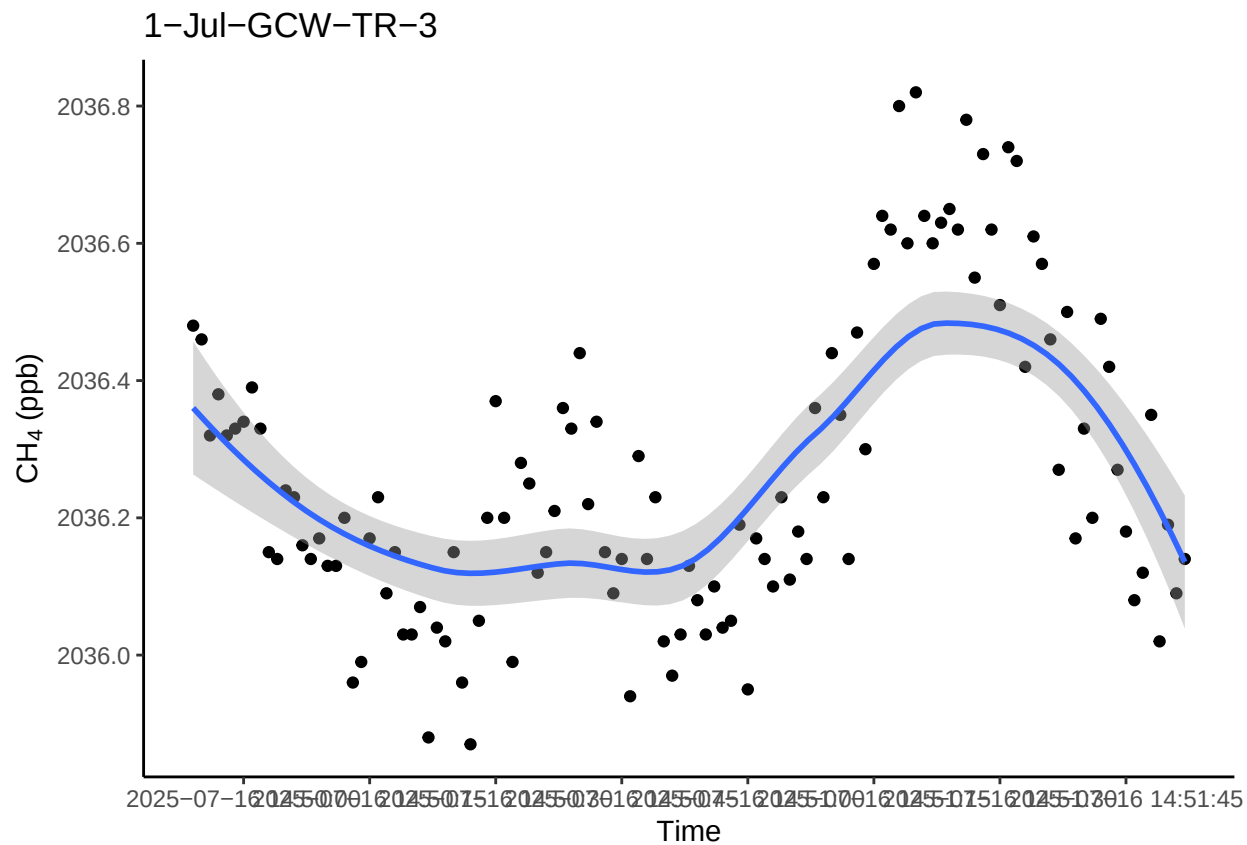


```
## Take a look at the fluxes <0.9 and a flux estimate below 1
```

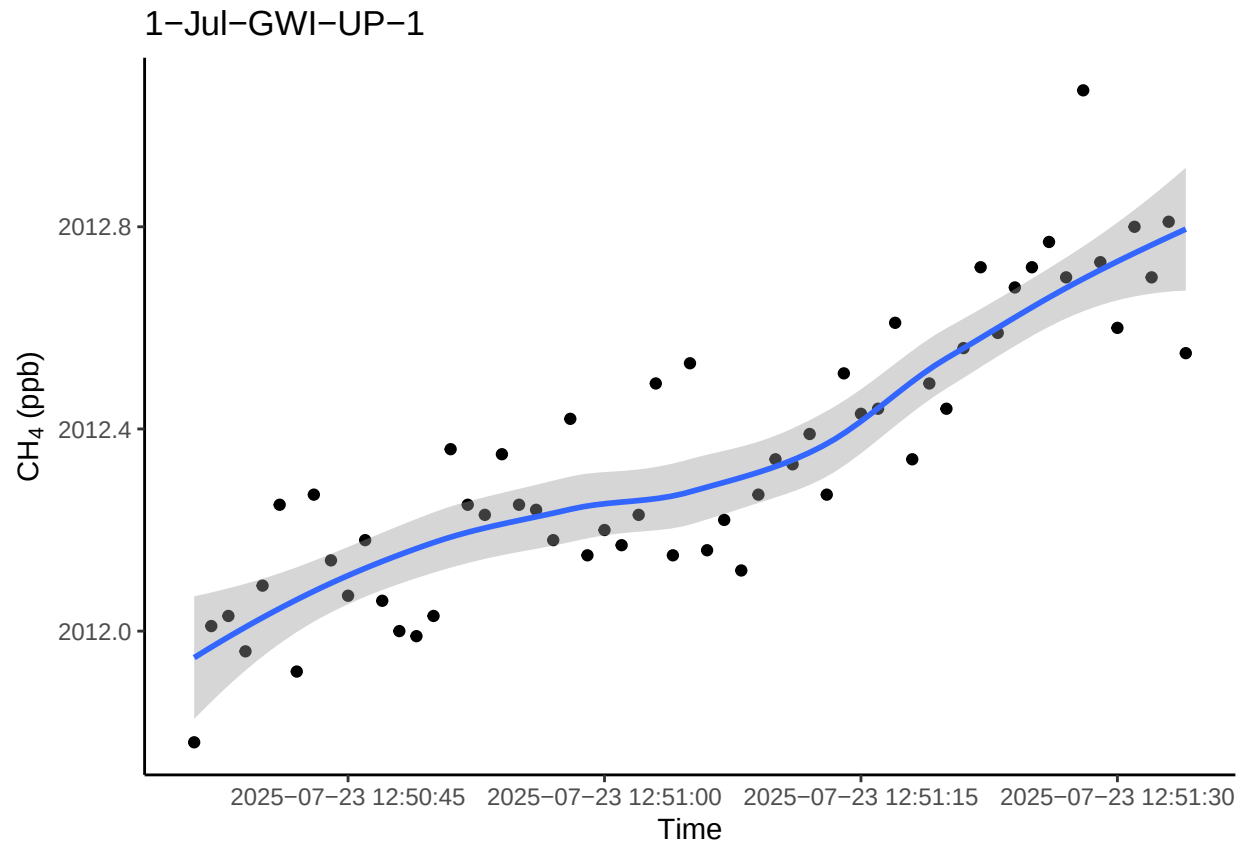
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



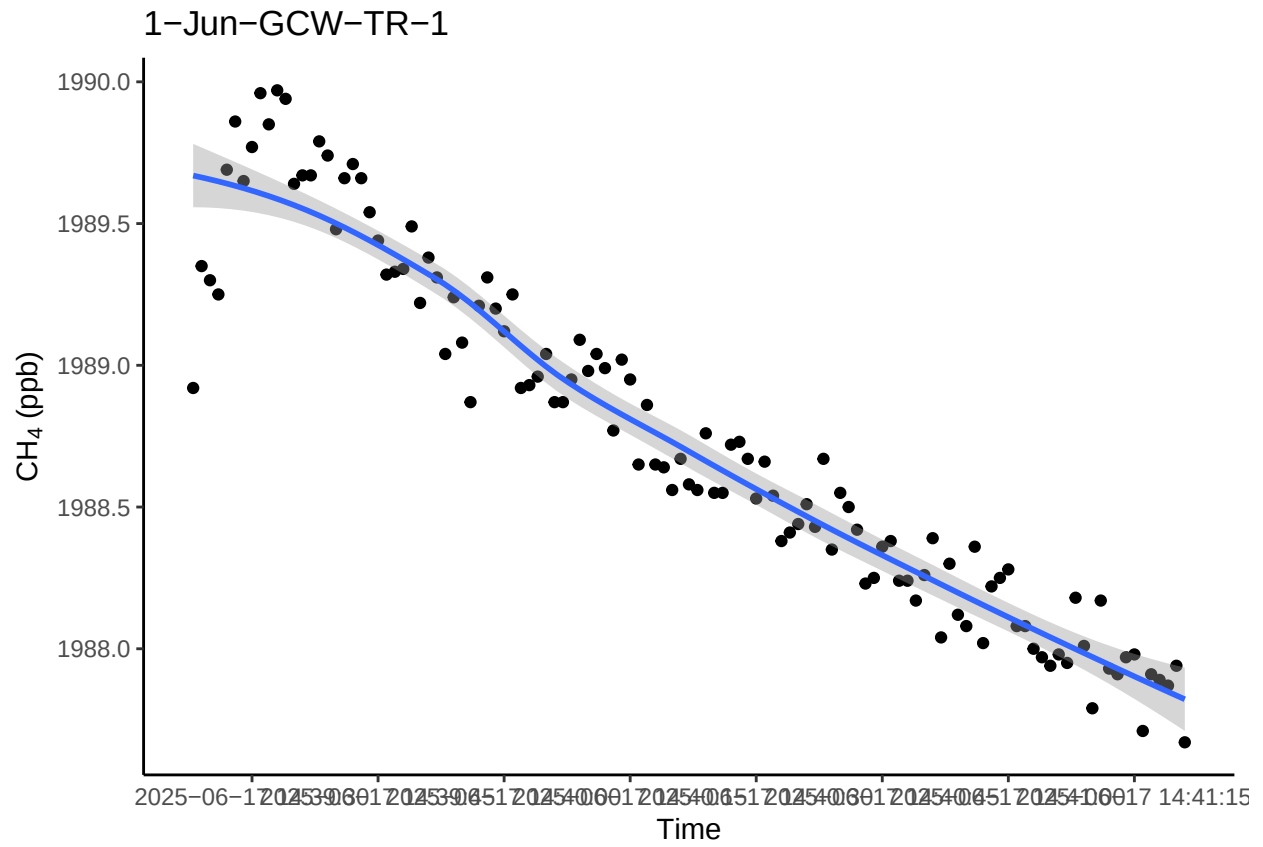
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



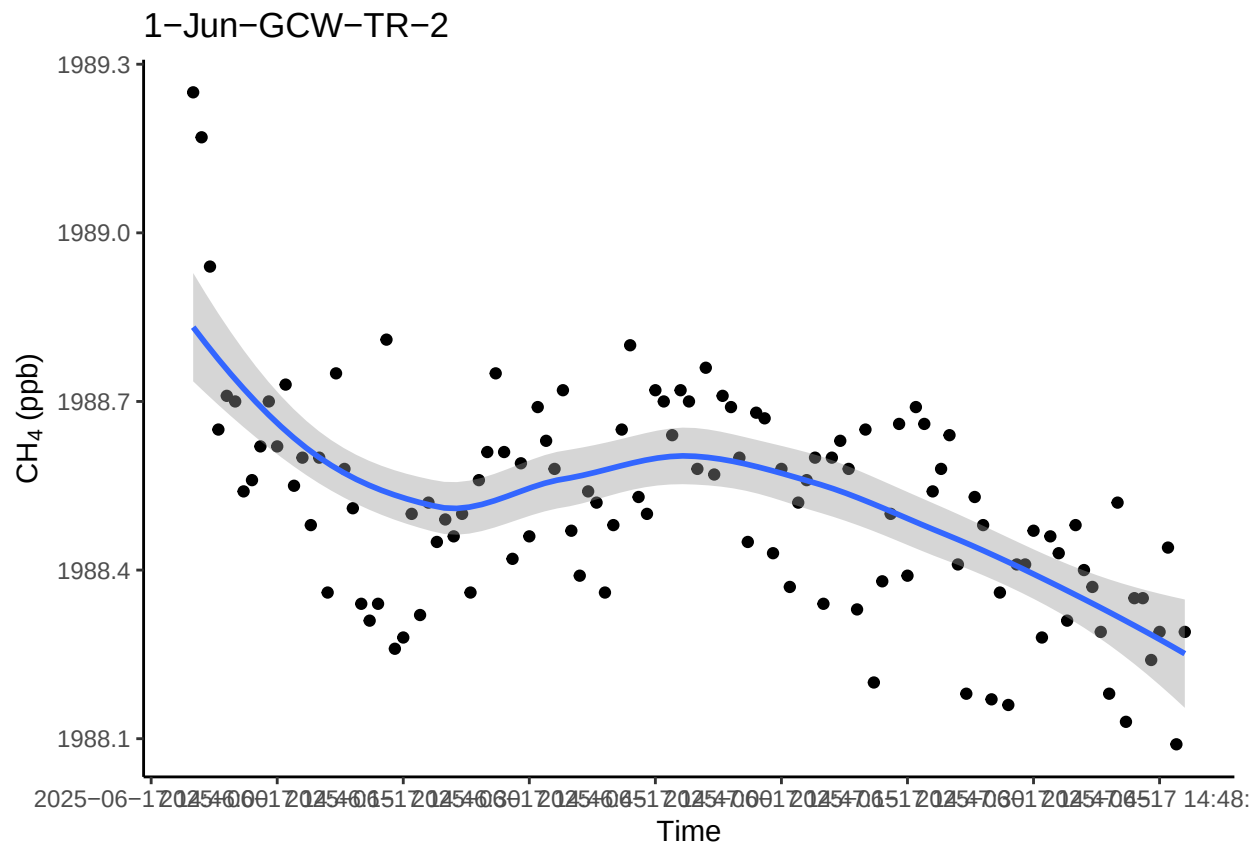
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



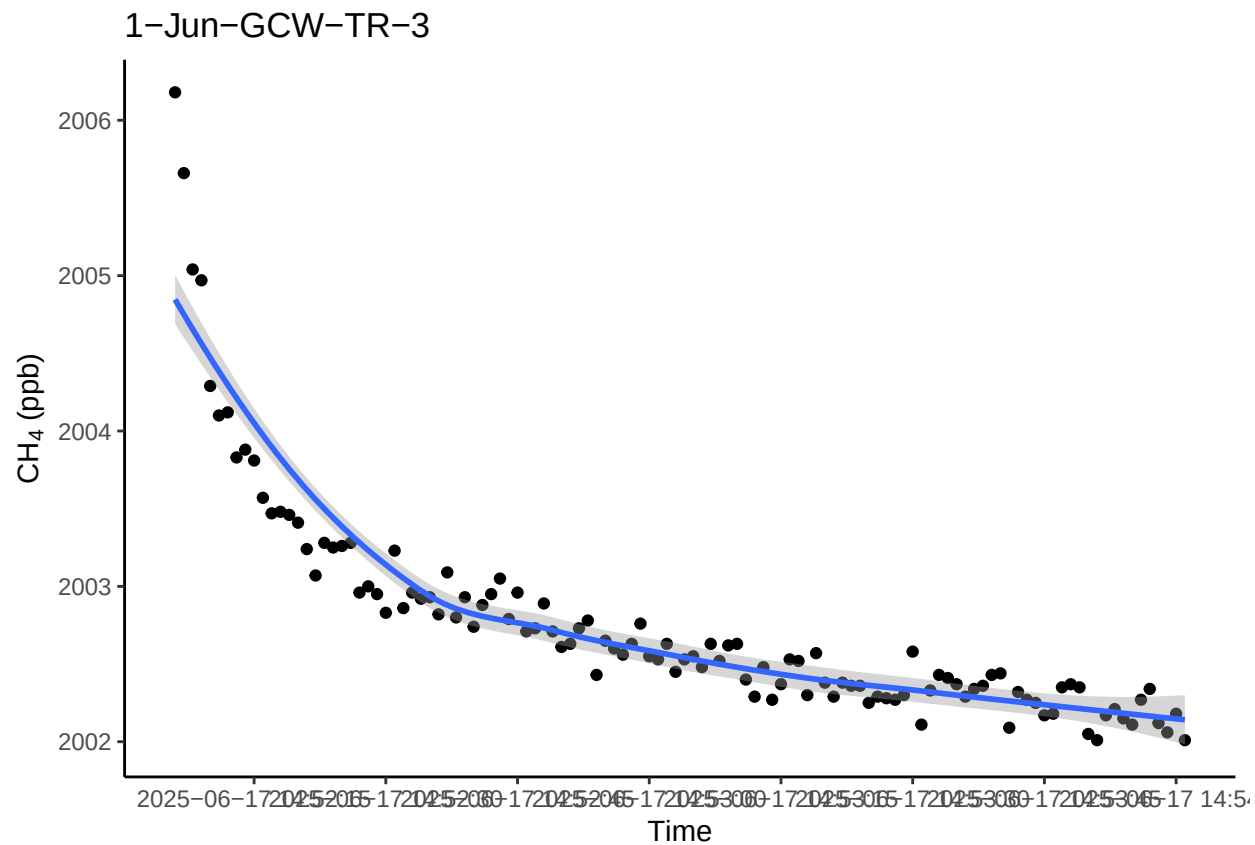
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



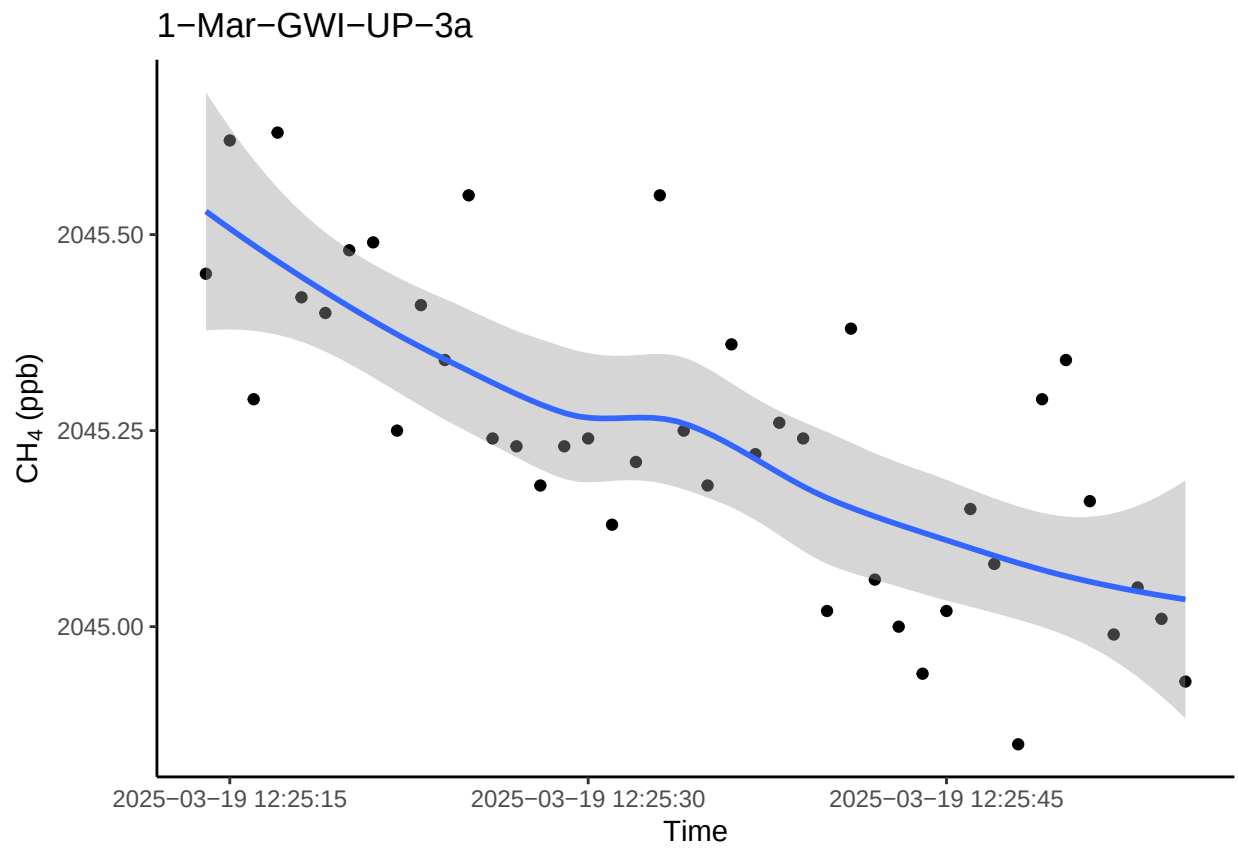
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



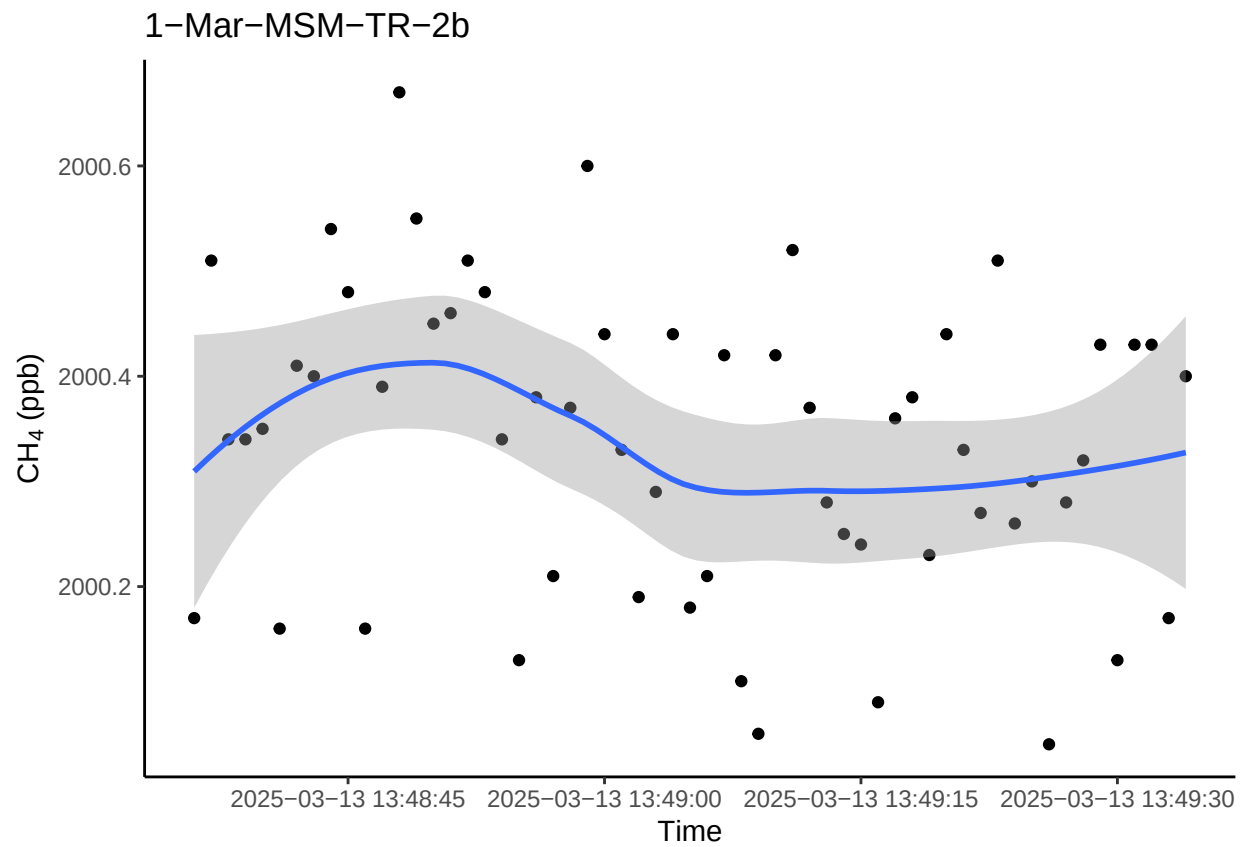
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



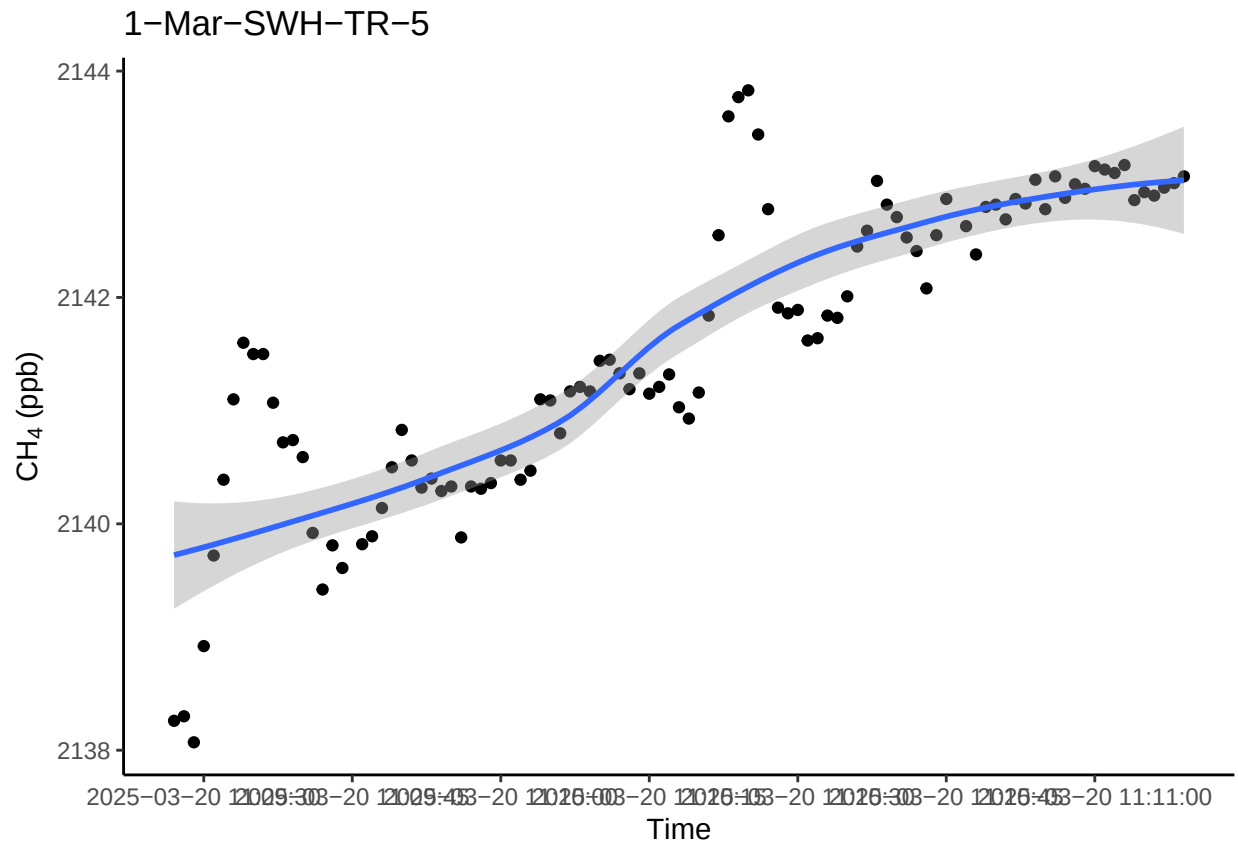
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

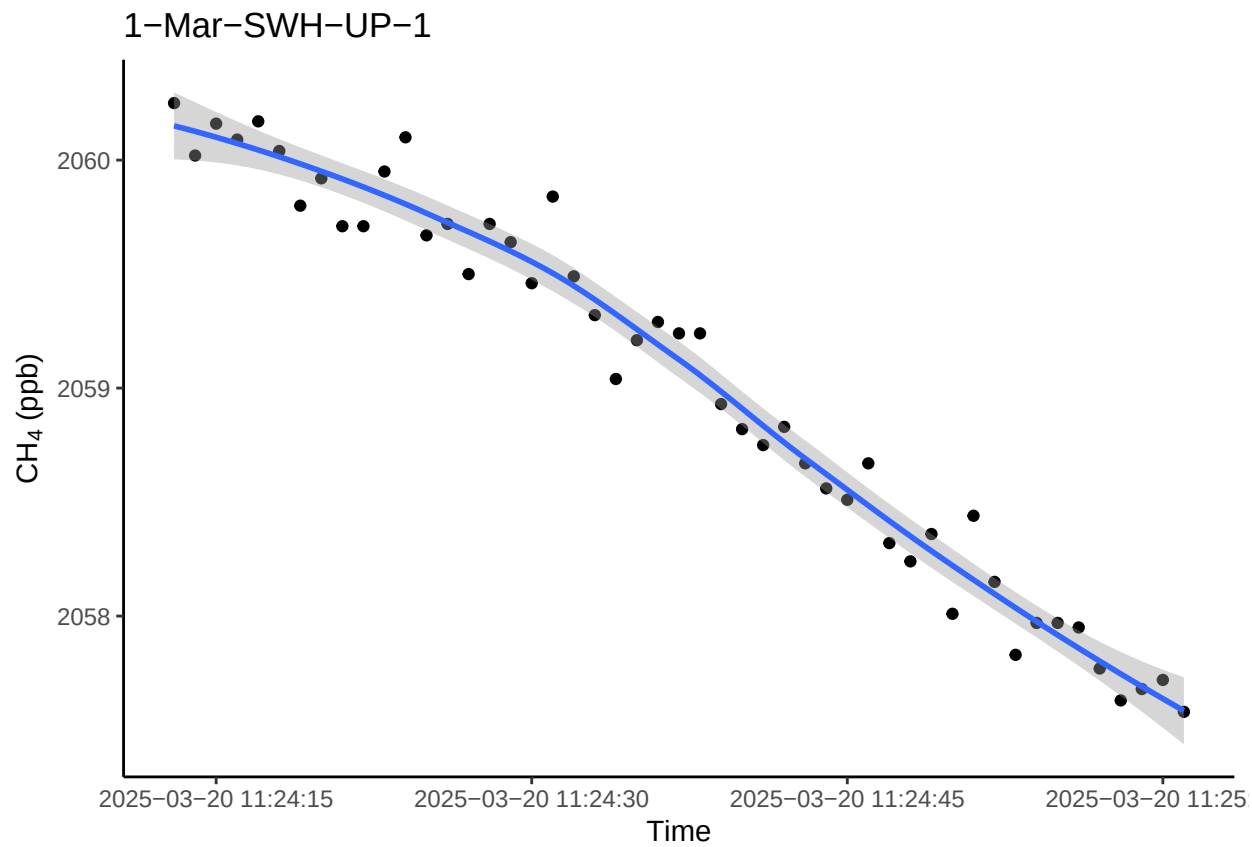
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



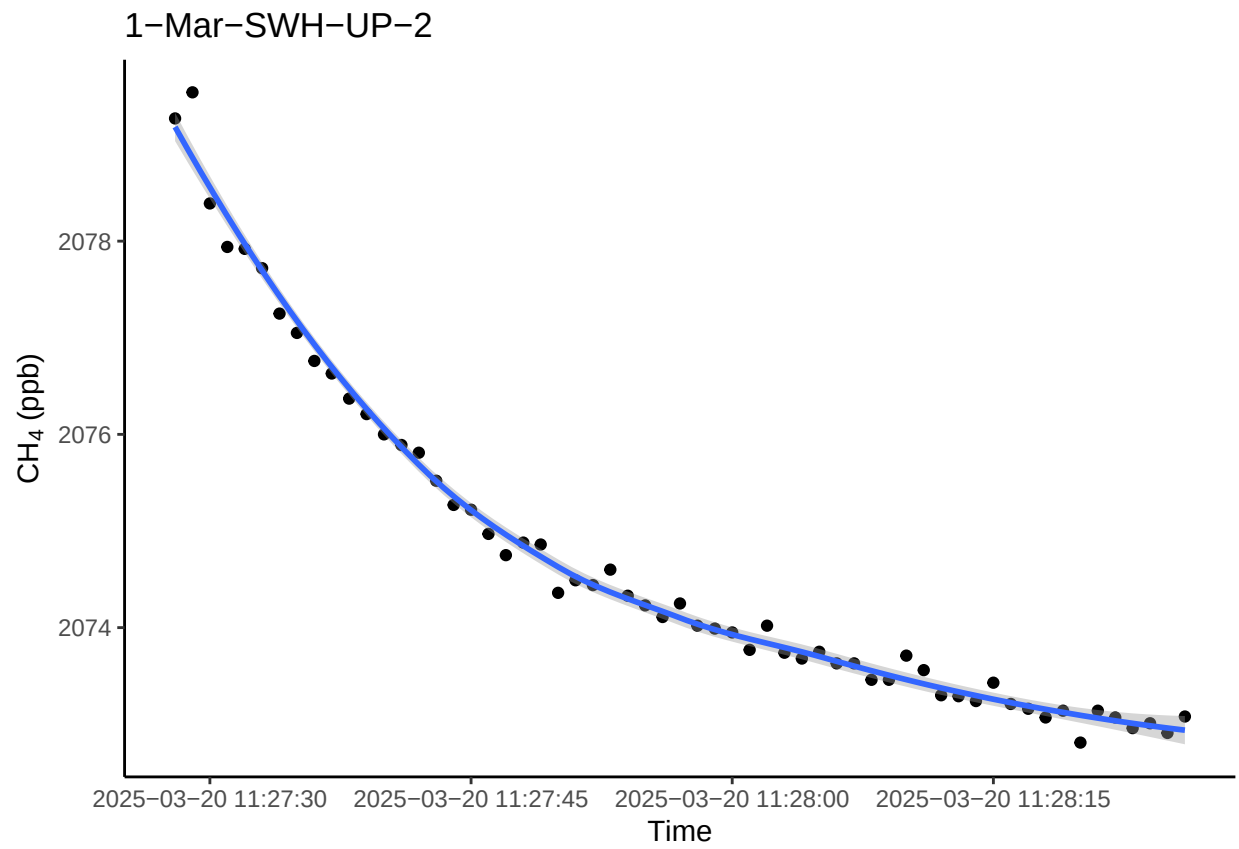
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



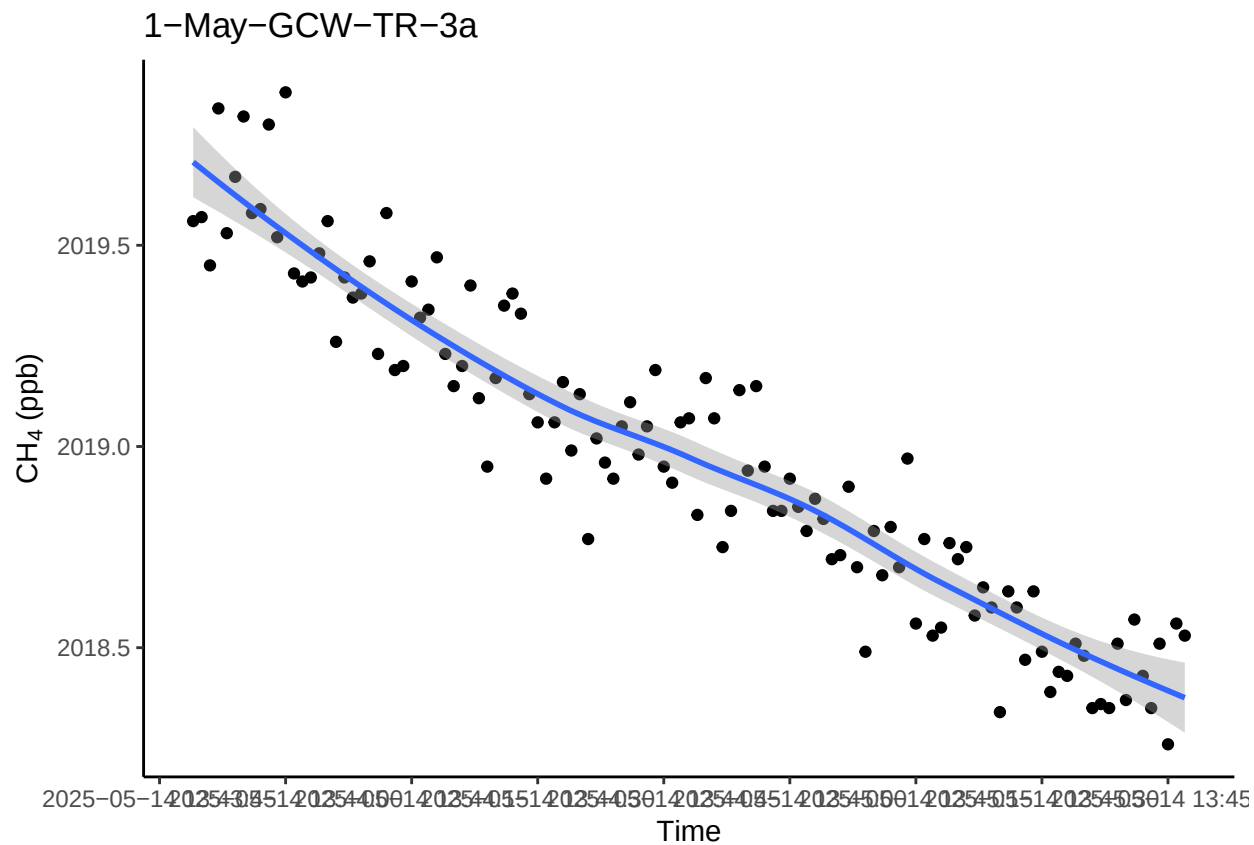
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



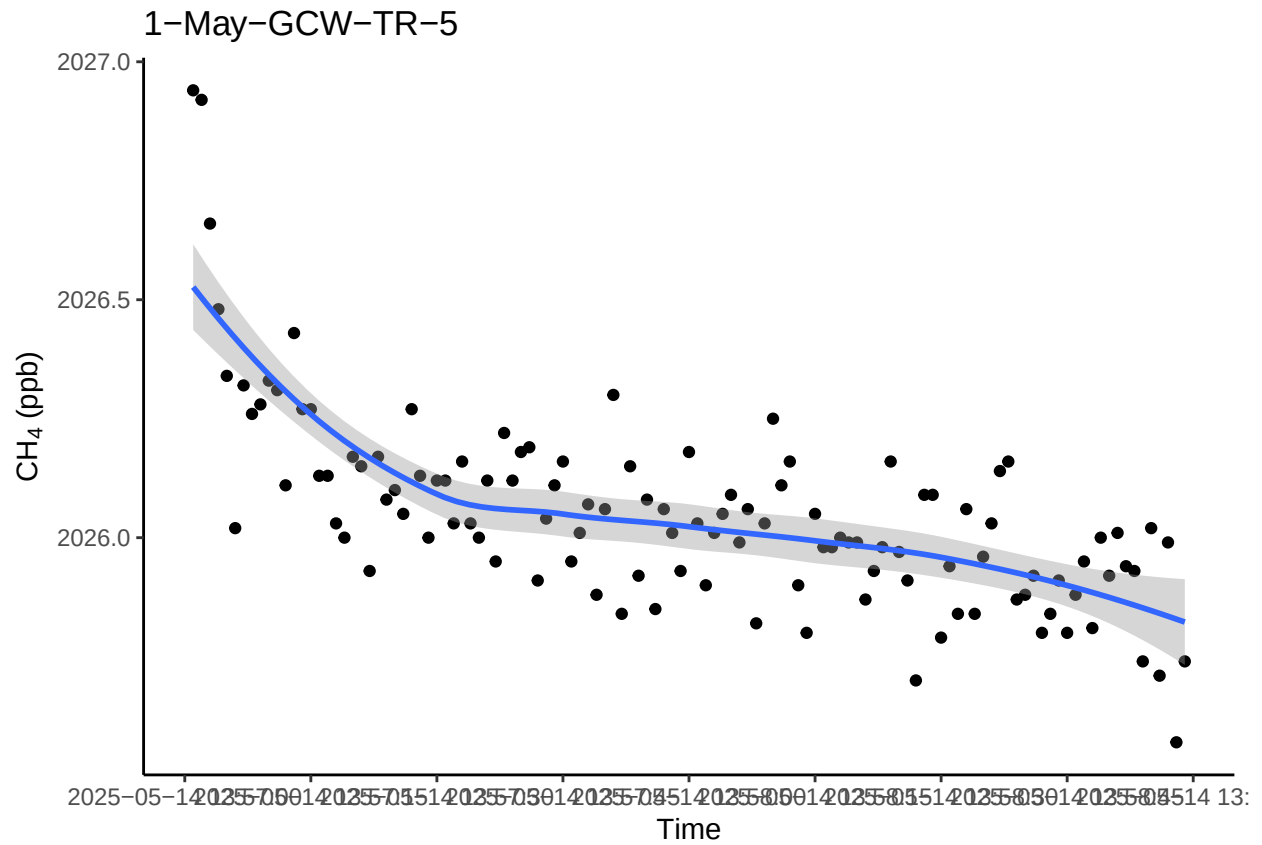
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



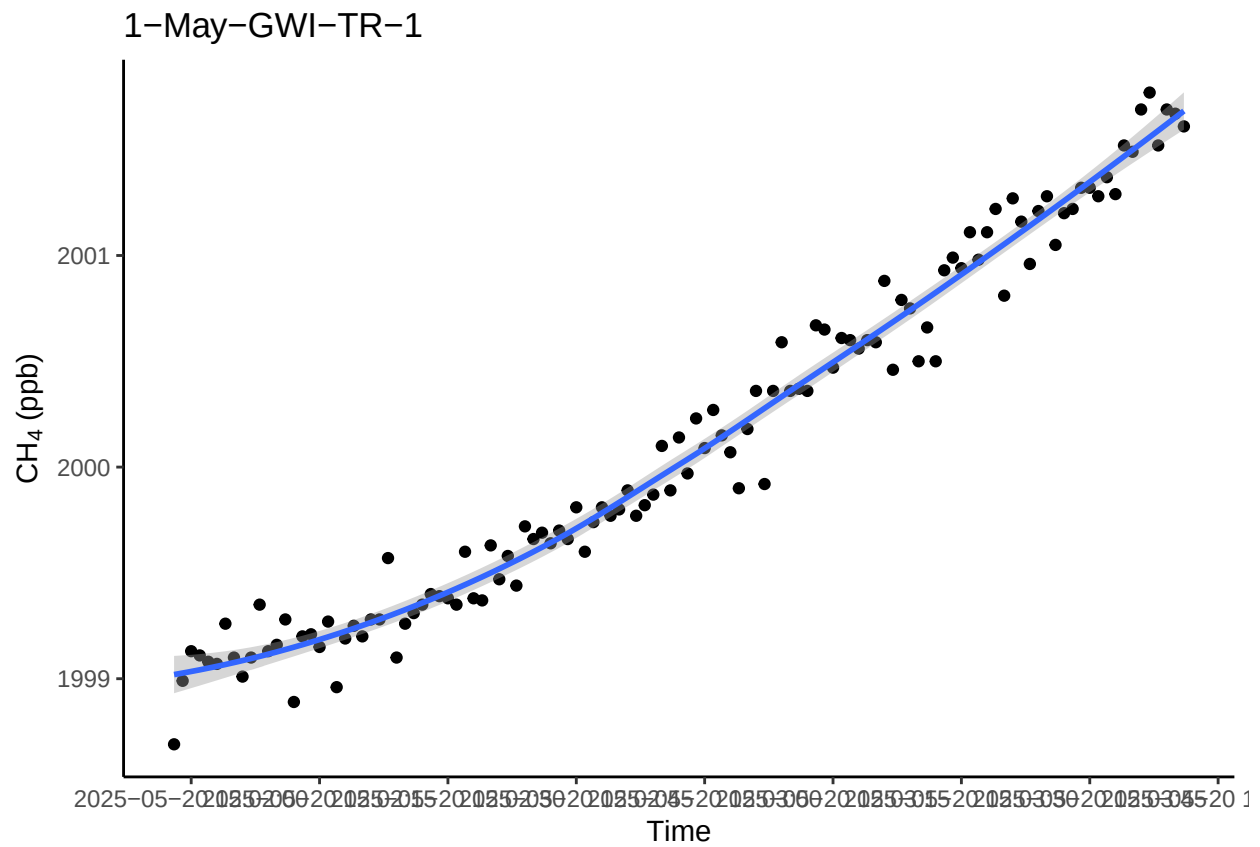
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

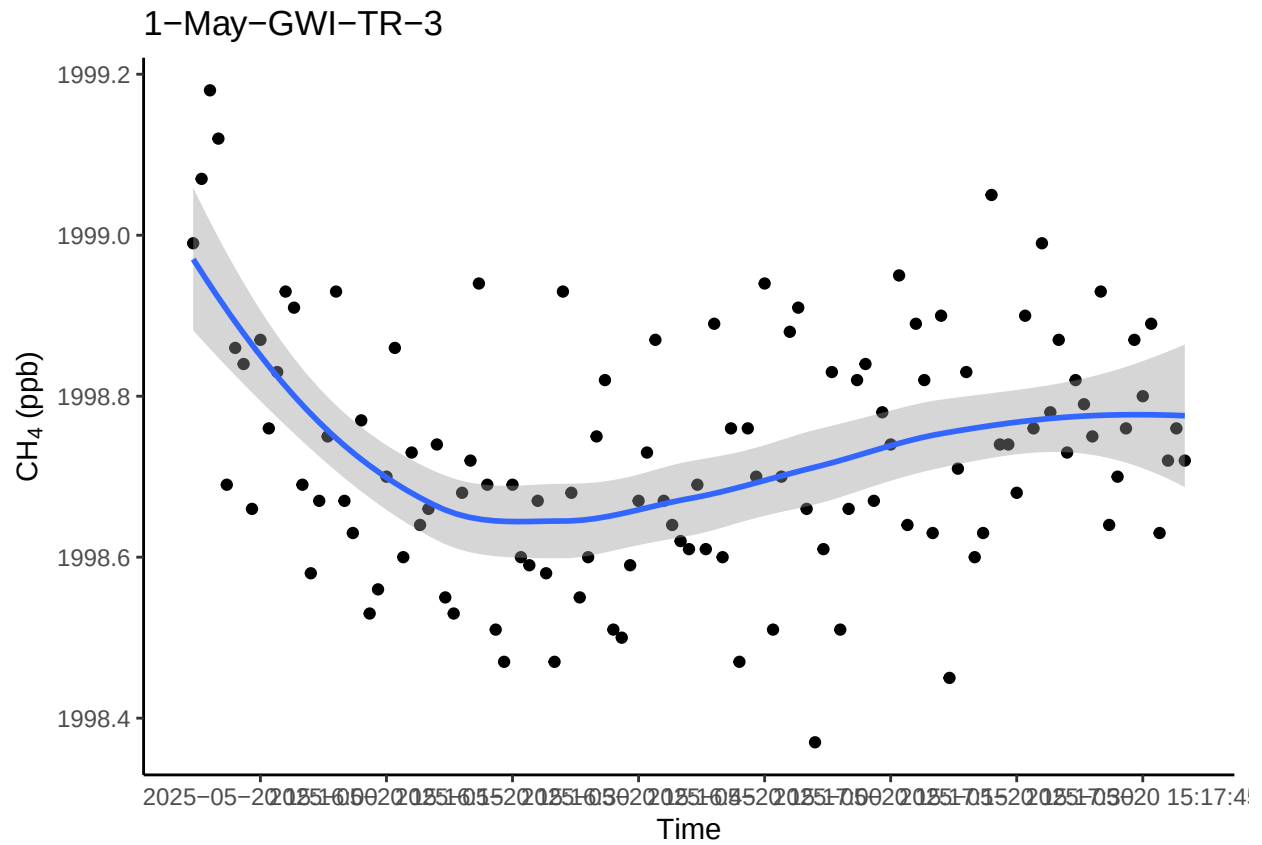
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



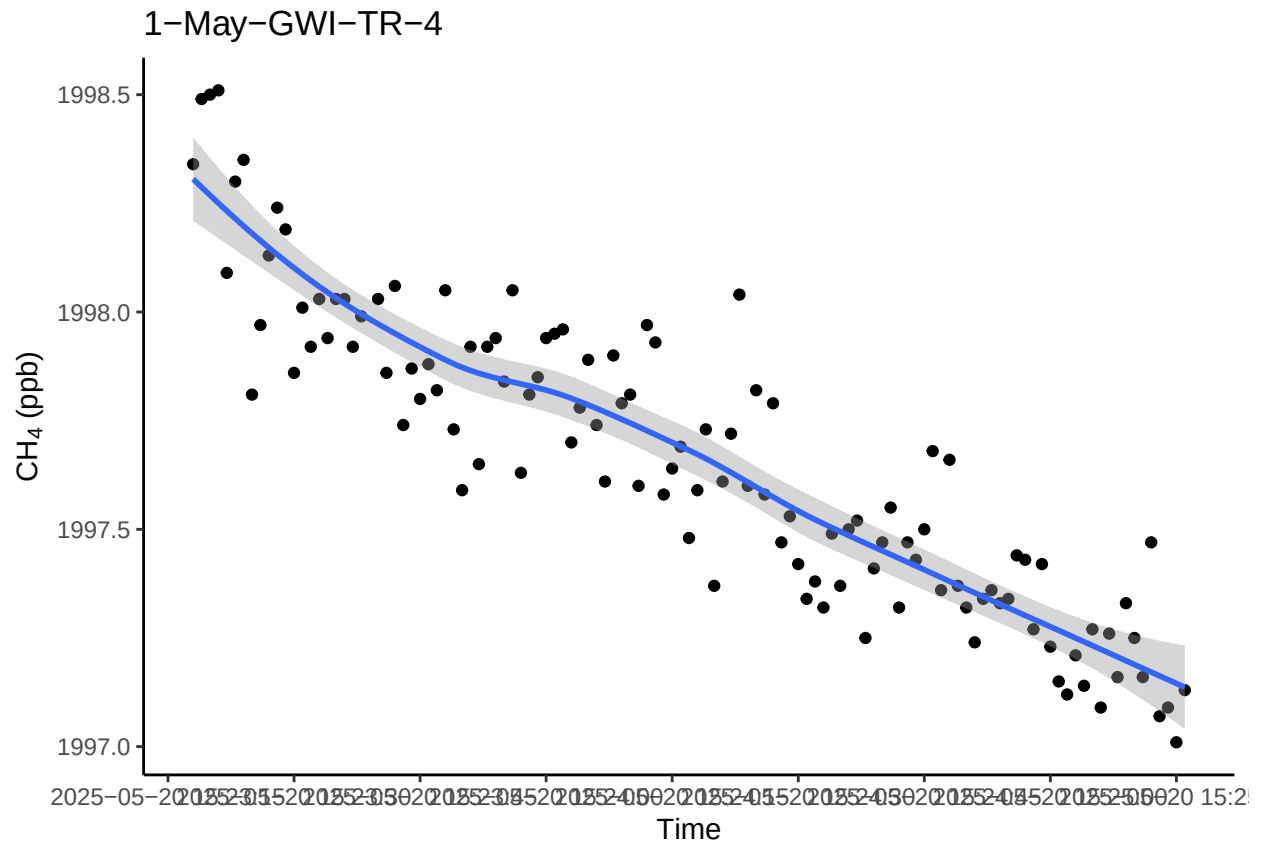
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

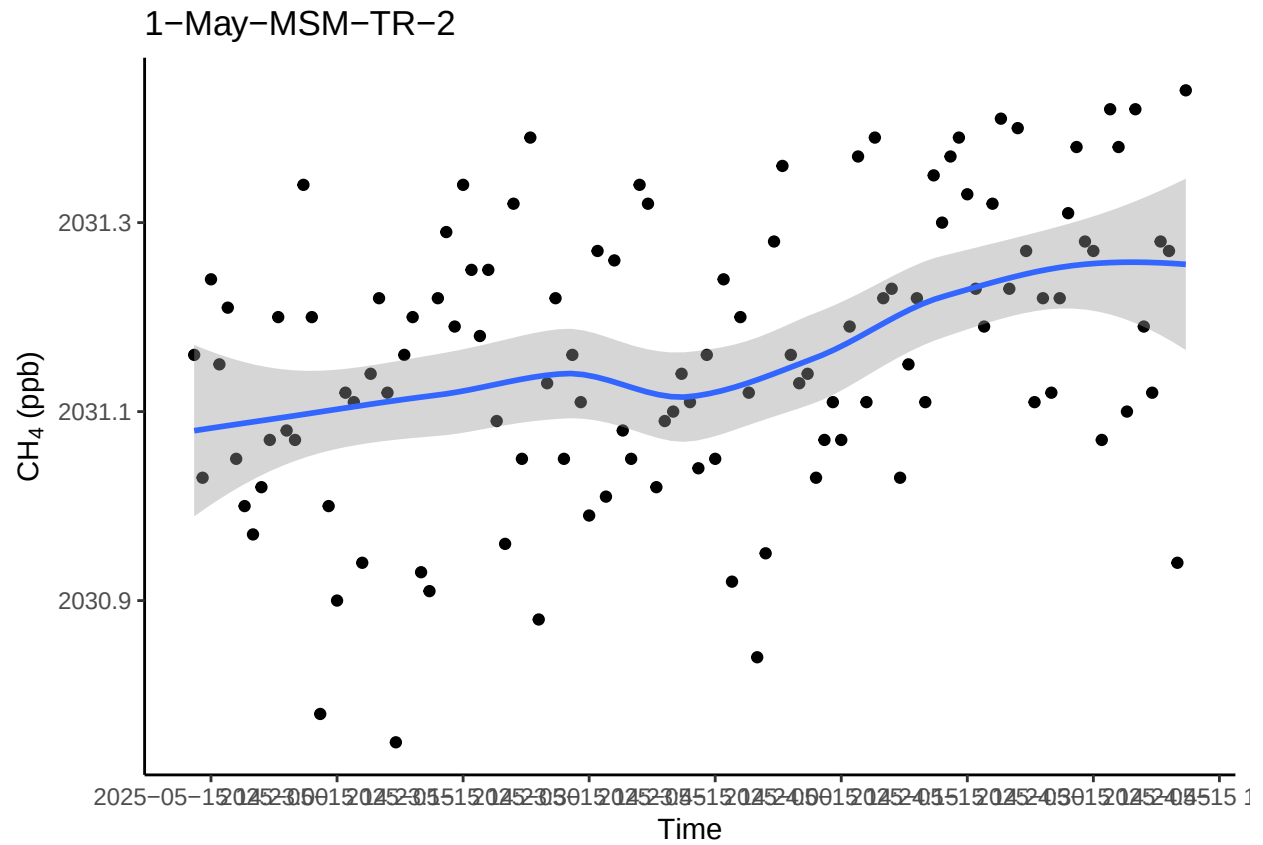
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

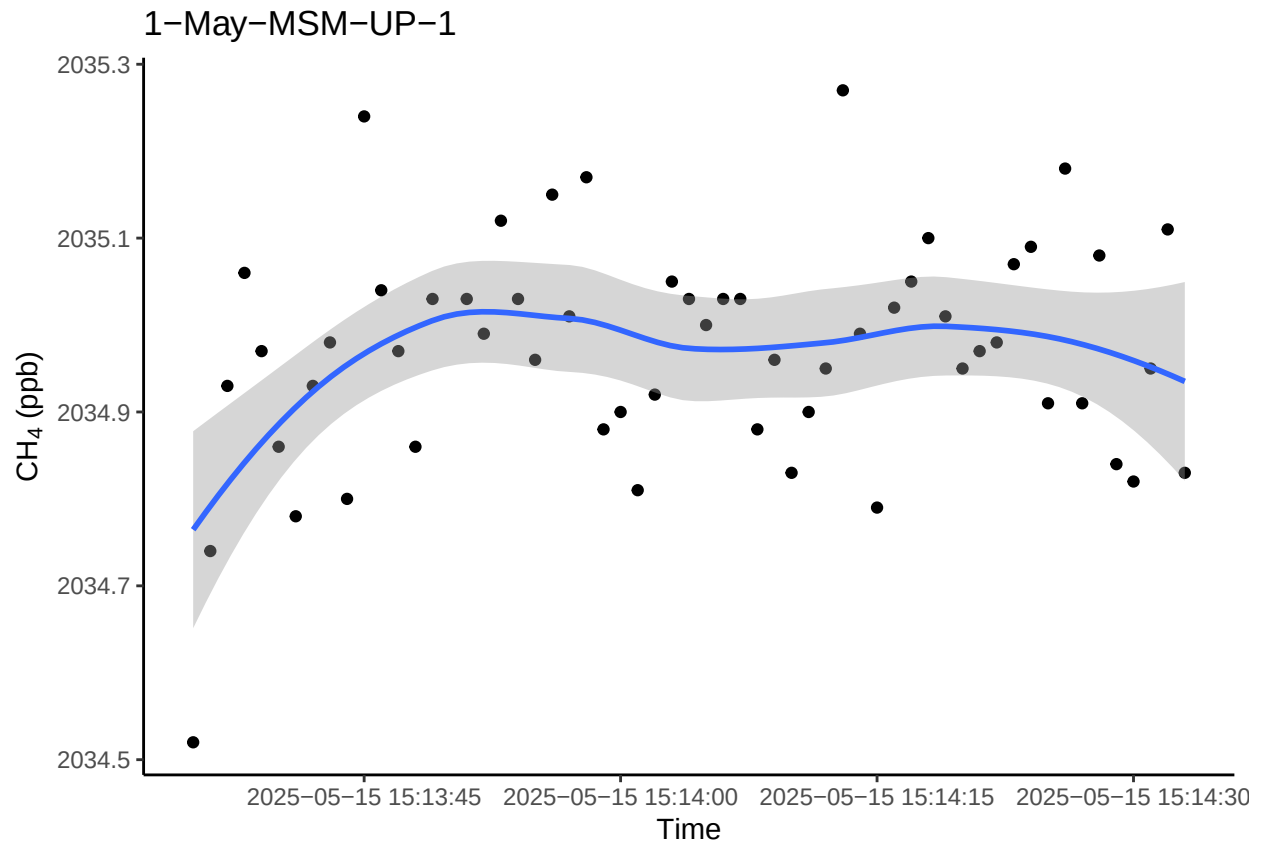


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

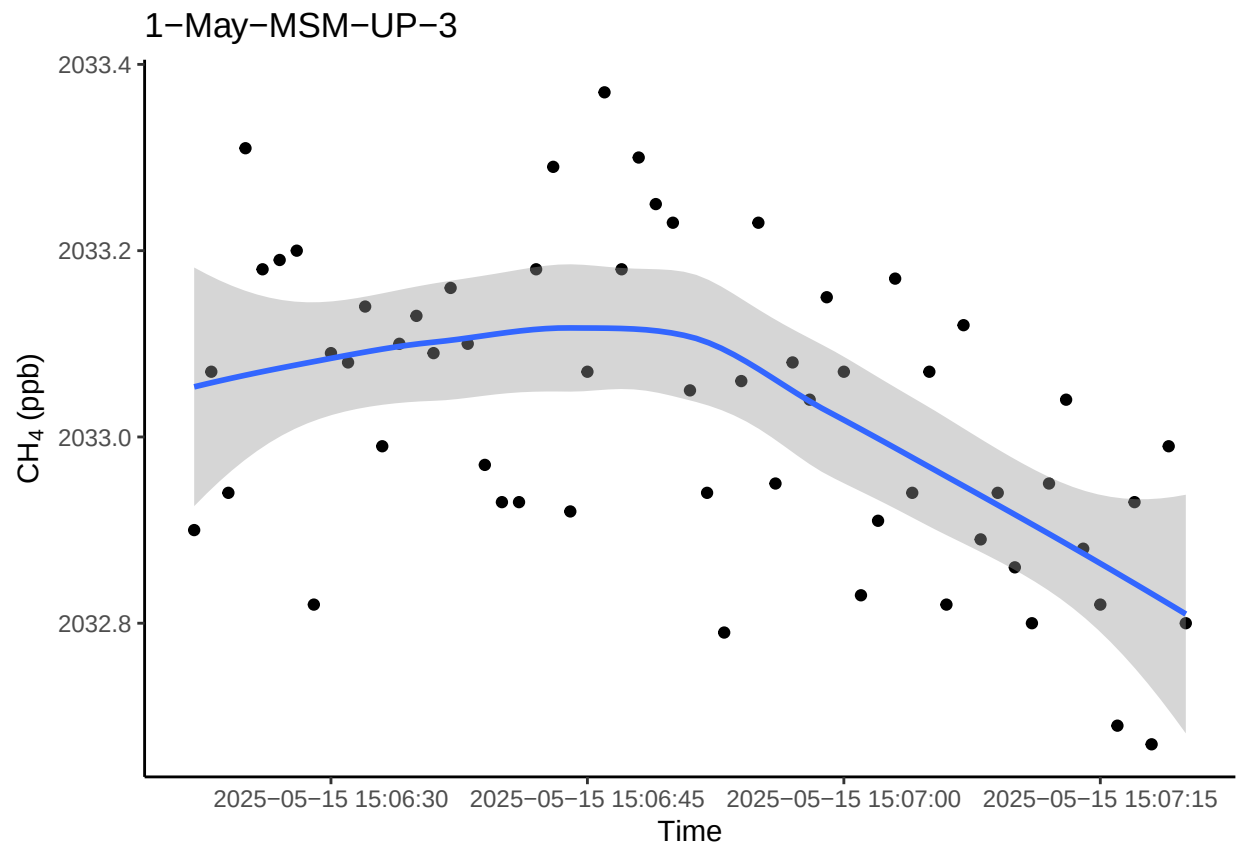


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

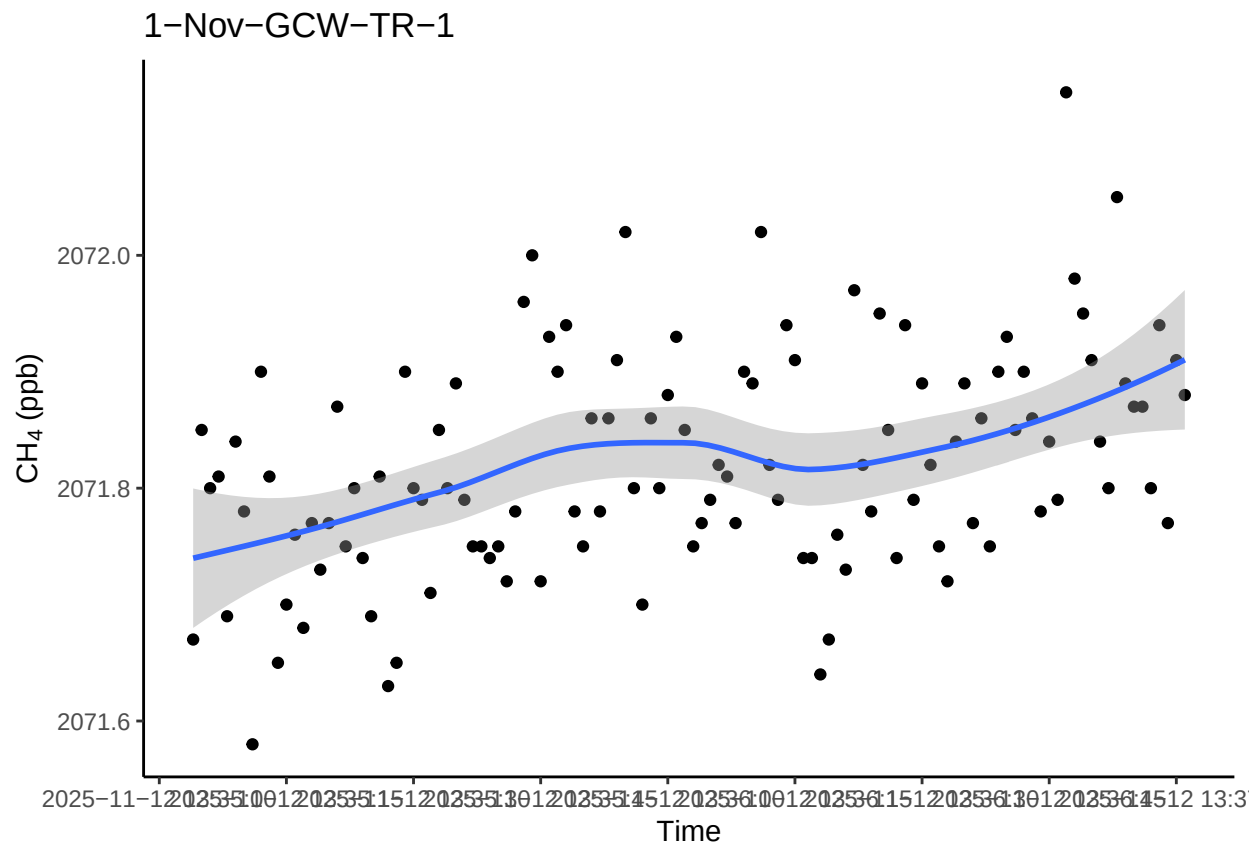
29



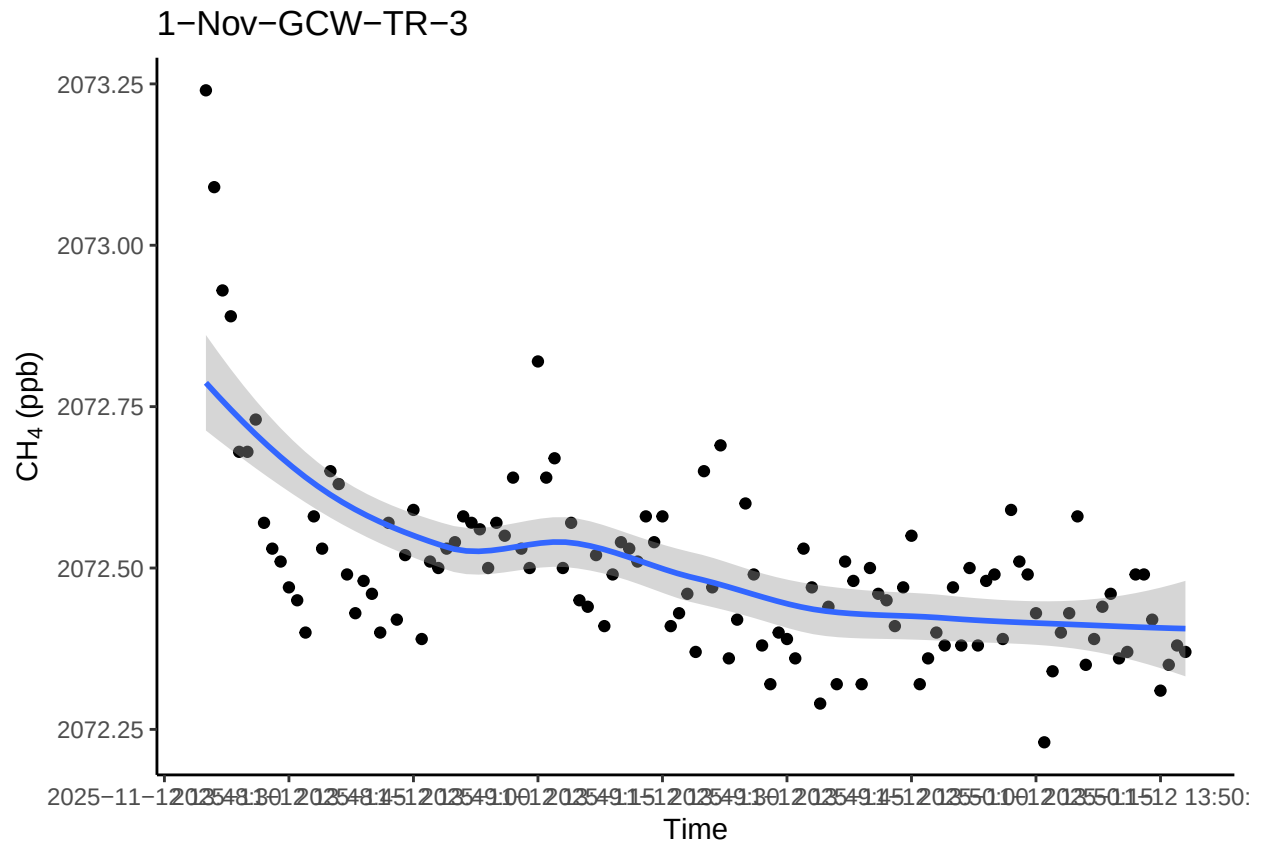
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



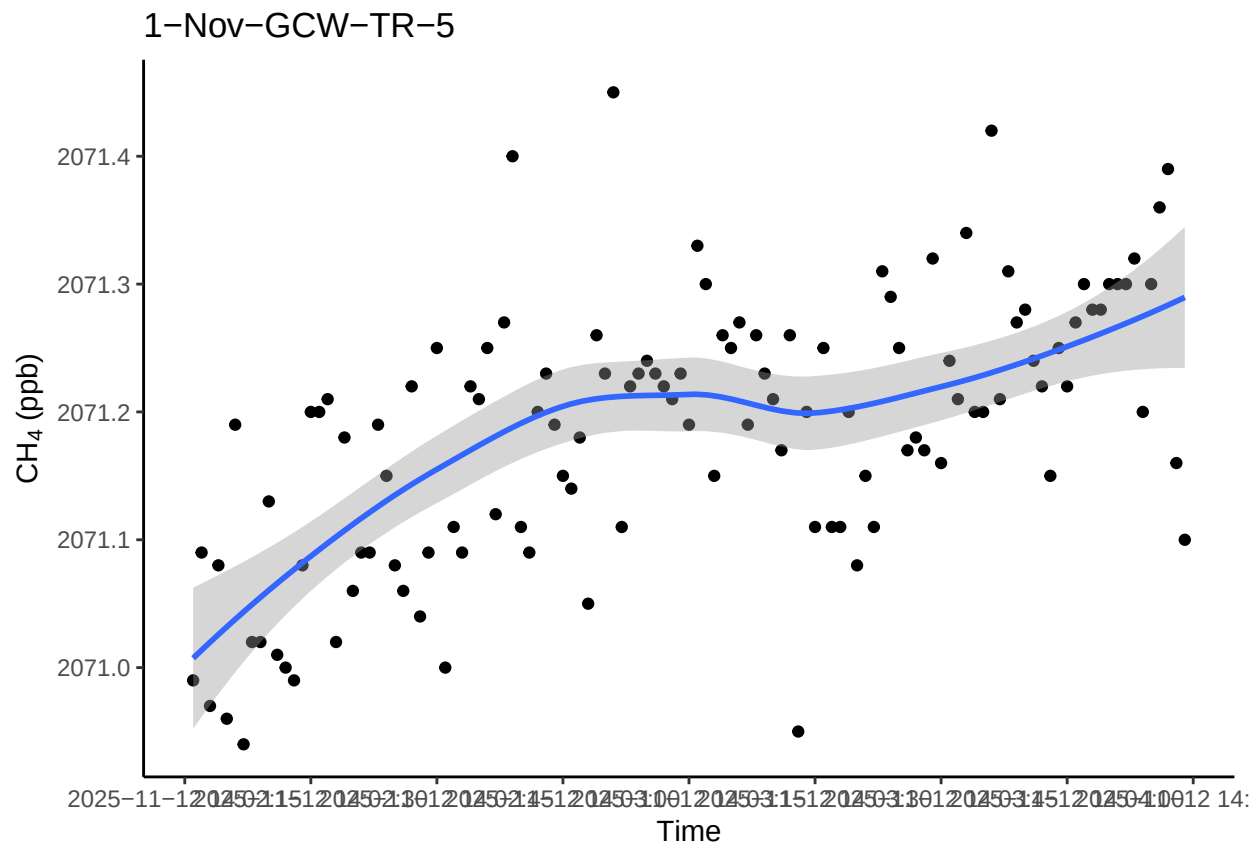
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



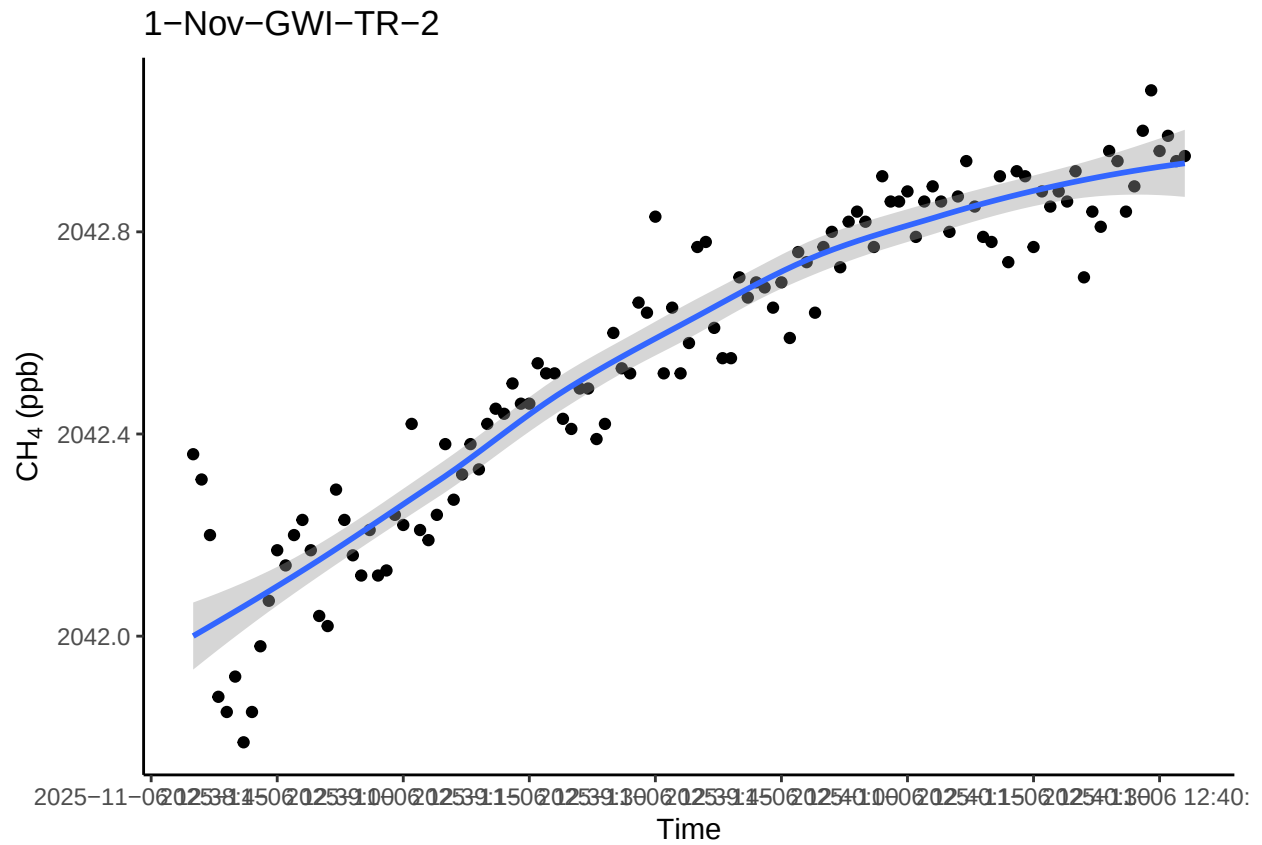
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

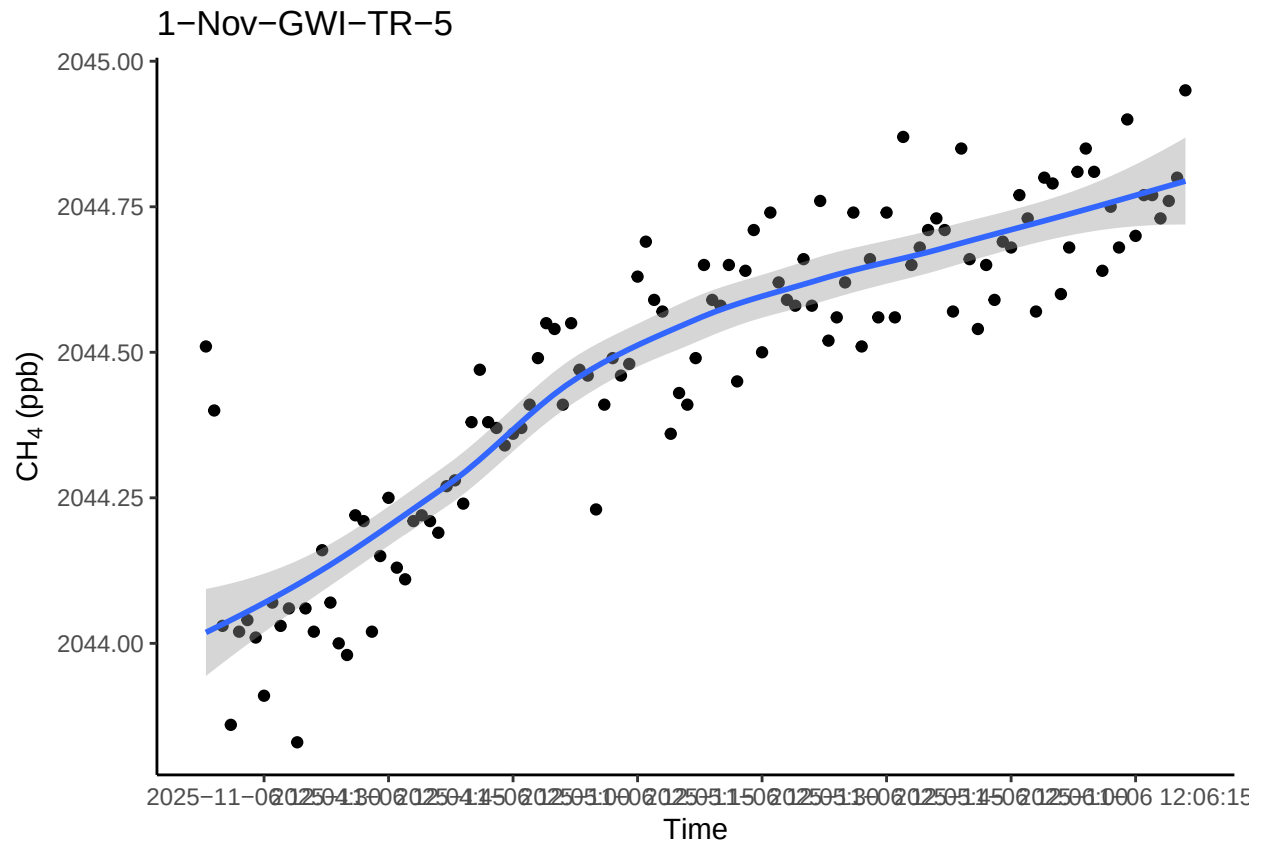
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



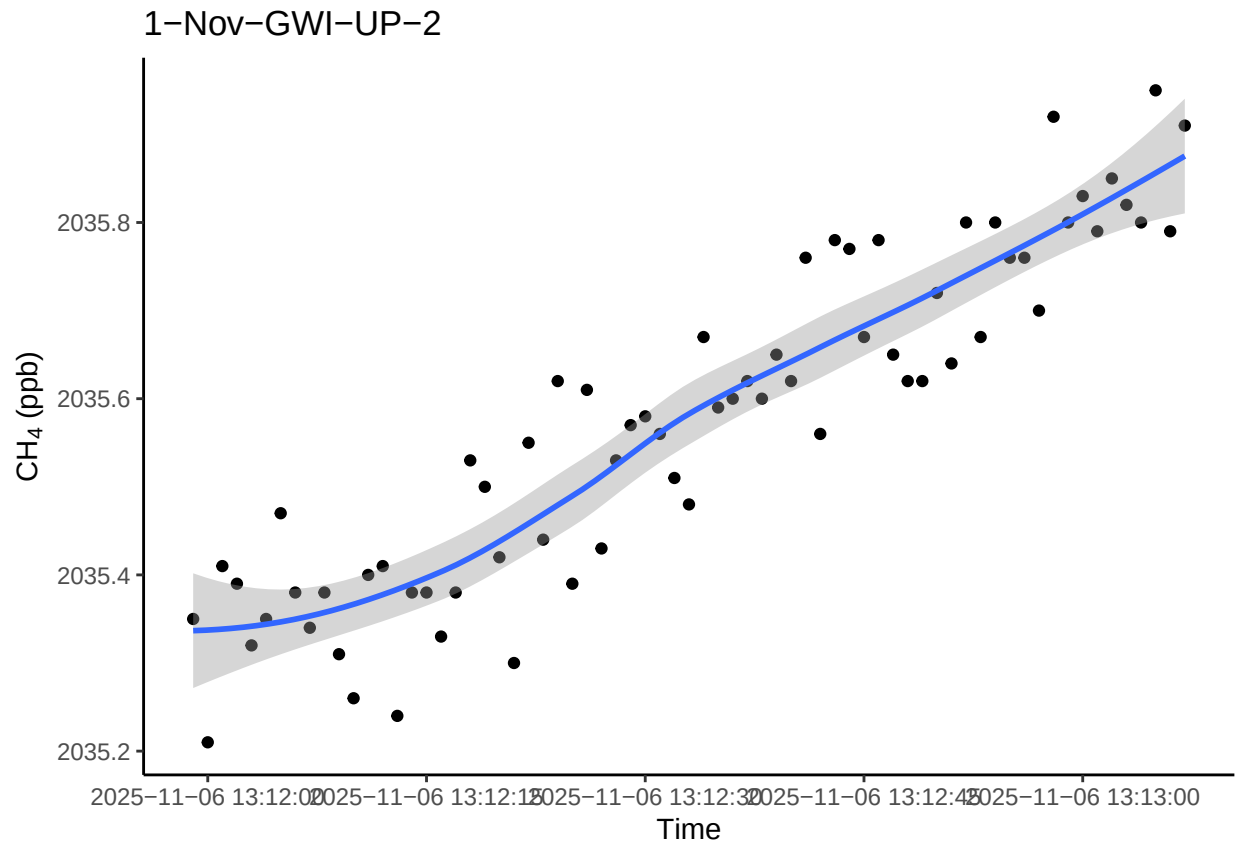
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



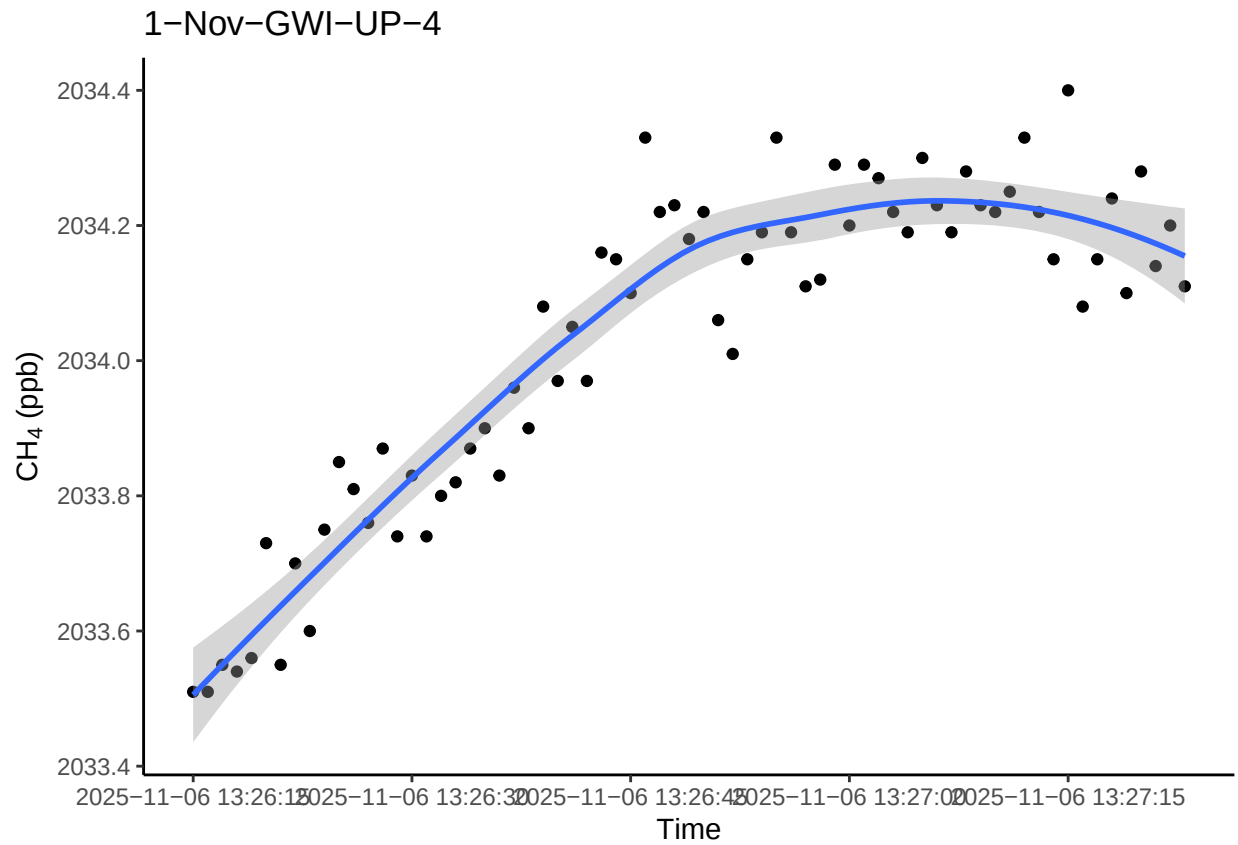
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



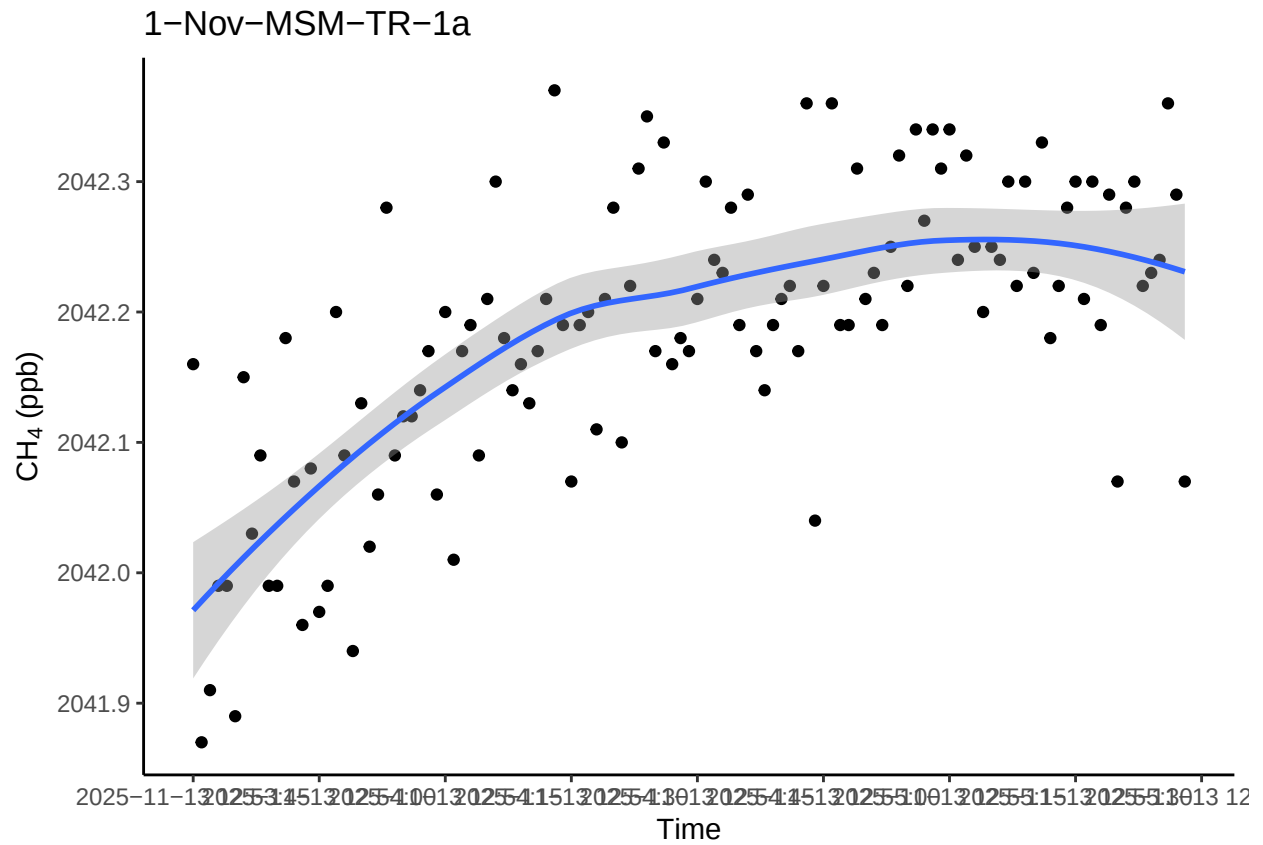
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



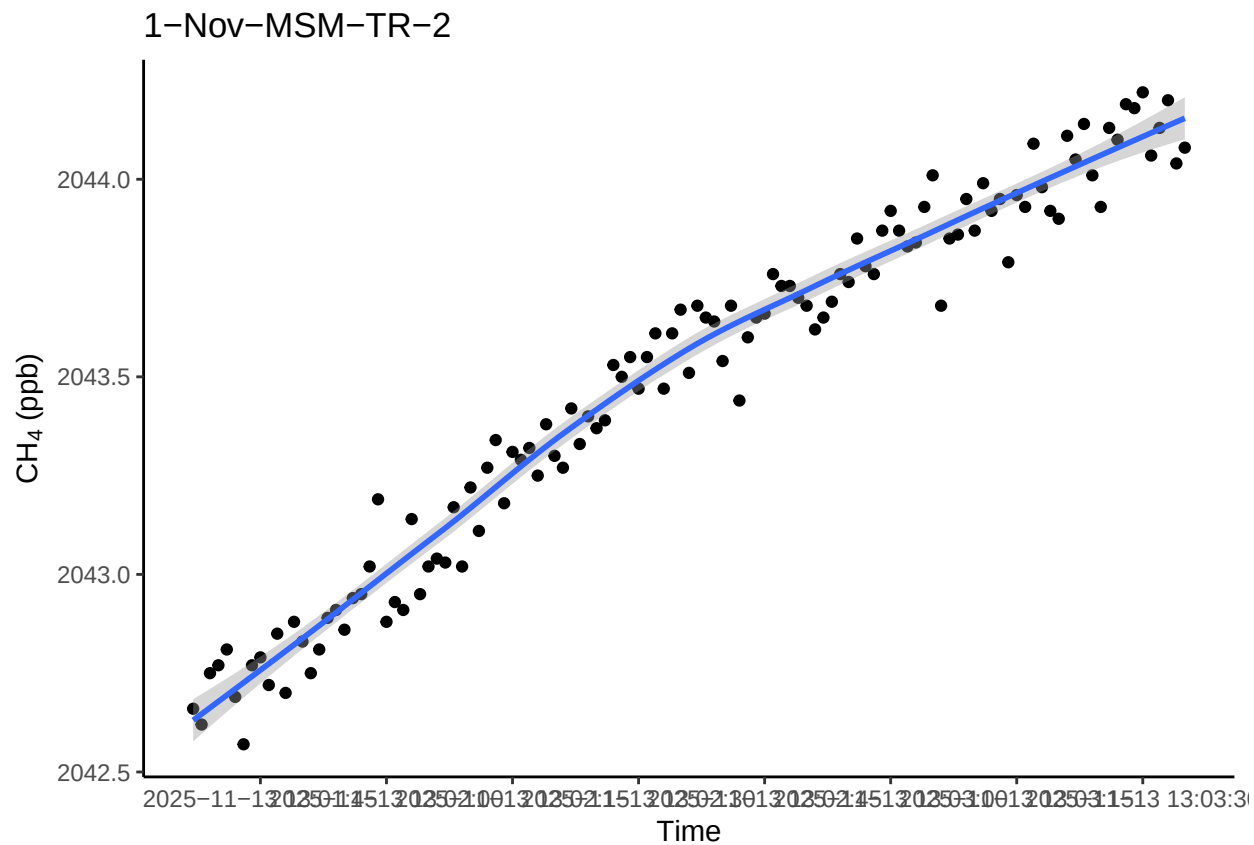
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



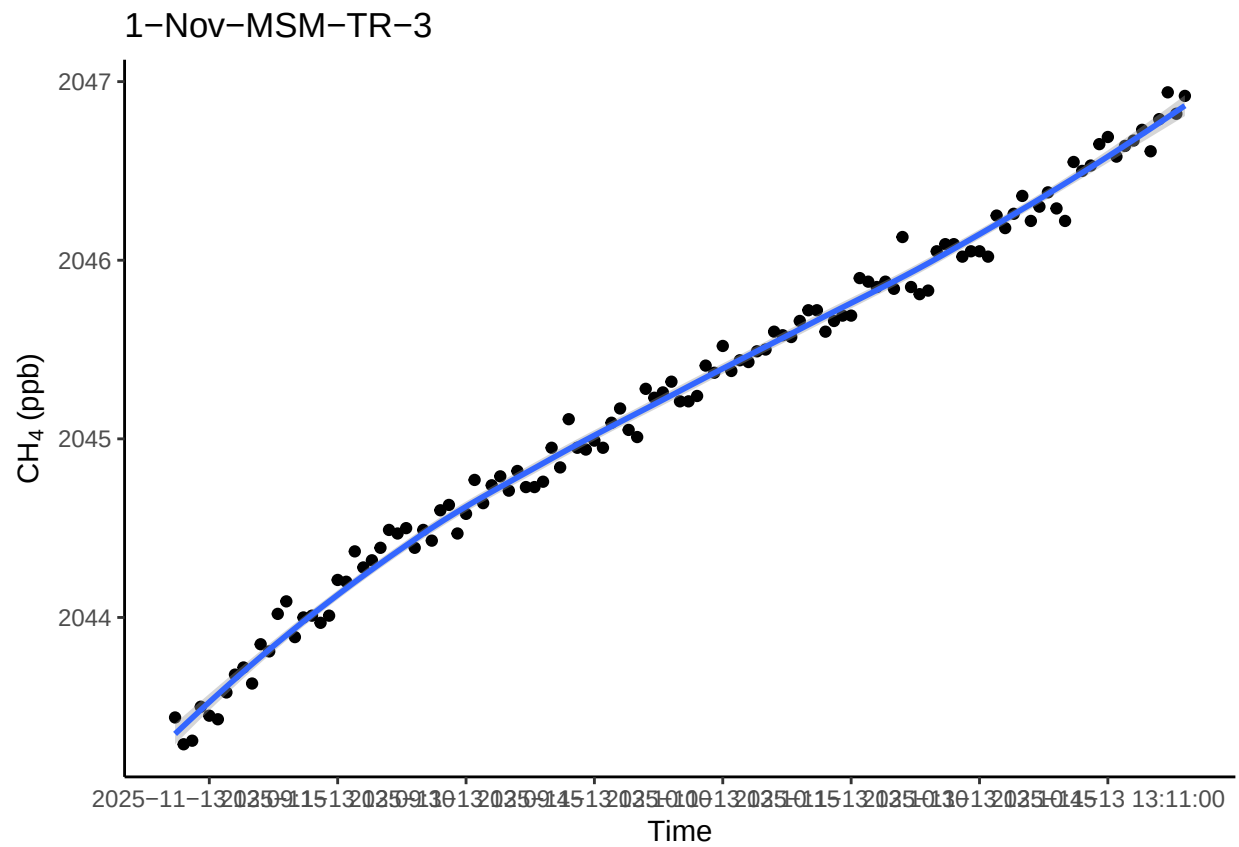
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



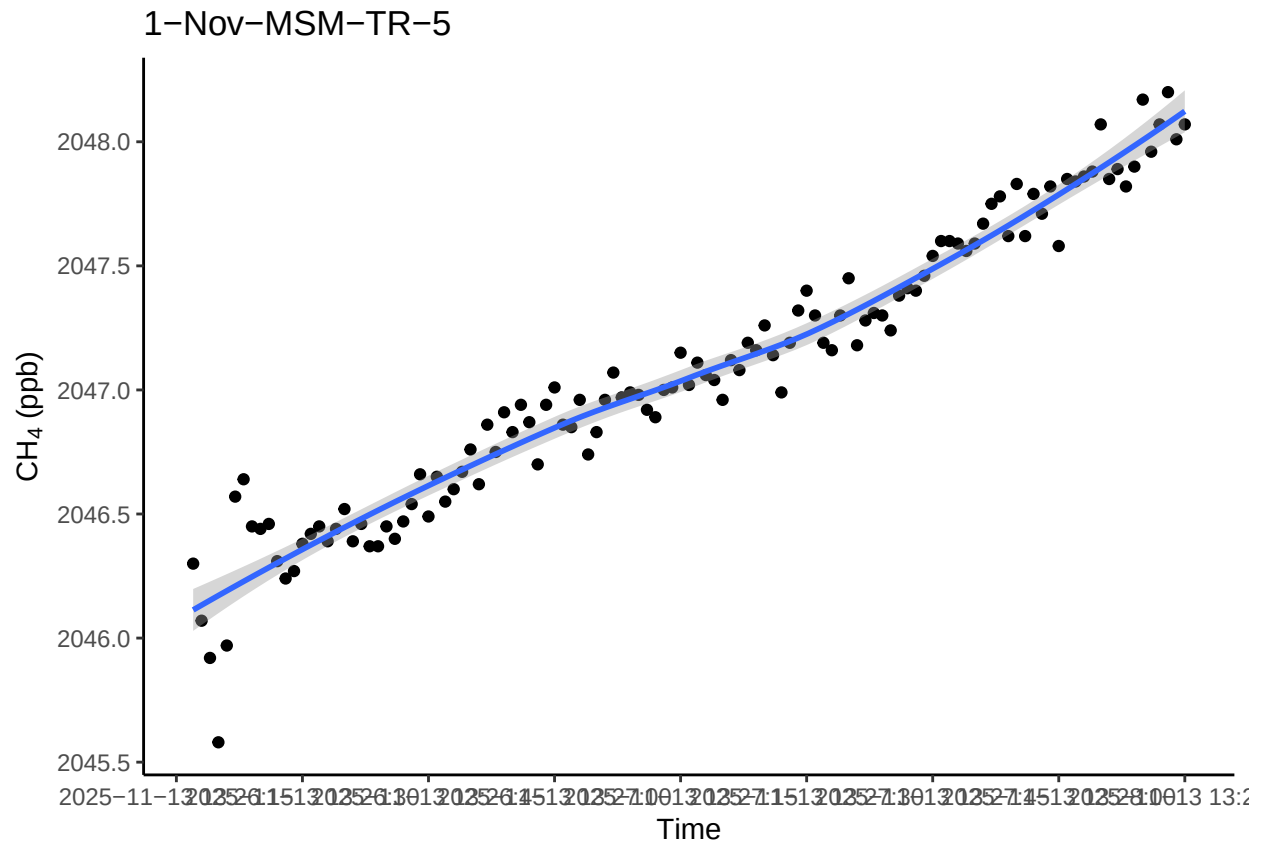
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



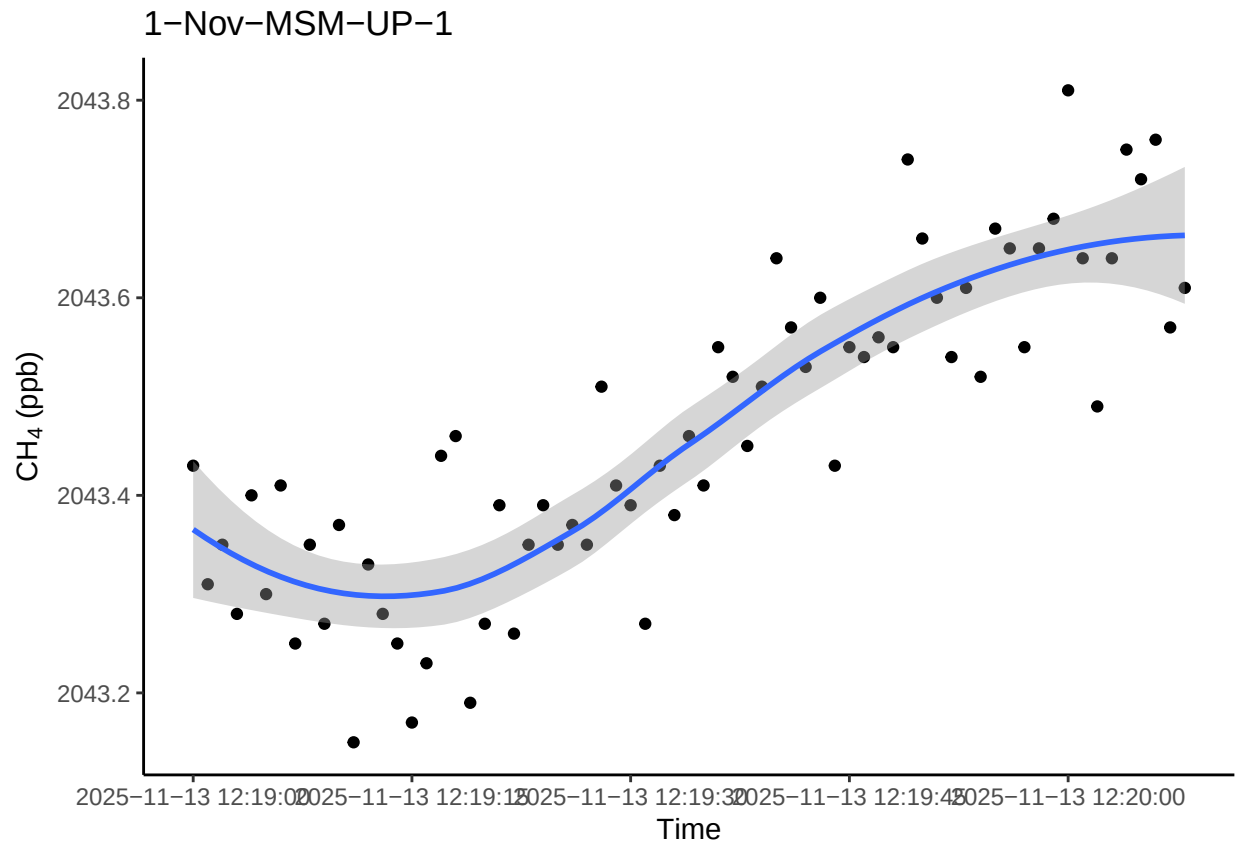
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

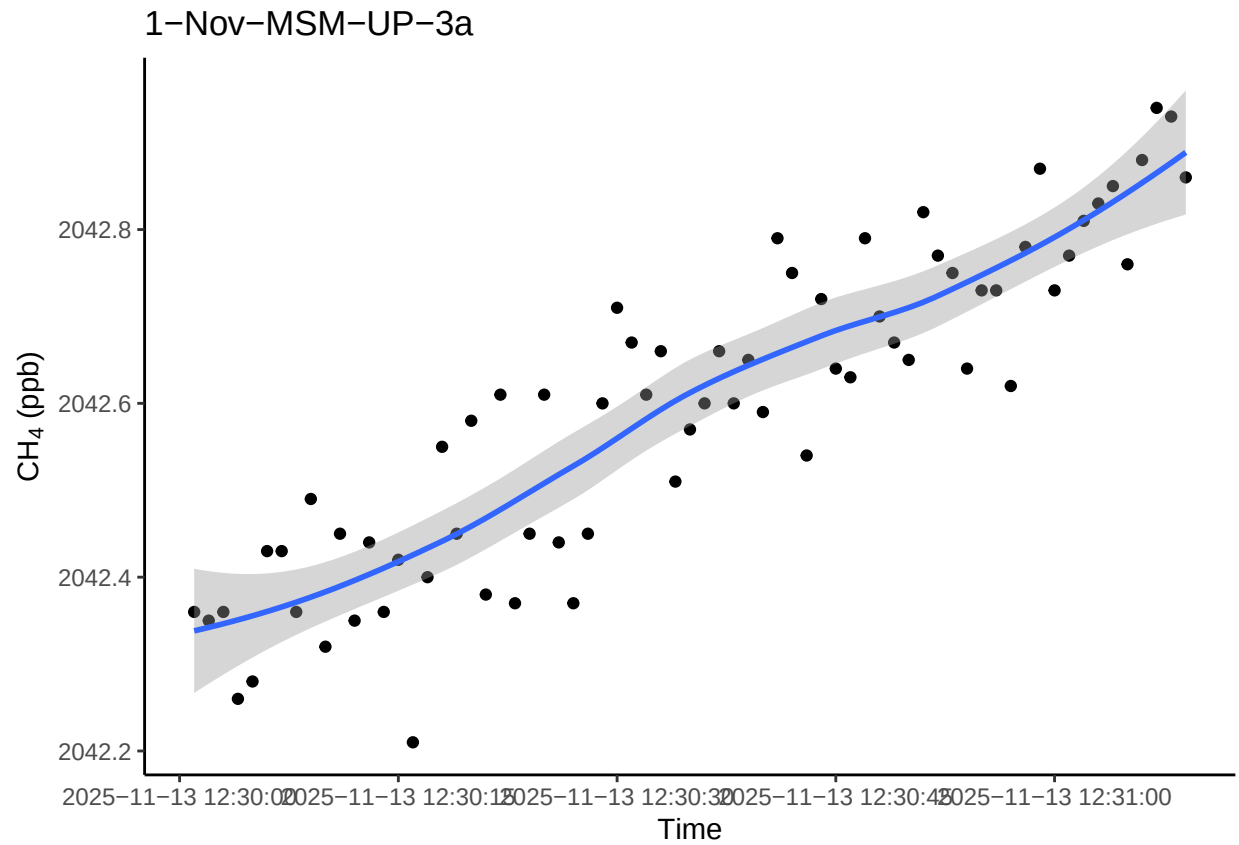
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

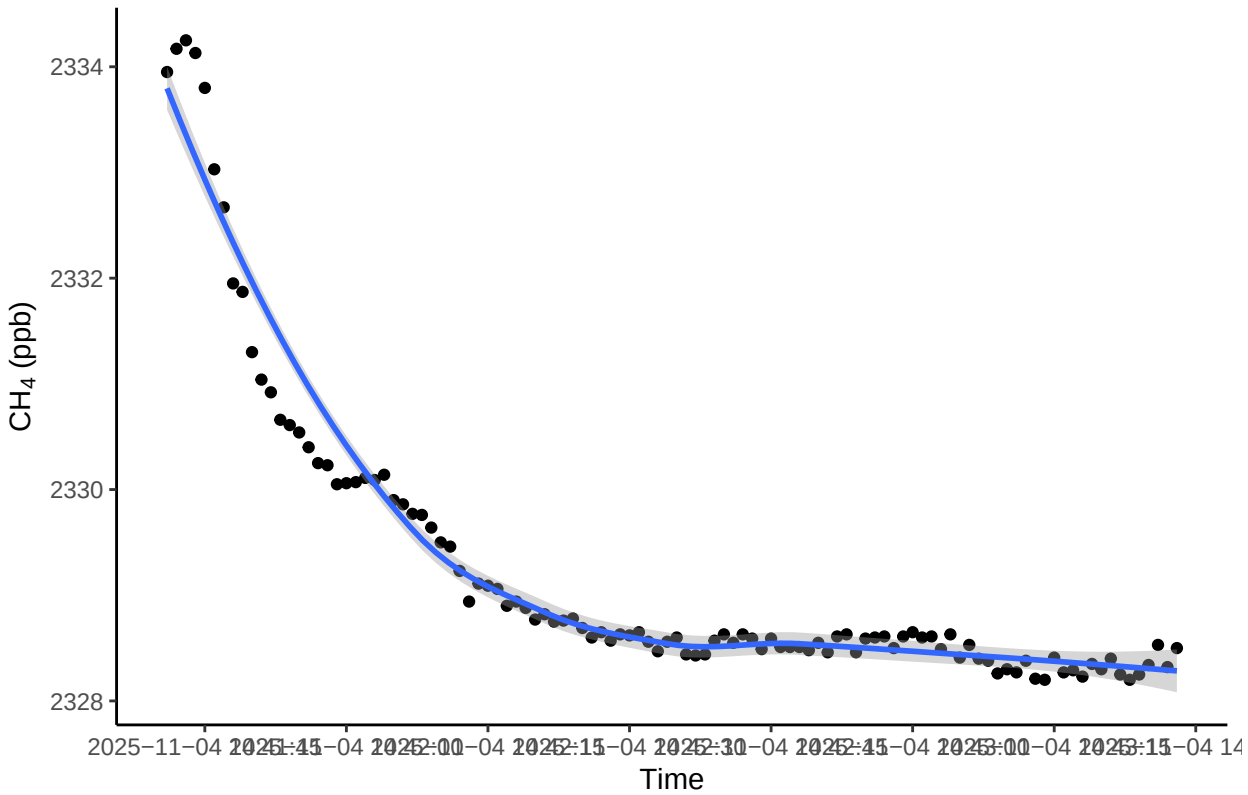


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

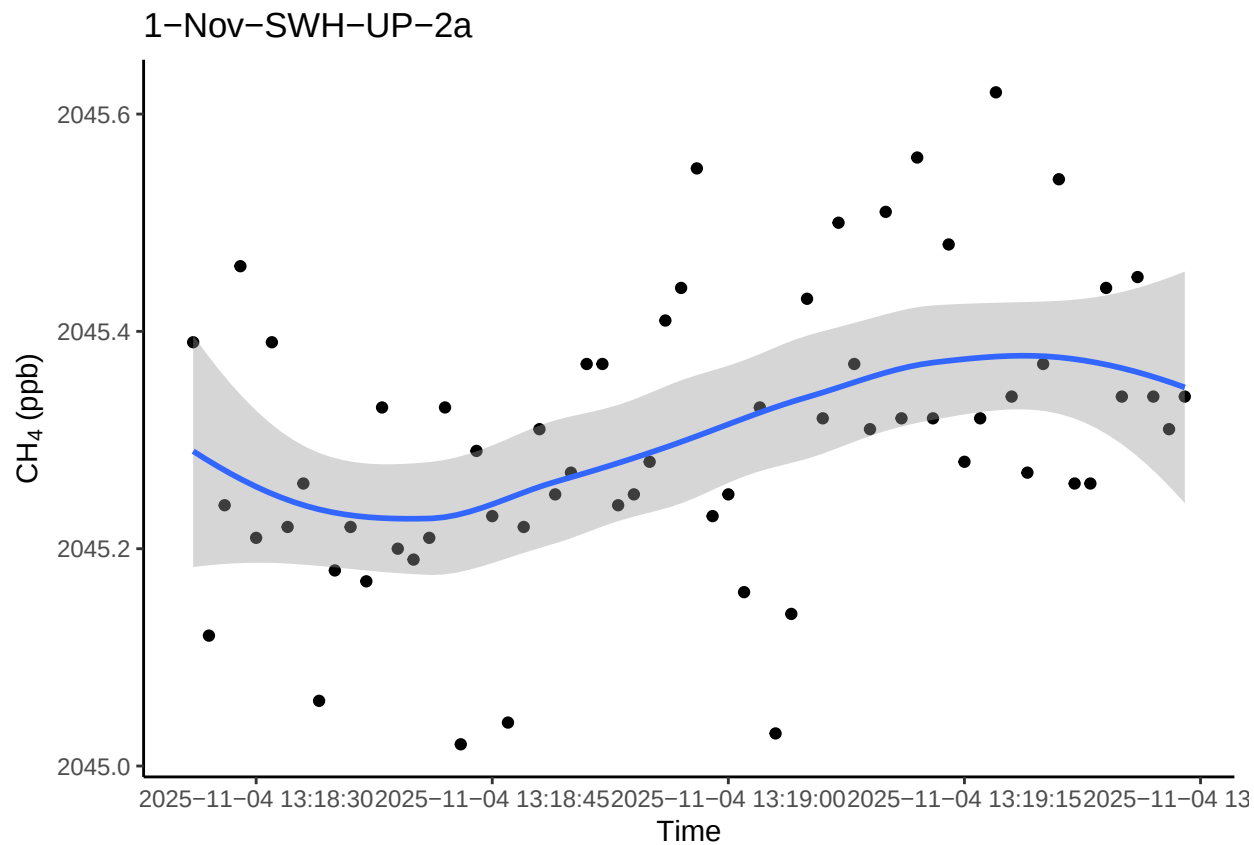


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

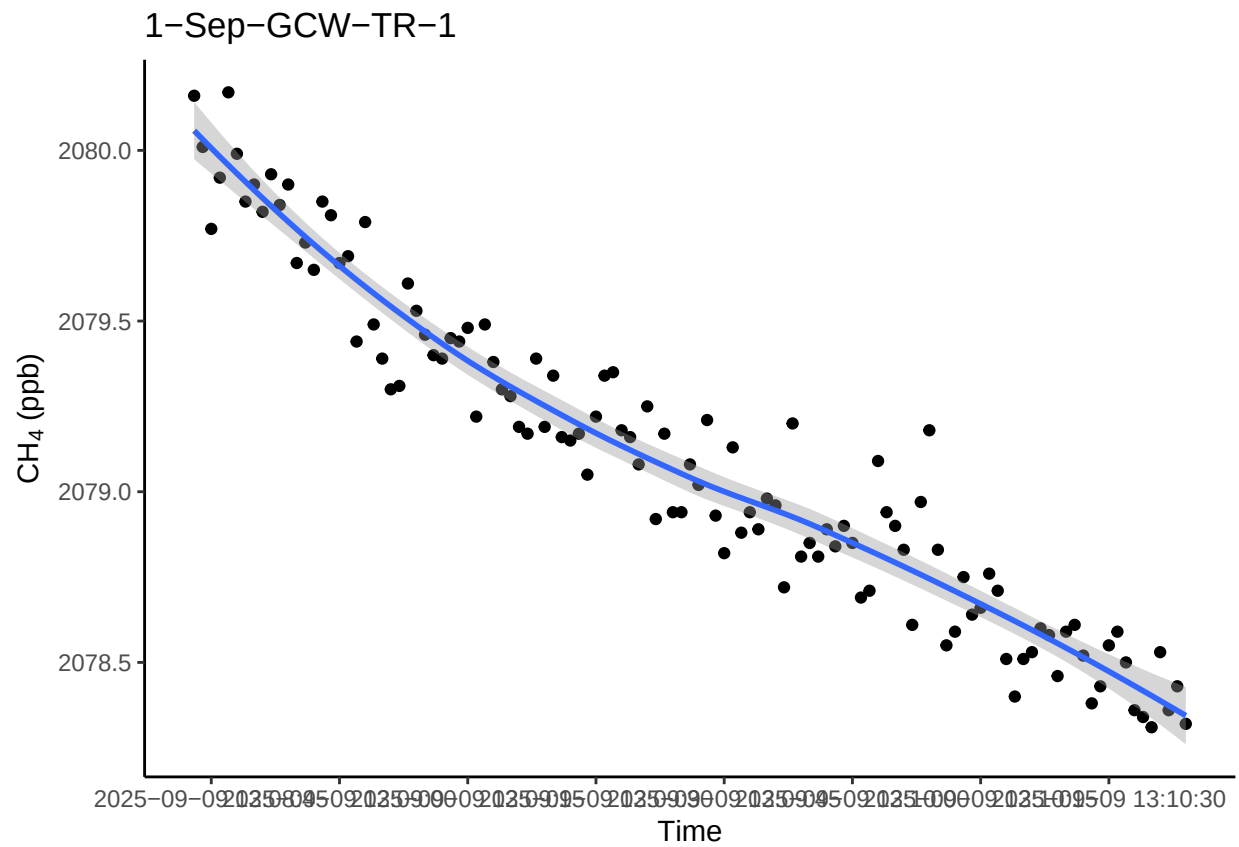
1-Nov-SWH-SWAMP-3b



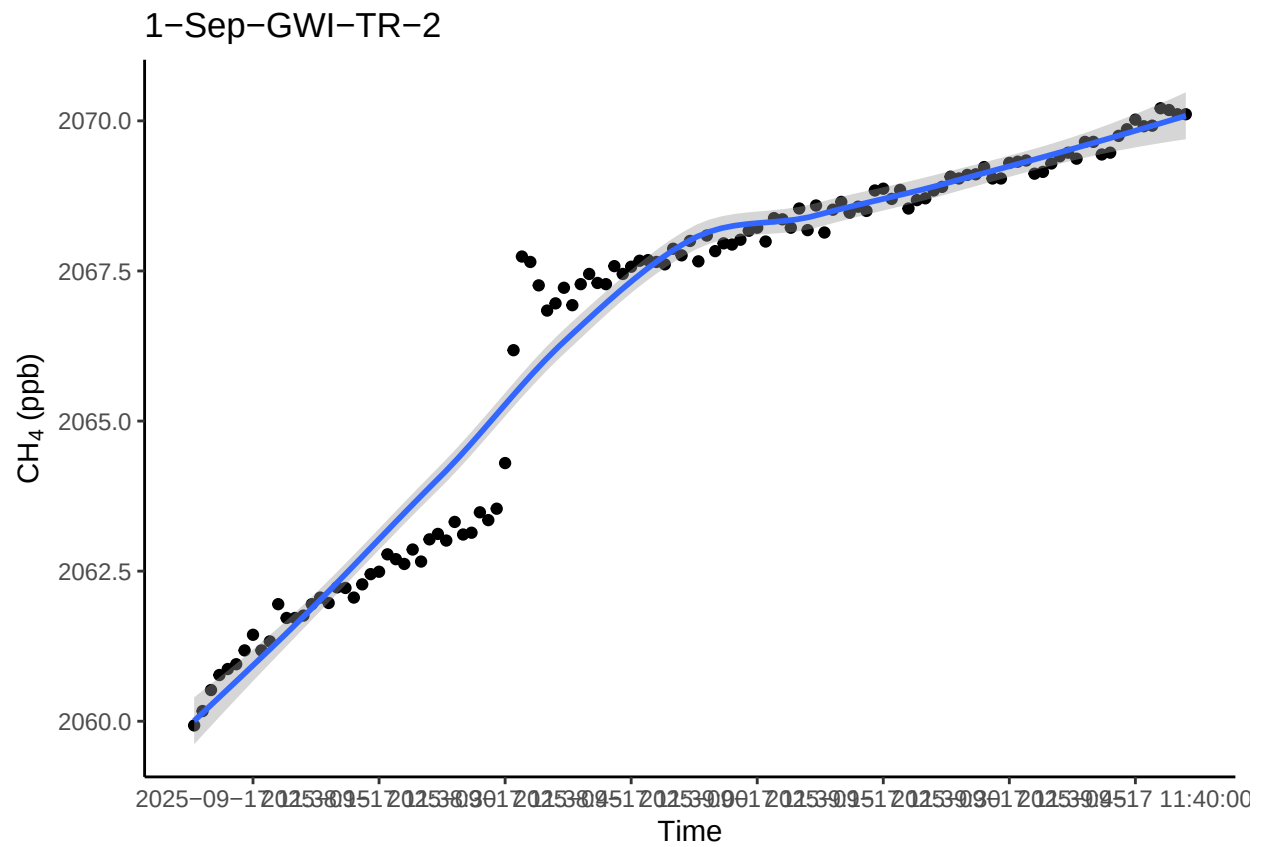
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



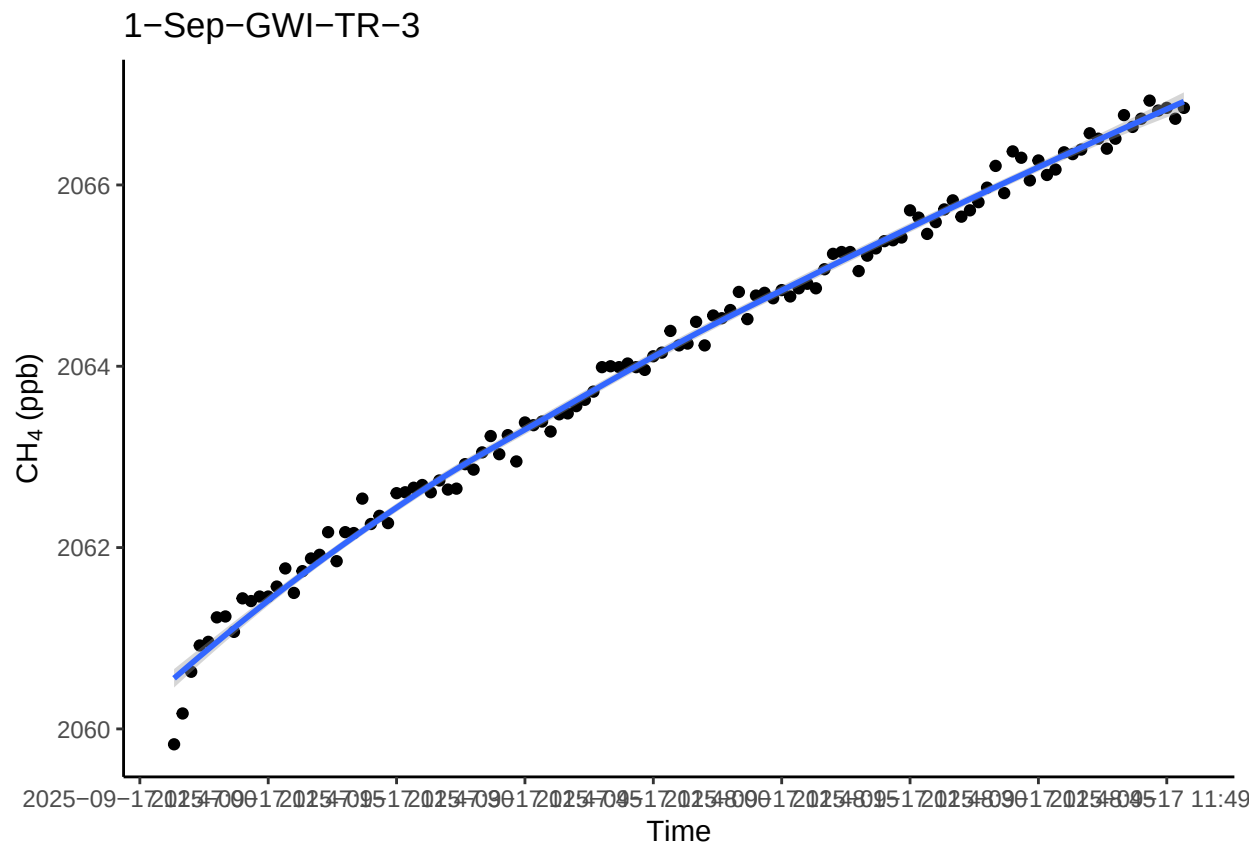
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



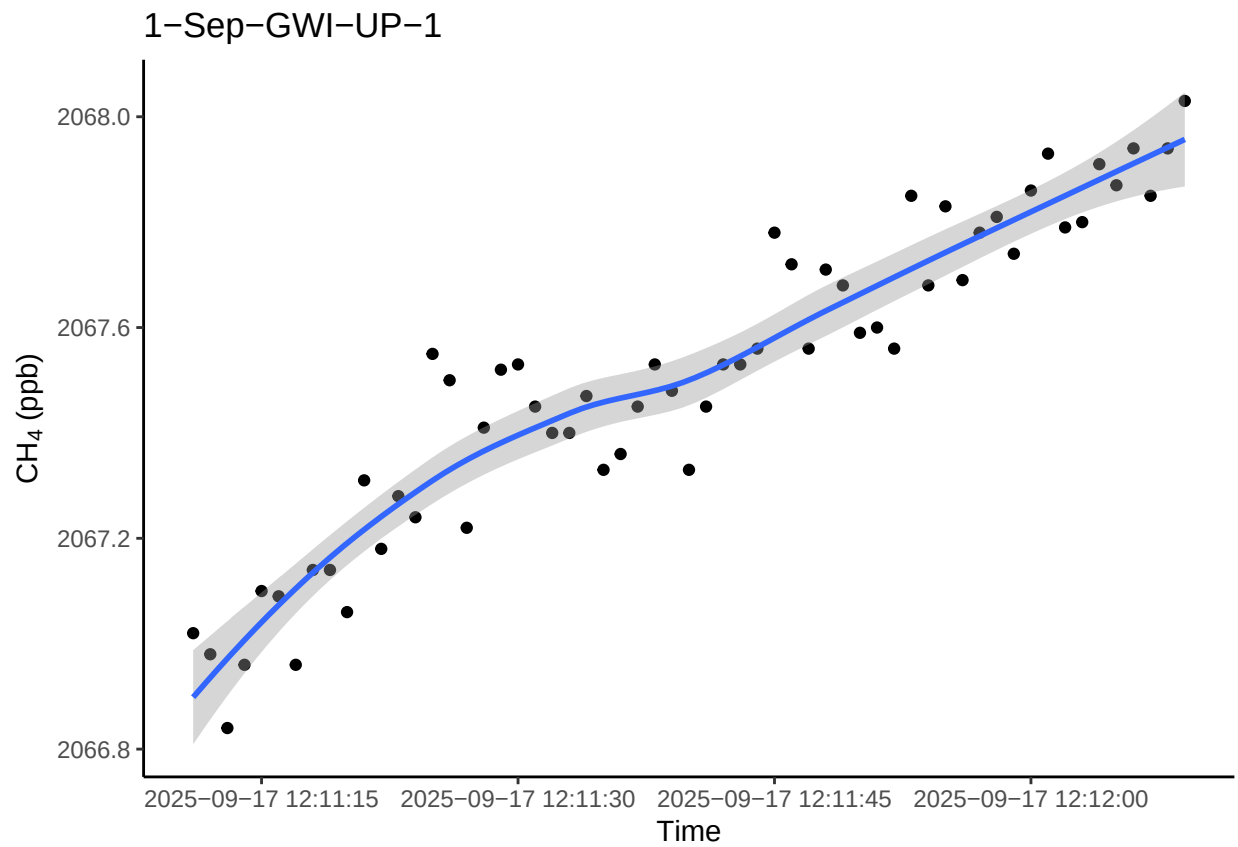
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



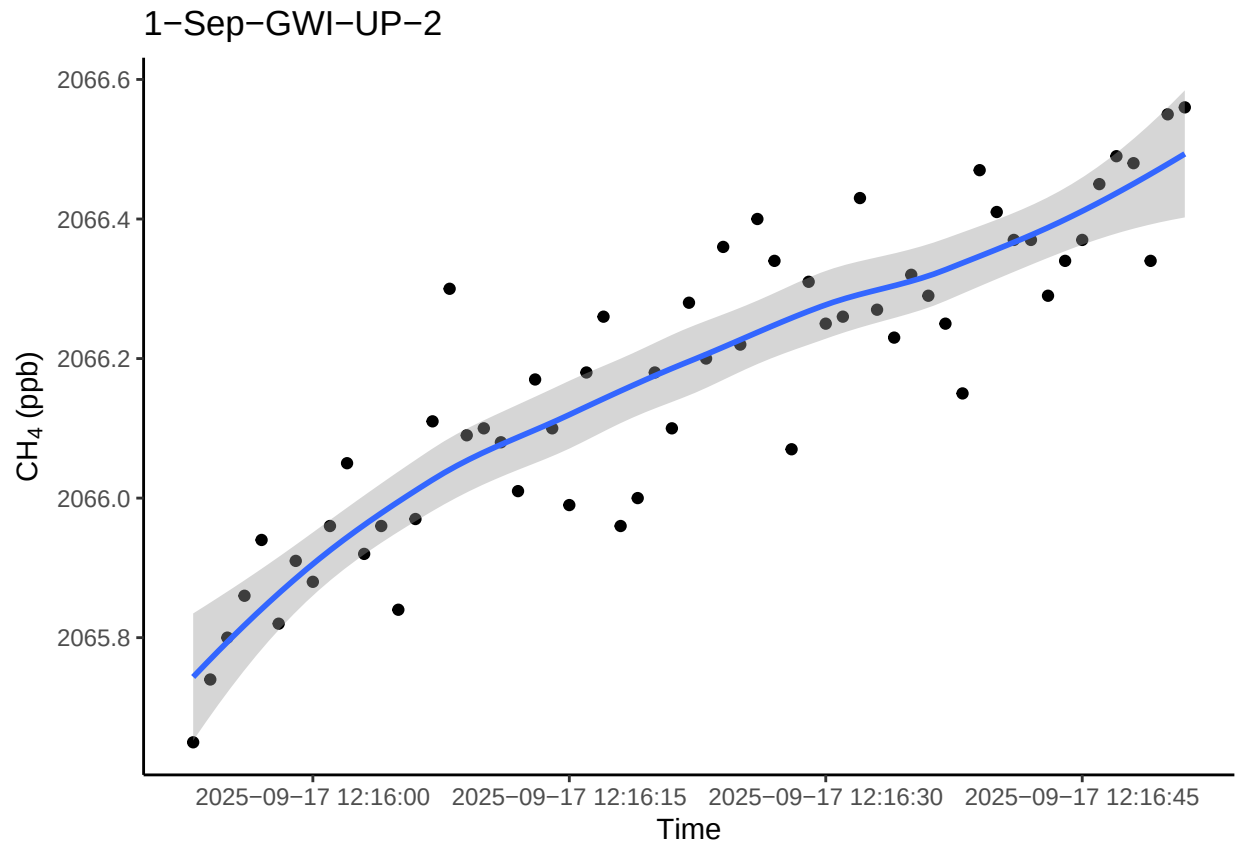
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

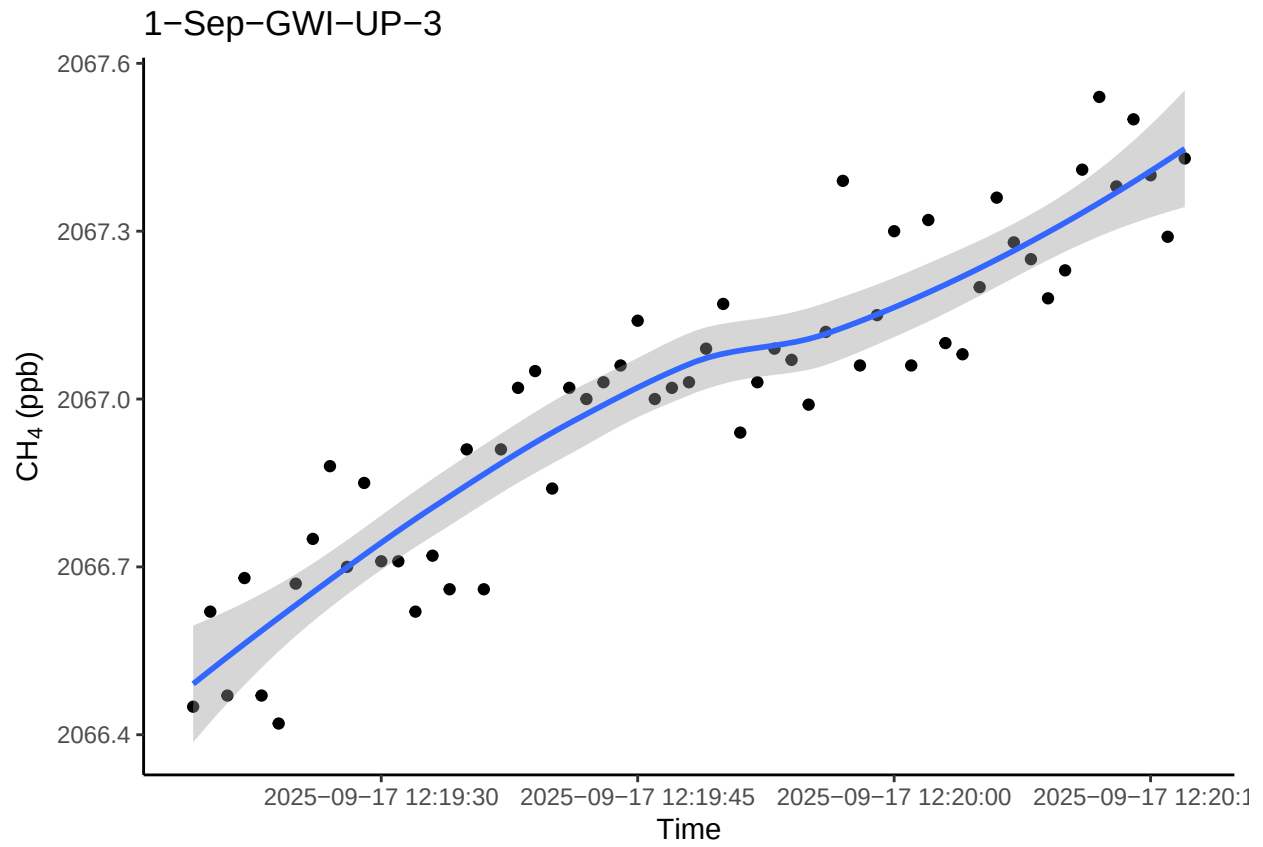
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



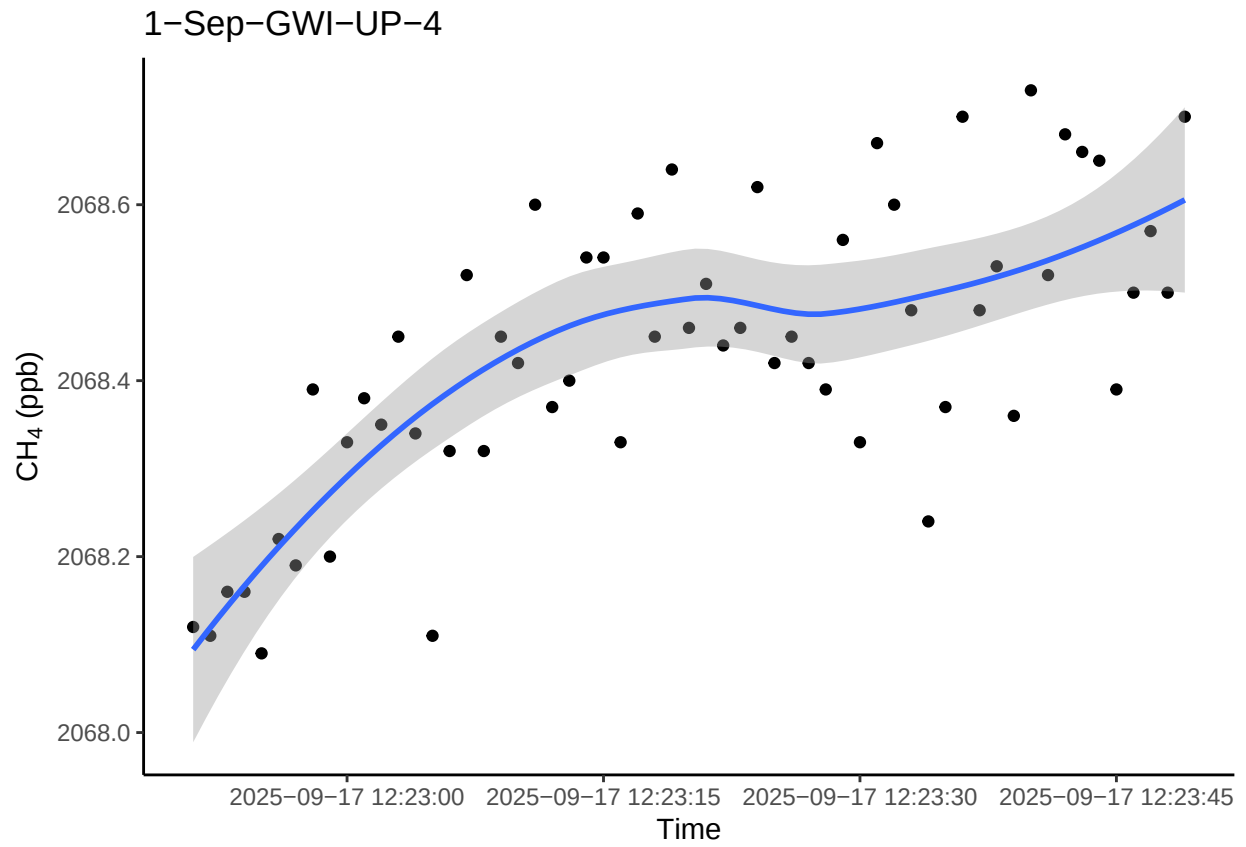
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



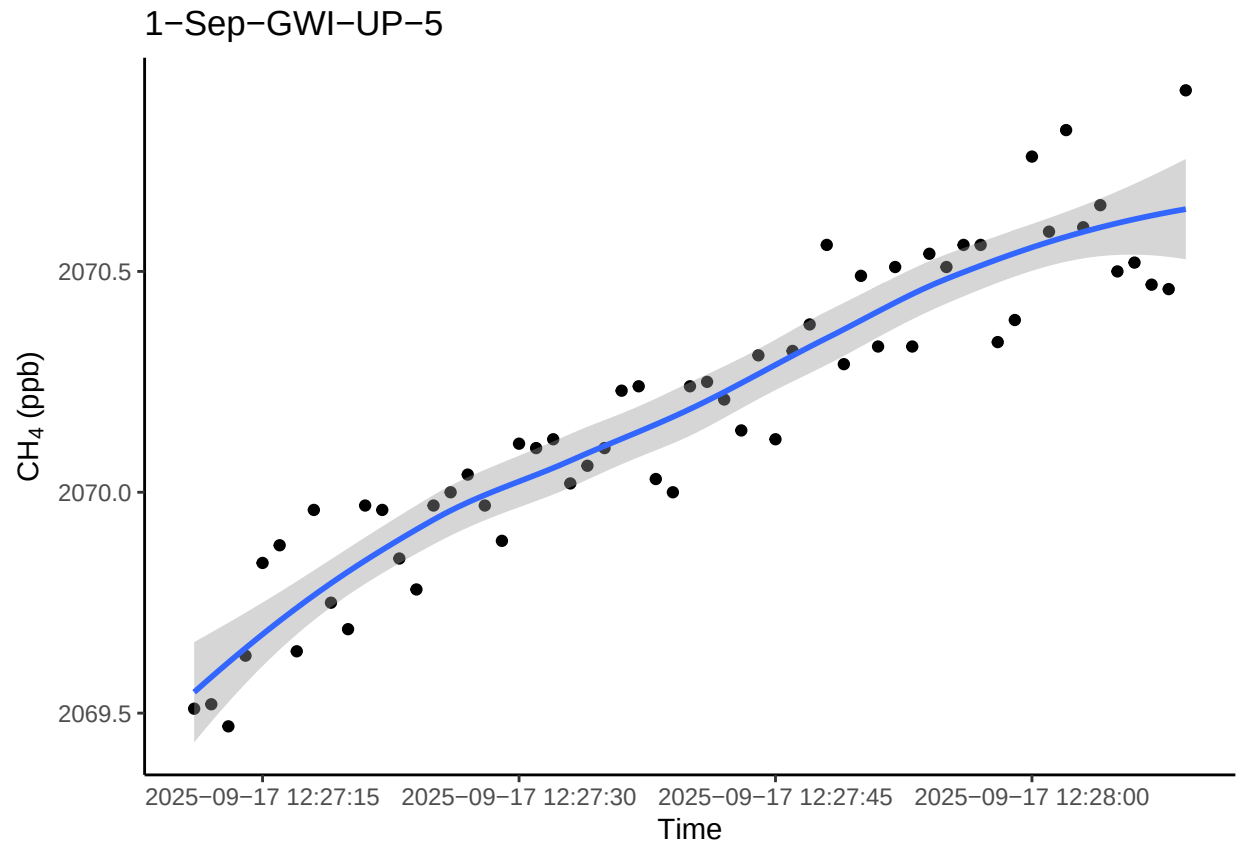
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



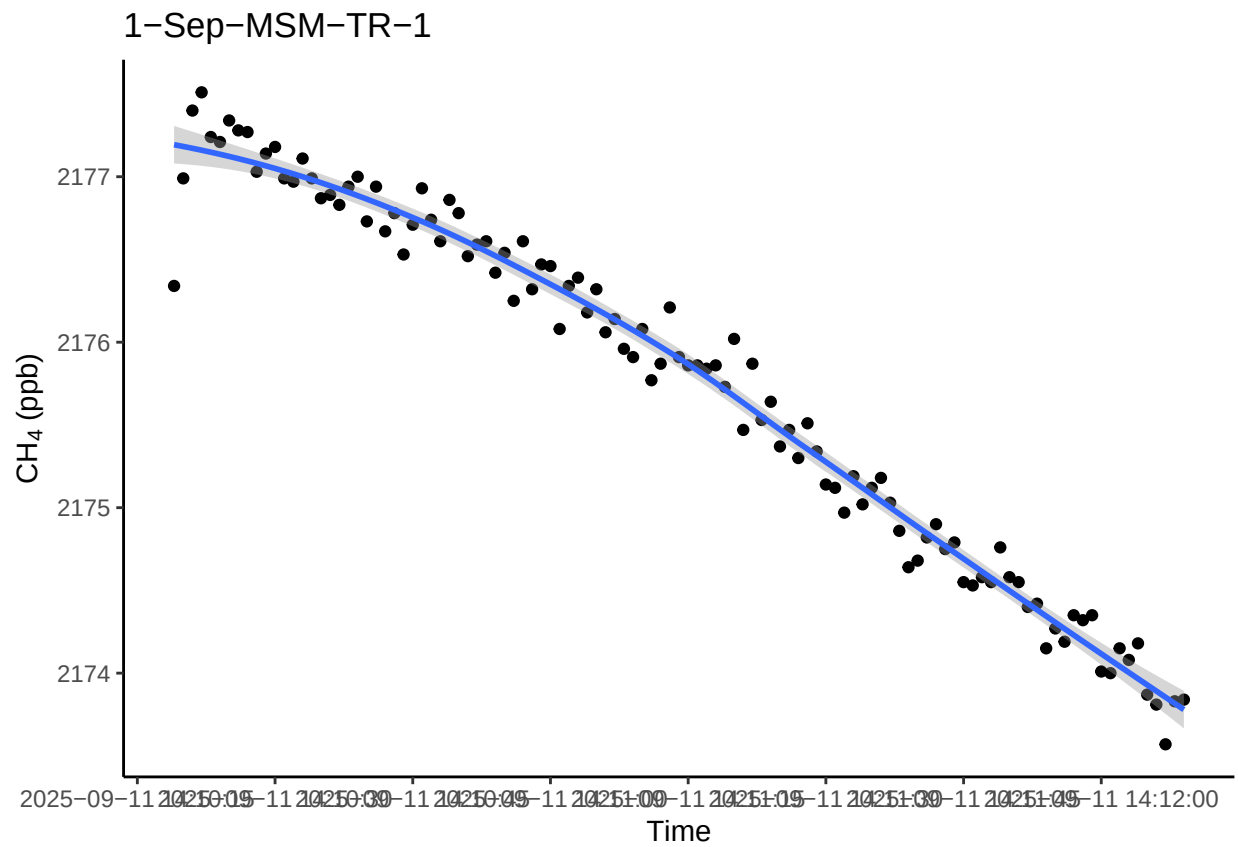
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



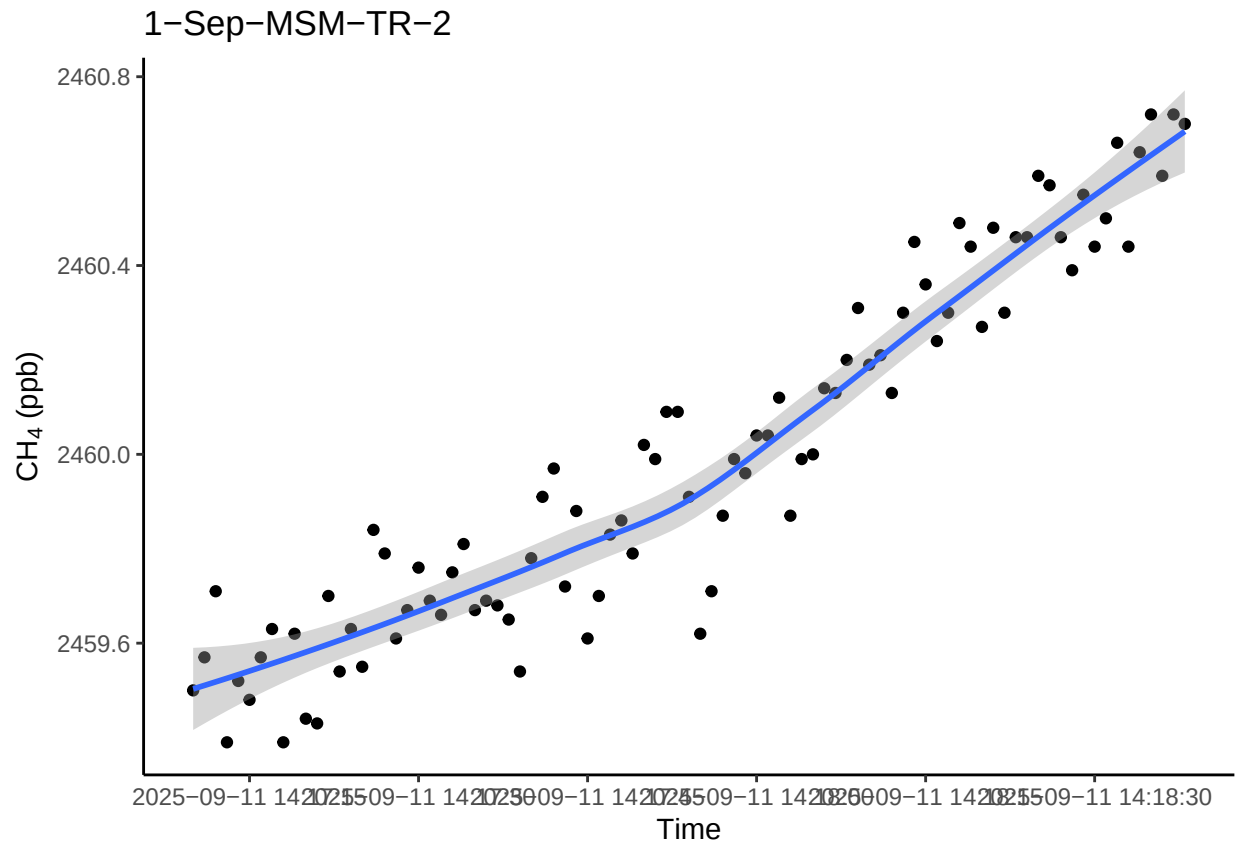
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



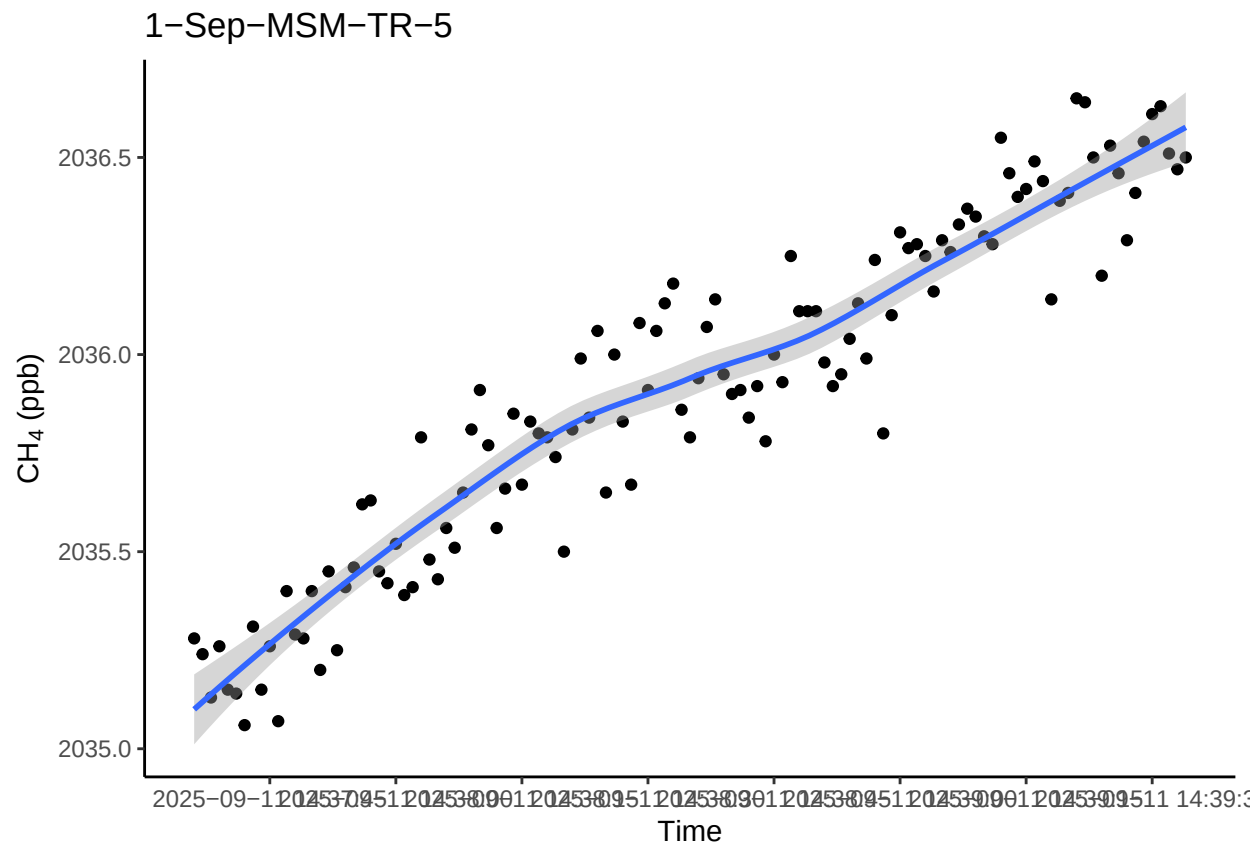
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



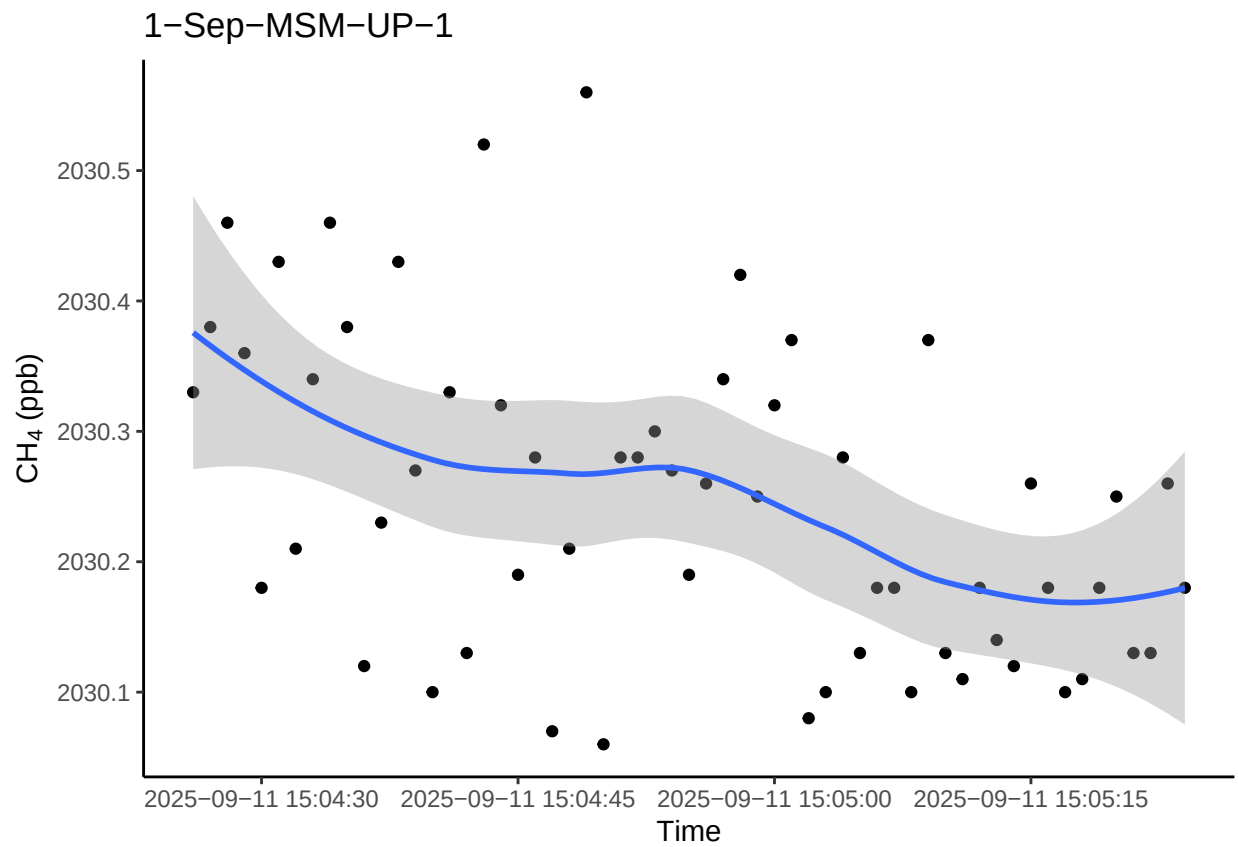
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



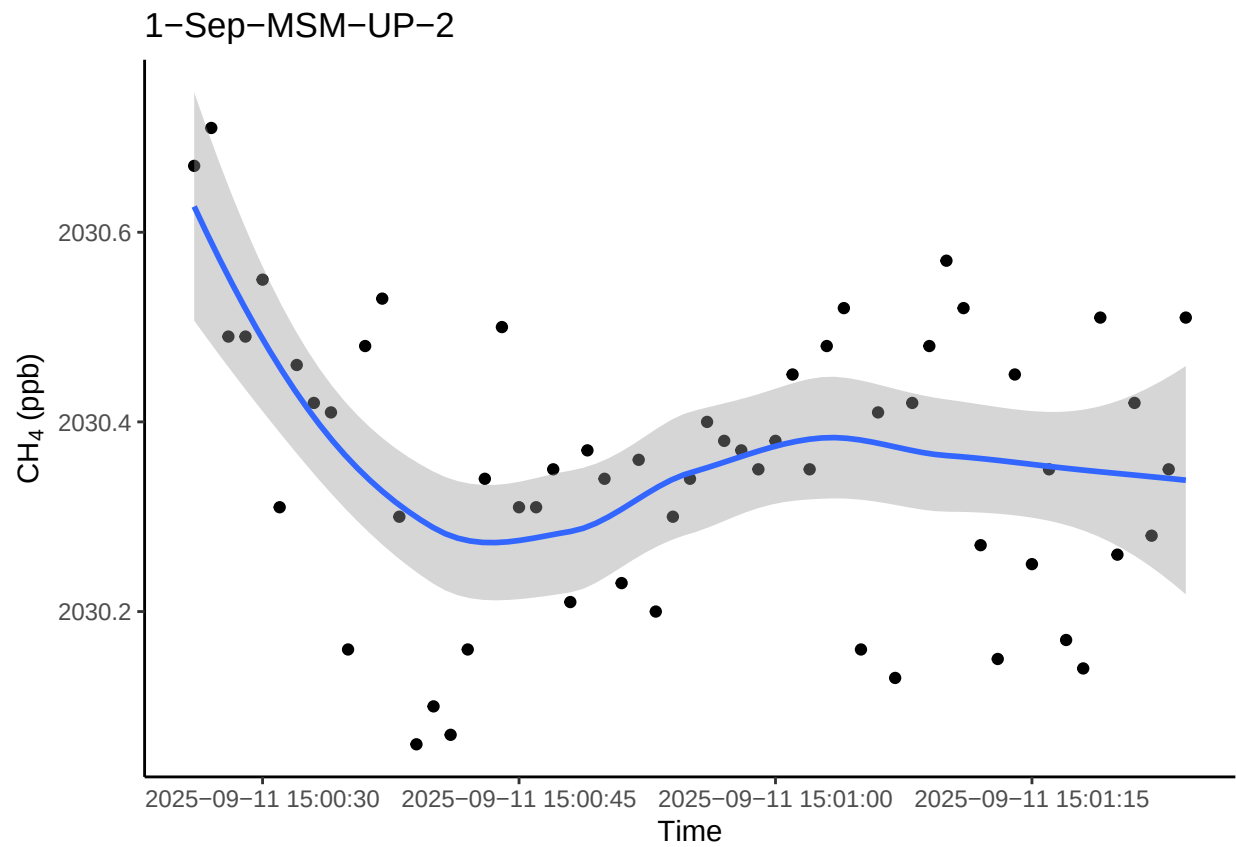
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

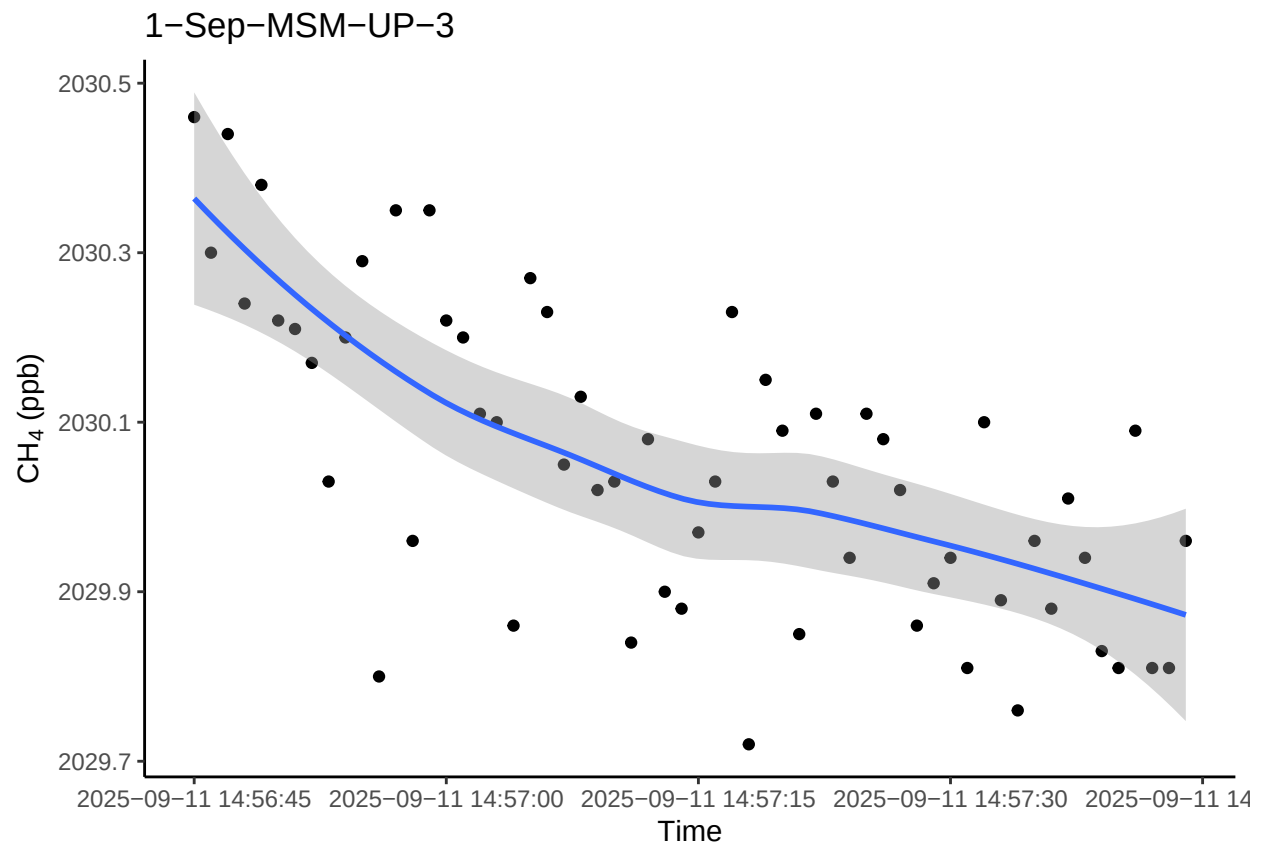
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



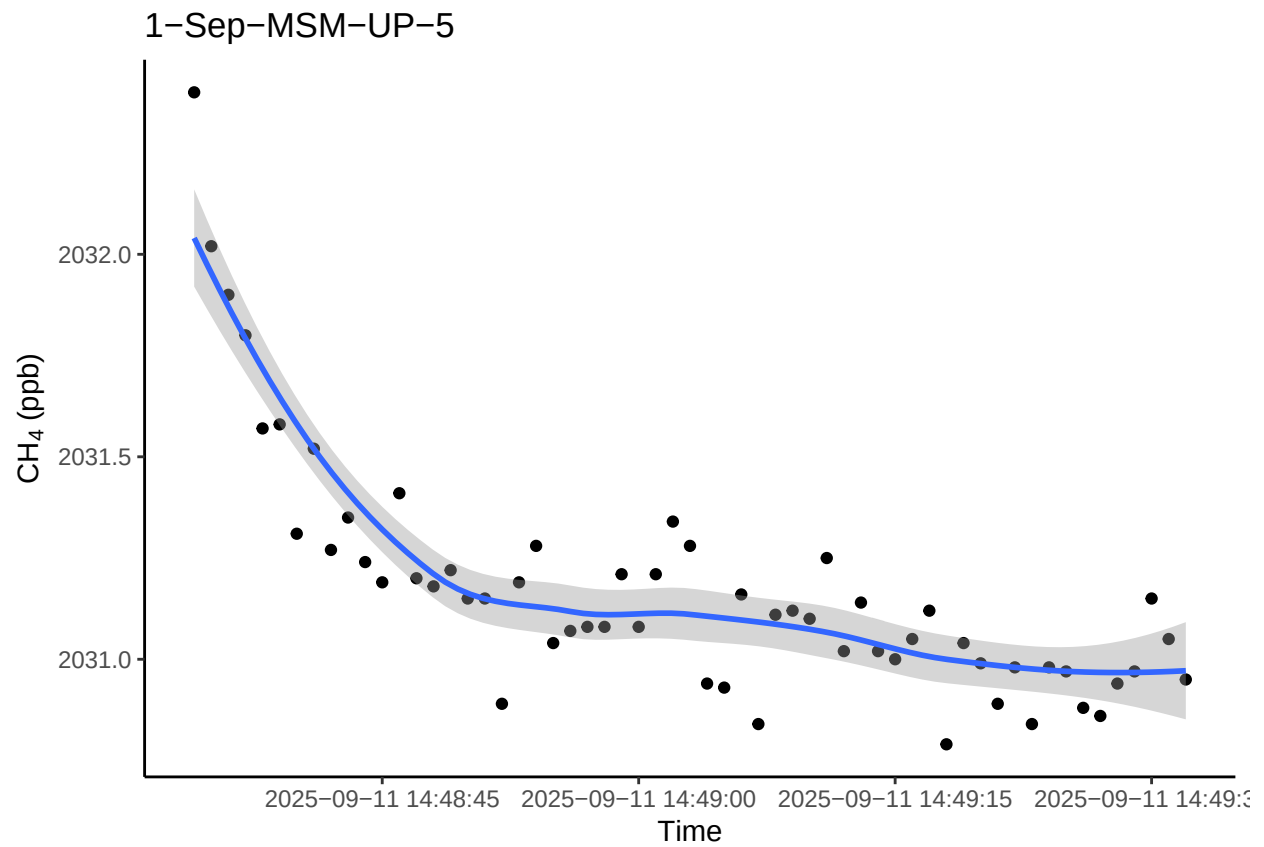
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



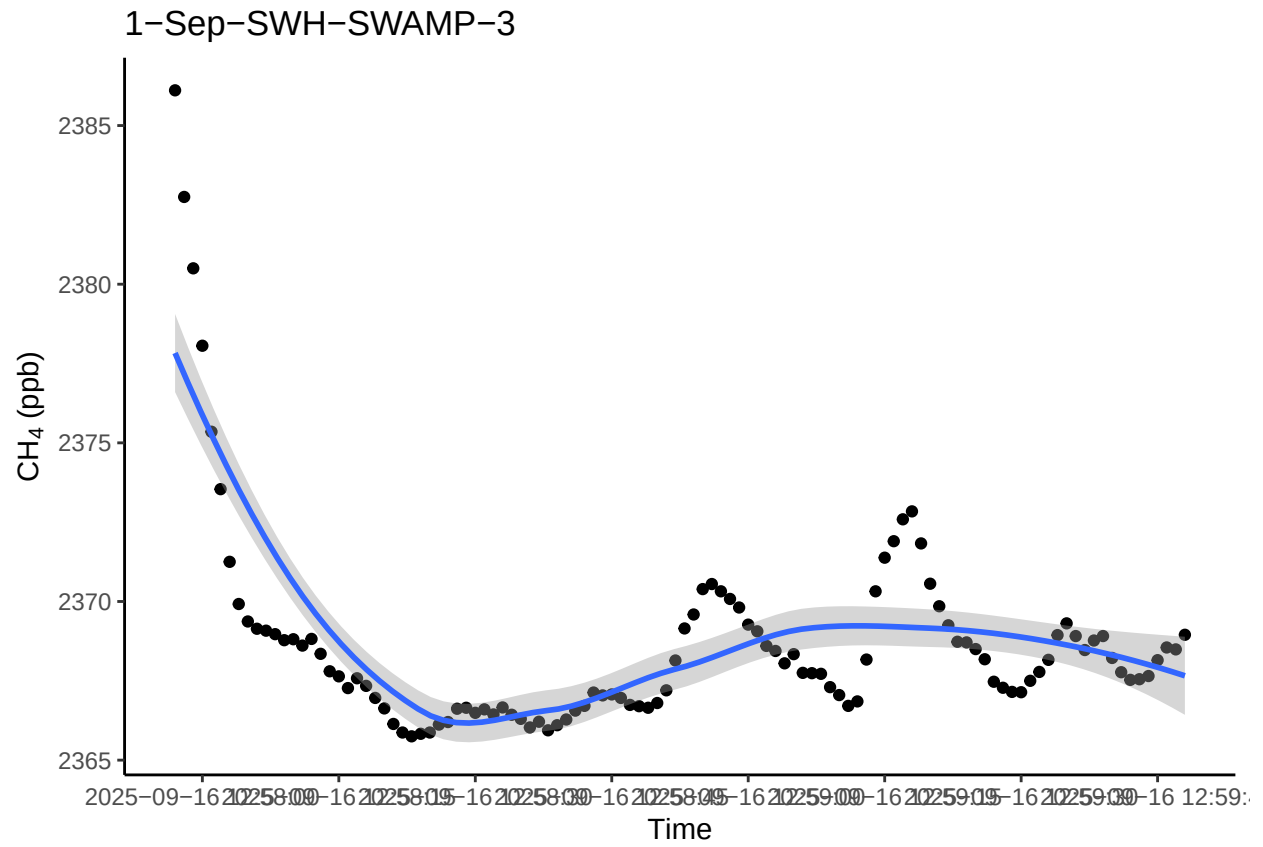
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



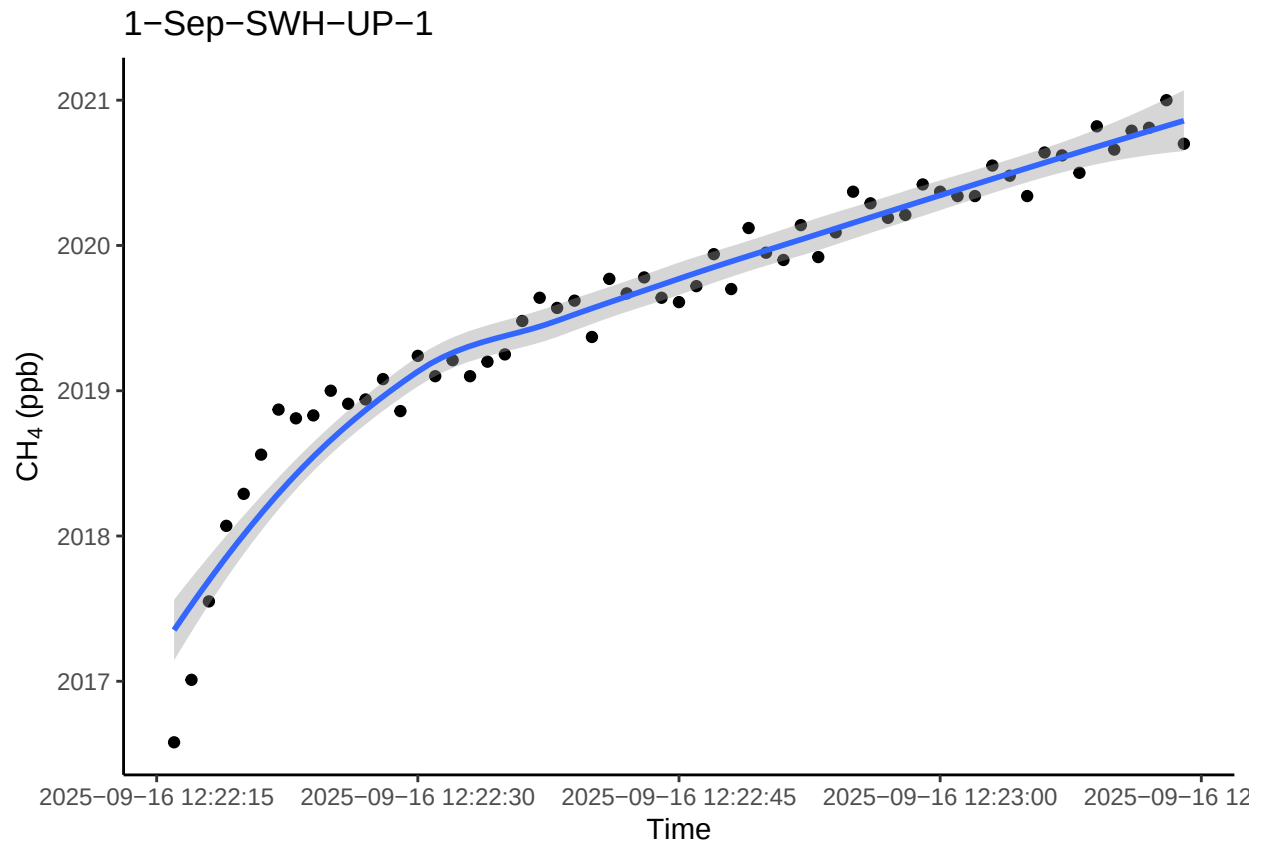
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



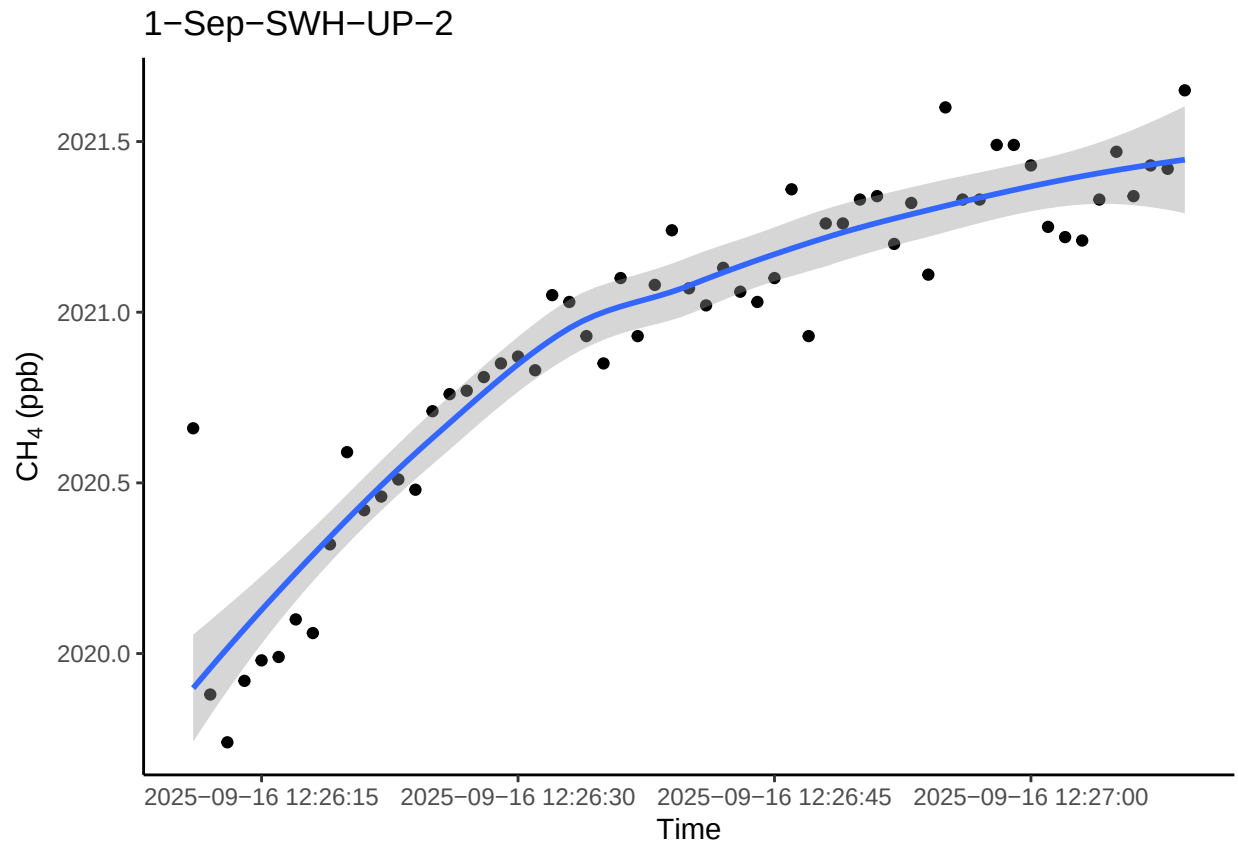
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



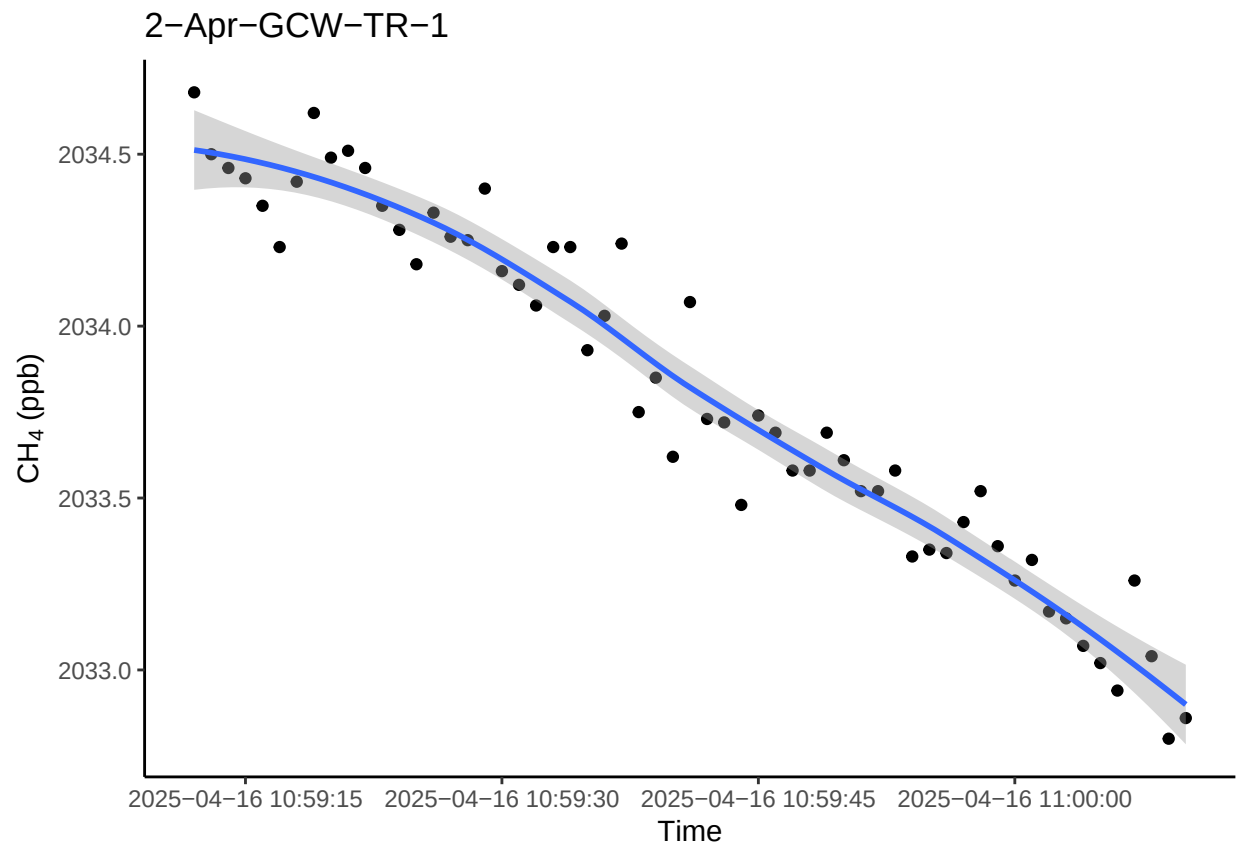
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



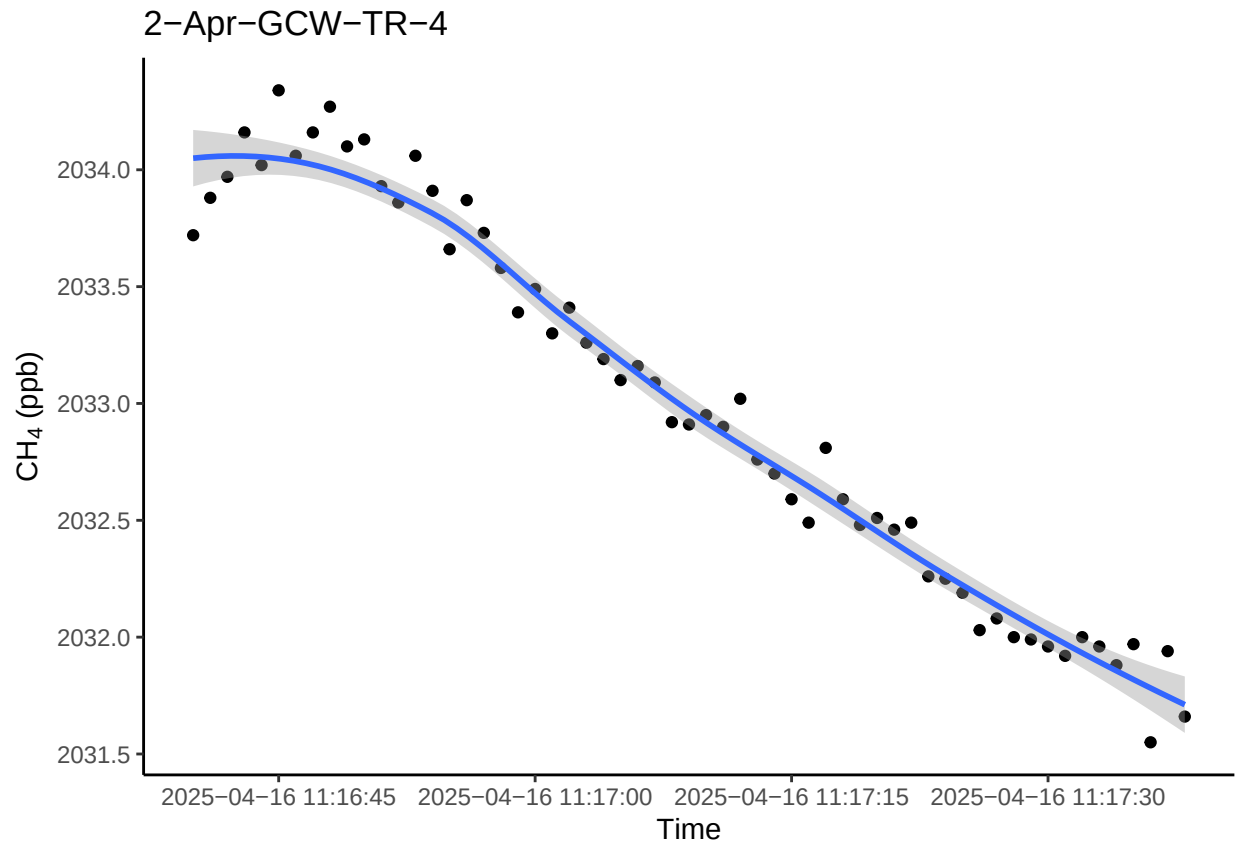
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



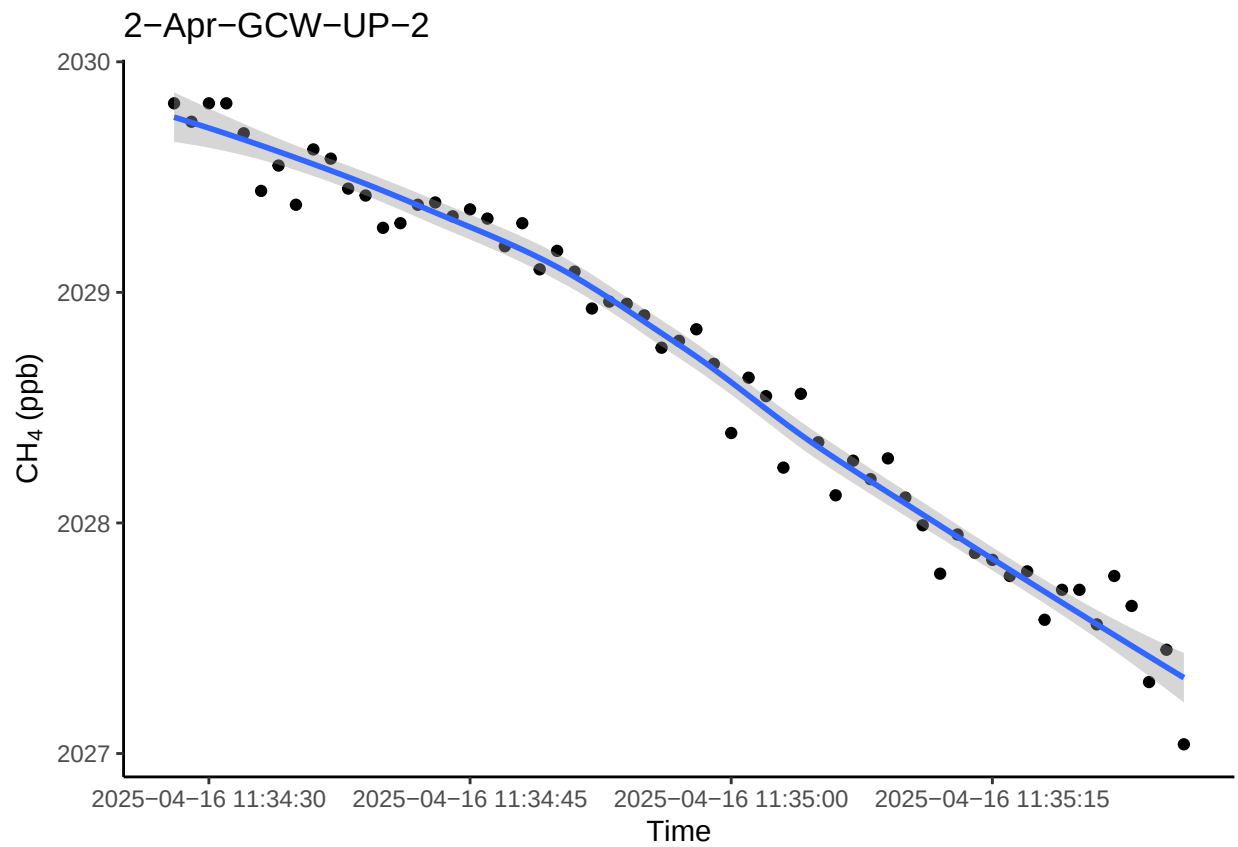
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

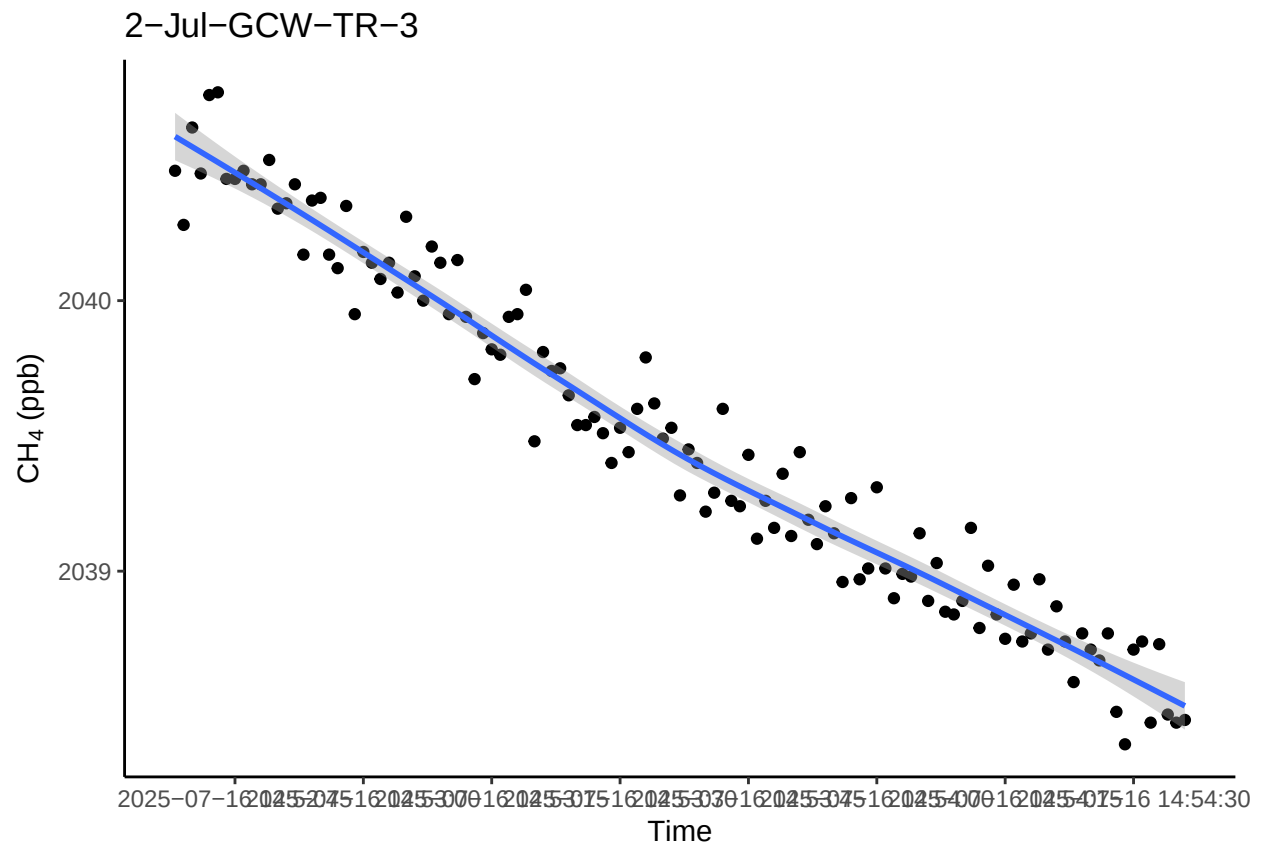
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



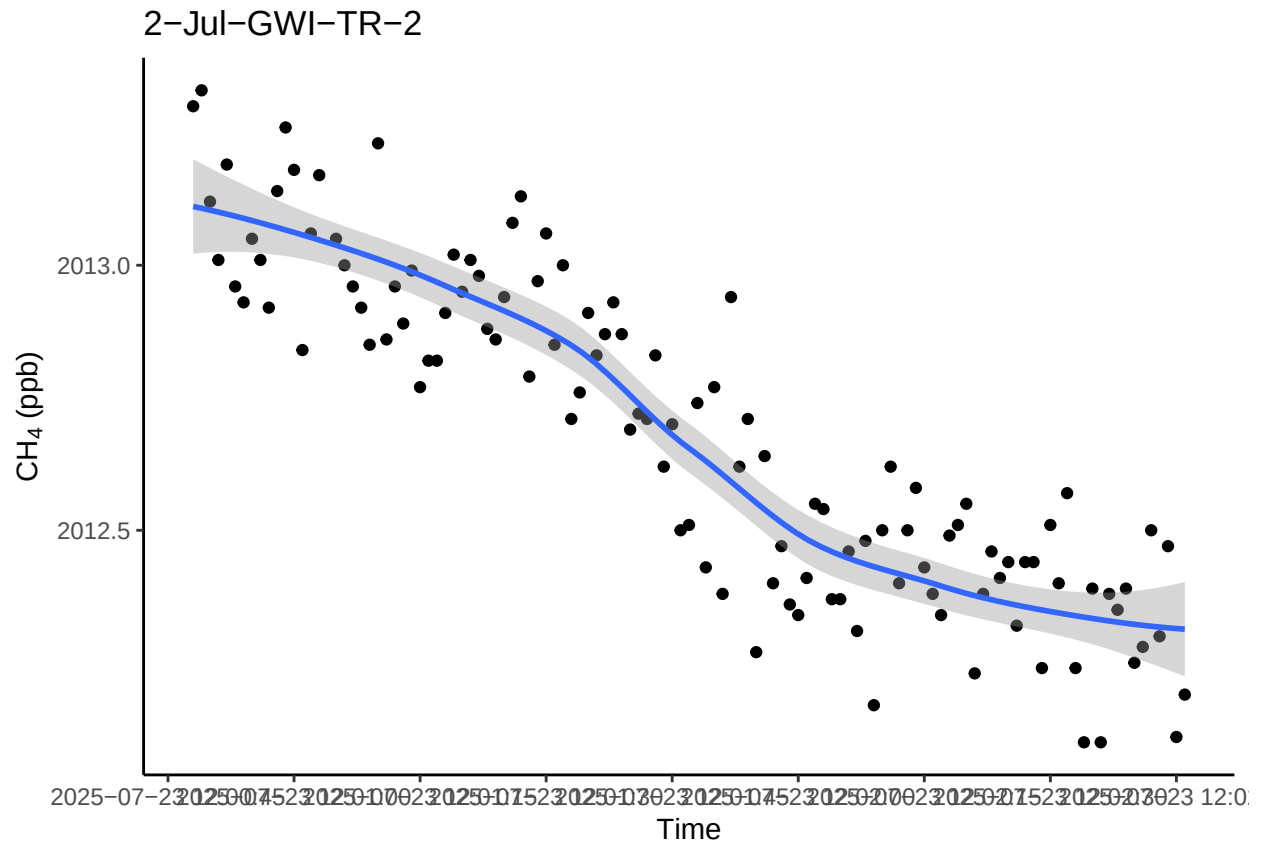
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



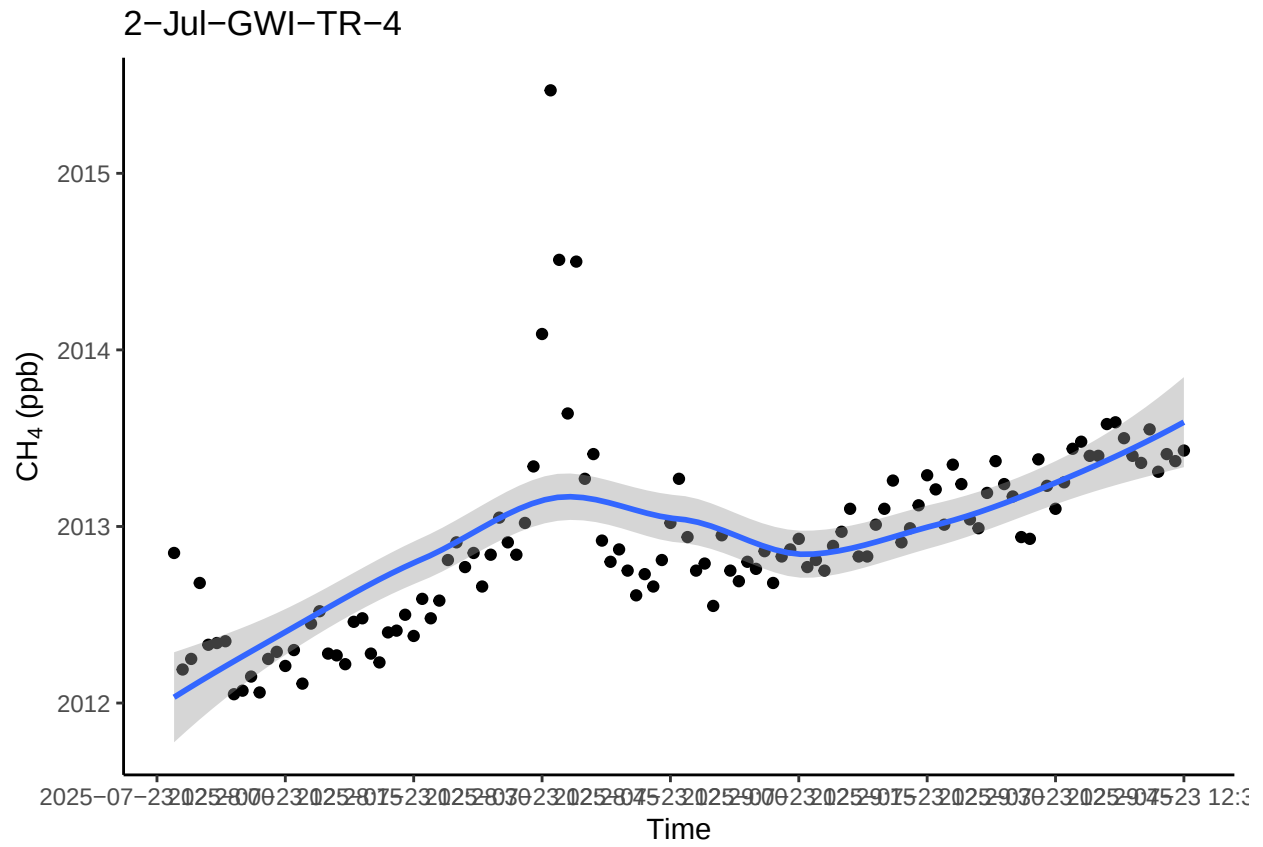
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



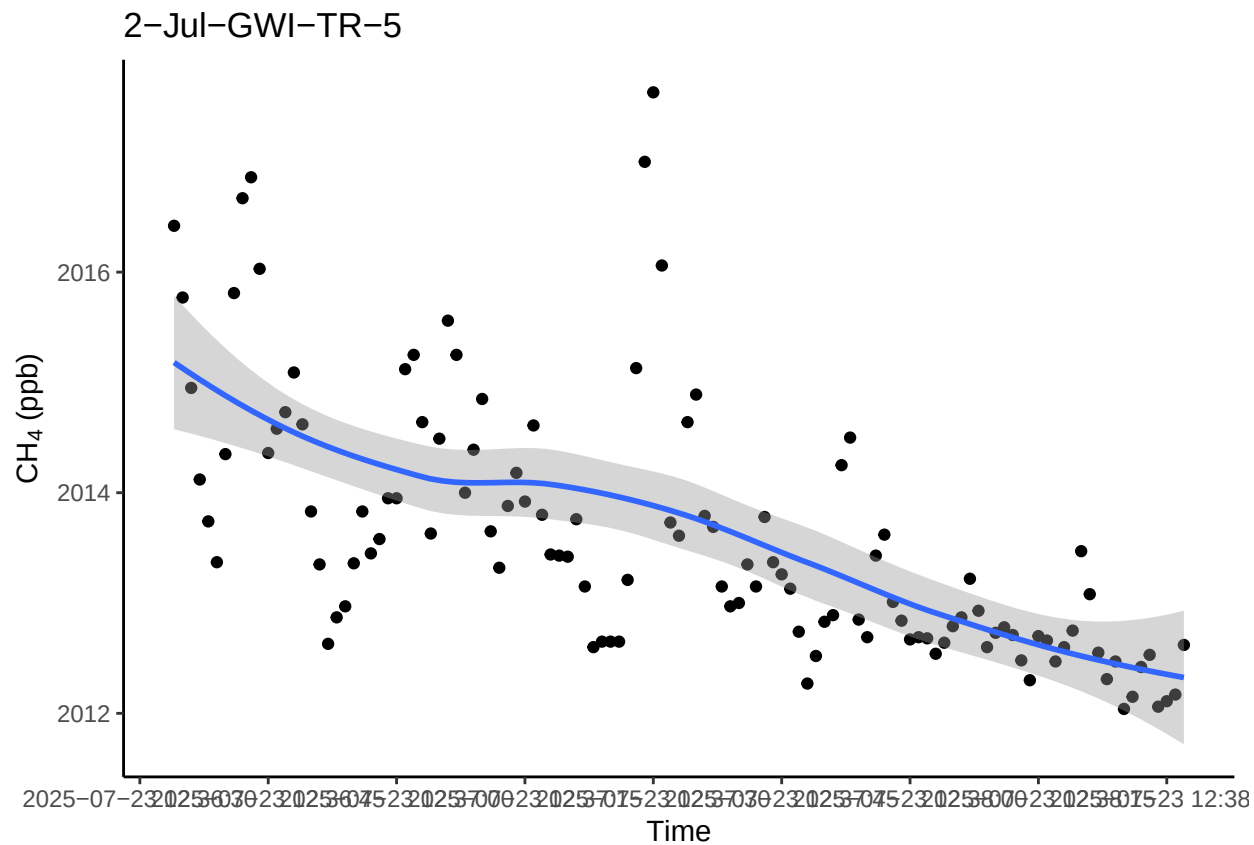
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



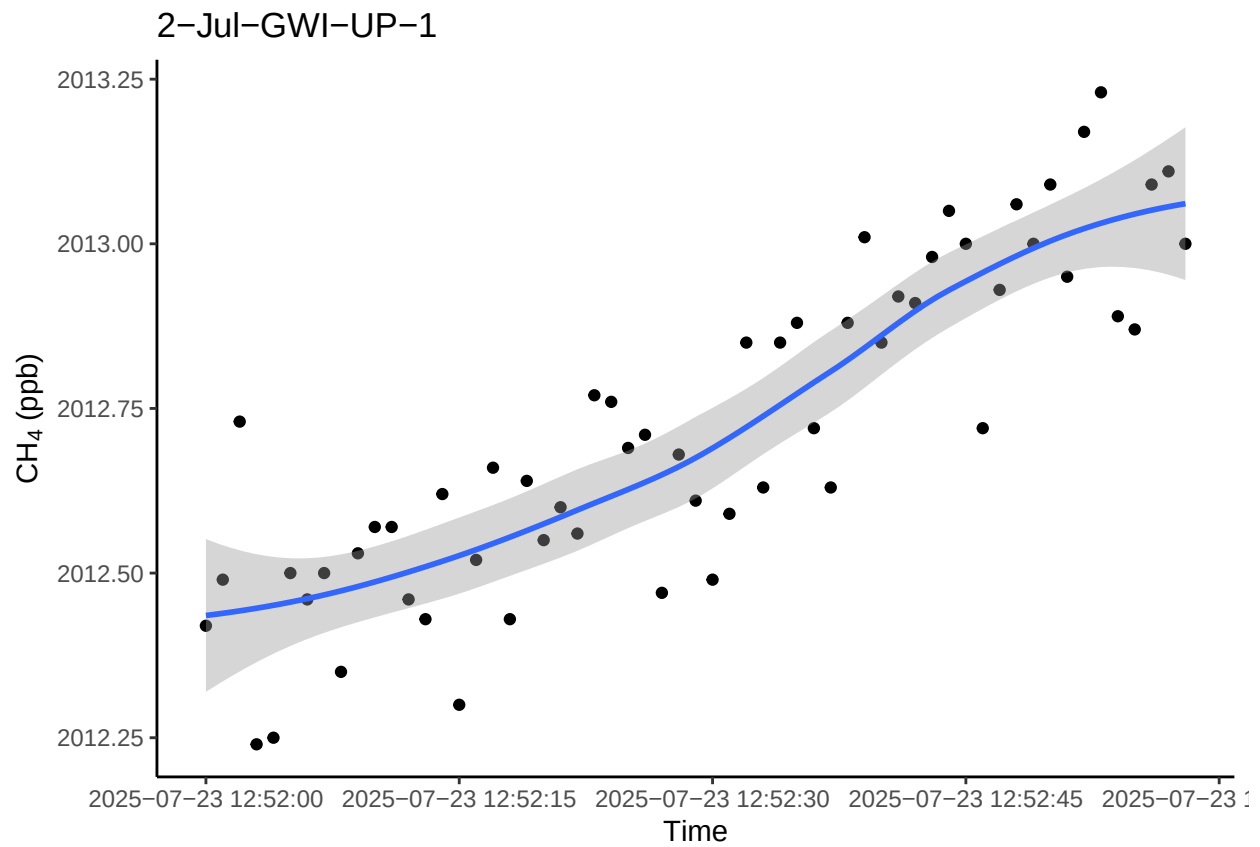
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



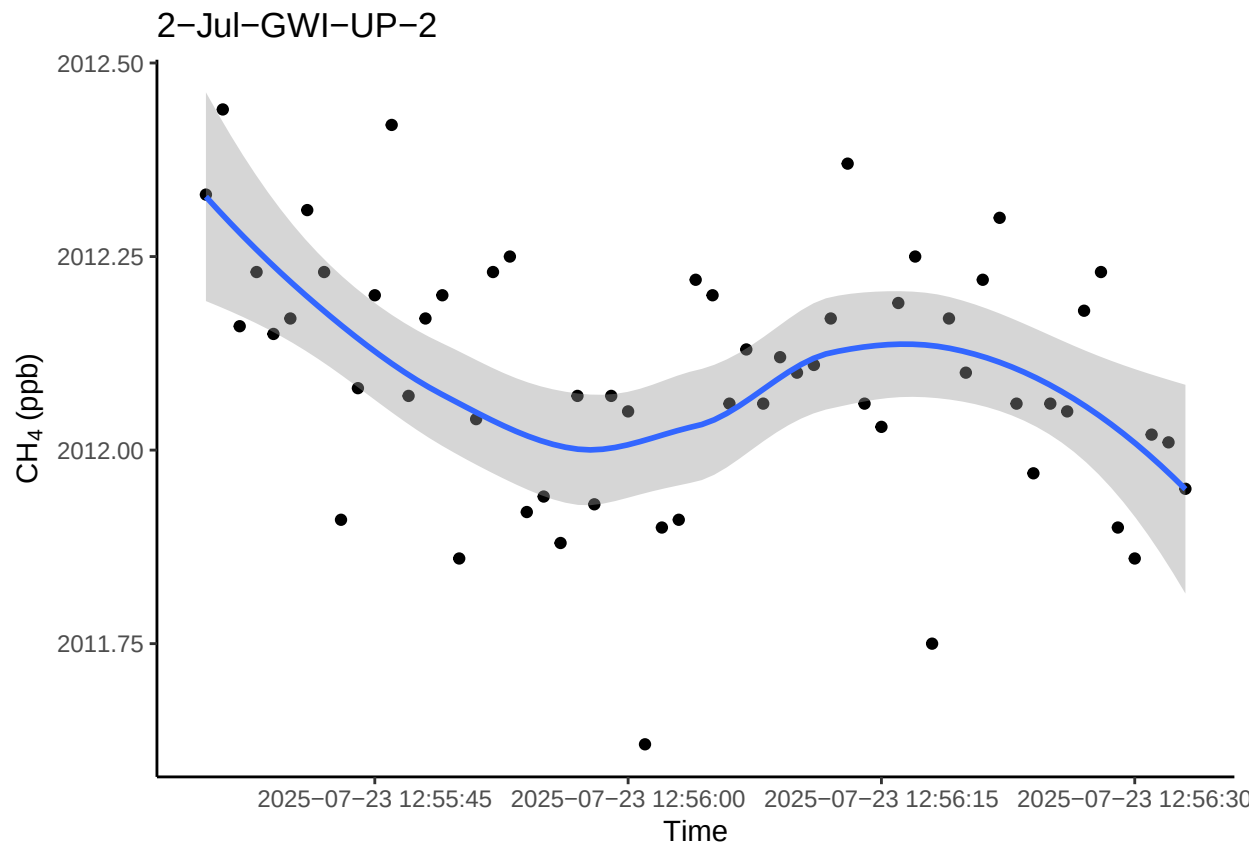
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



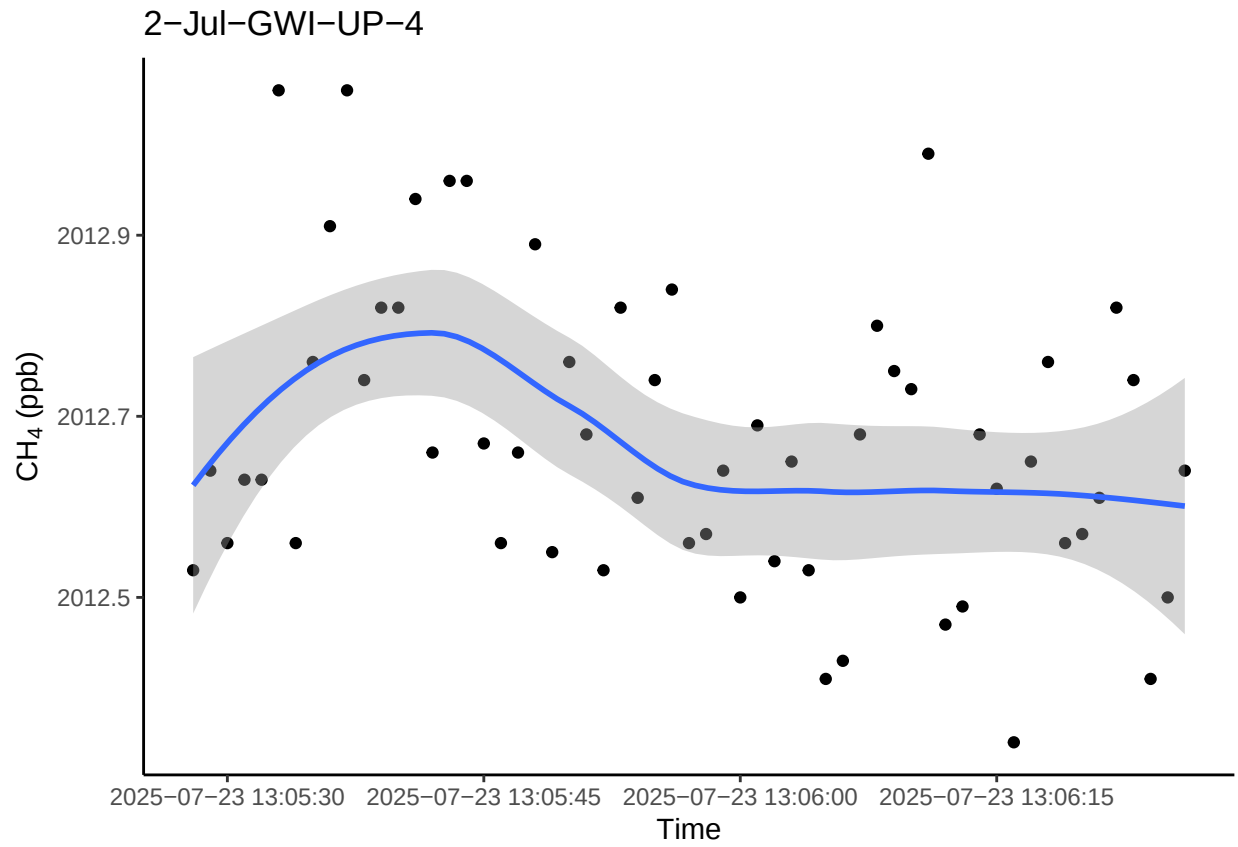
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



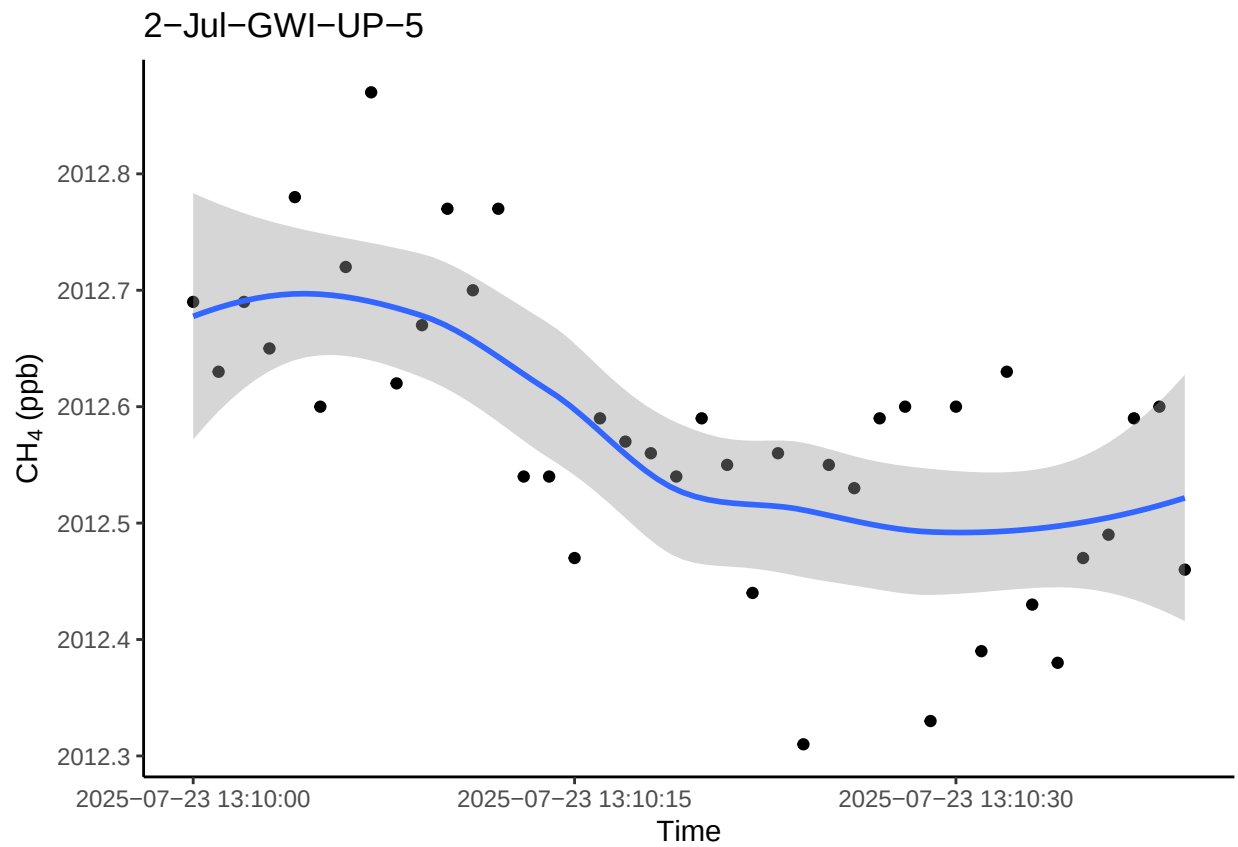
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

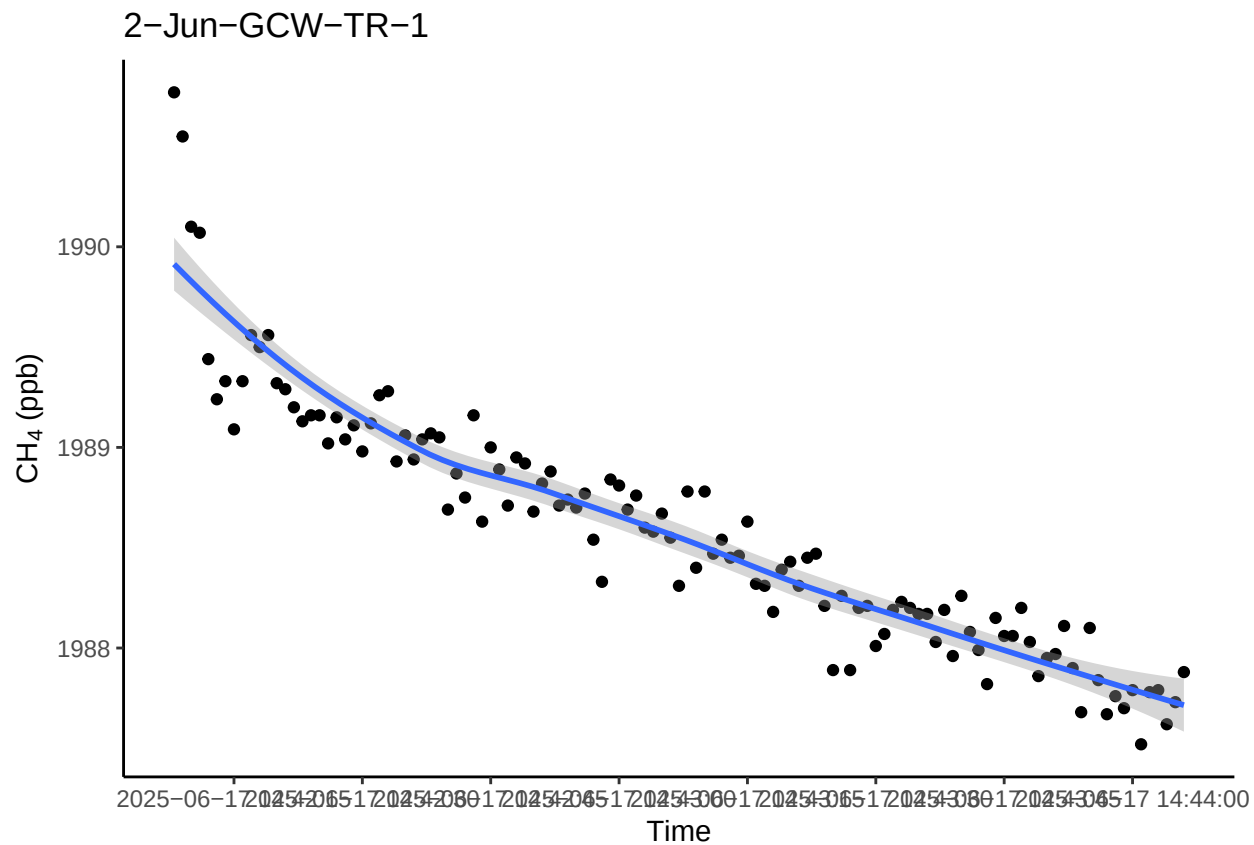
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



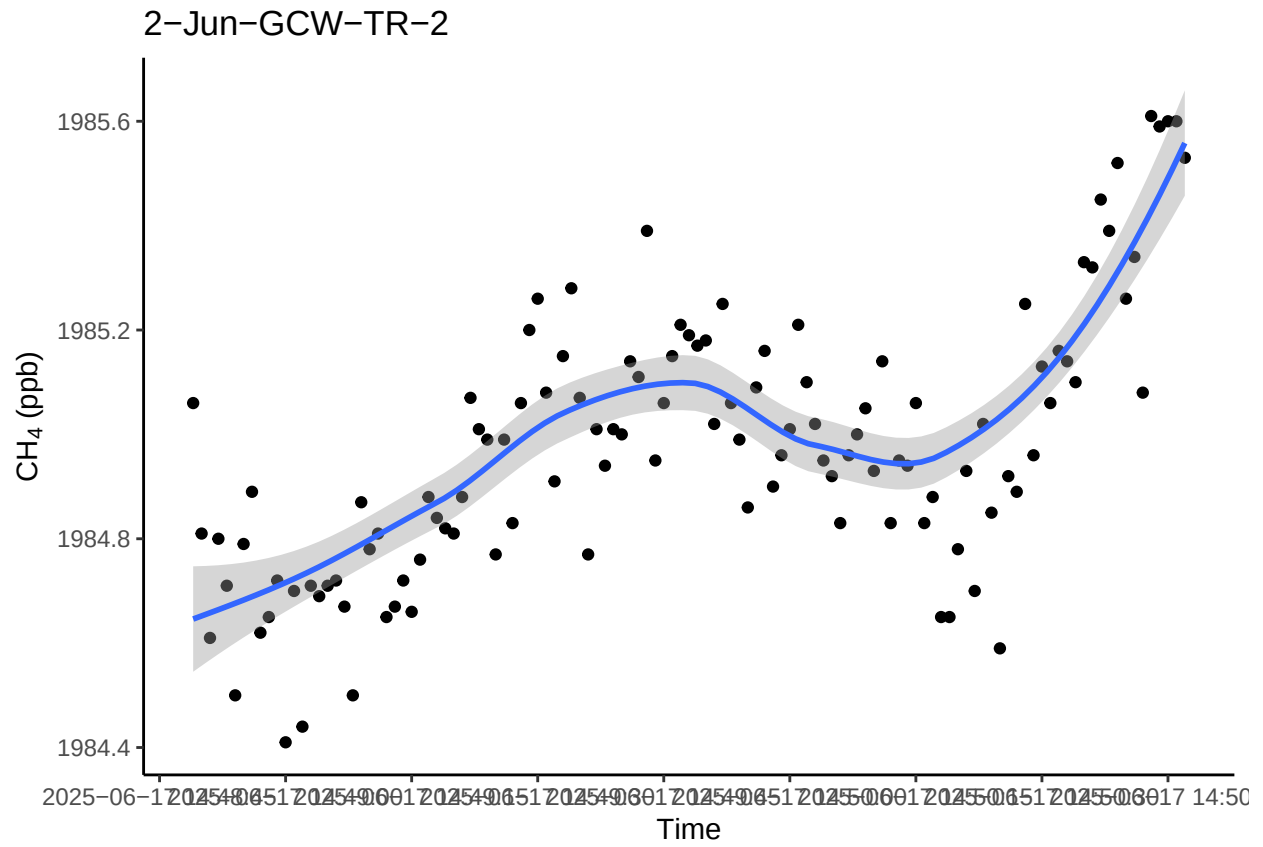
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



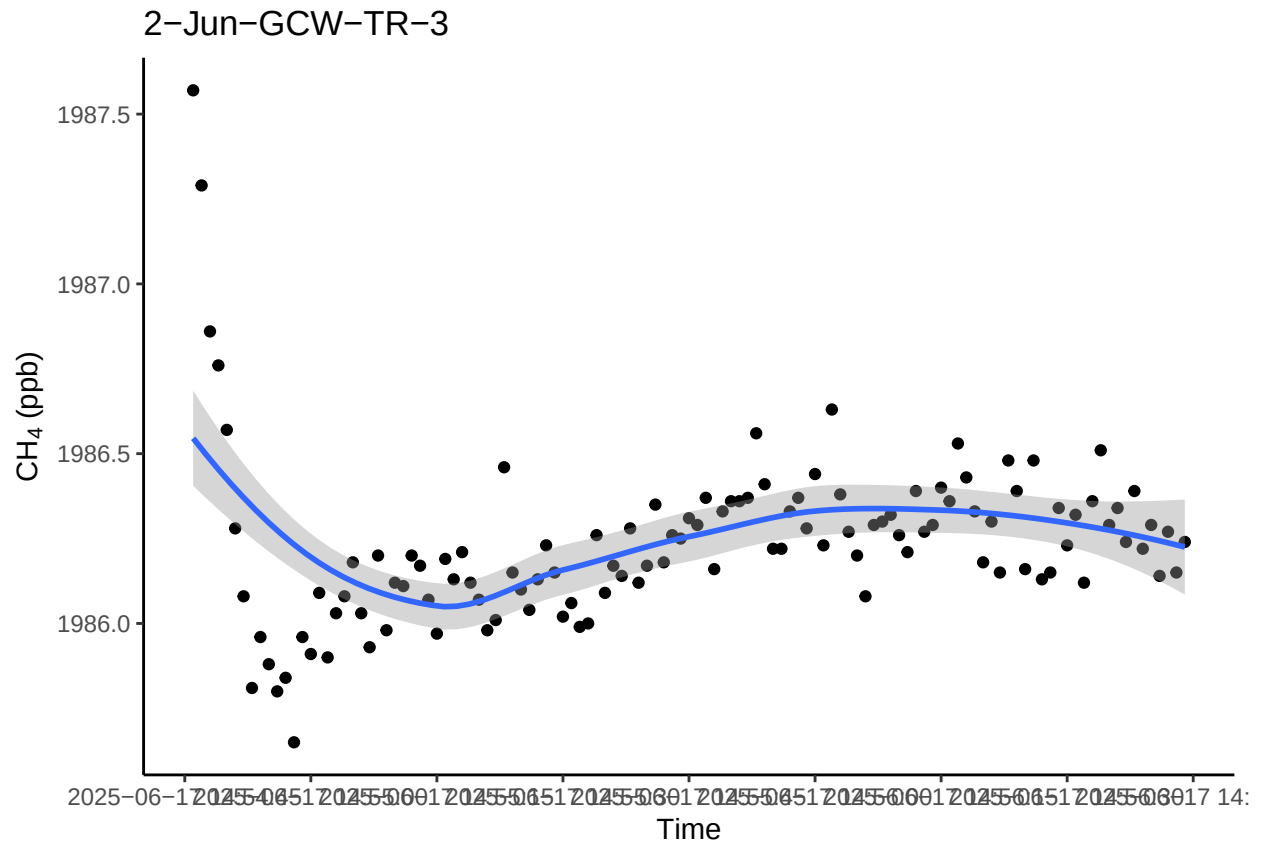
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



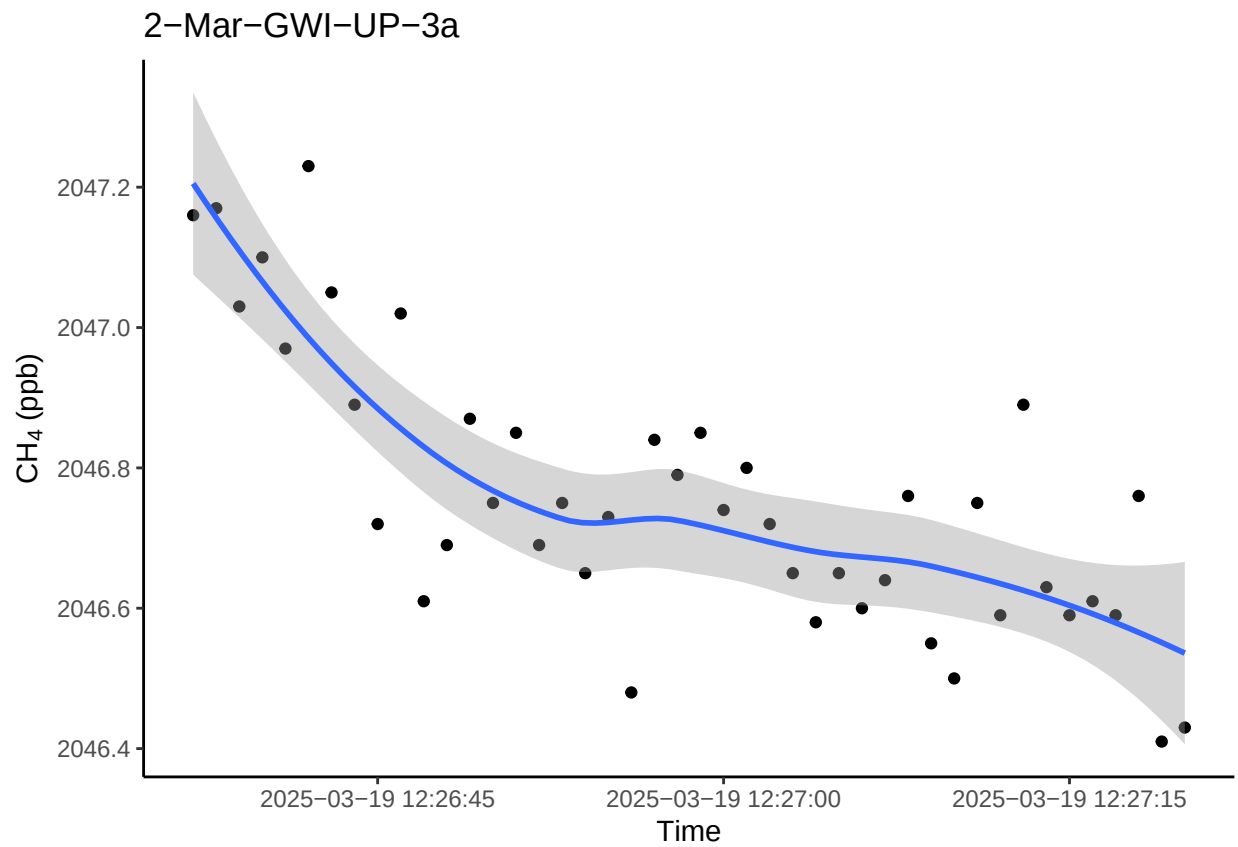
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



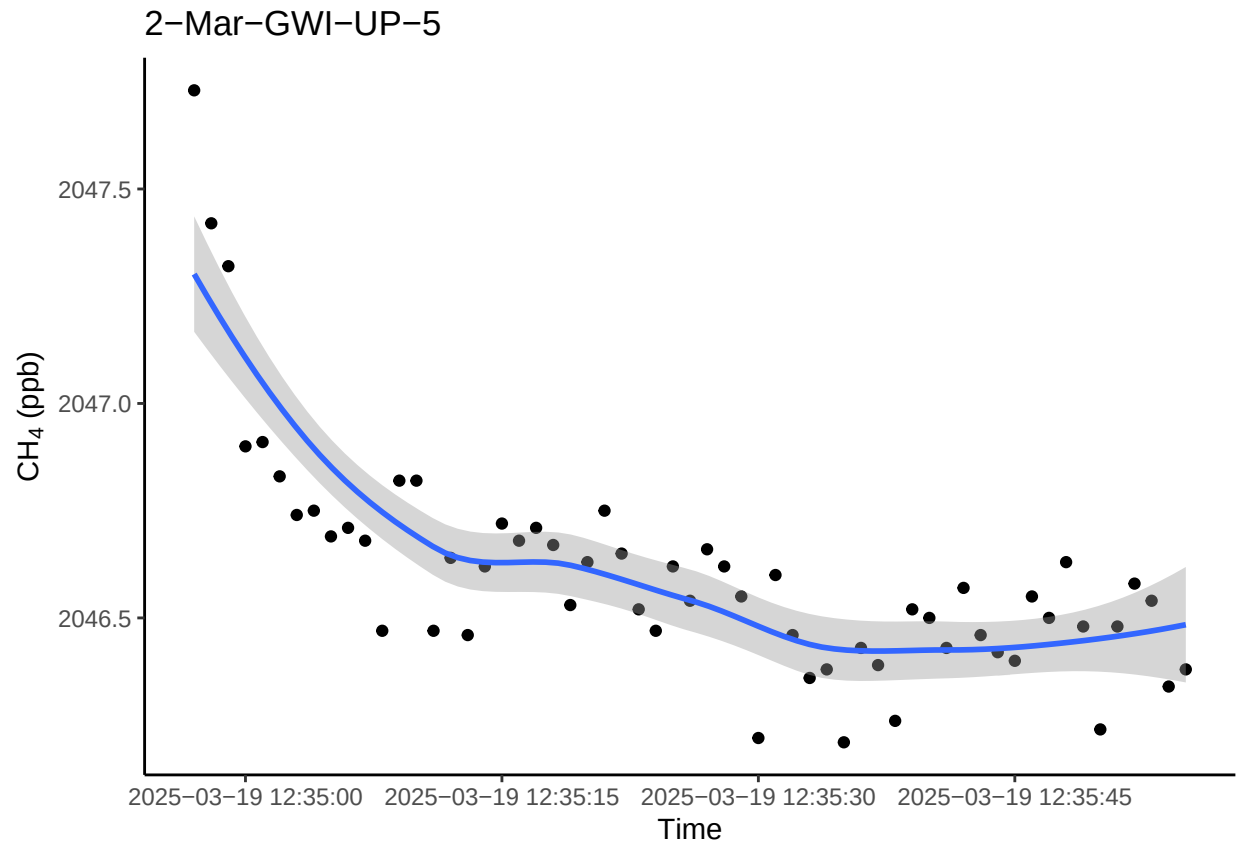
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



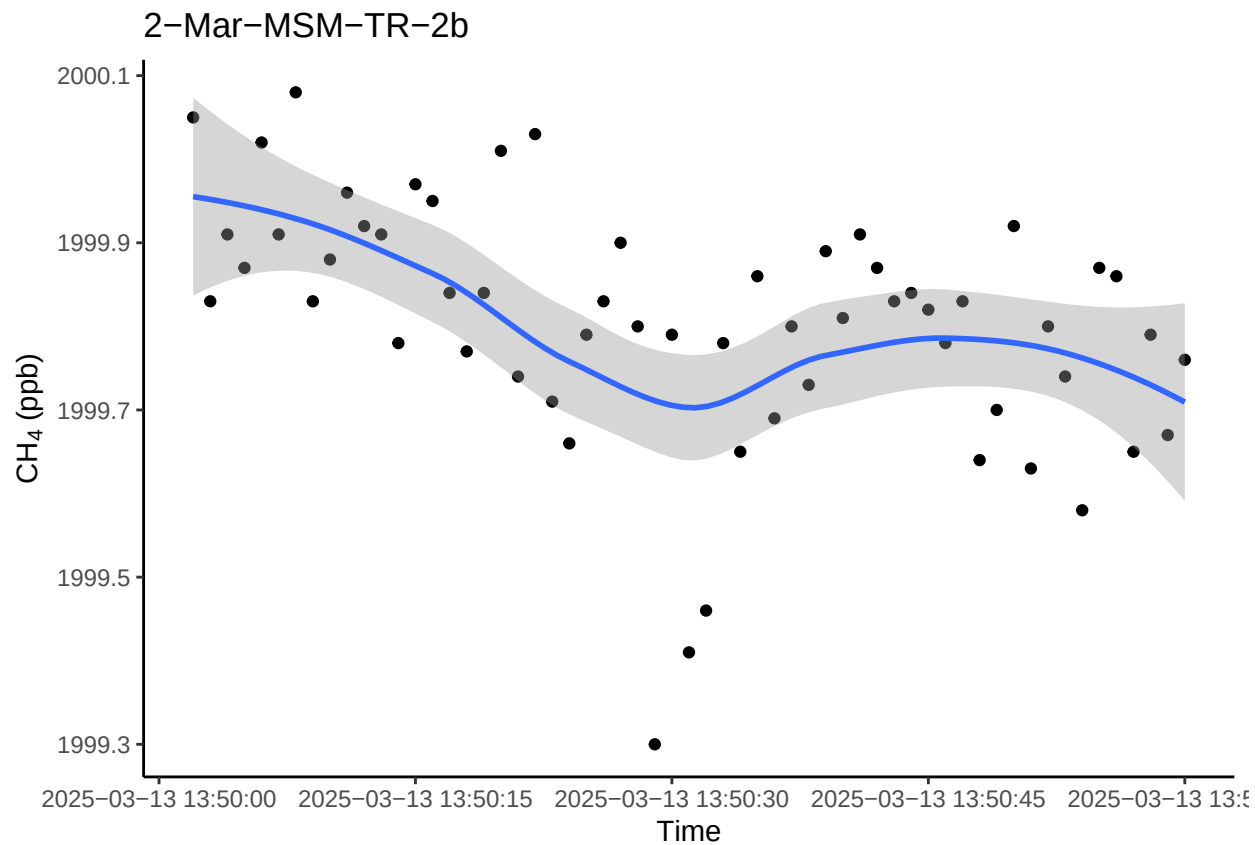
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



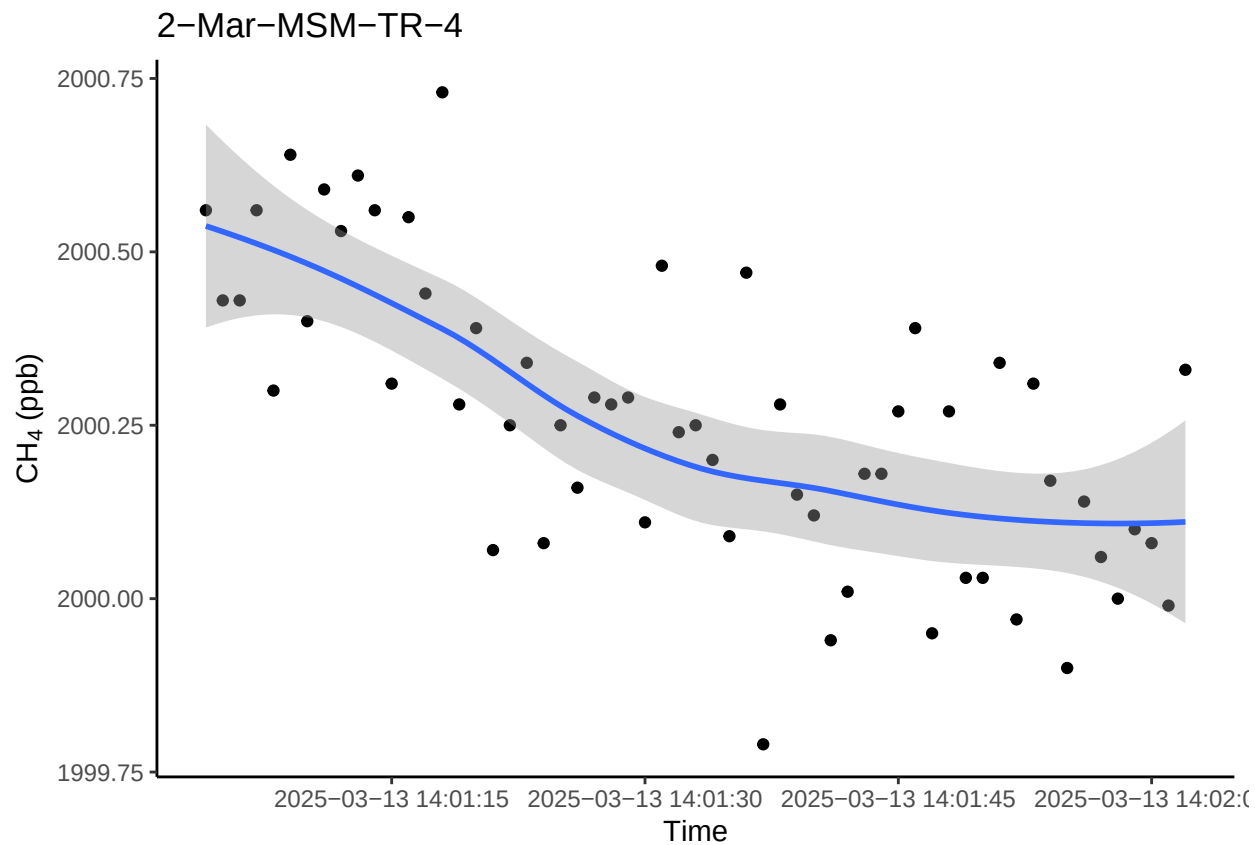
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



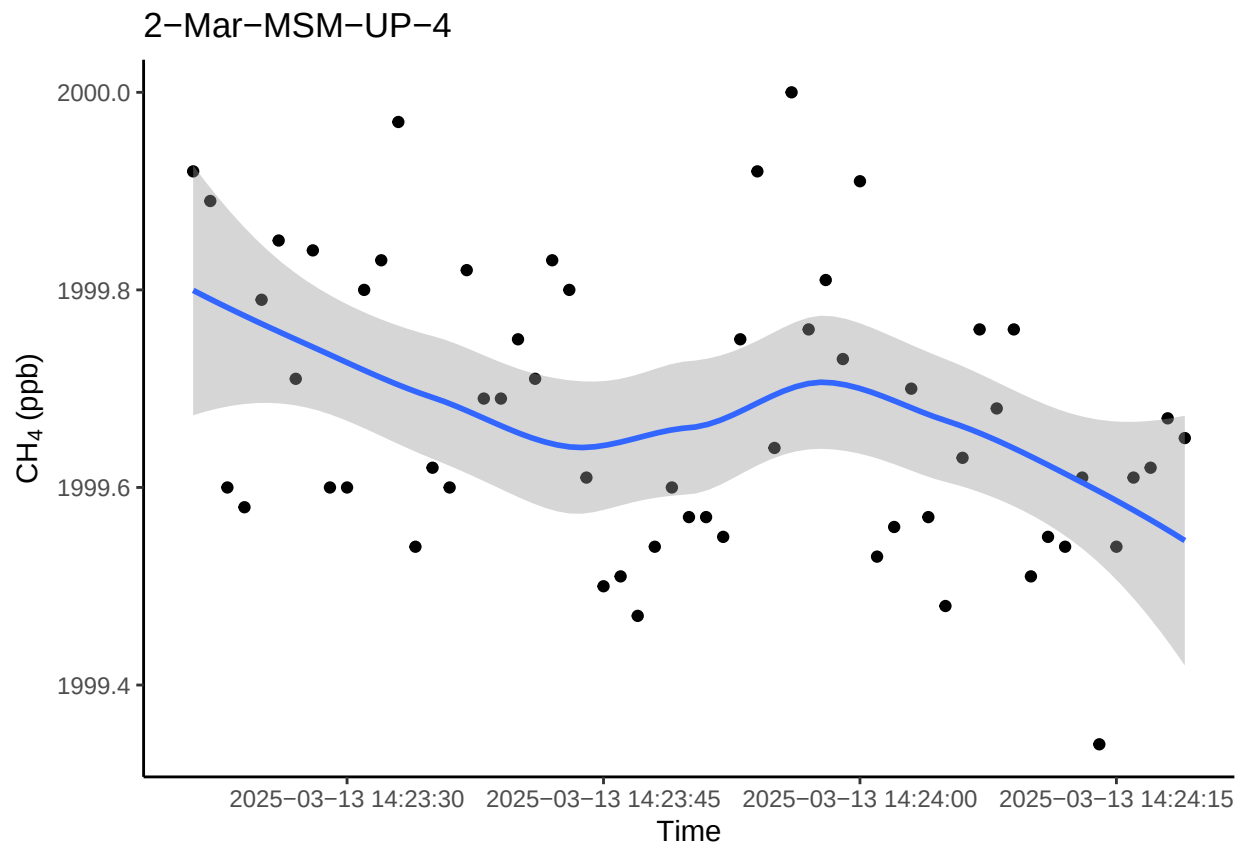
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

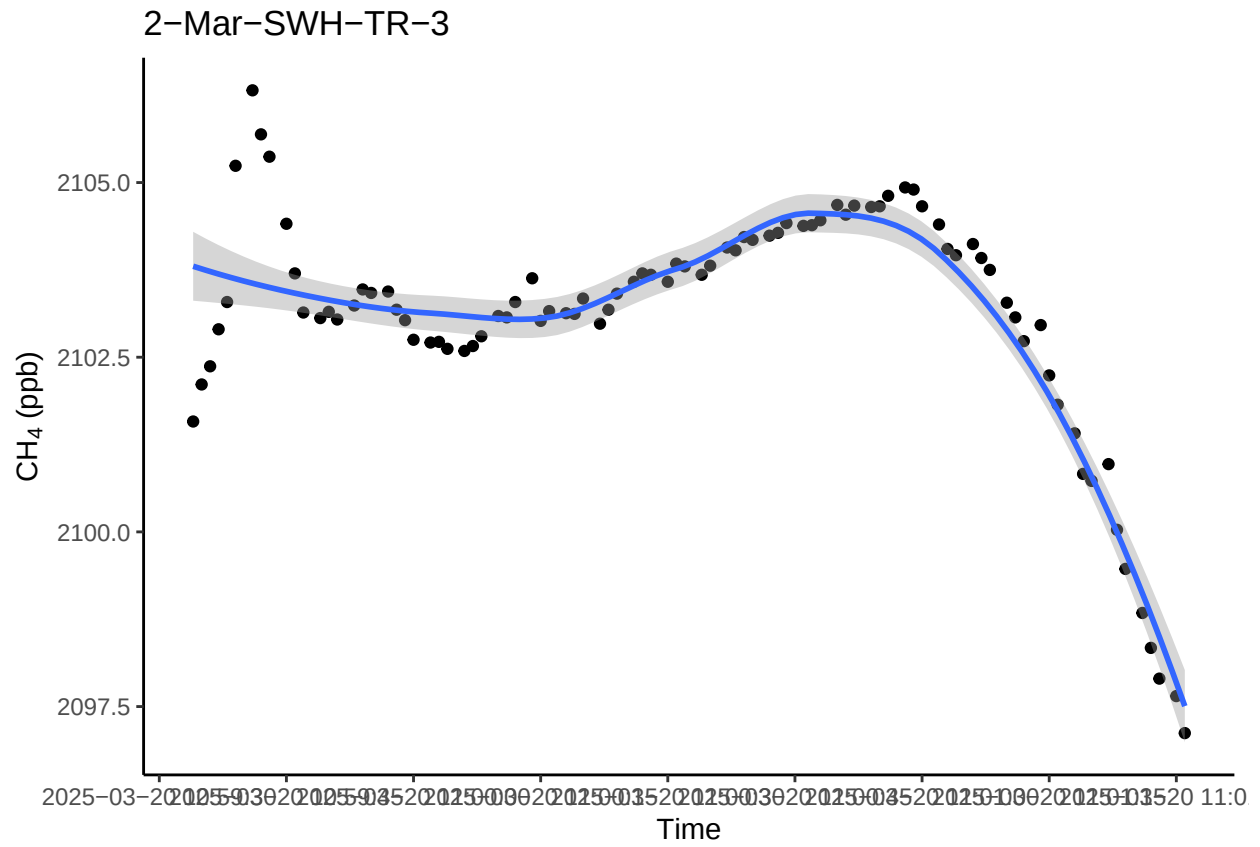
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



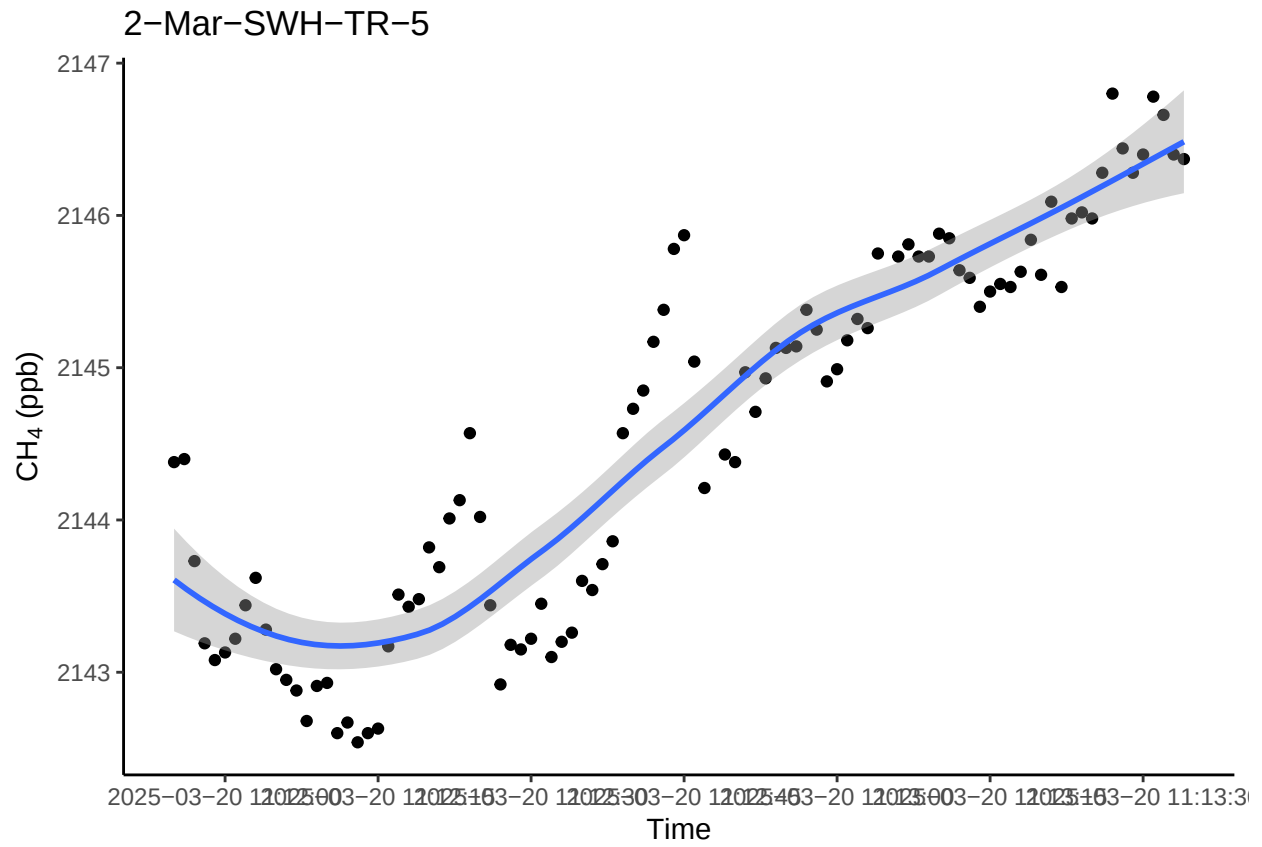
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



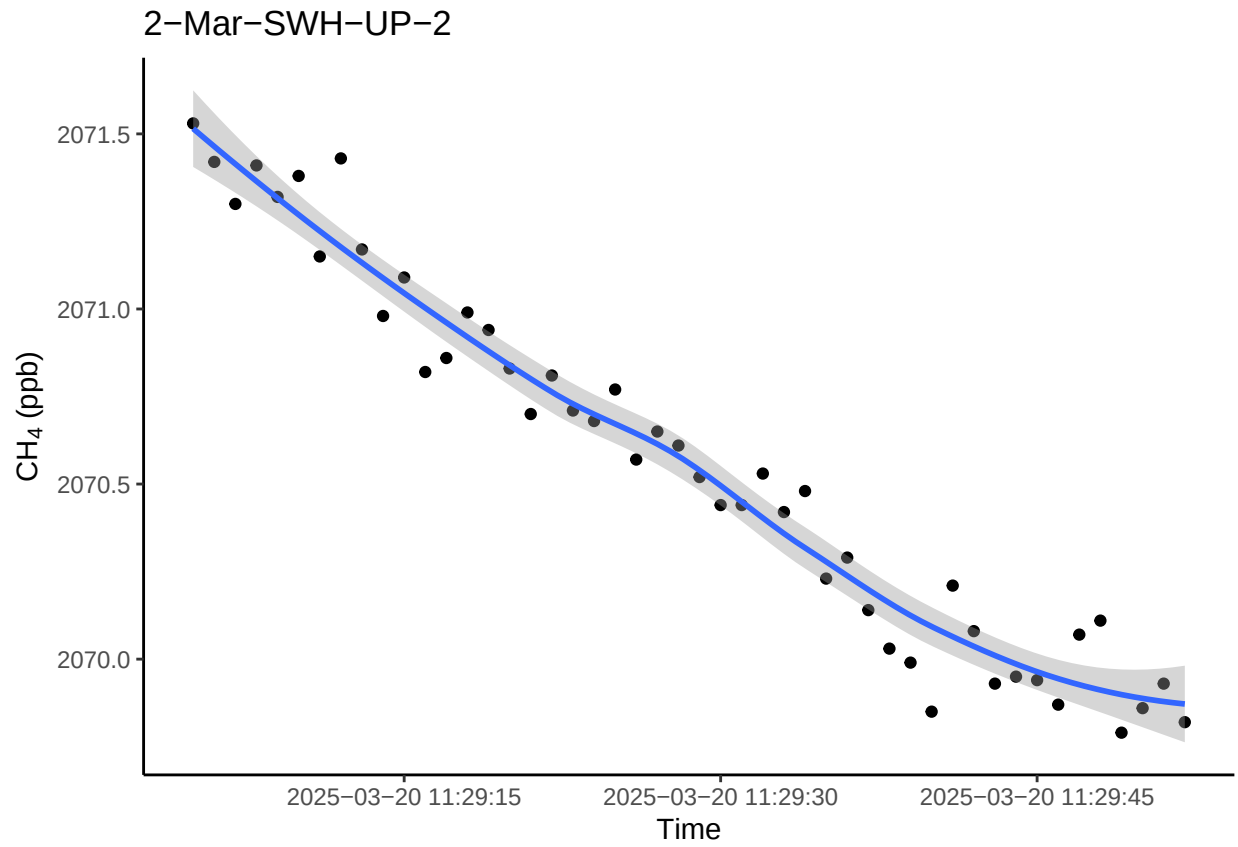
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



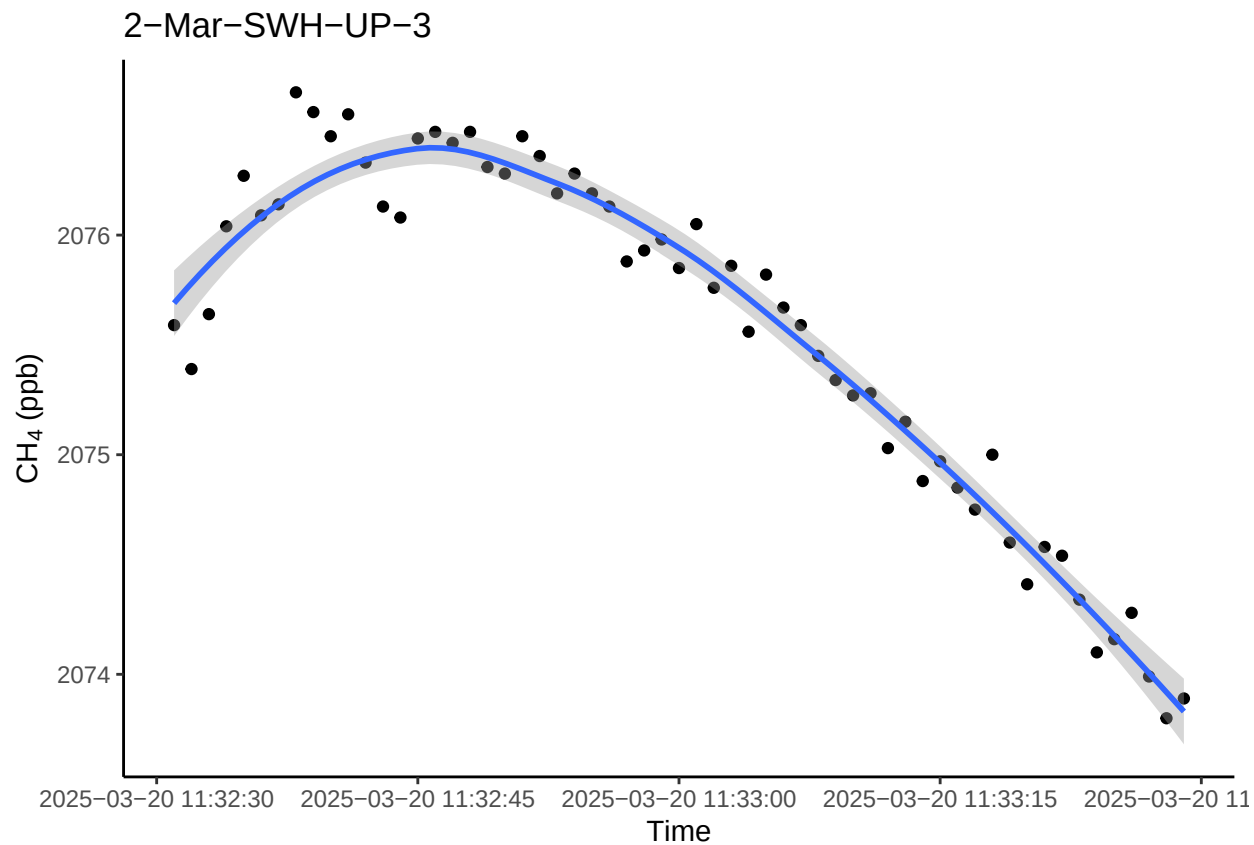
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



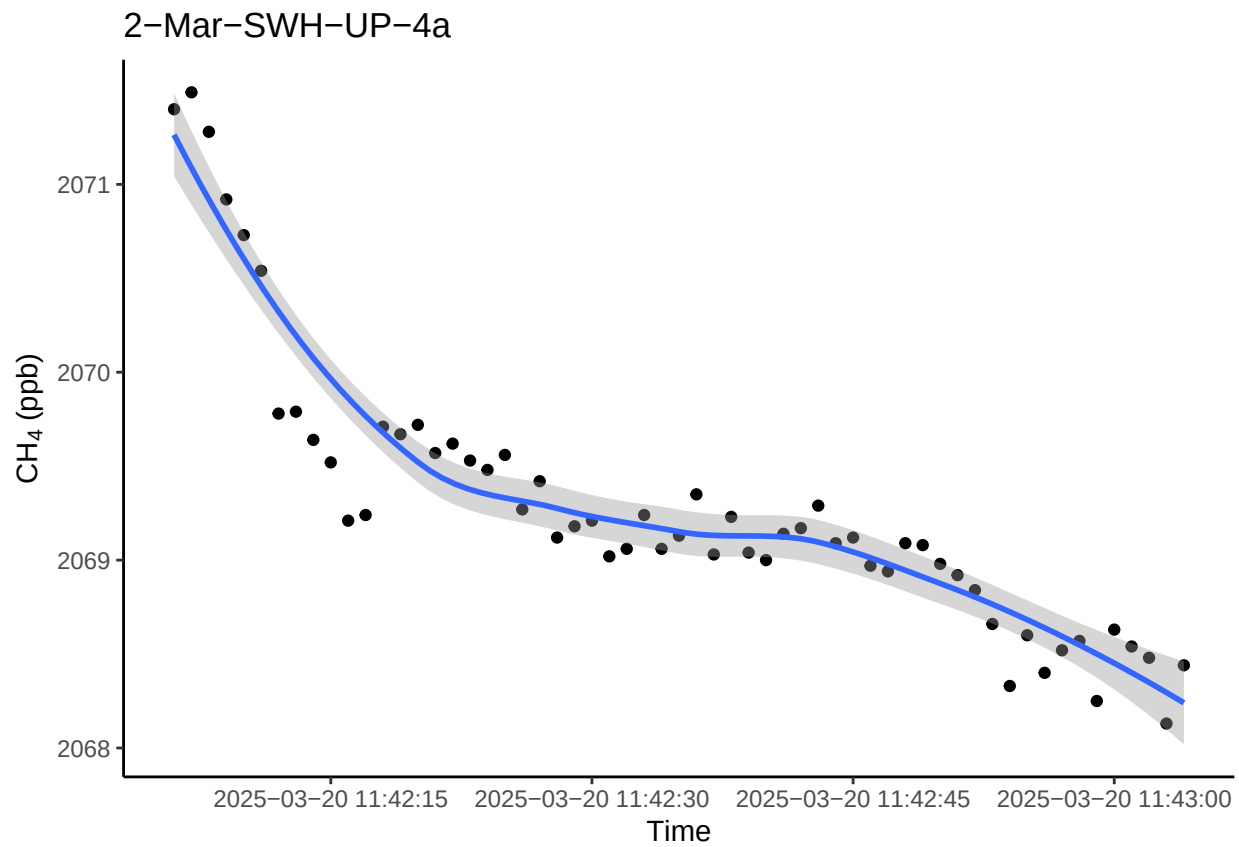
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



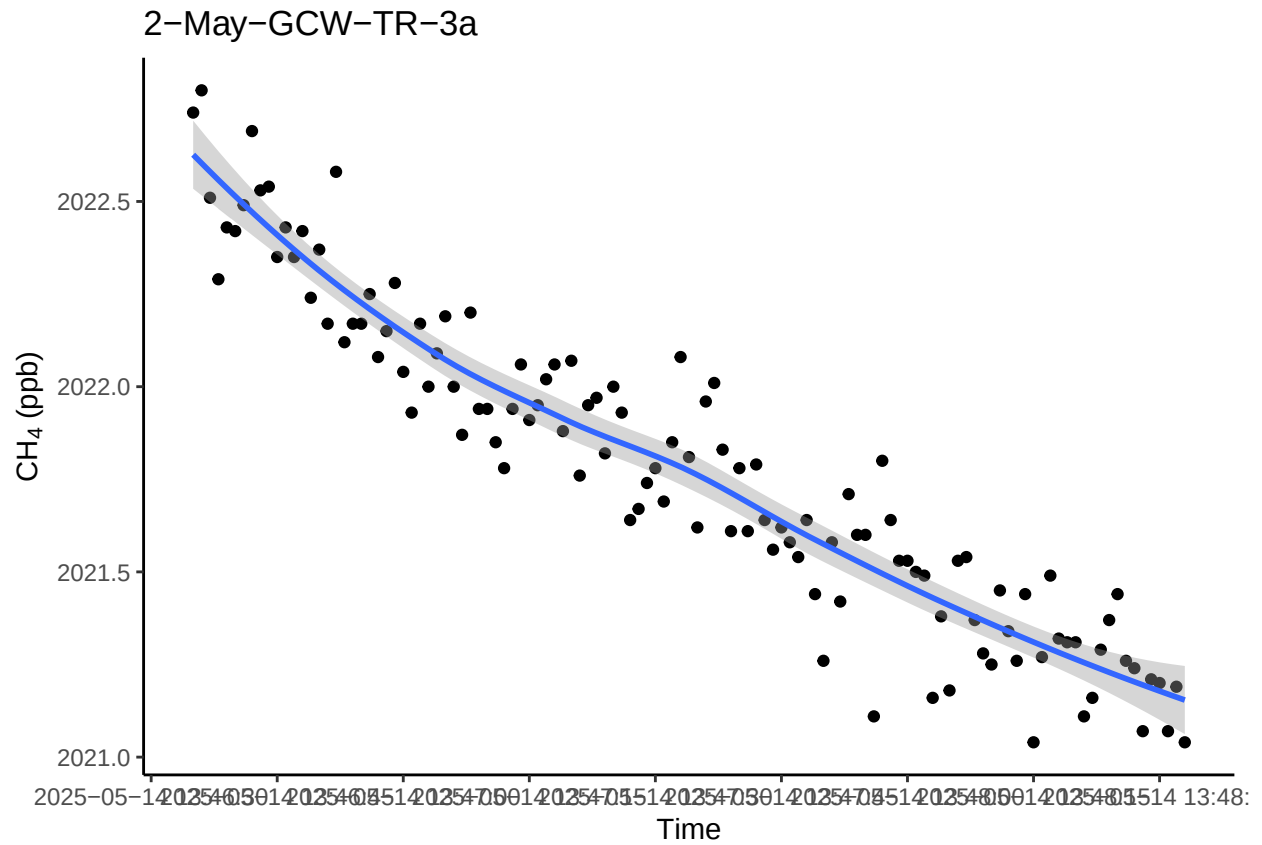
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



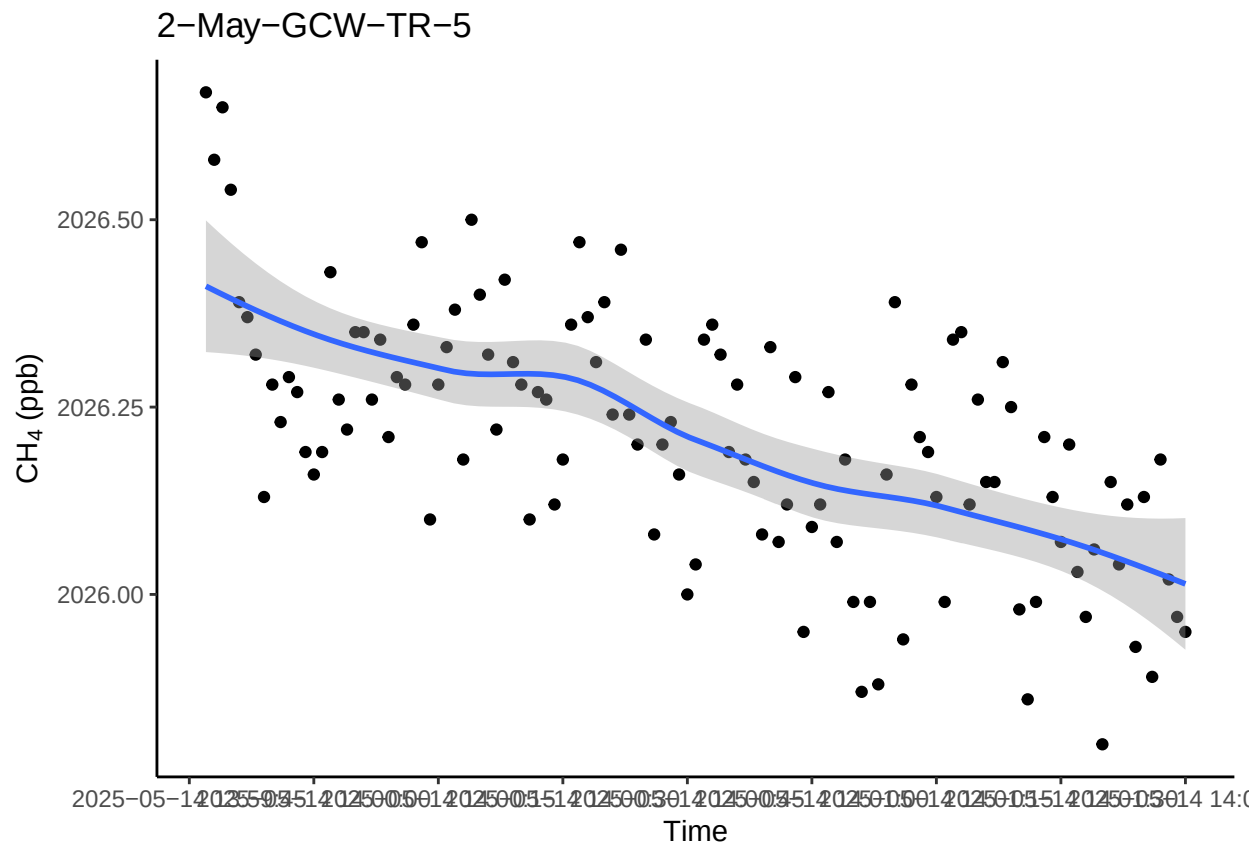
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



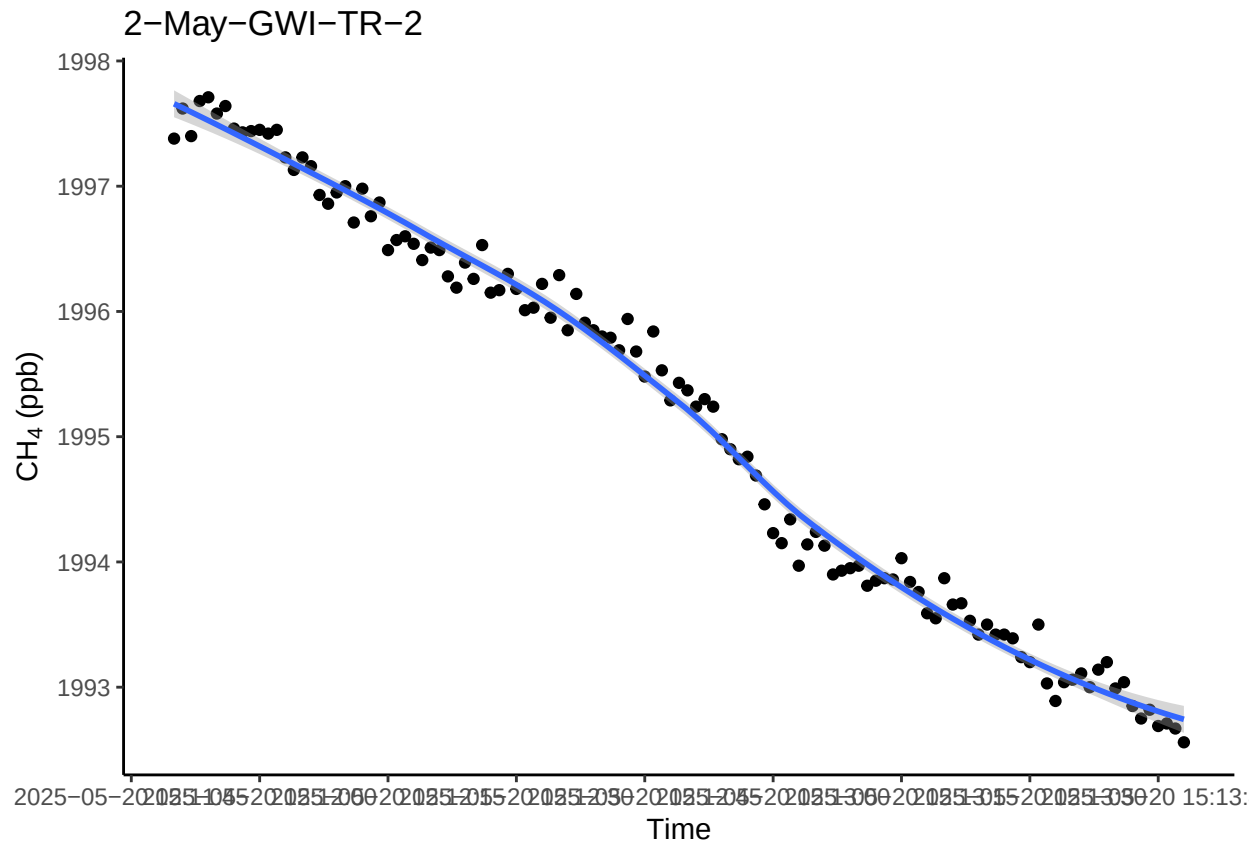
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

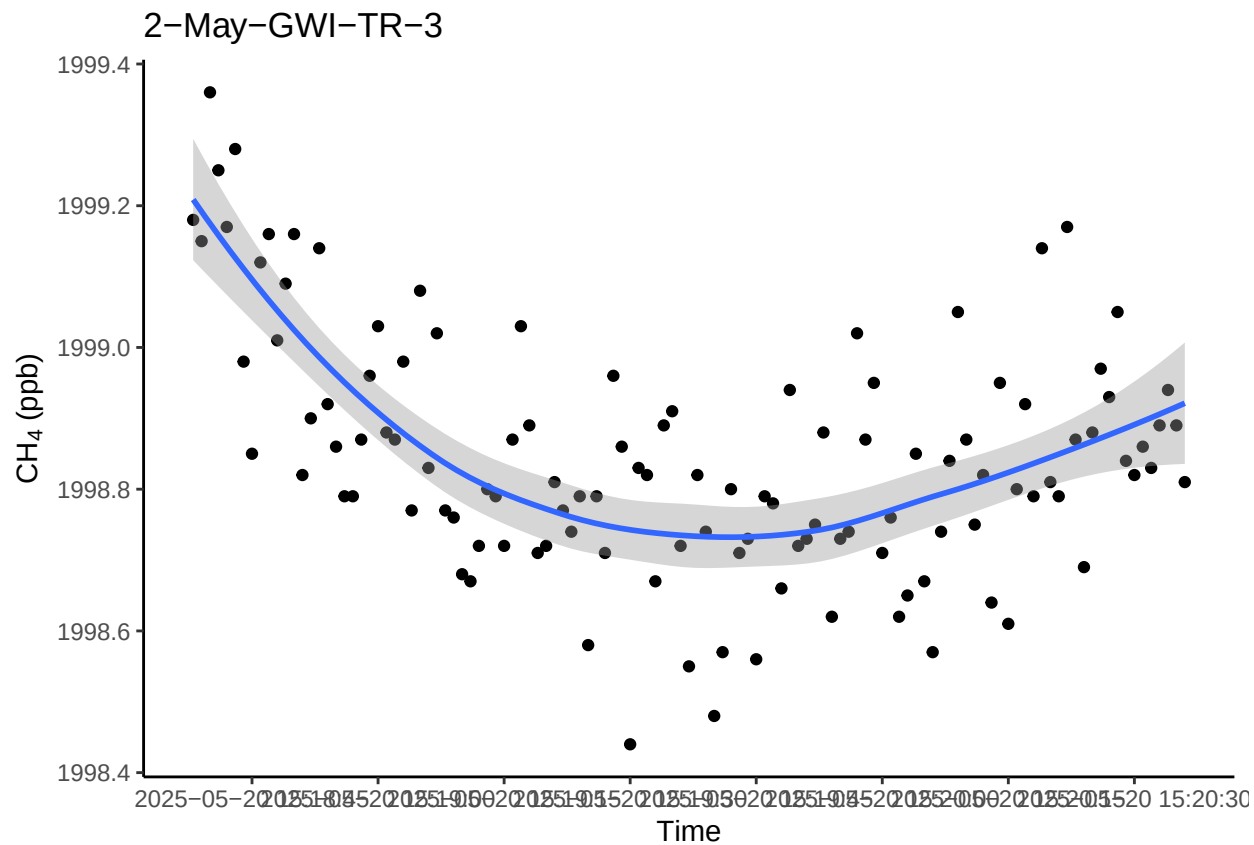
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



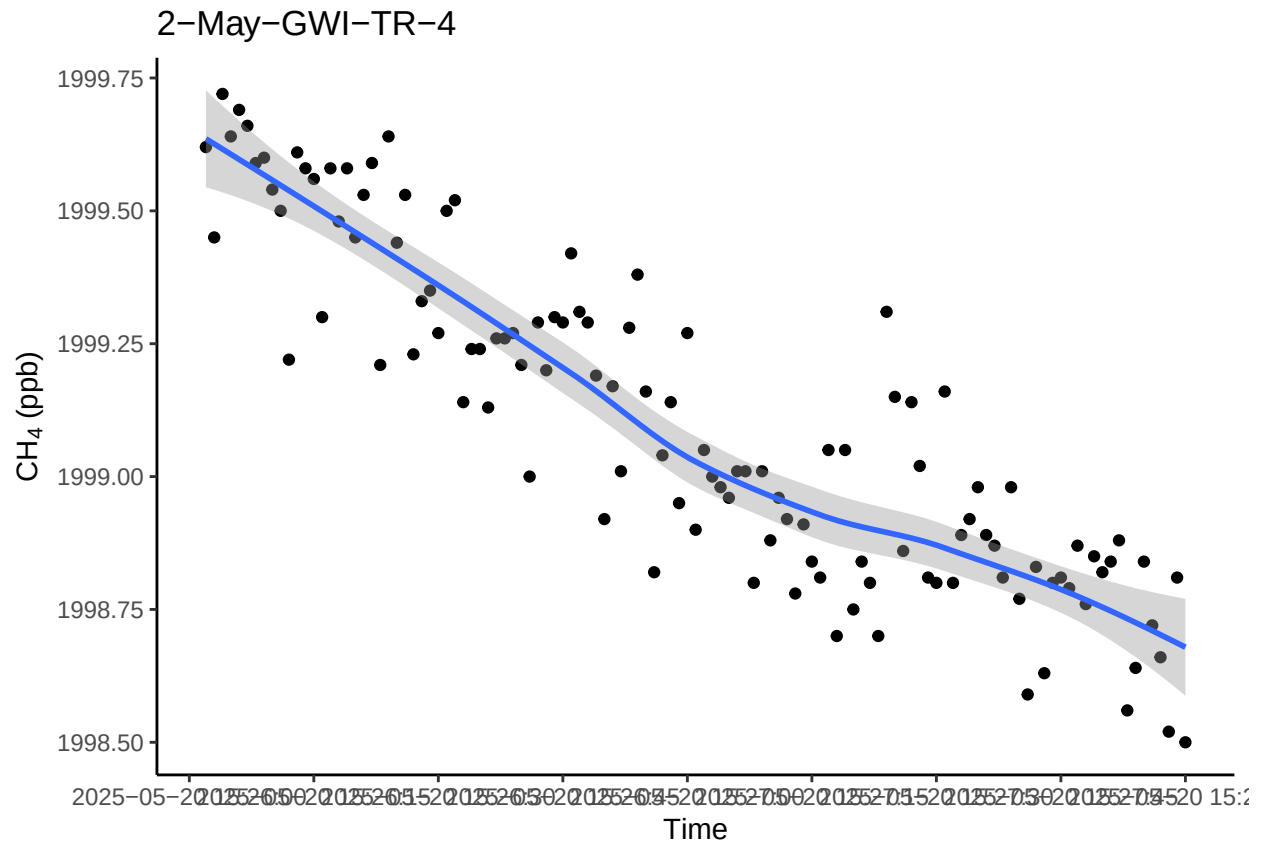
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



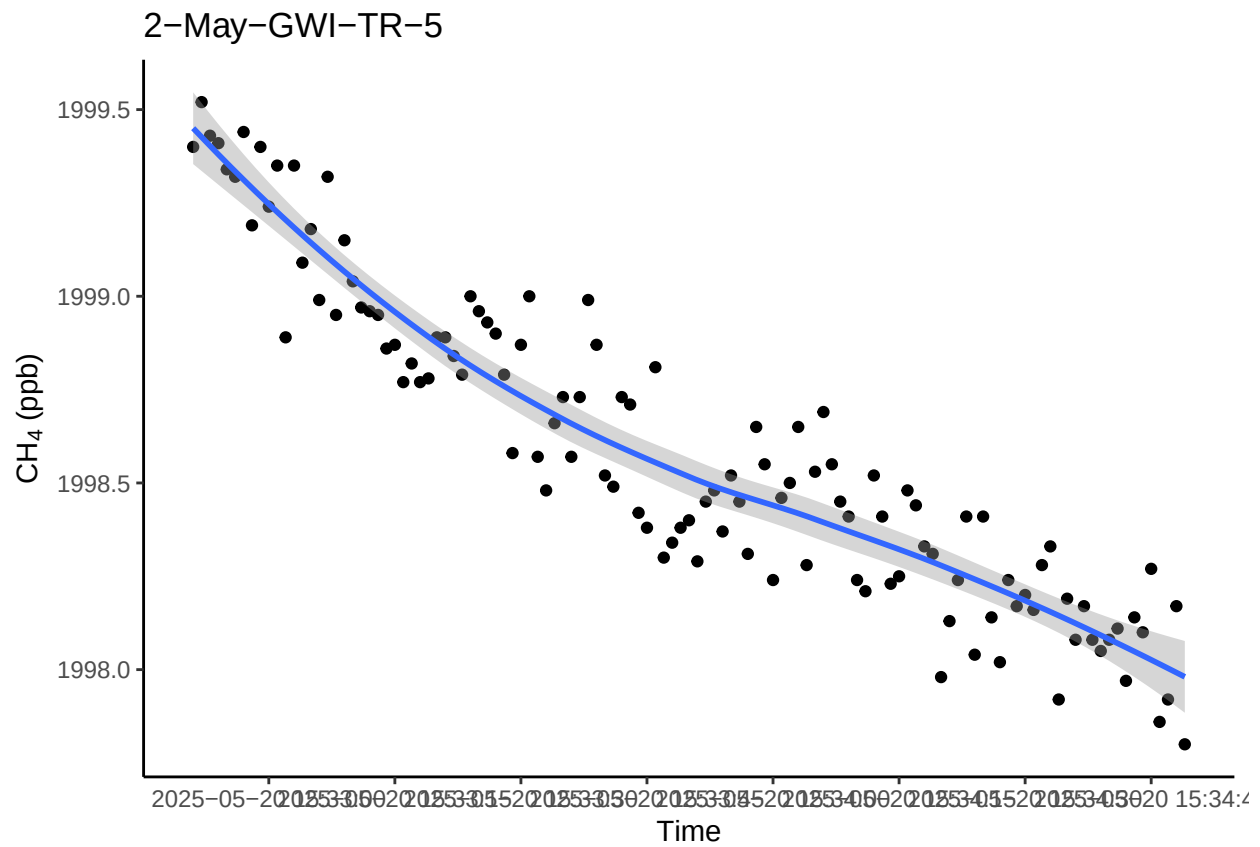
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



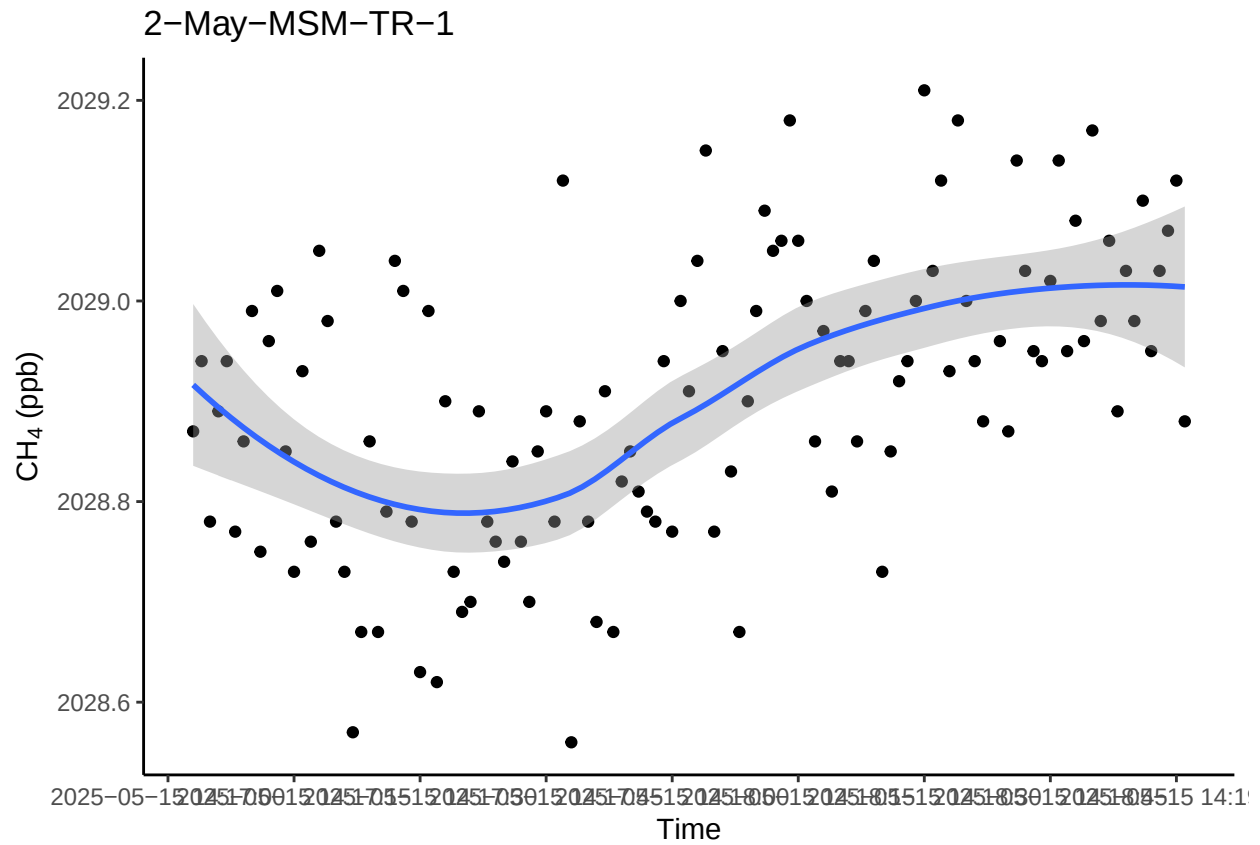
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



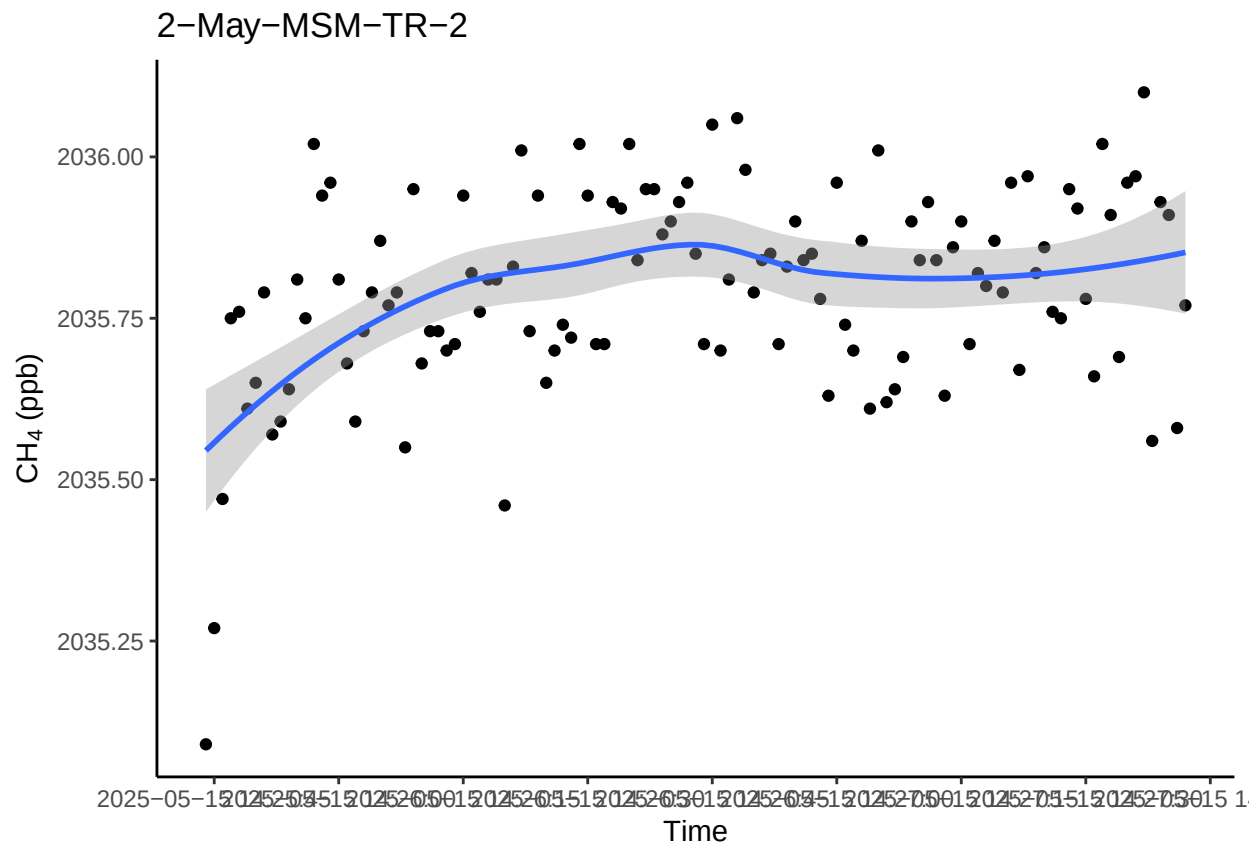
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



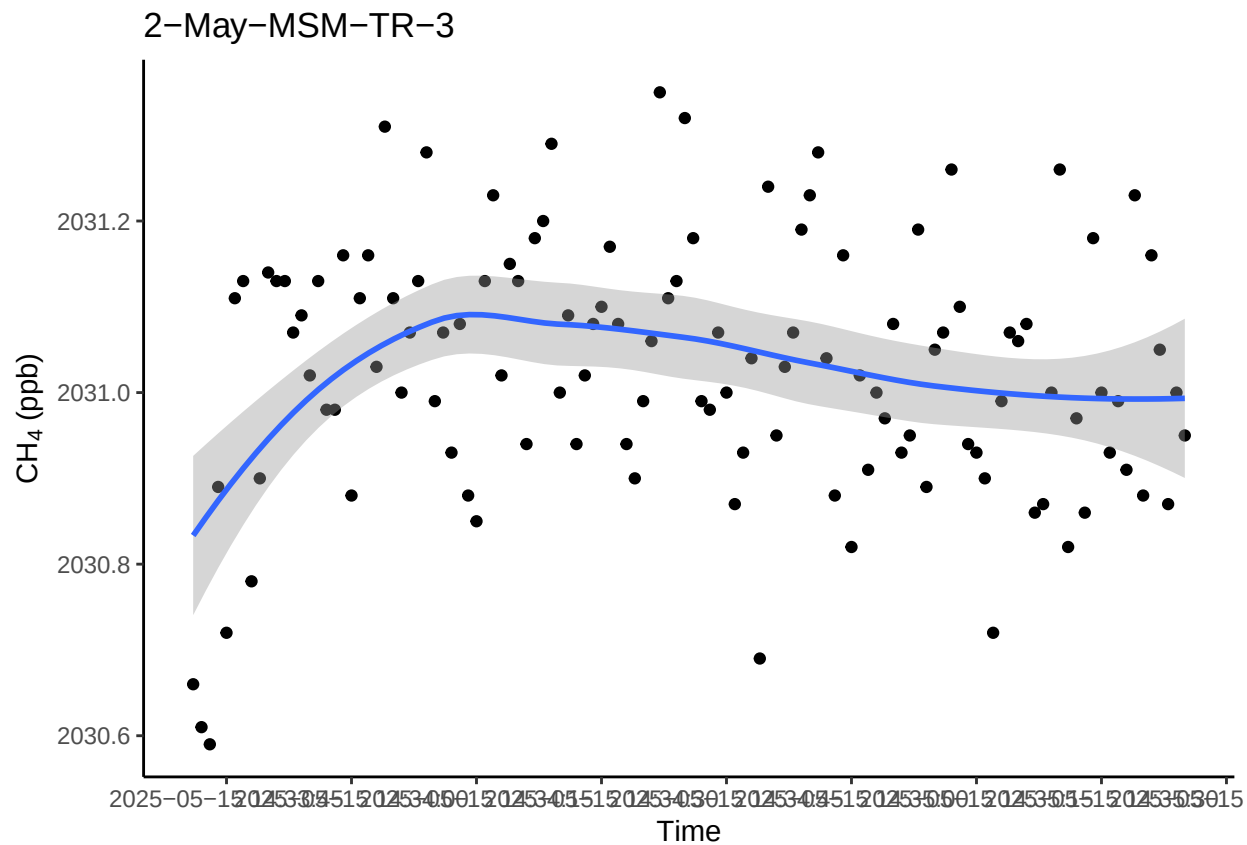
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



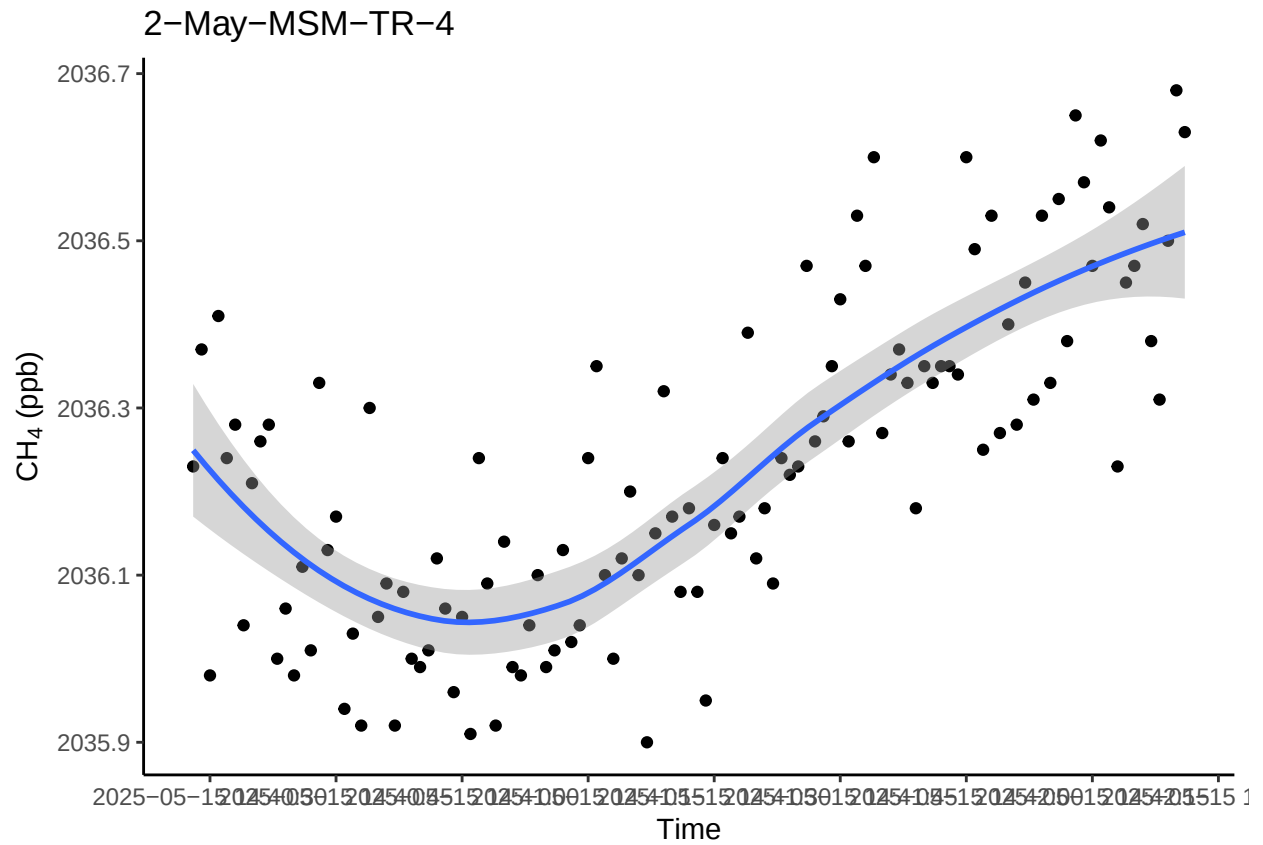
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



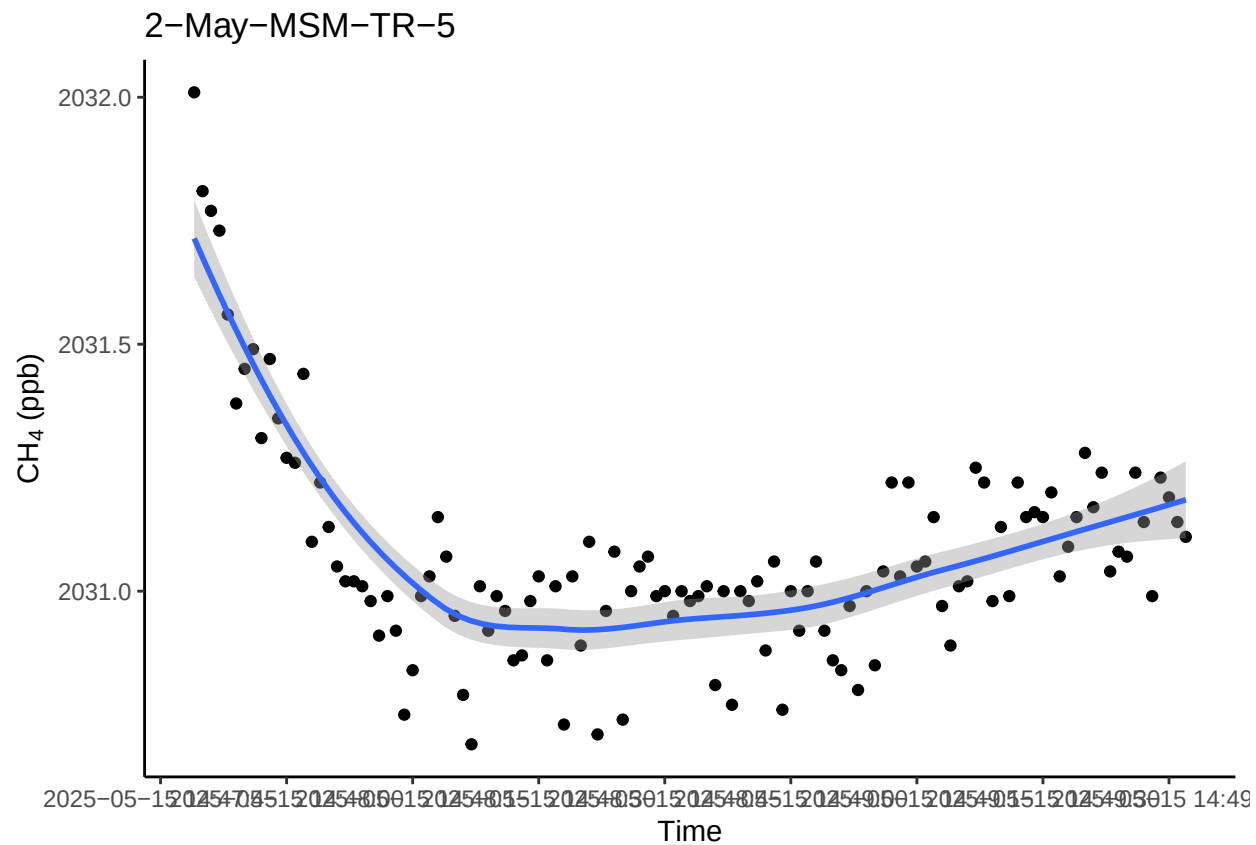
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

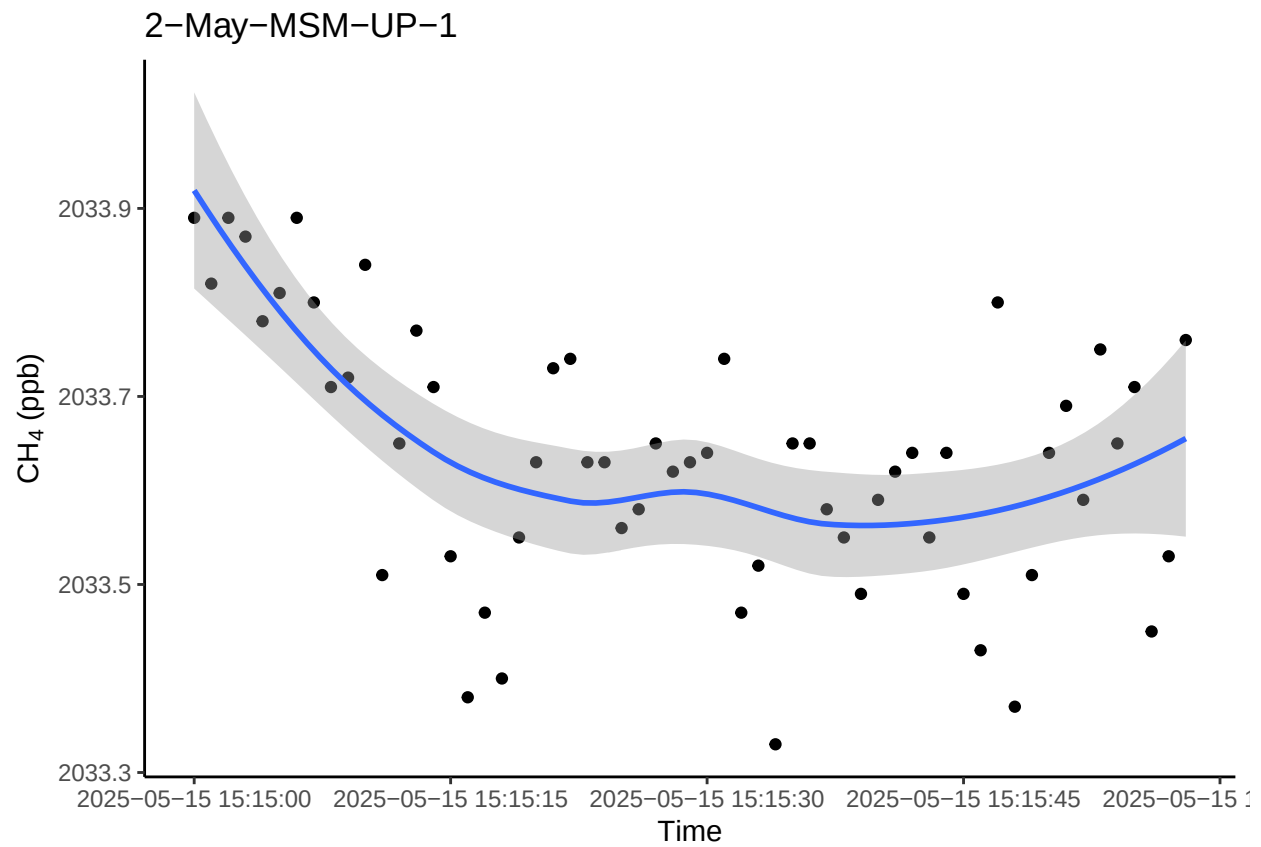
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



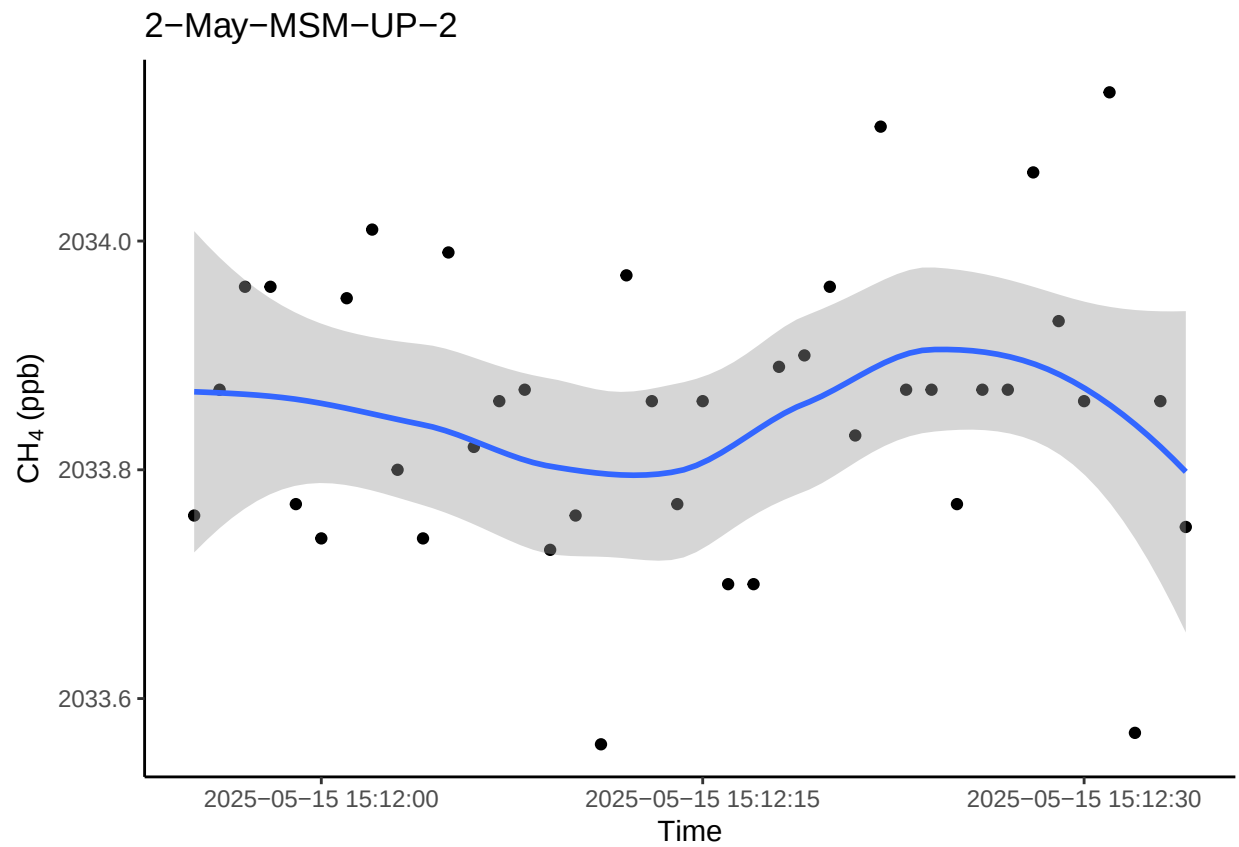
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



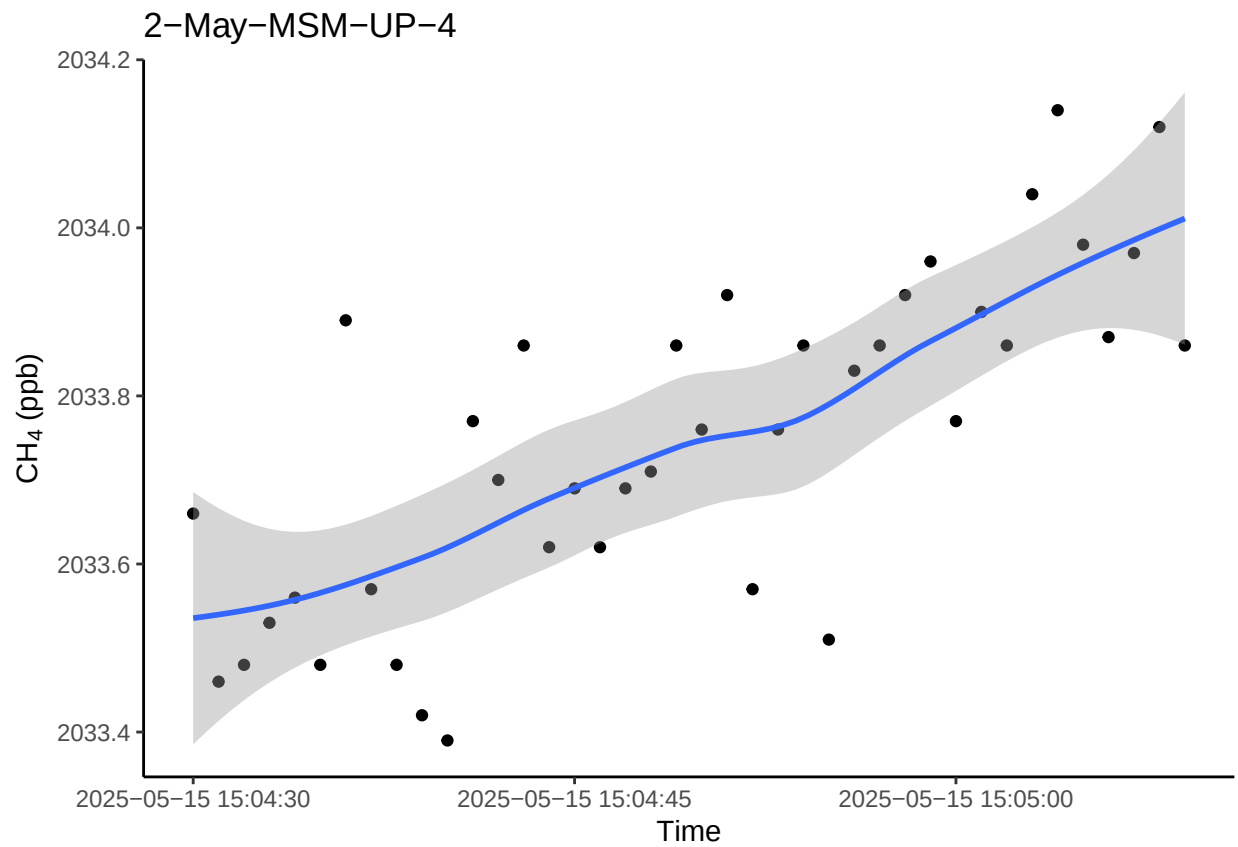
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



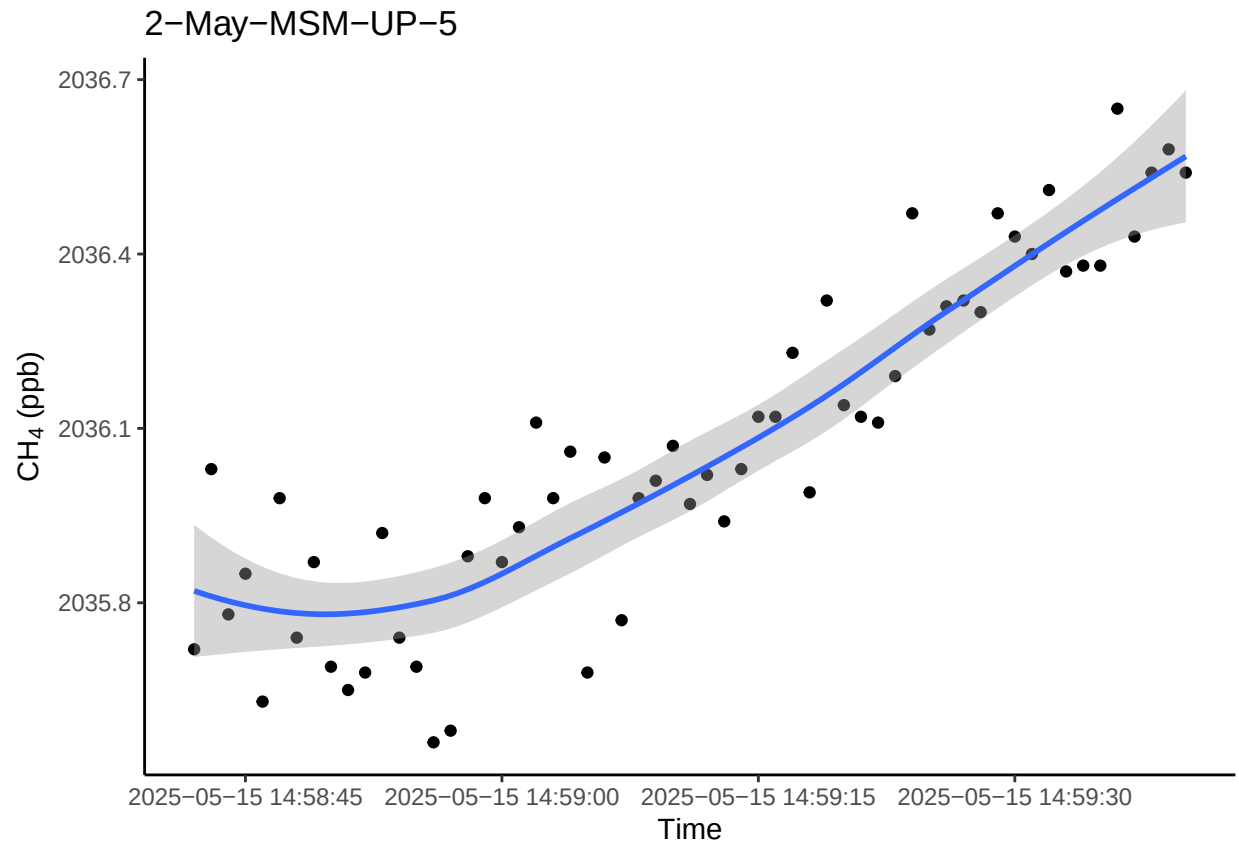
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



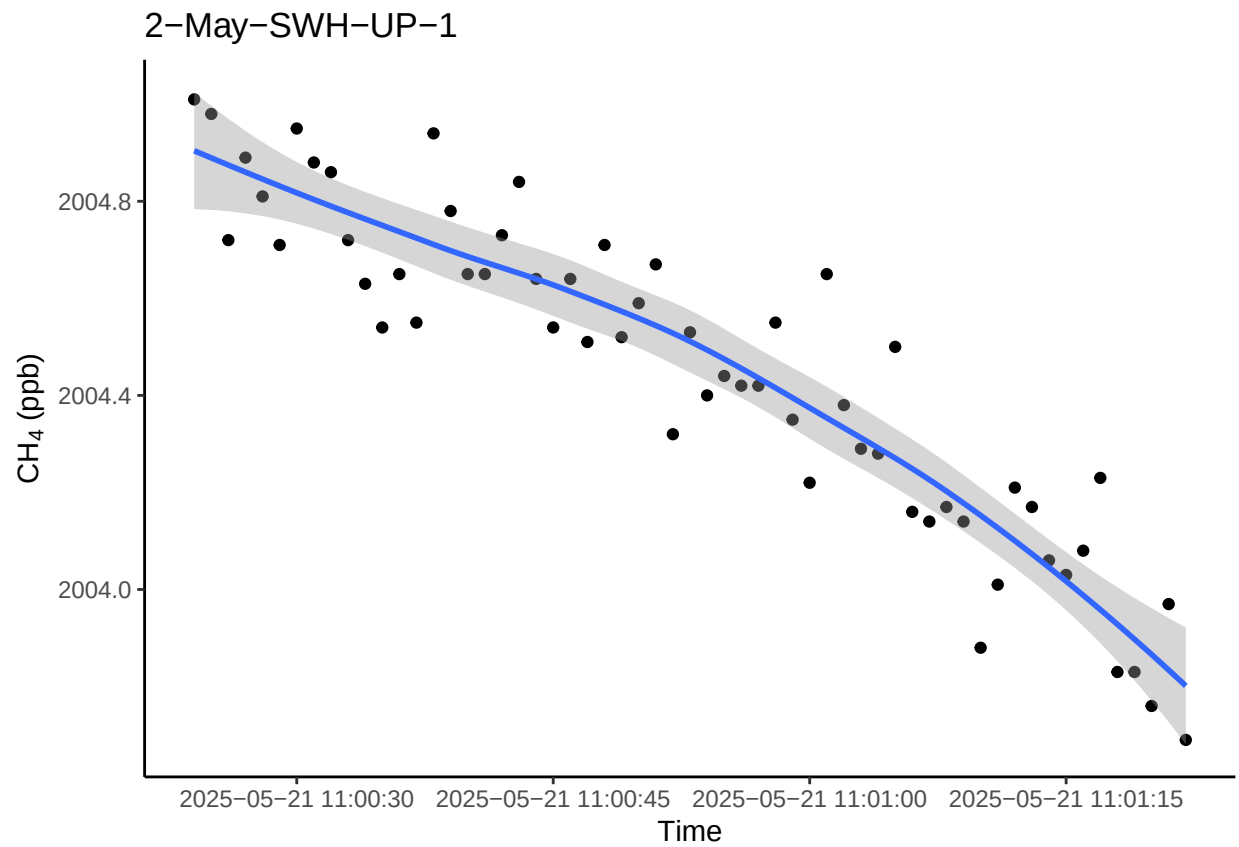
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



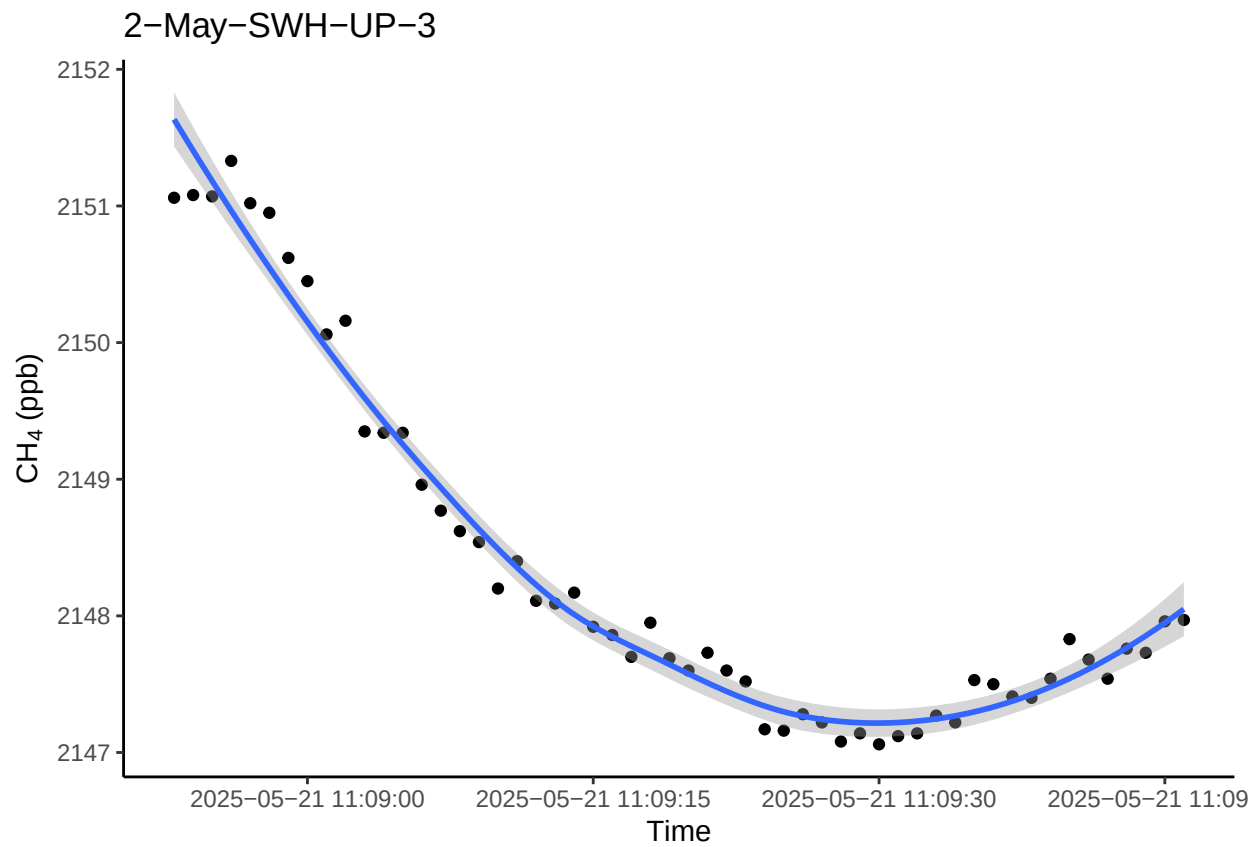
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



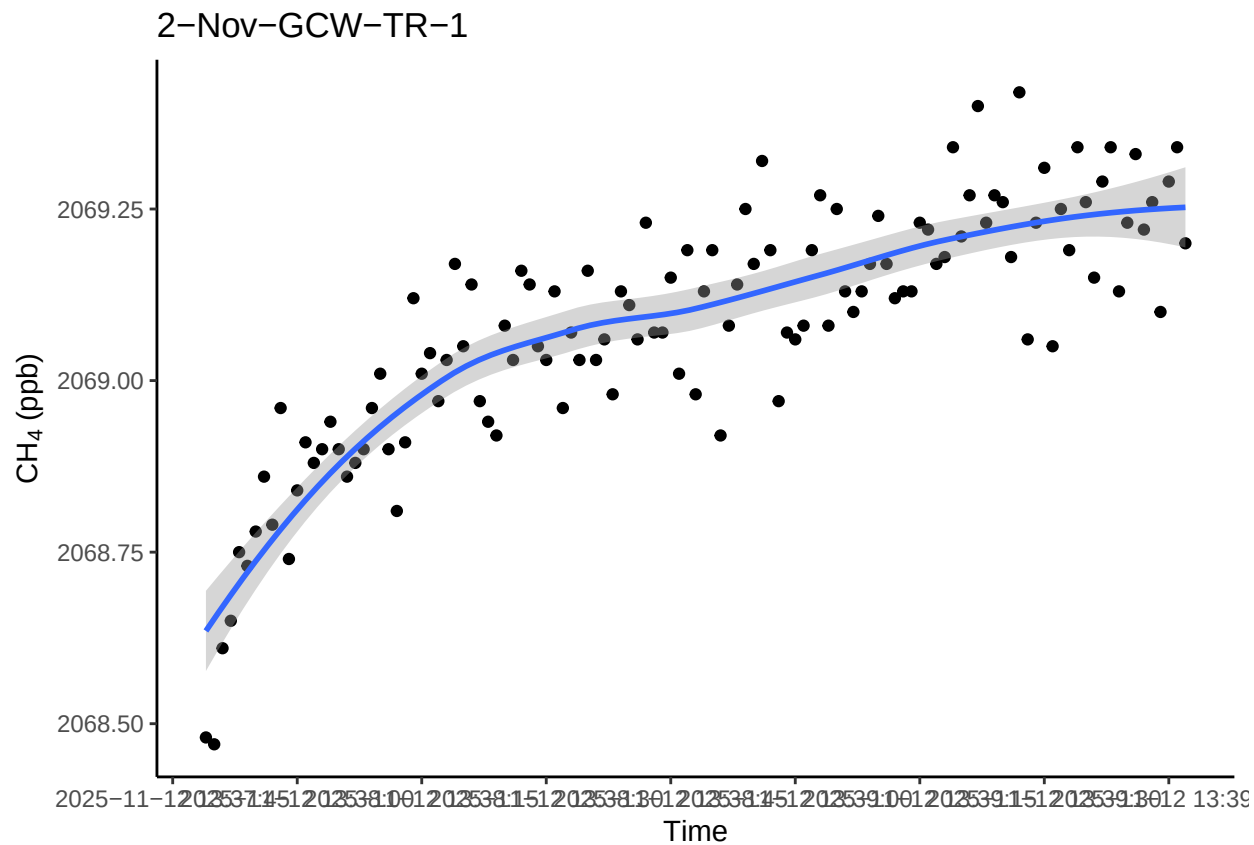
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



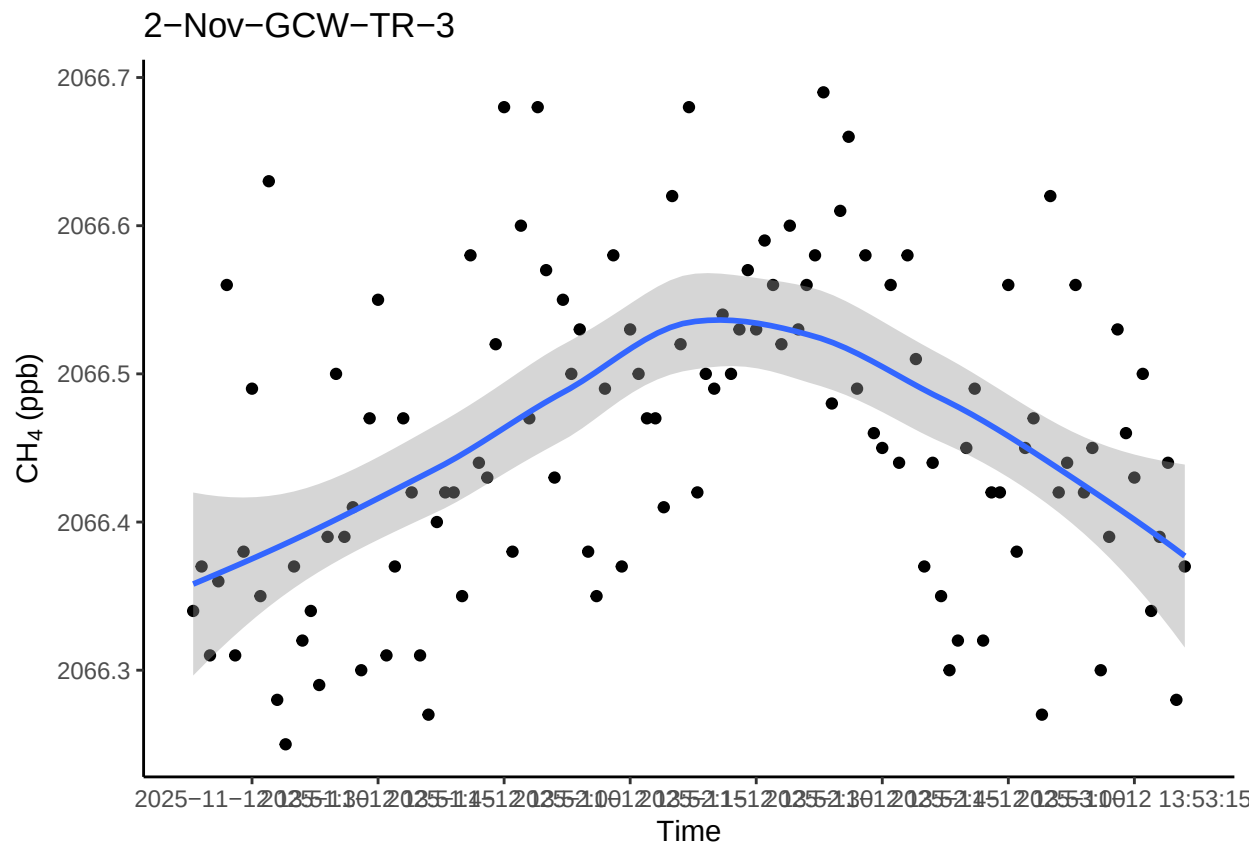
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

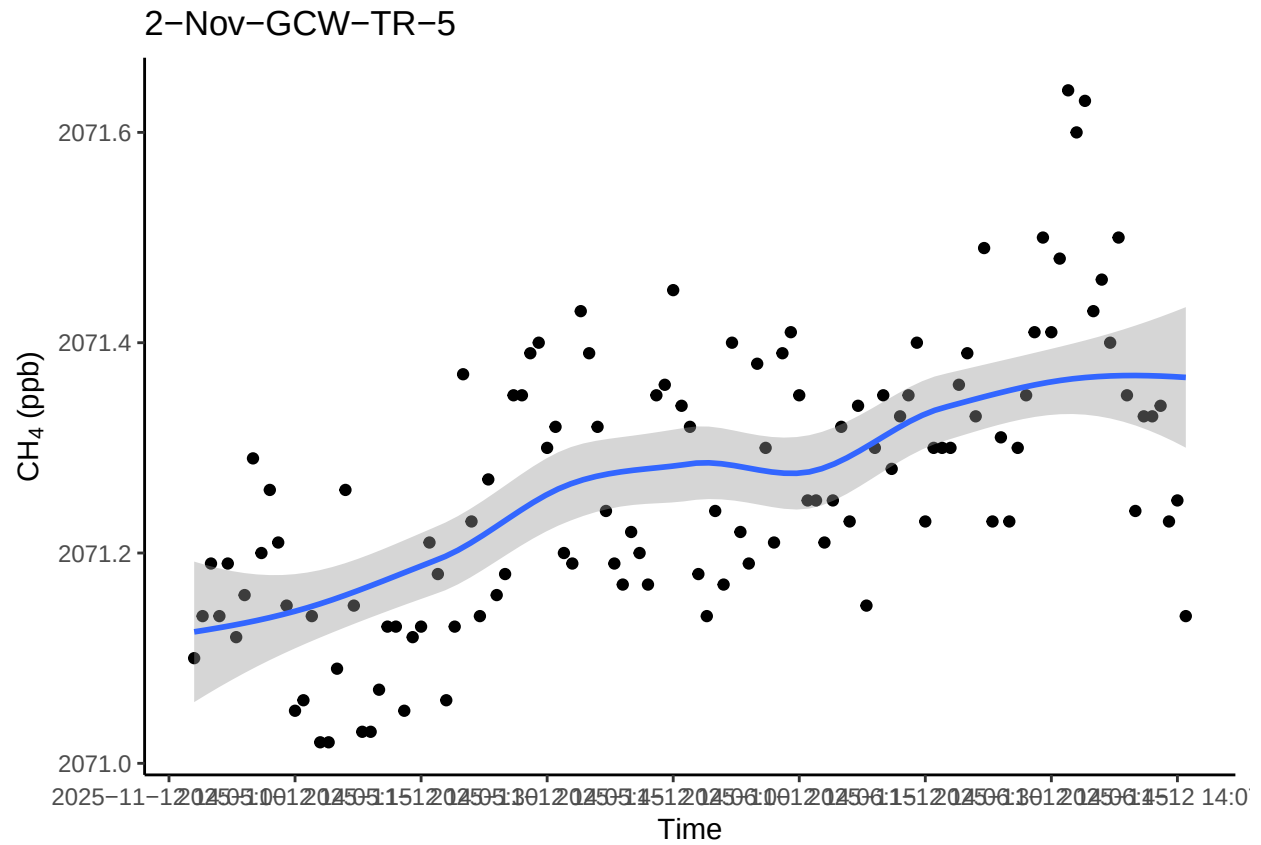
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



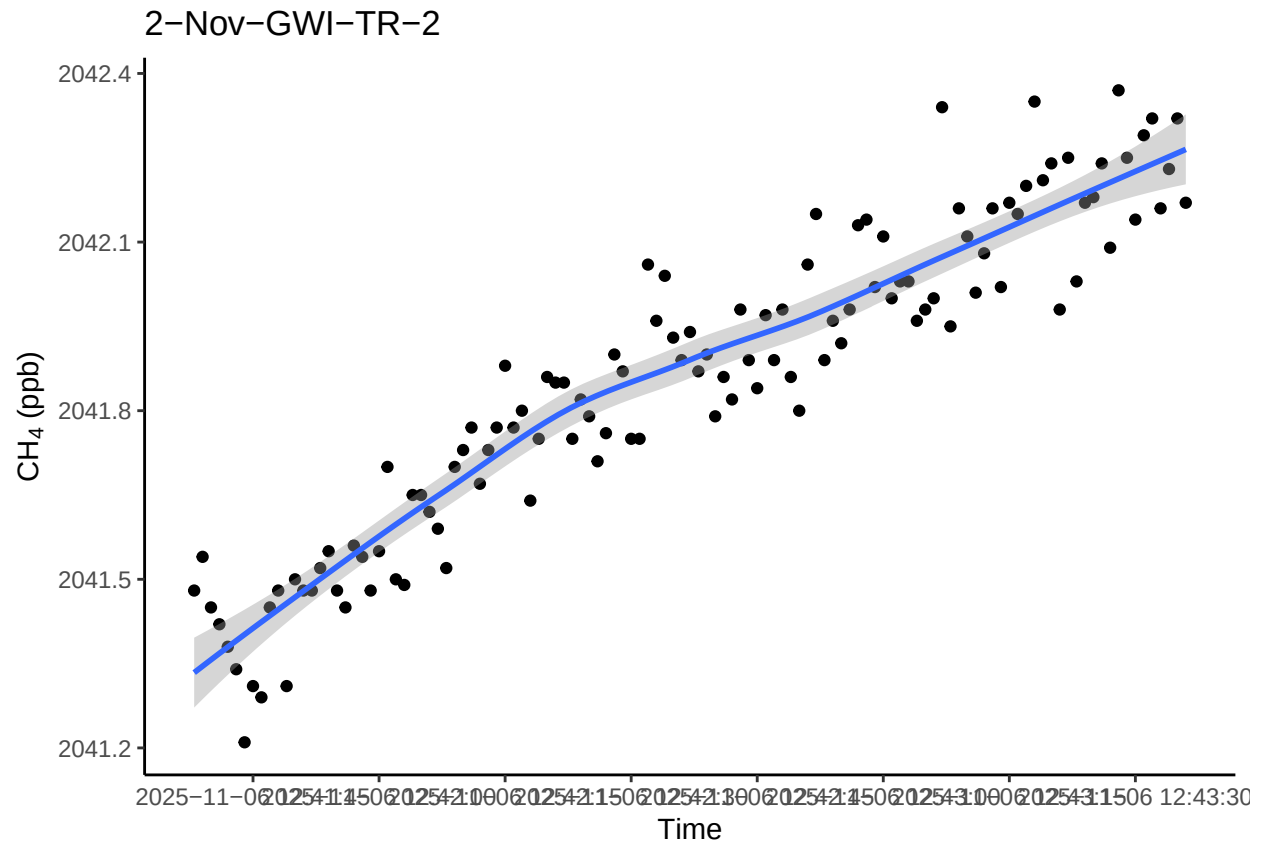
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



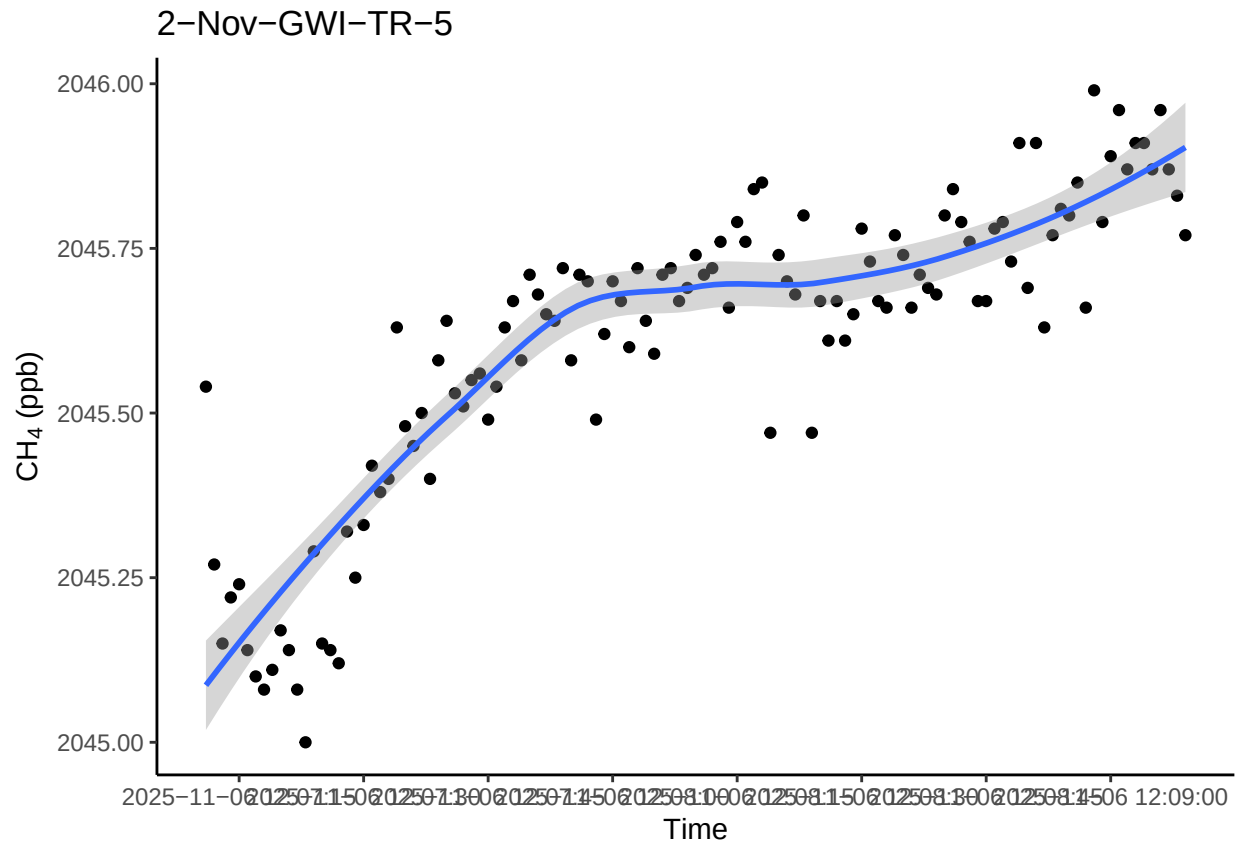
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



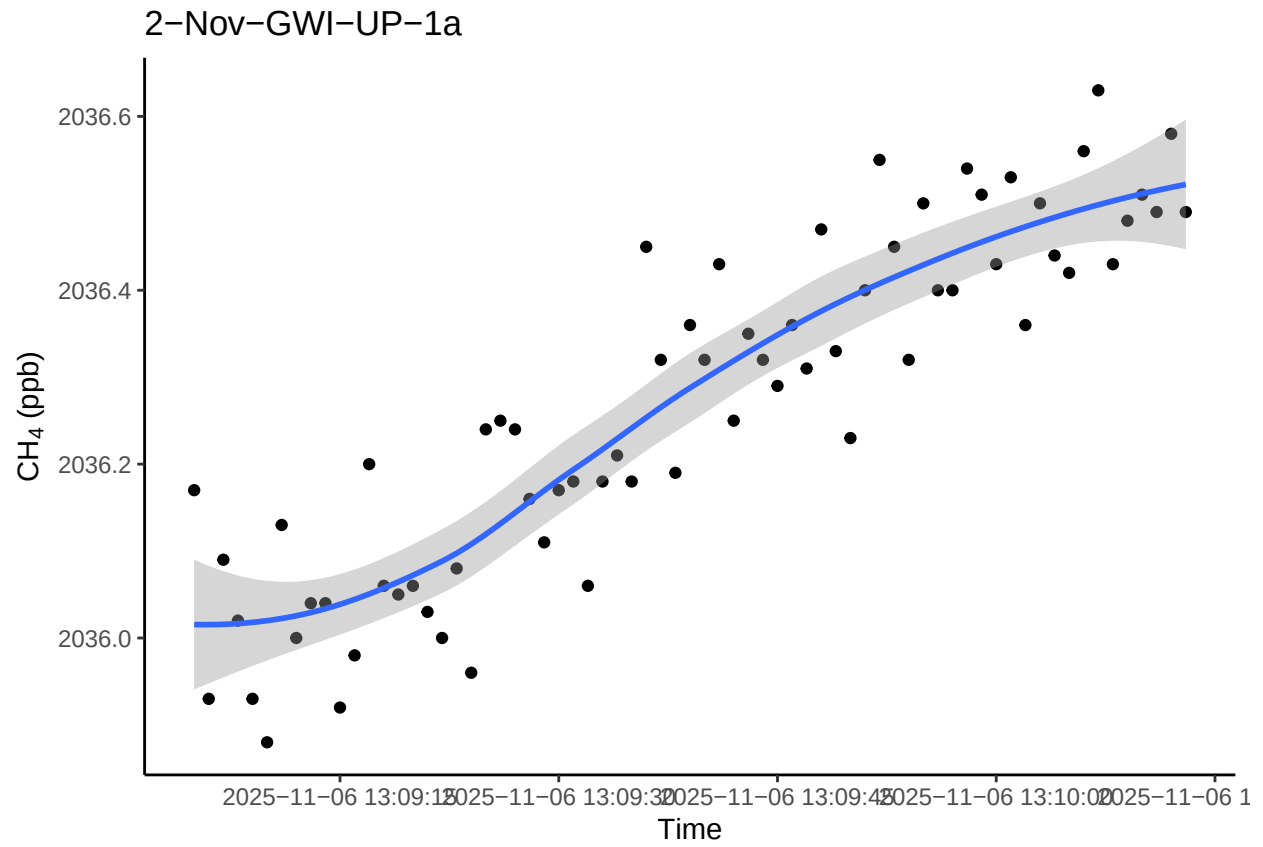
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



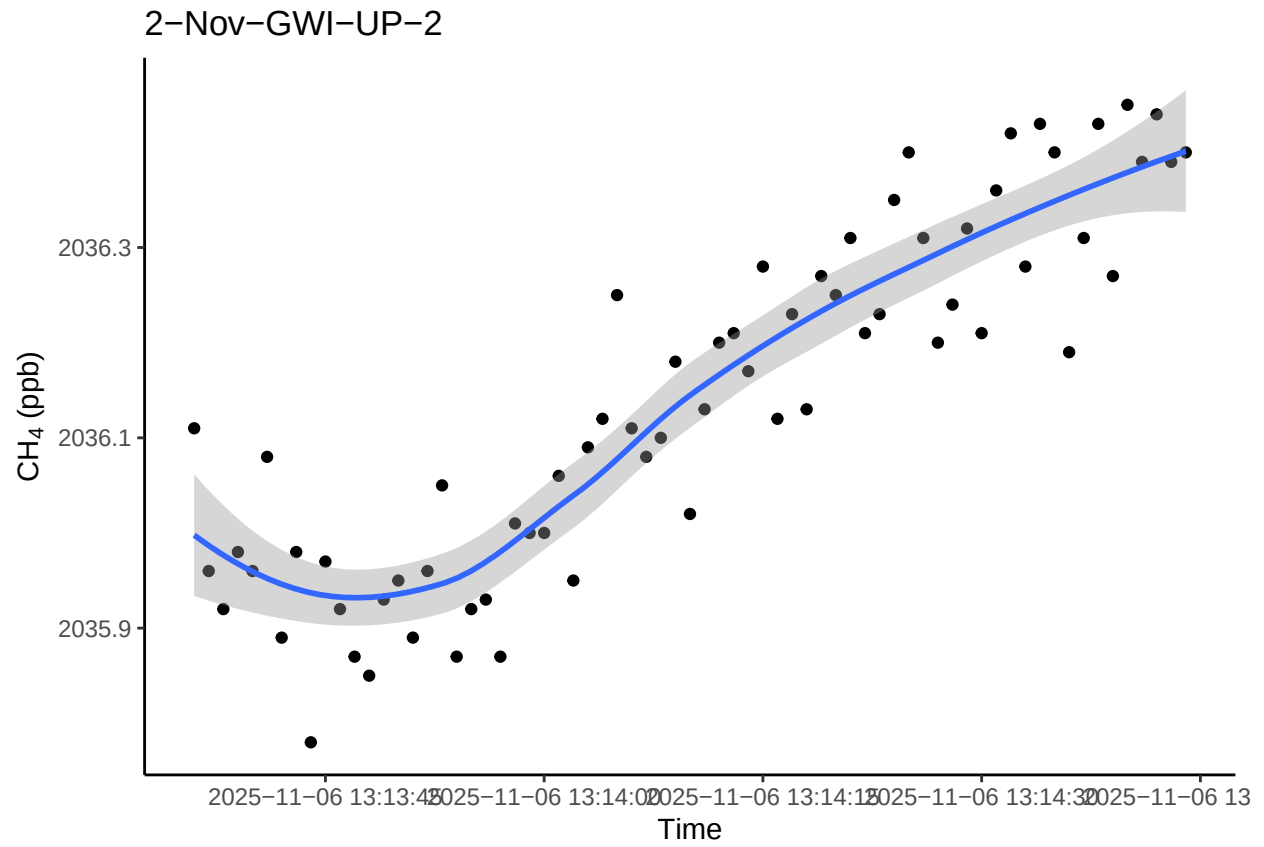
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



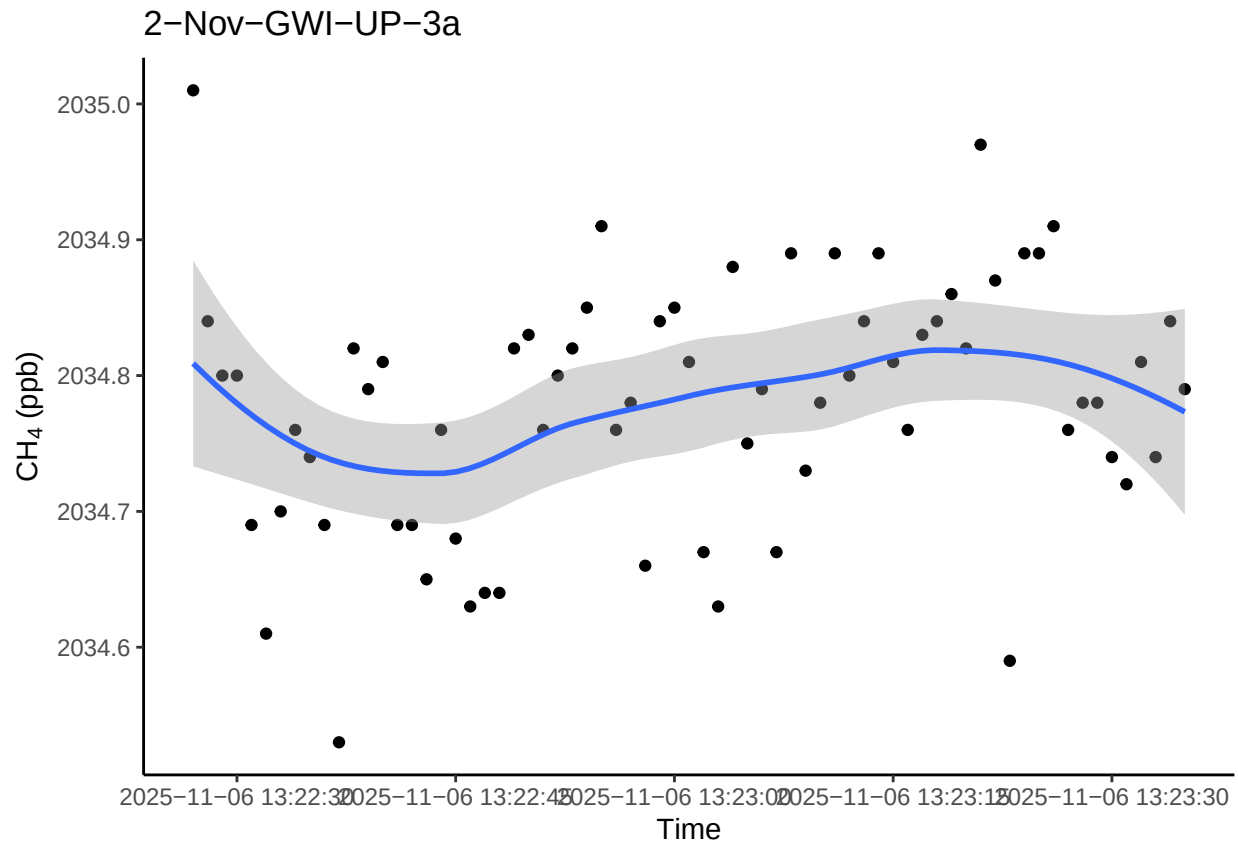
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



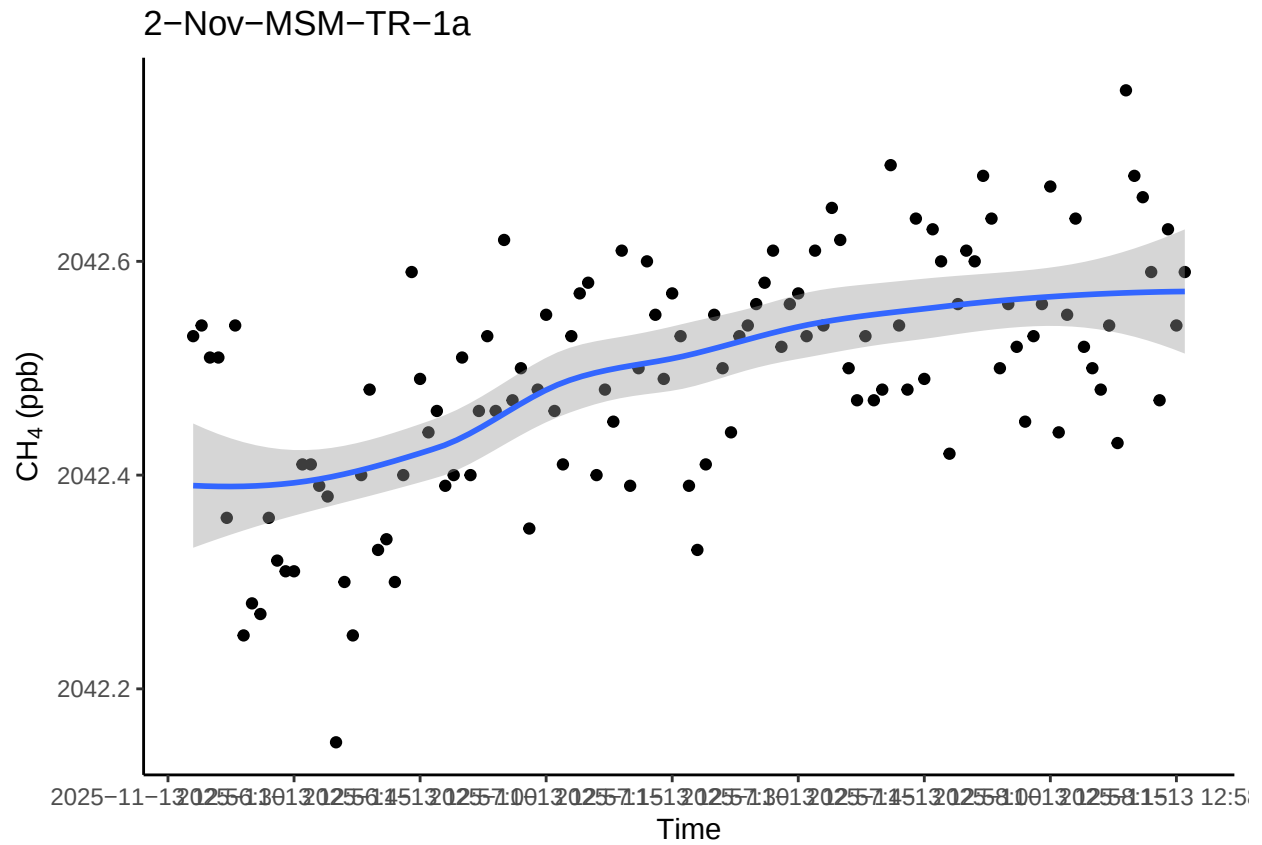
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



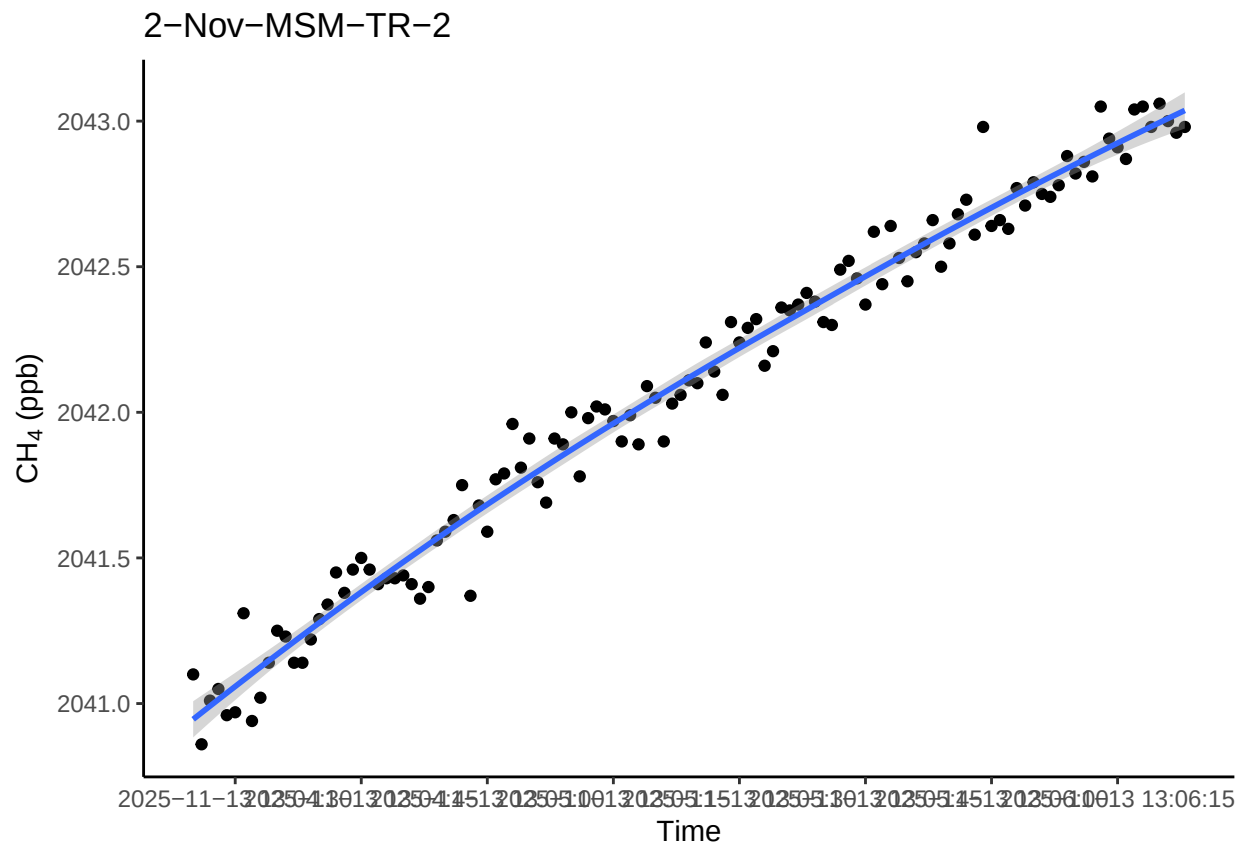
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

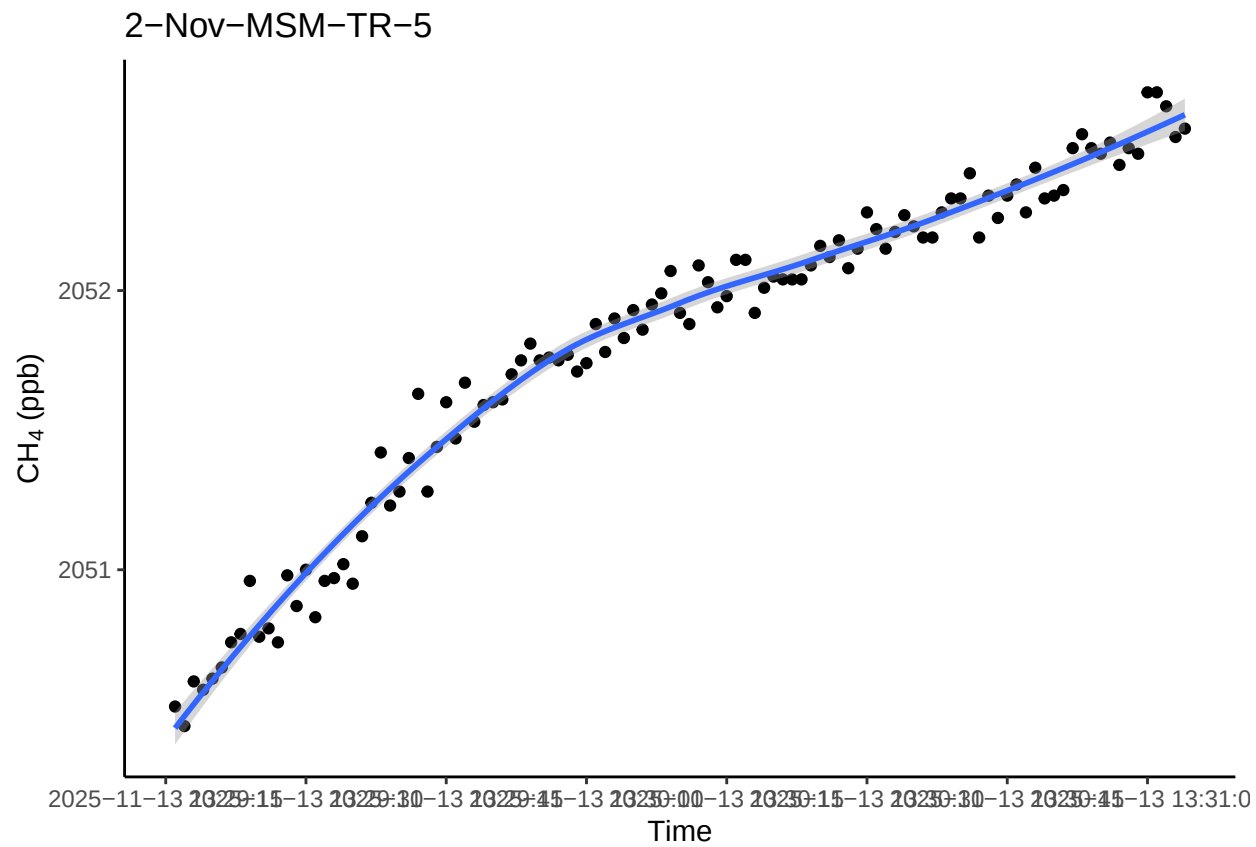
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



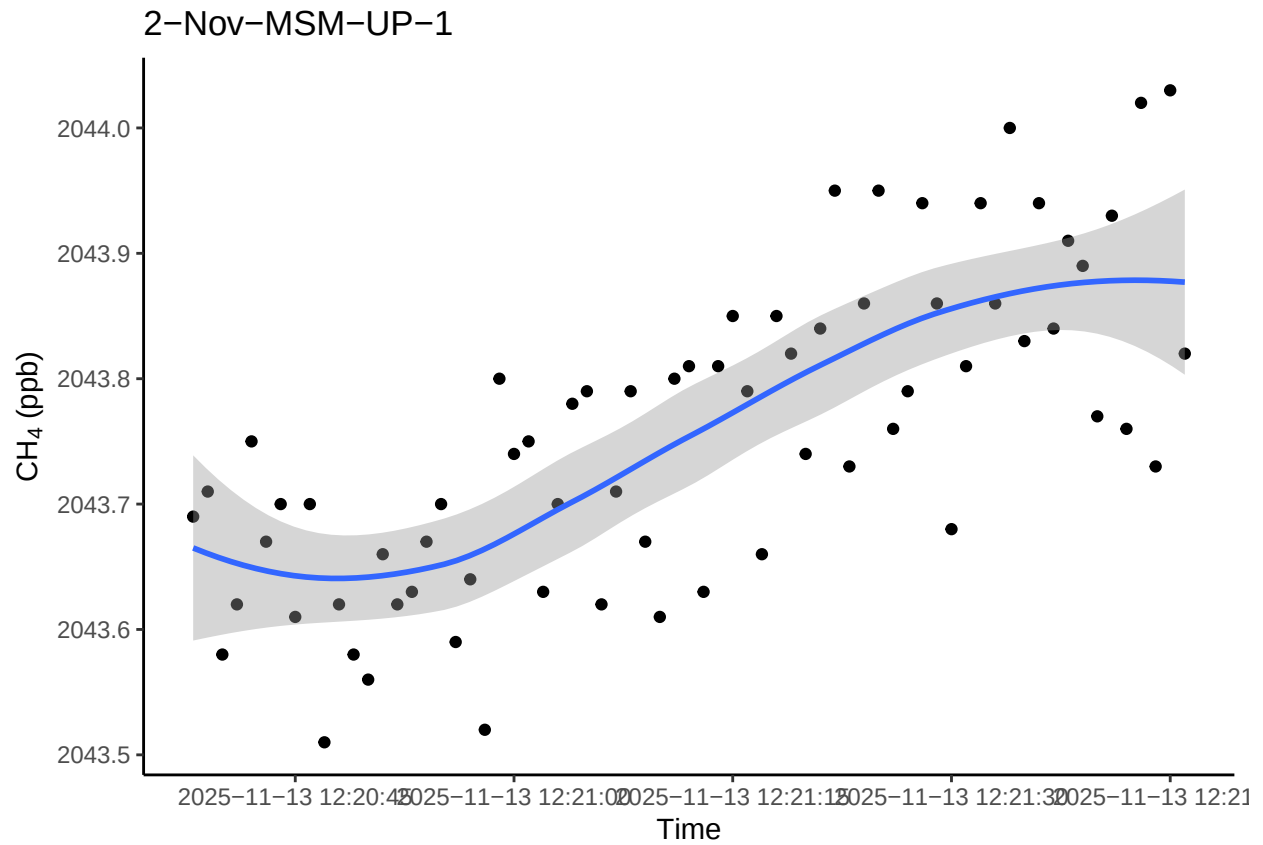
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



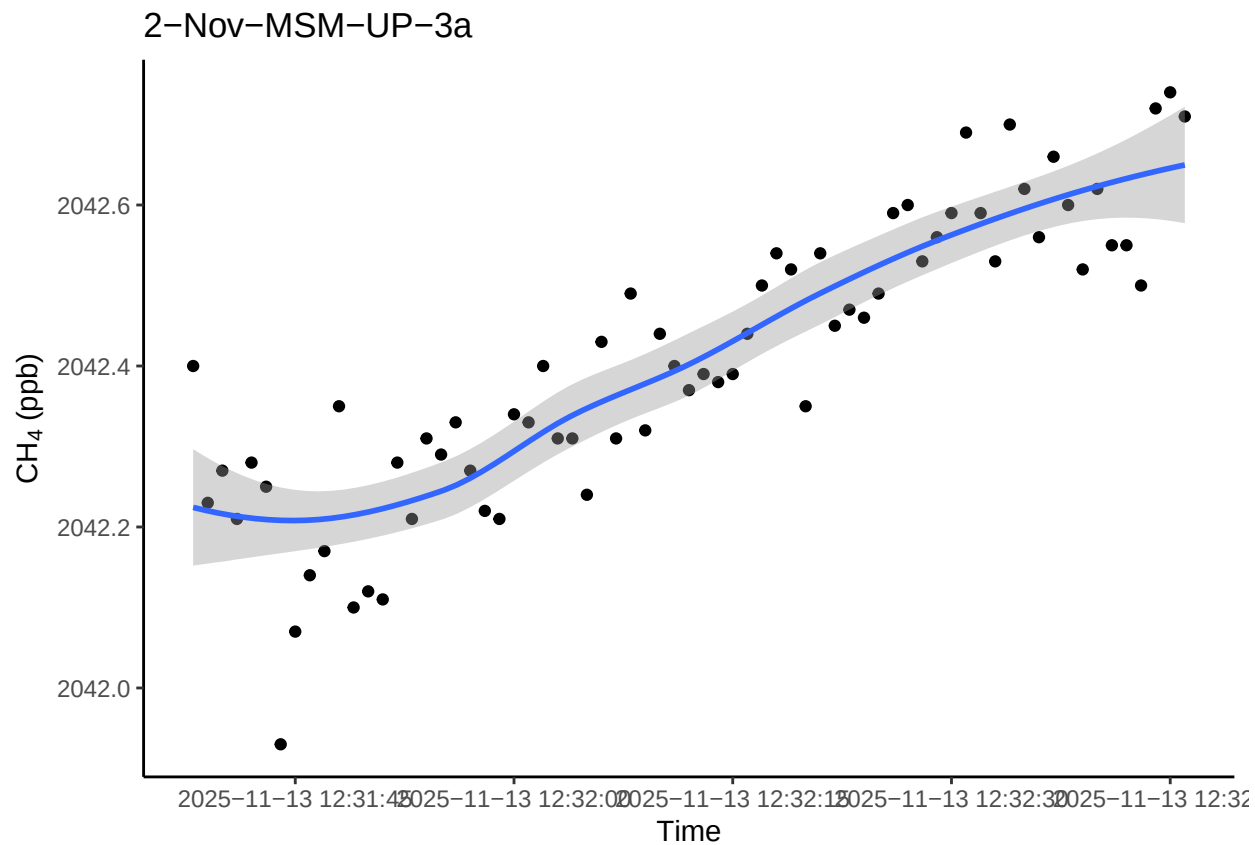
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



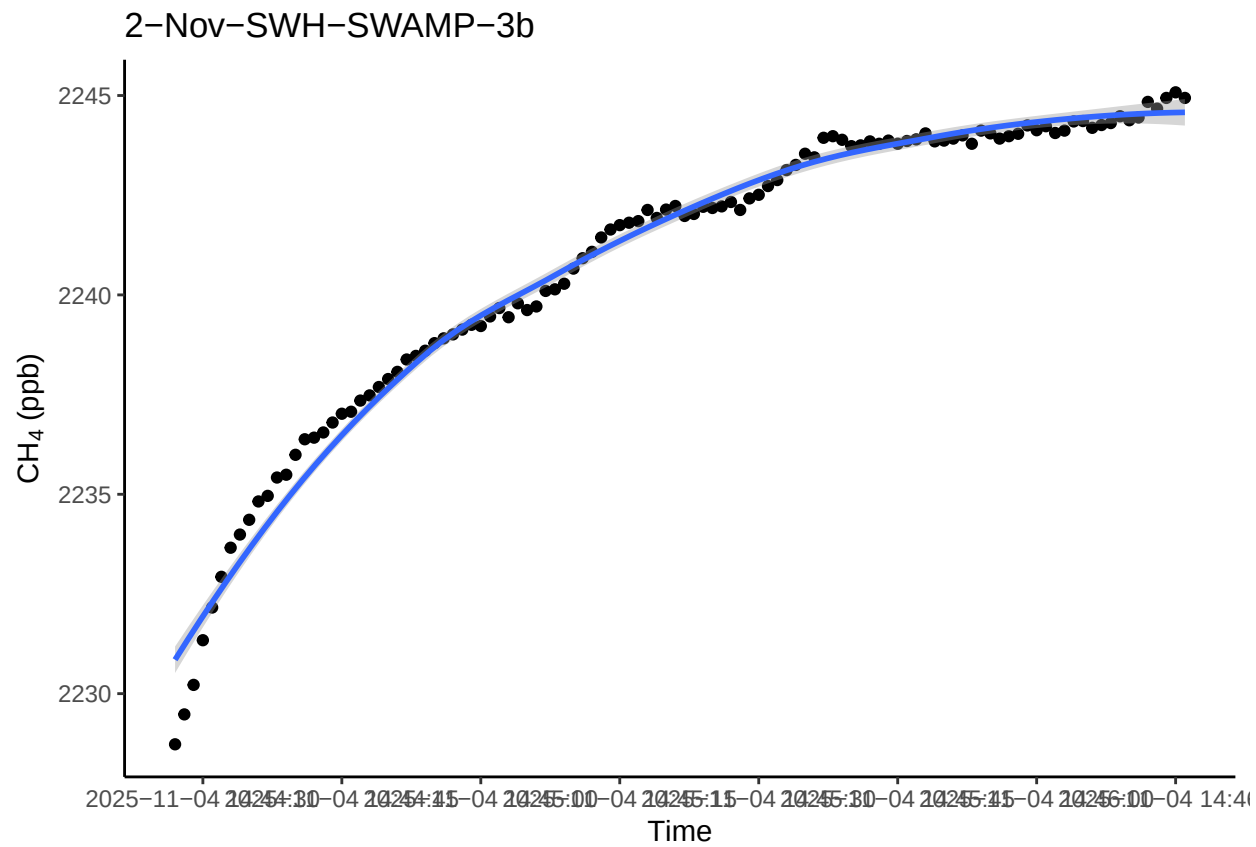
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



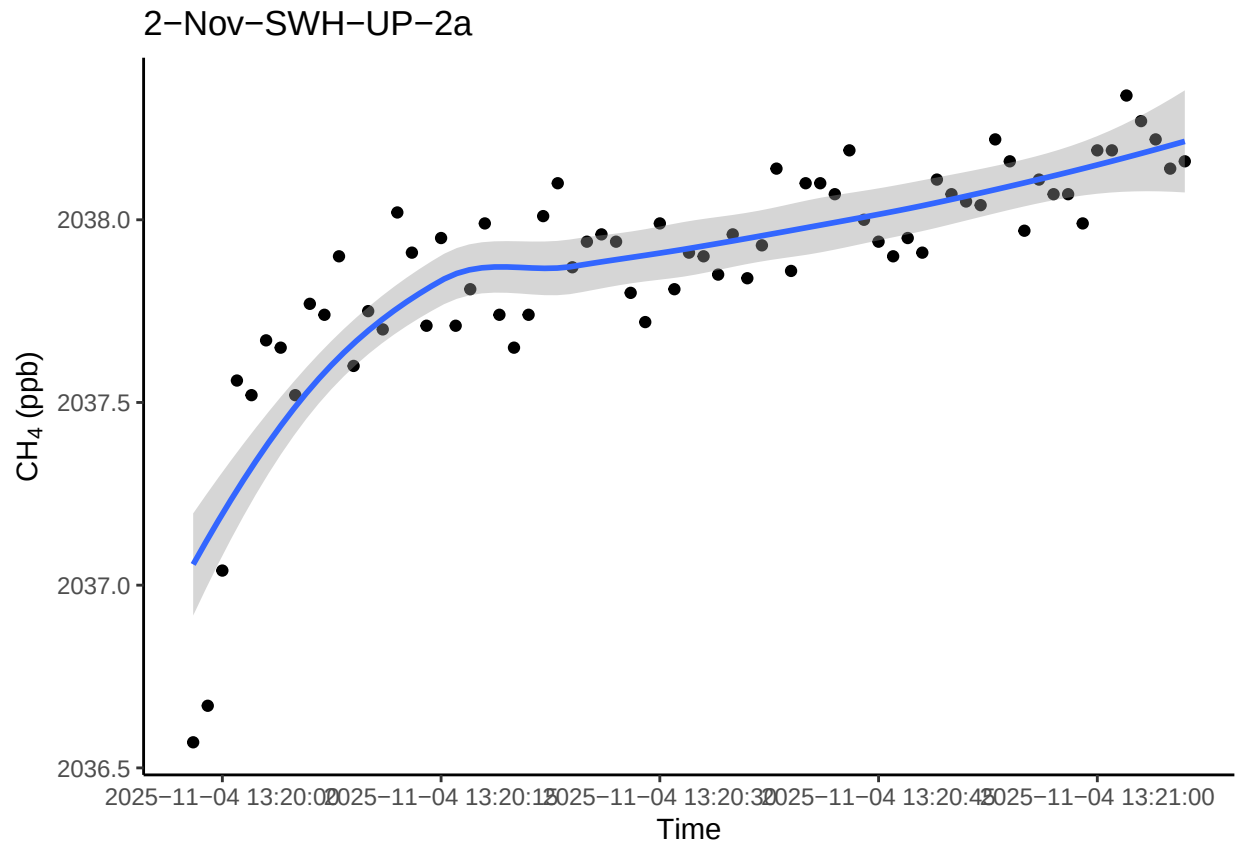
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



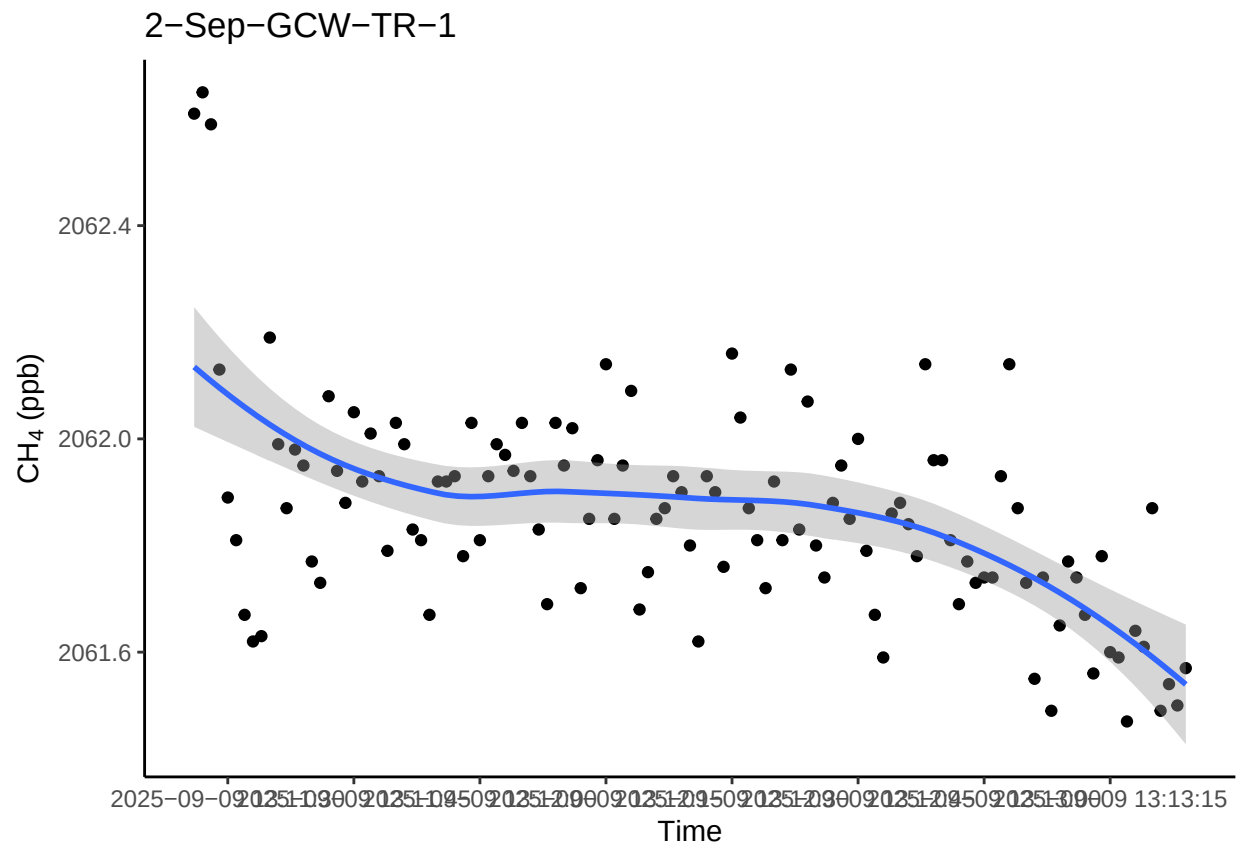
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



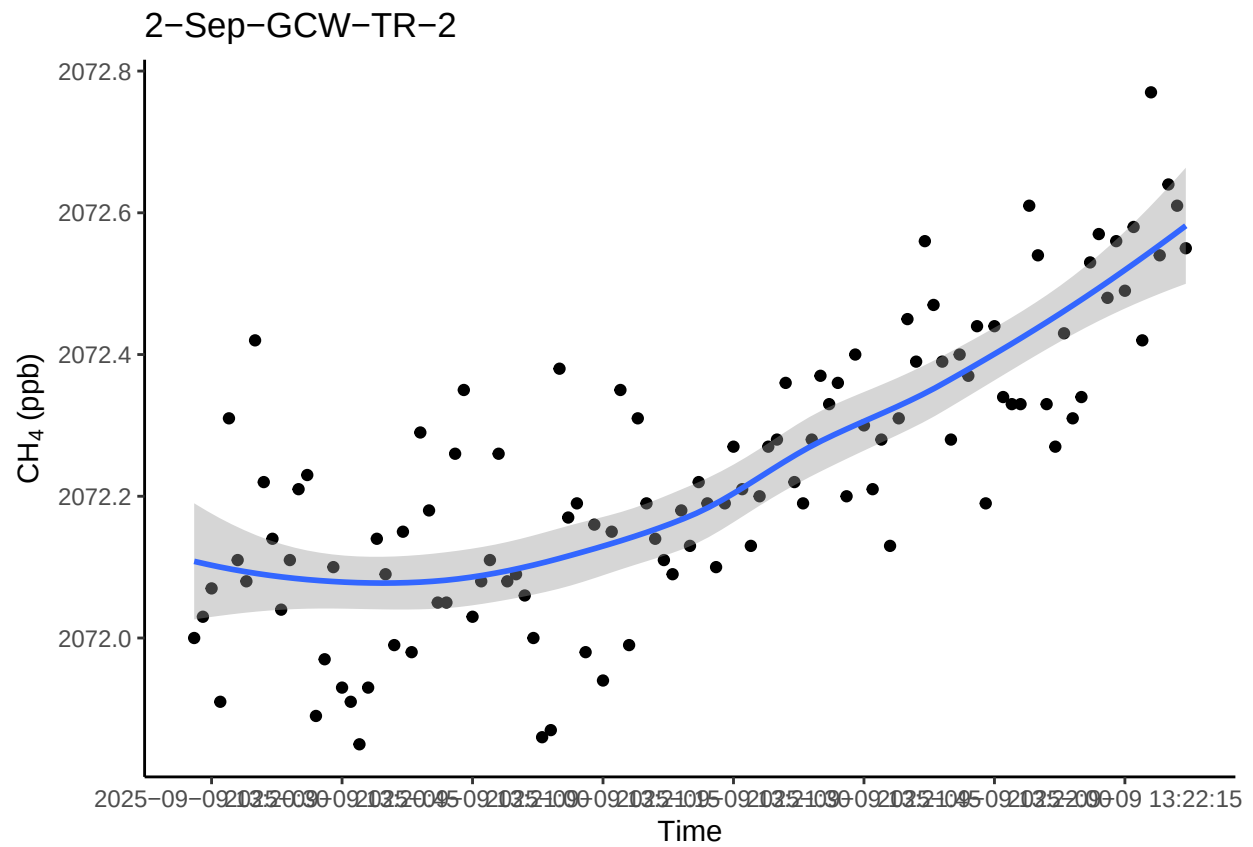
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



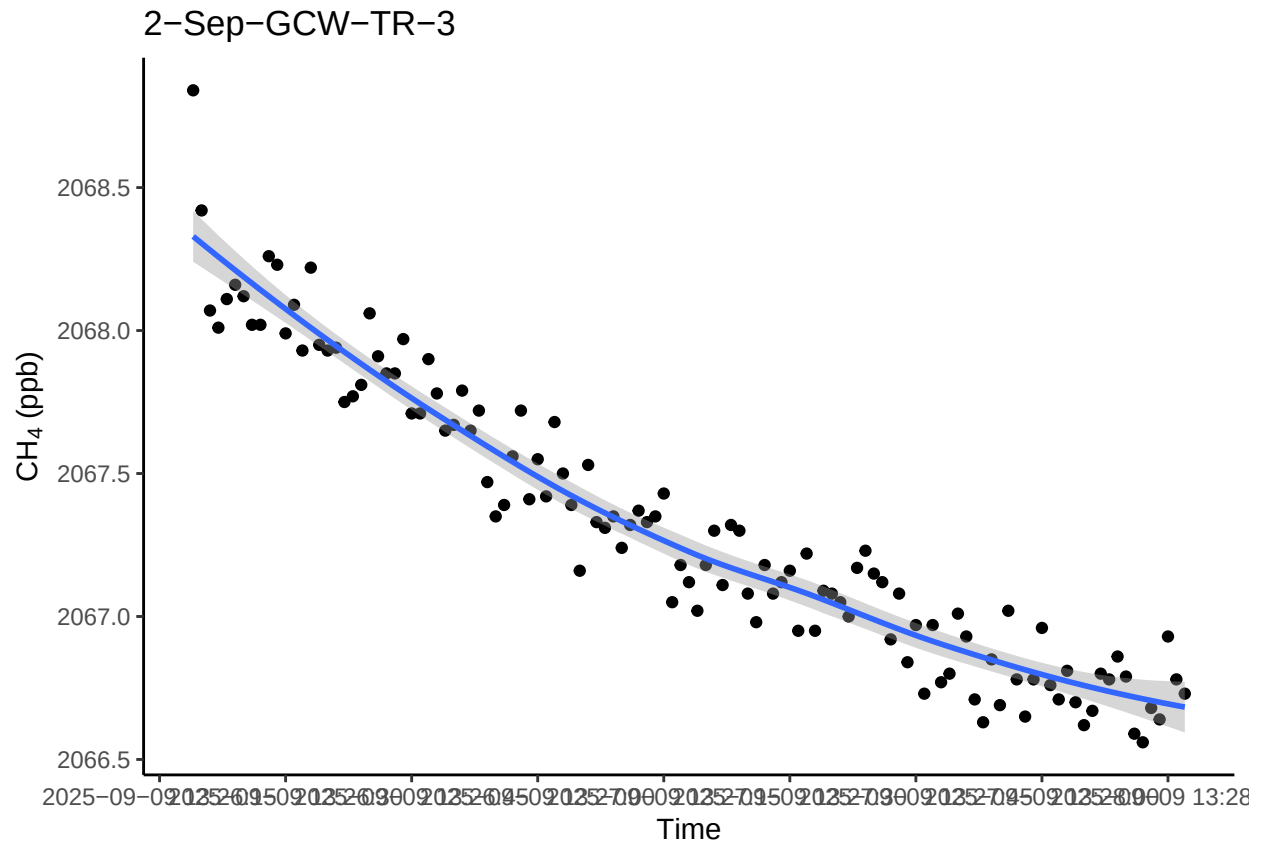
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

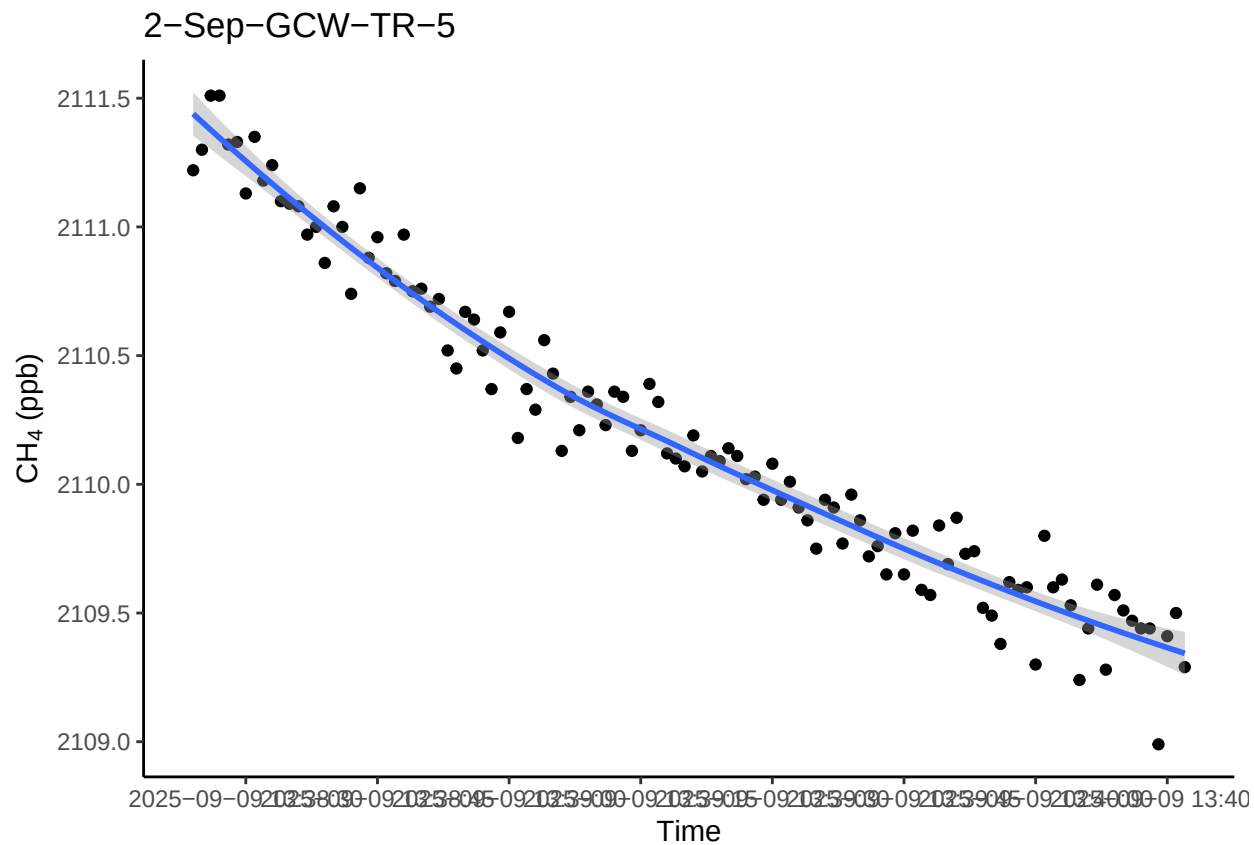
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



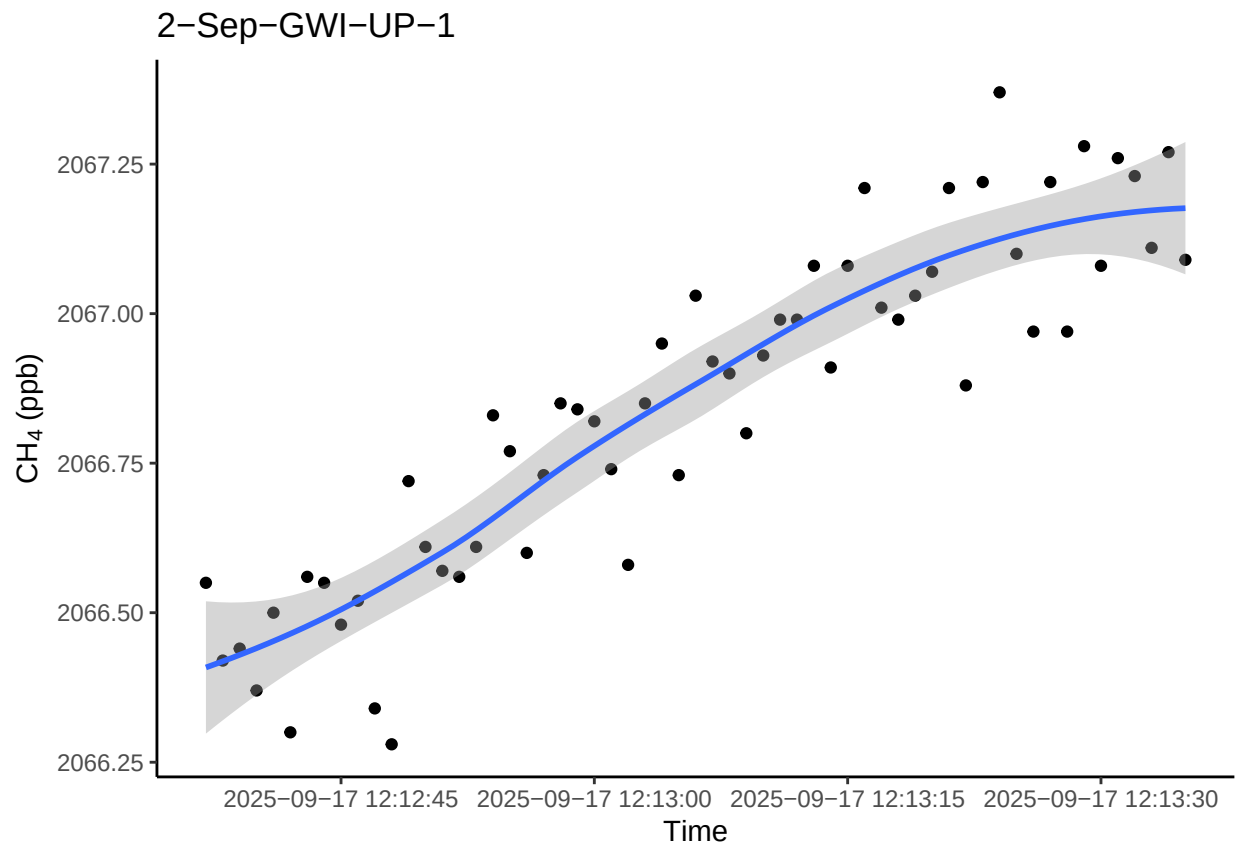
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



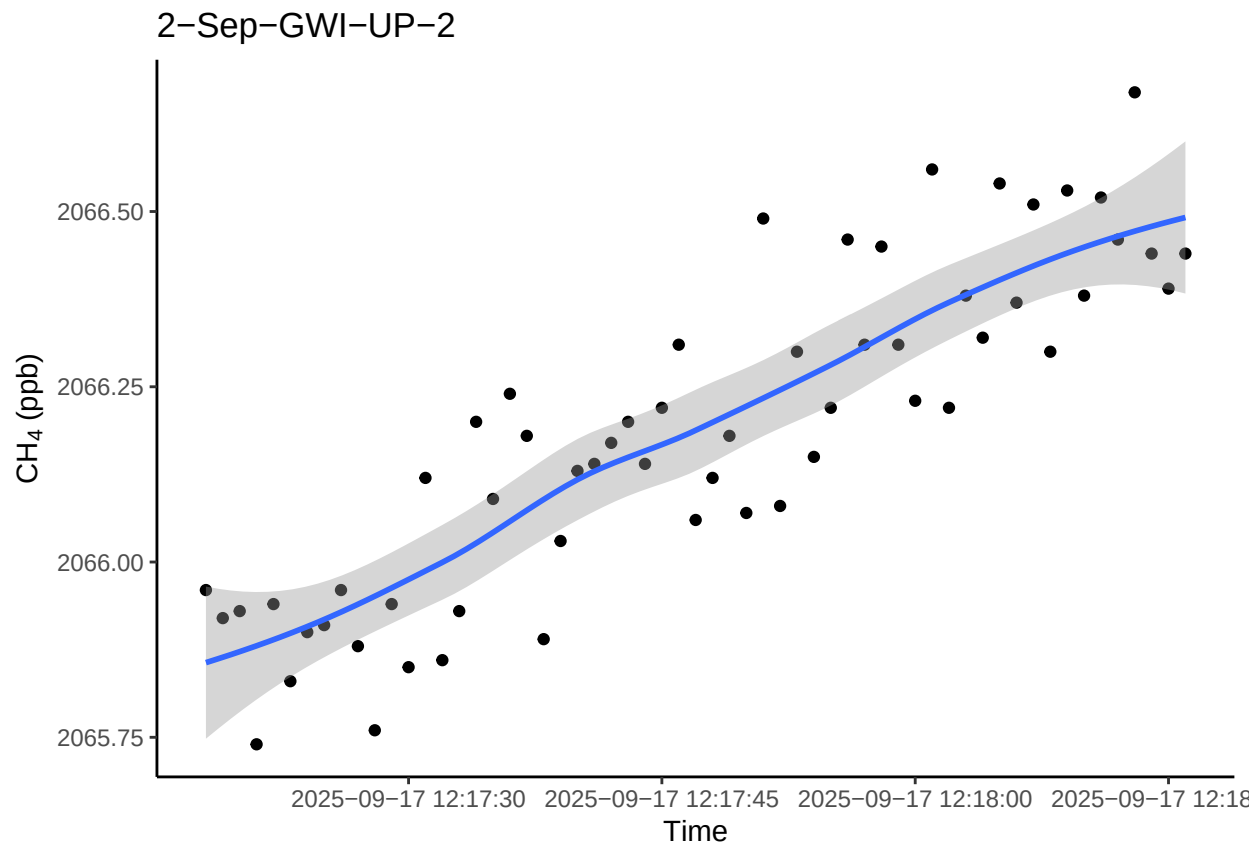
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



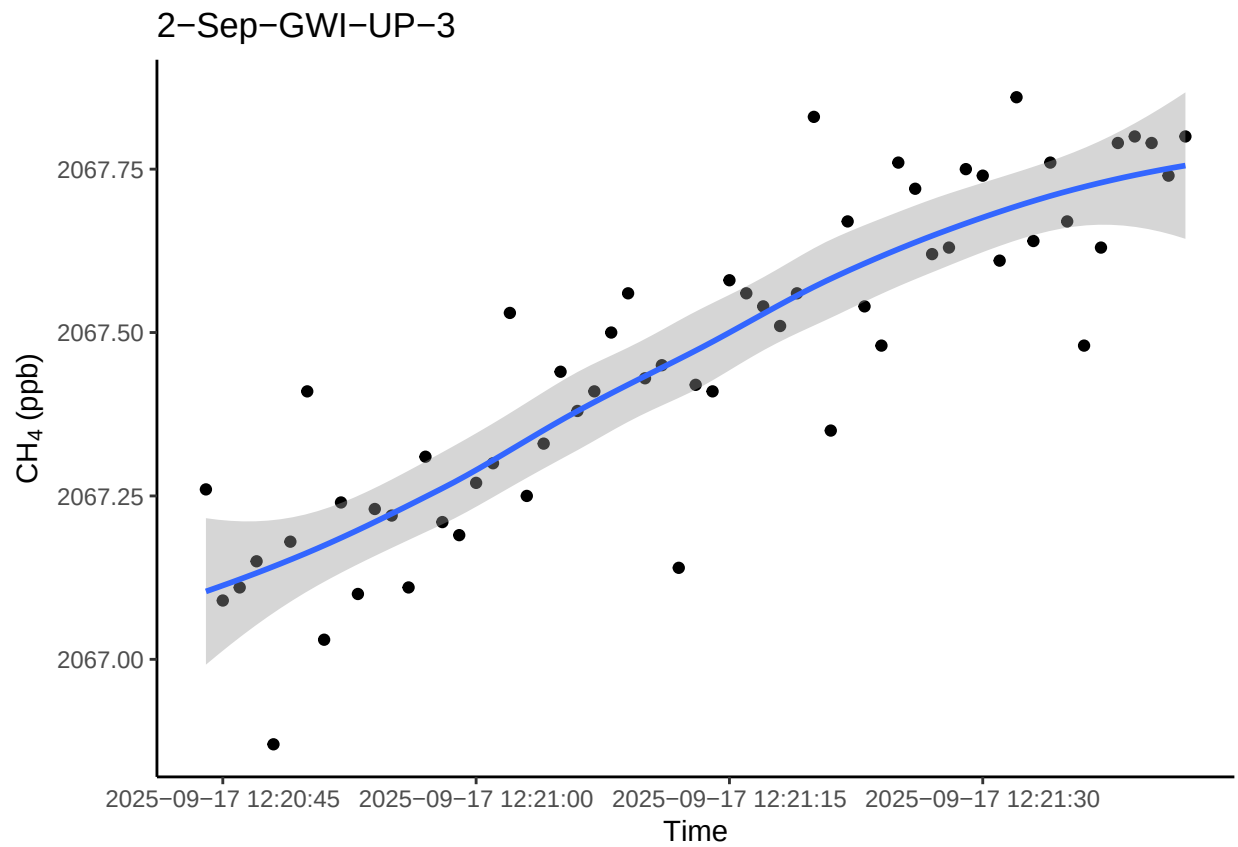
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



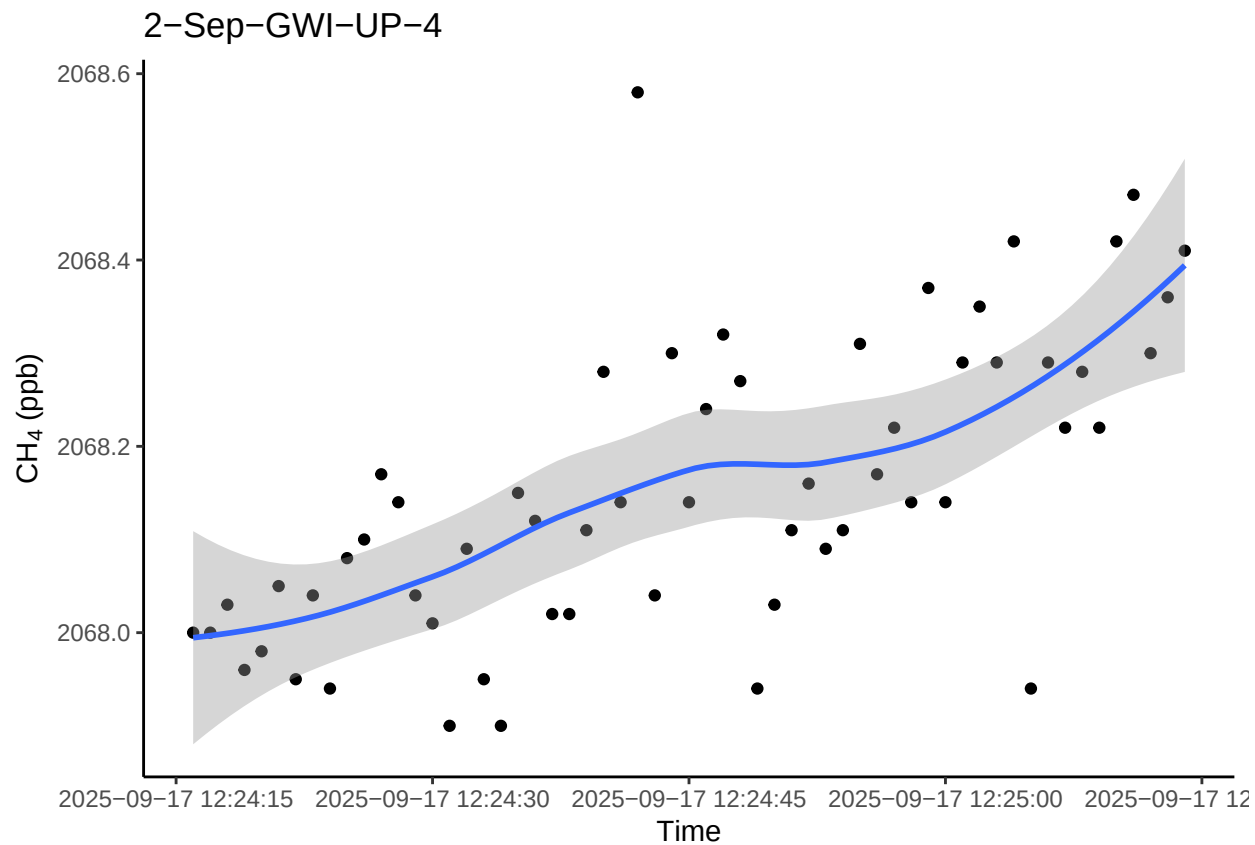
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



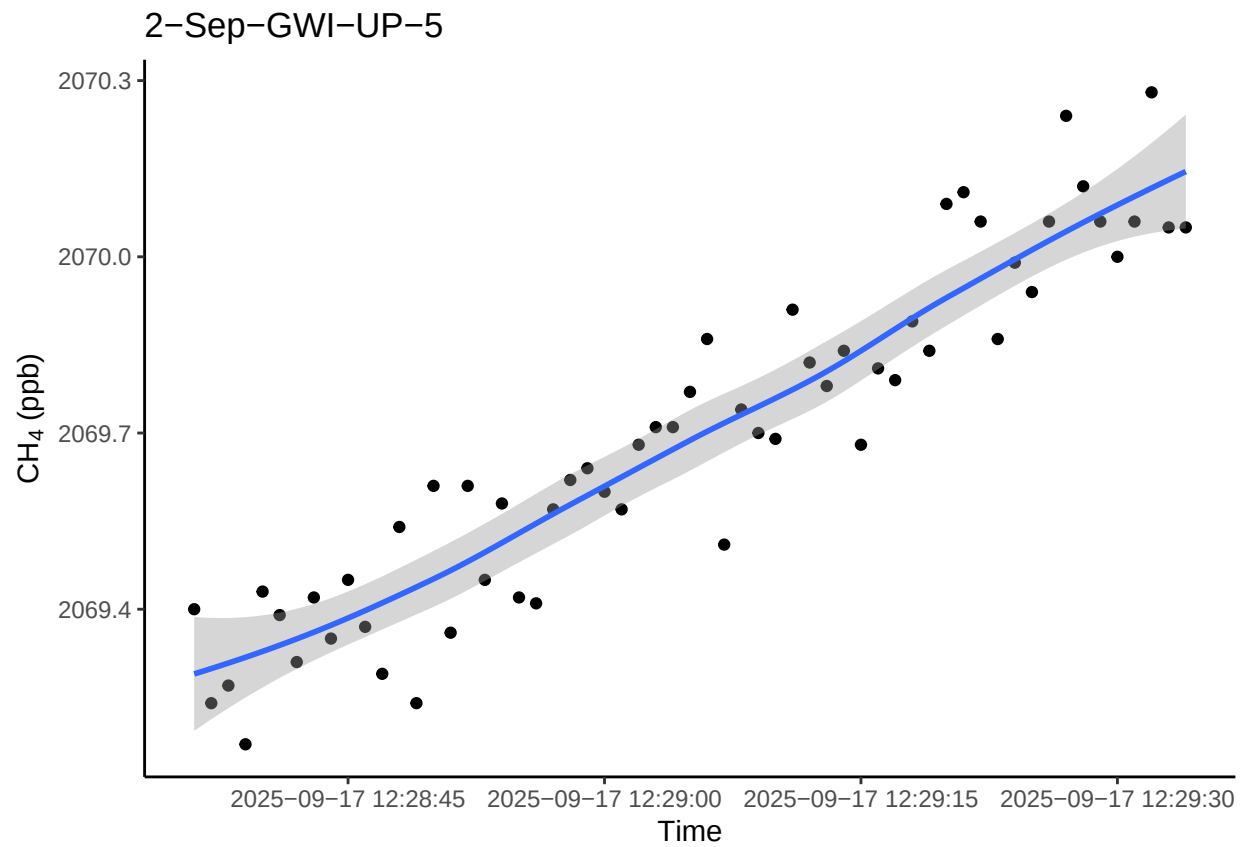
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



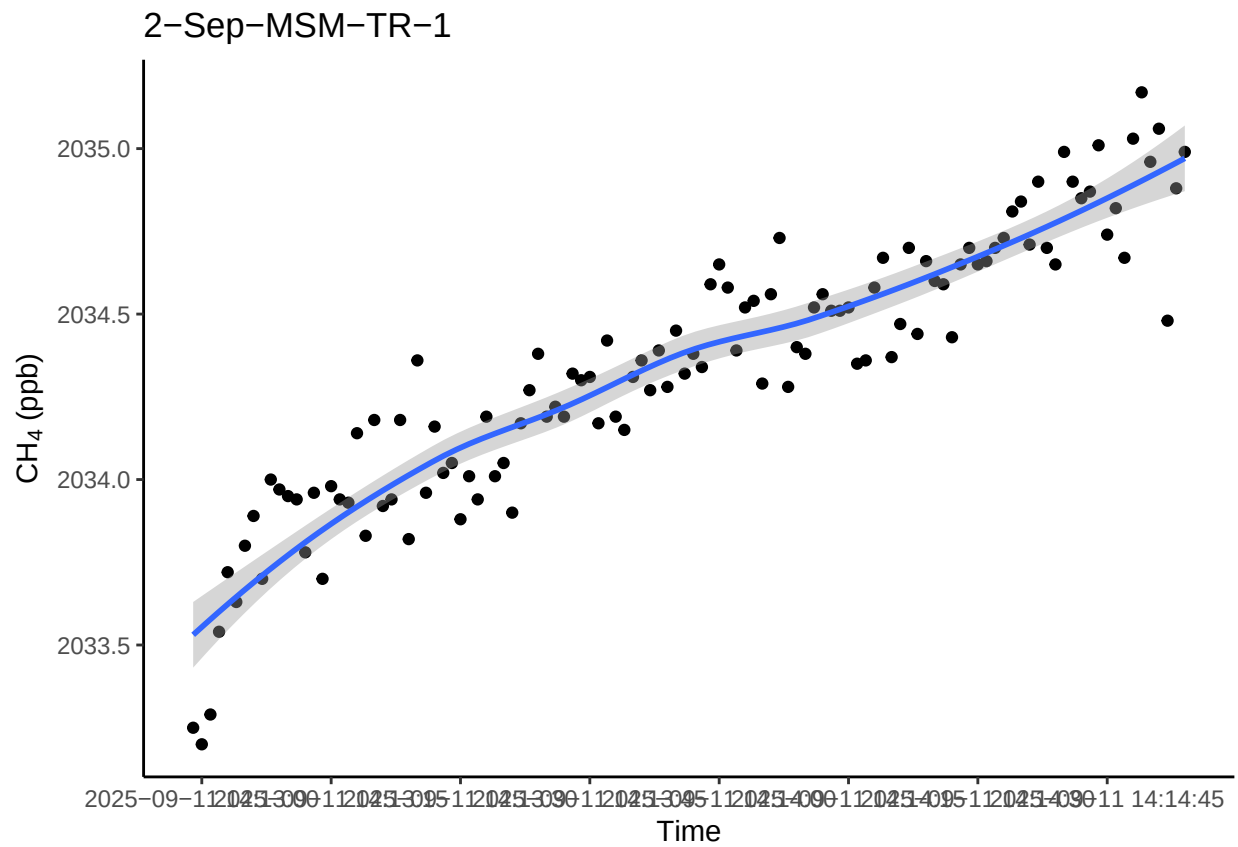
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



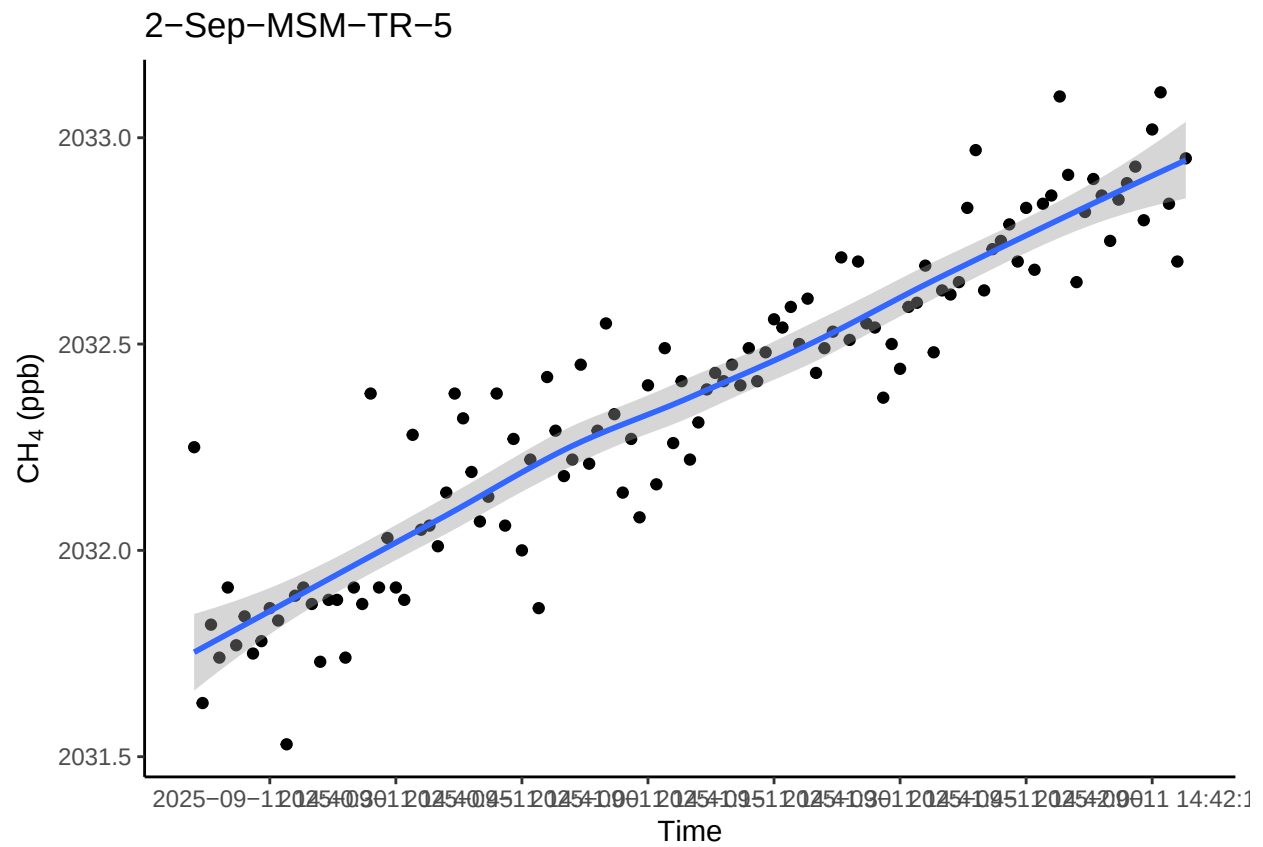
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

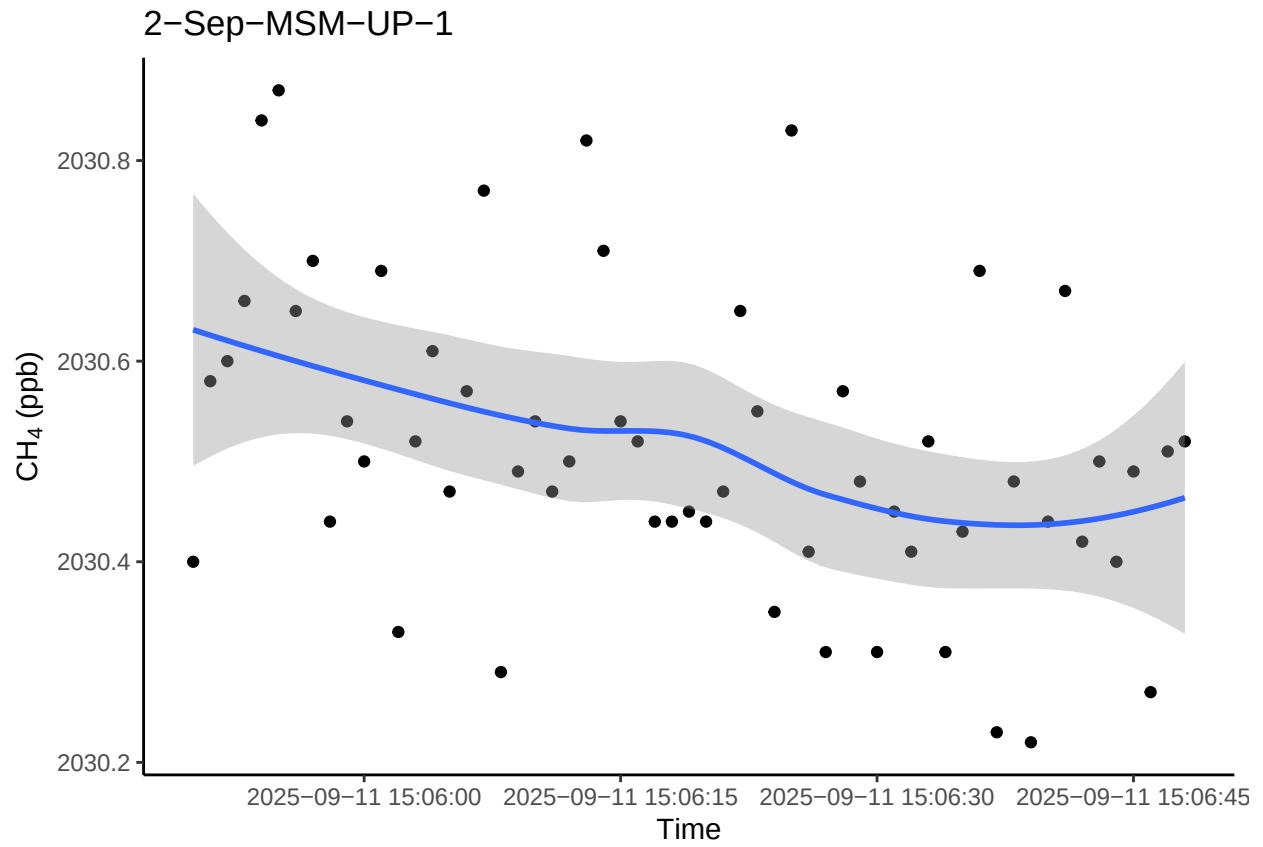
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



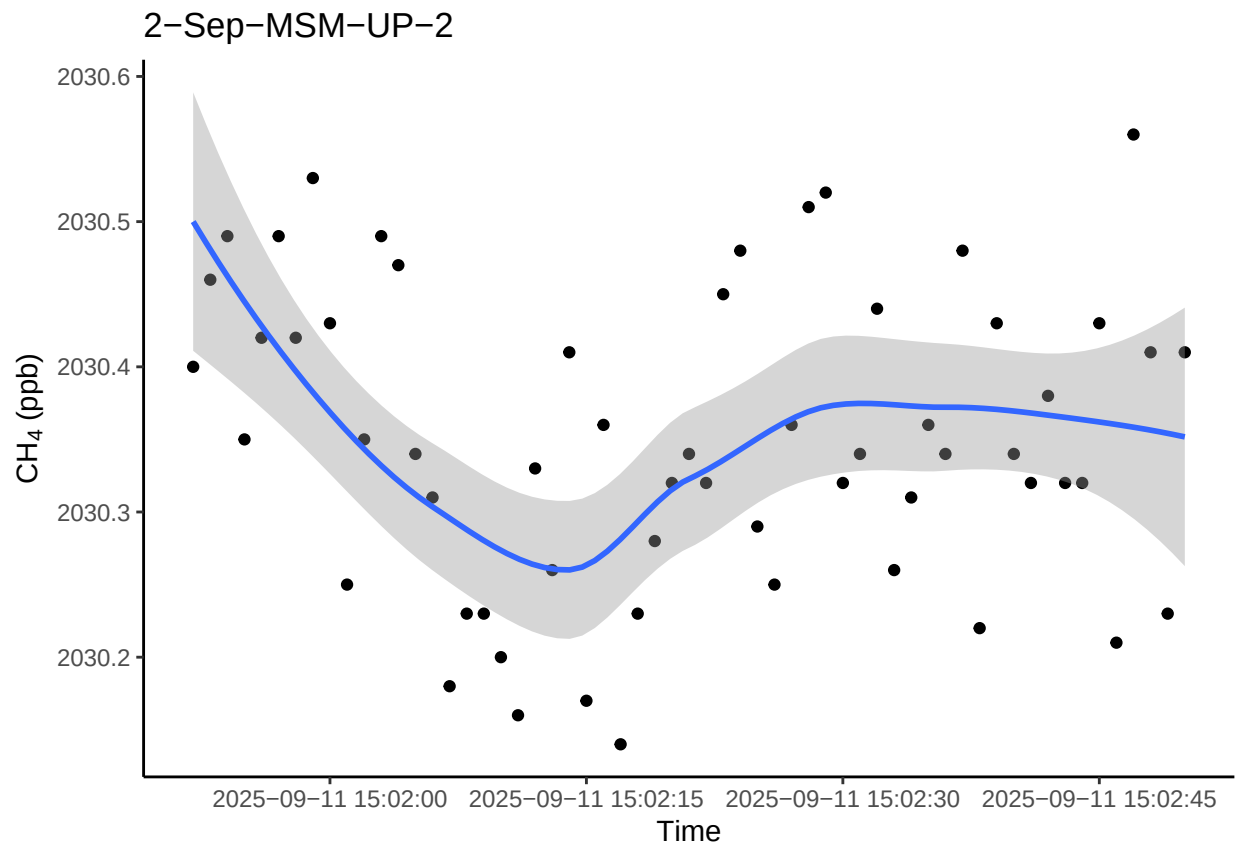
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



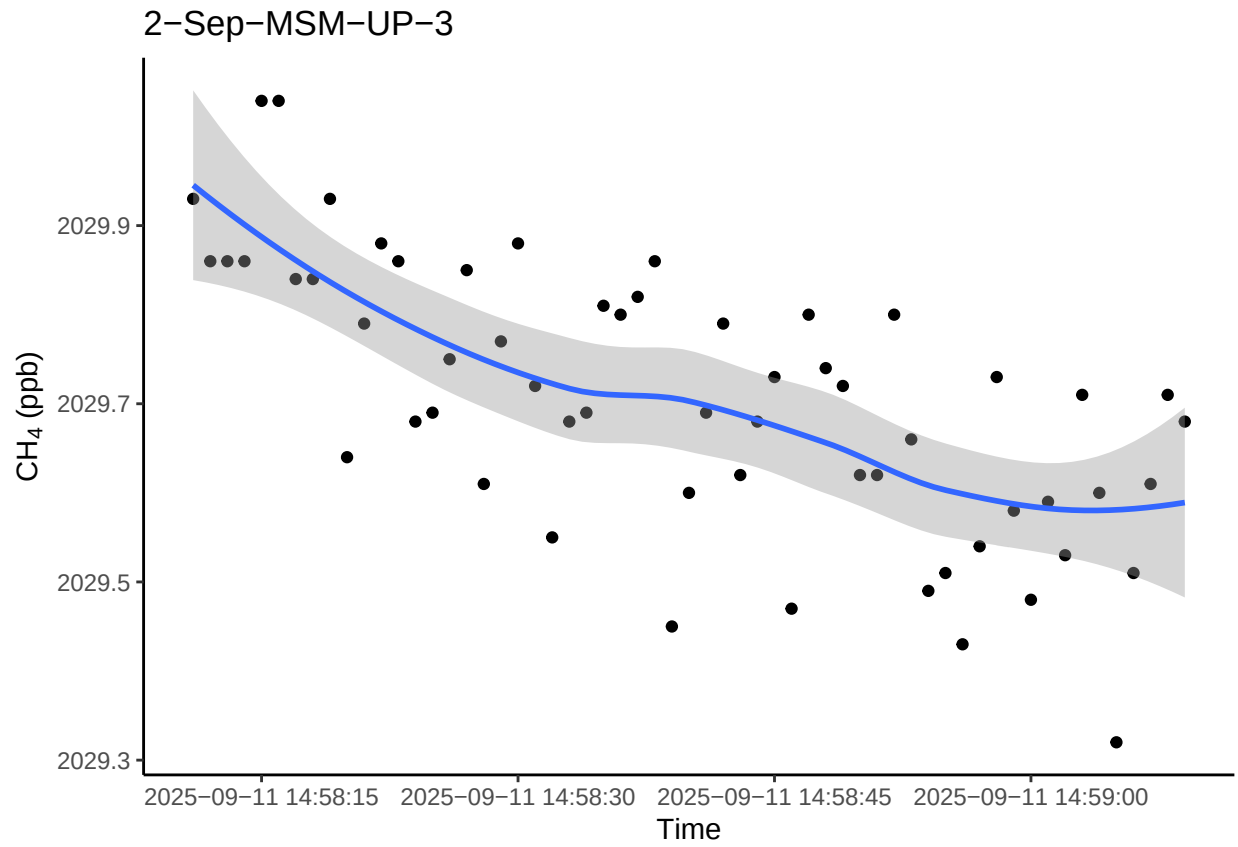
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



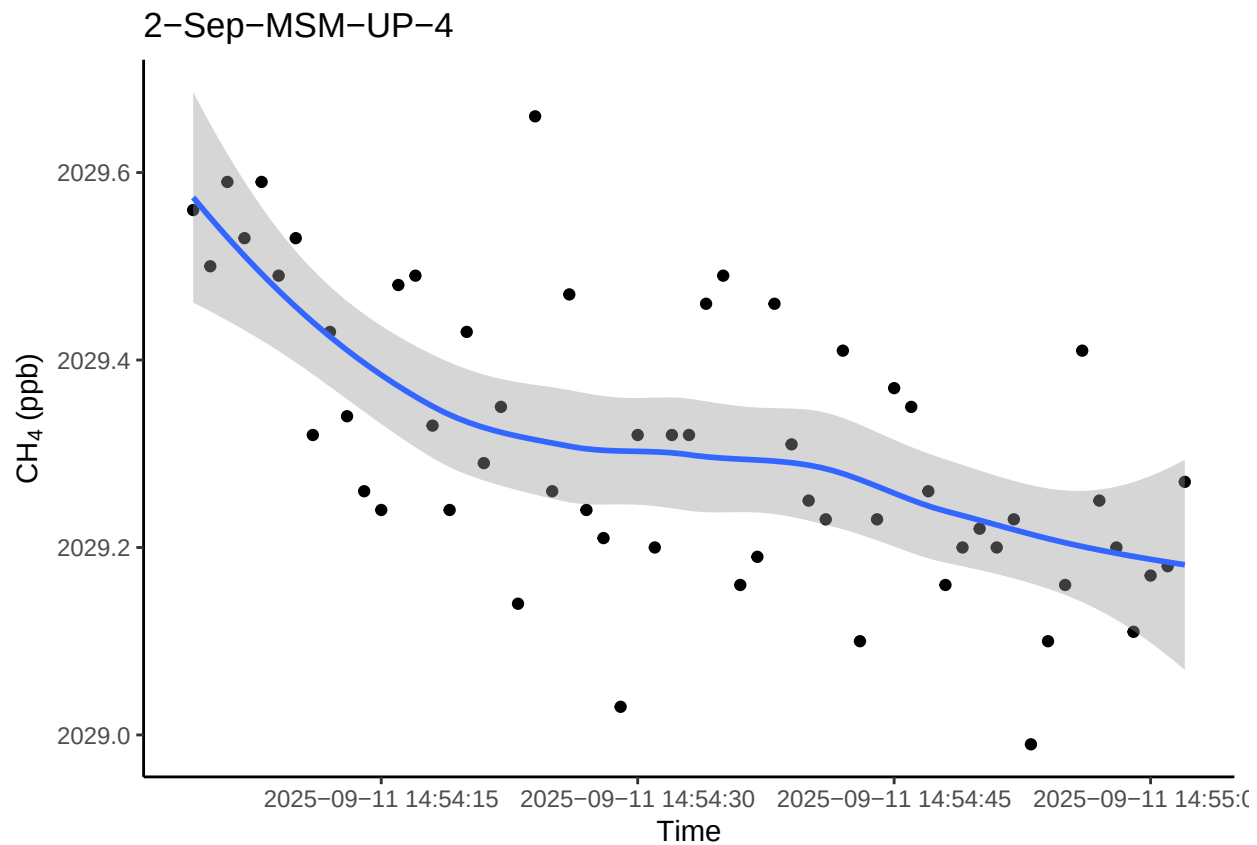
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



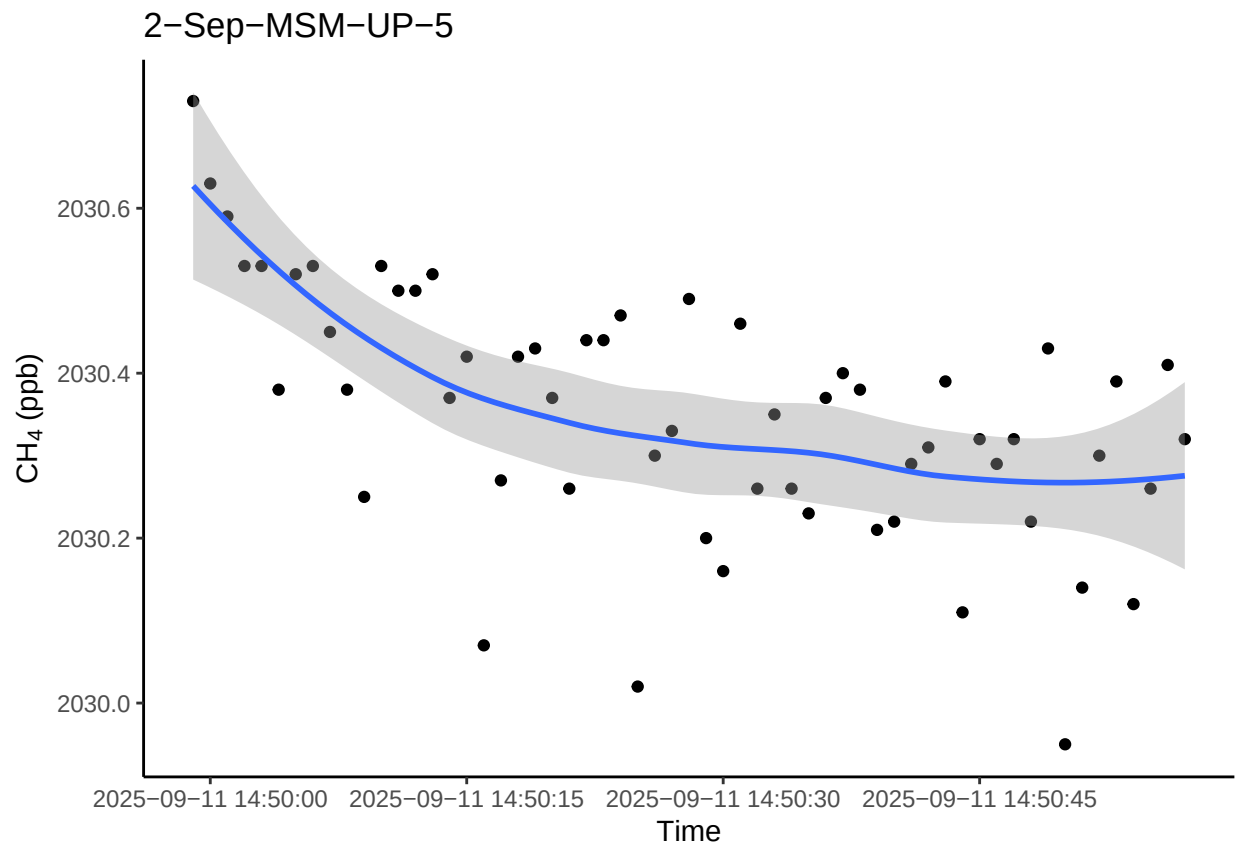
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



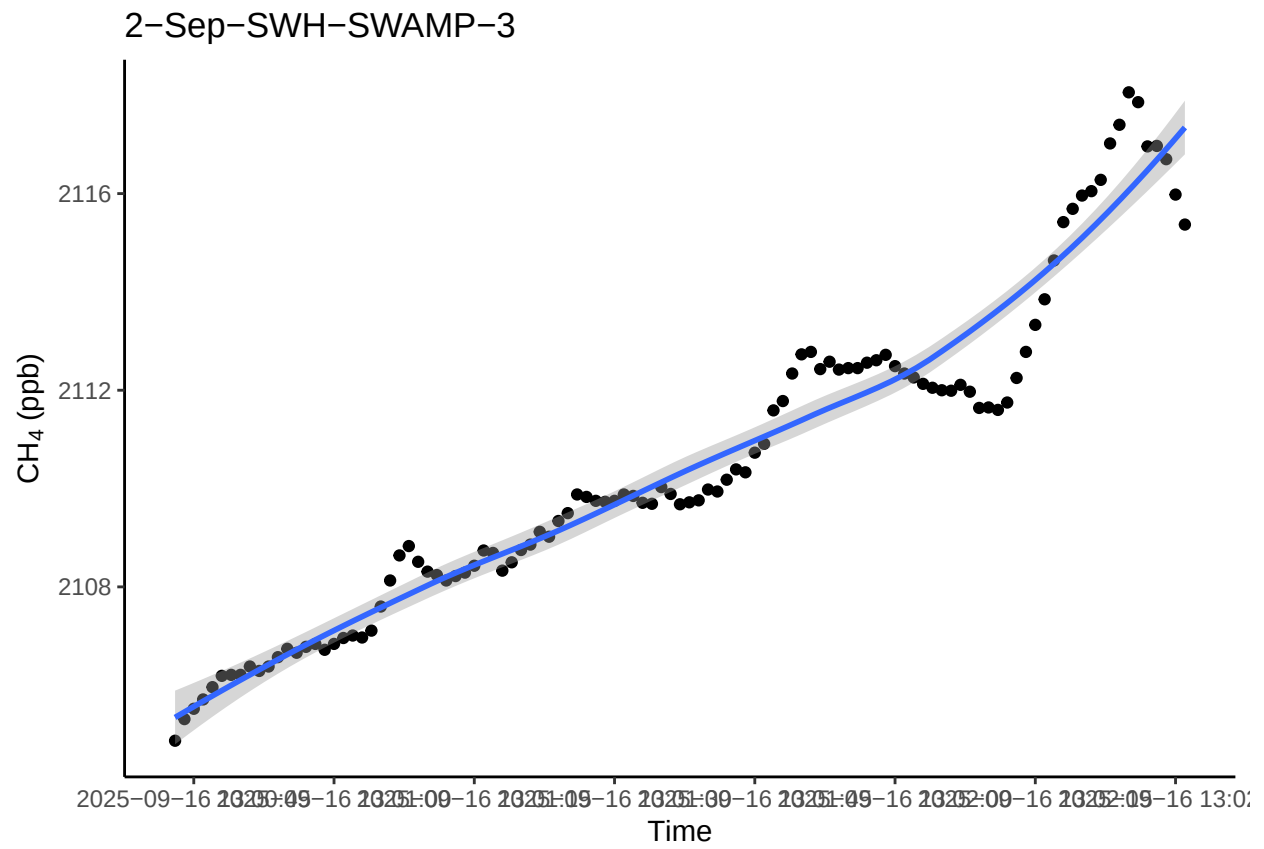
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



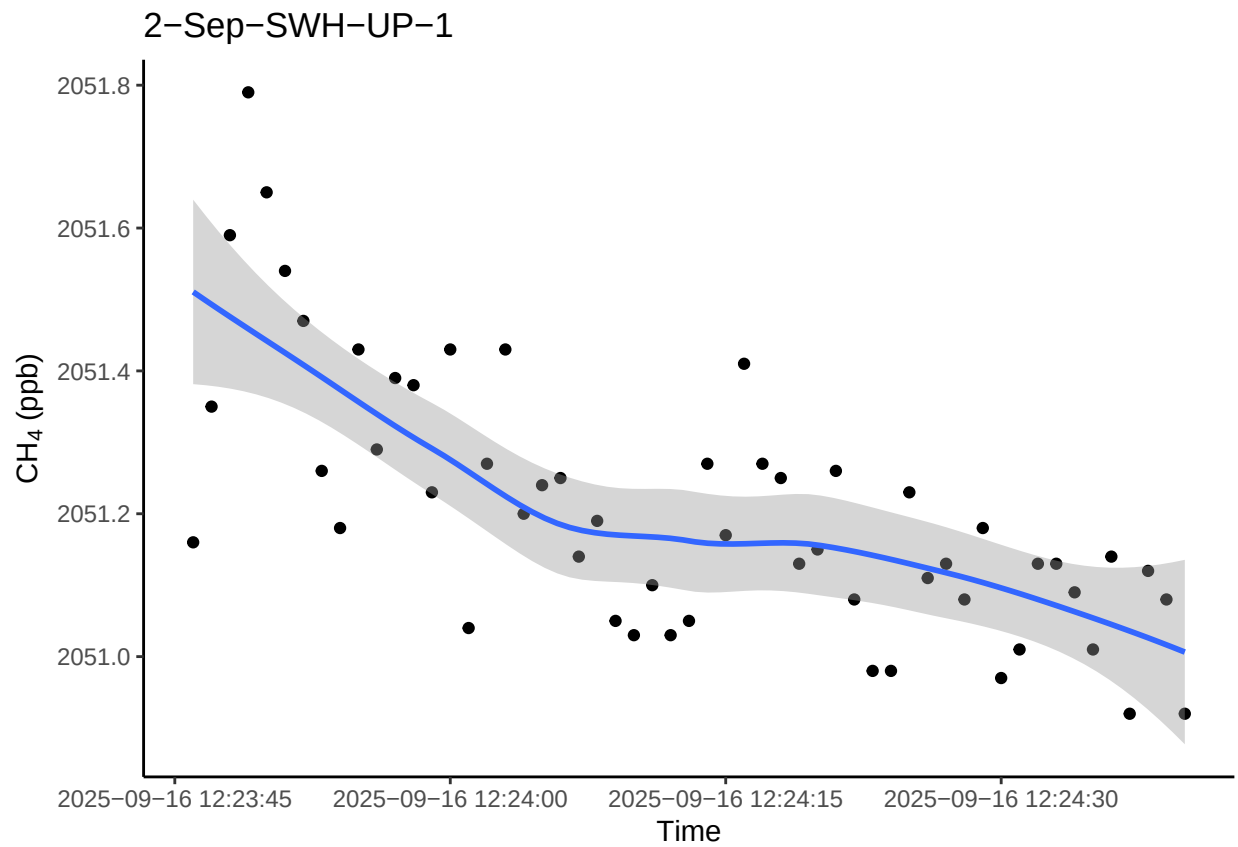
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



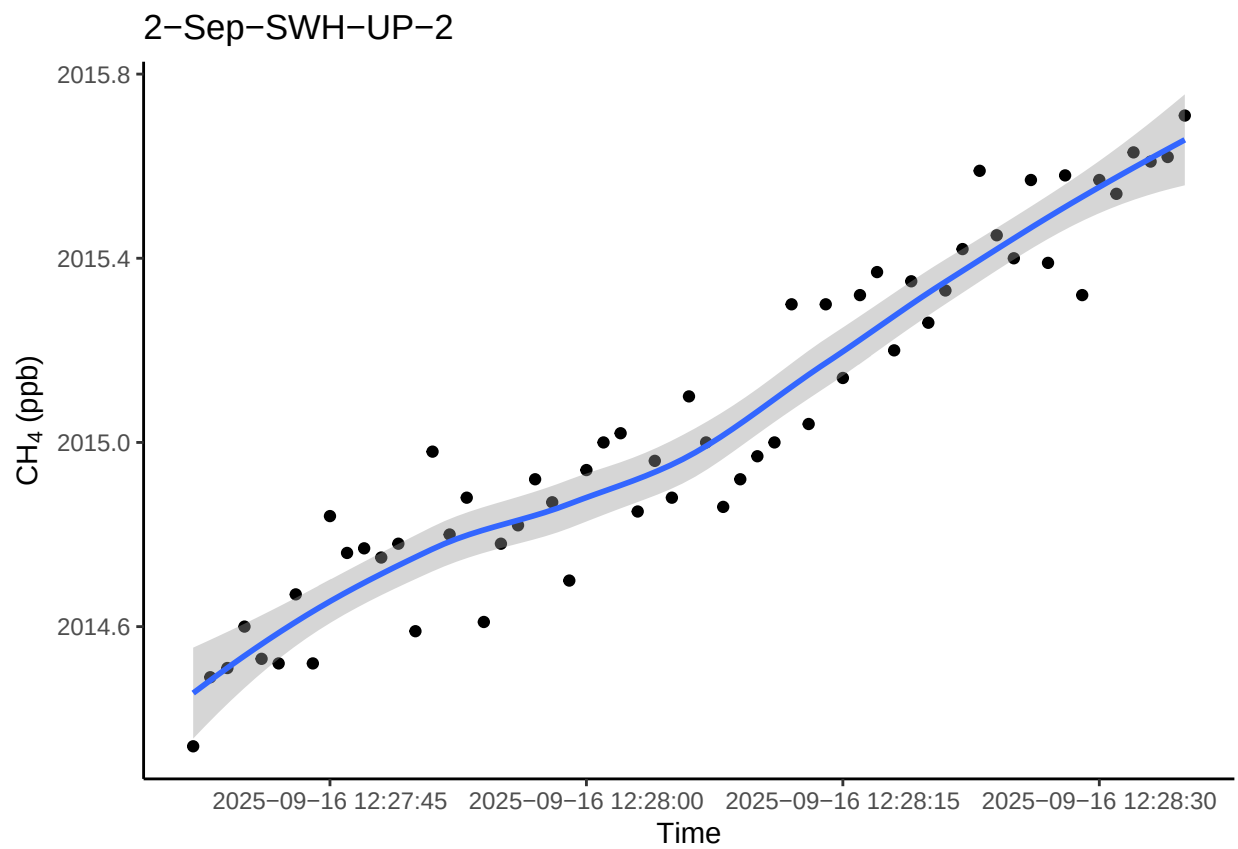
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

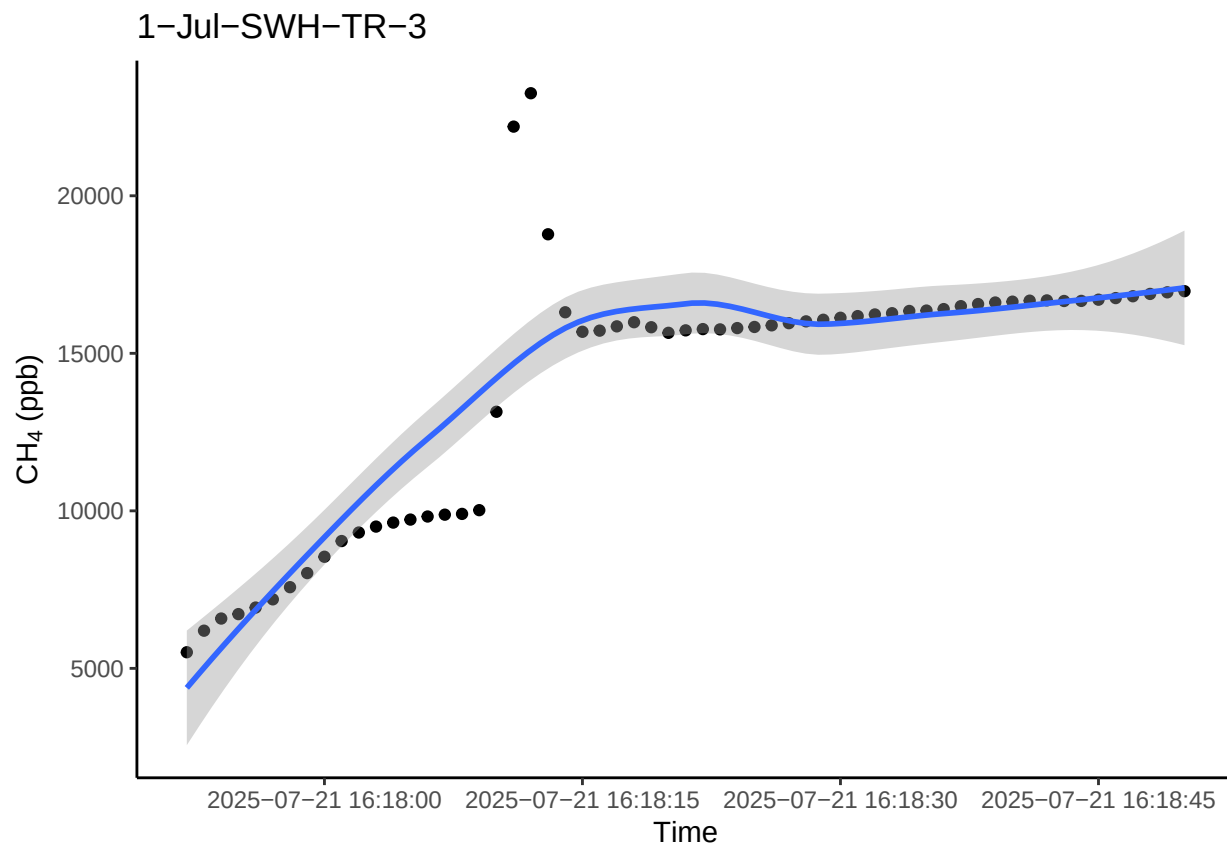


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

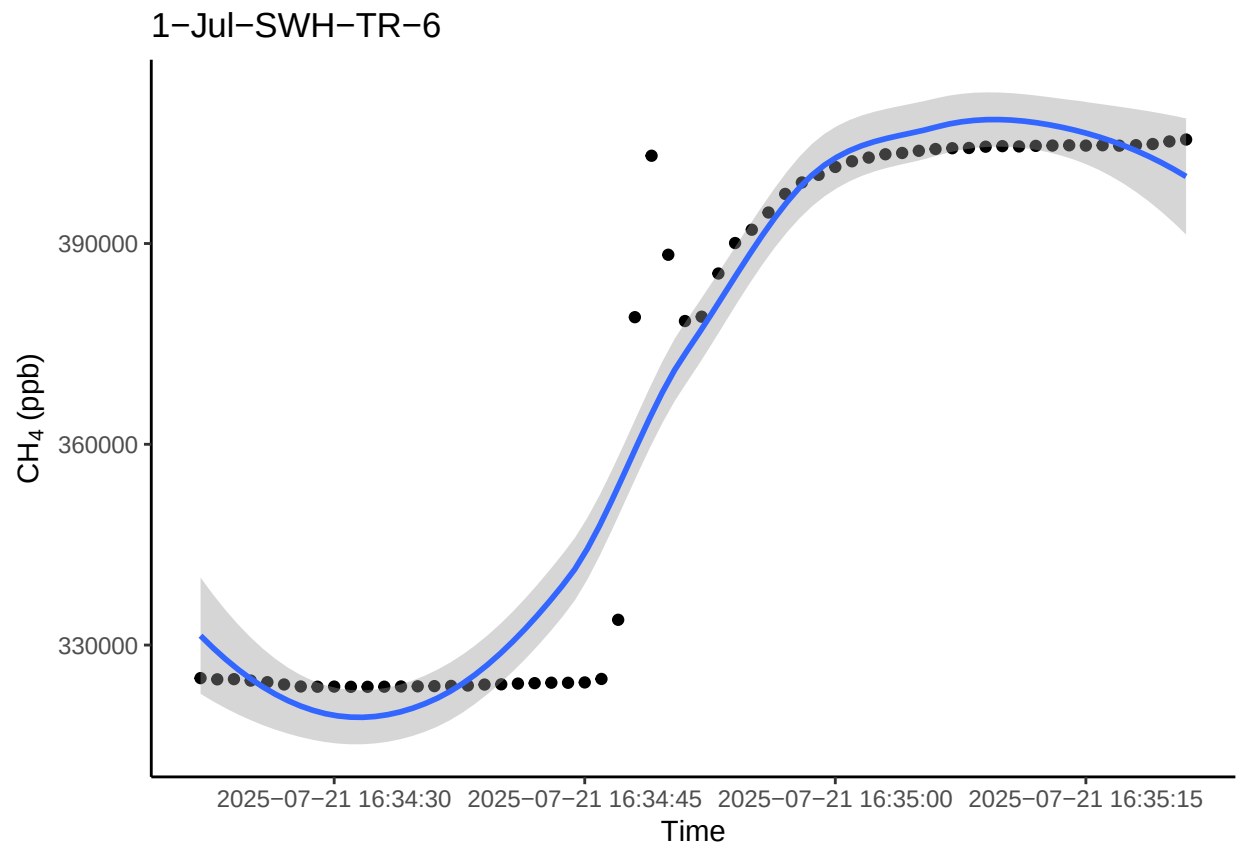


```
## Take a look at the bad fluxes <0.9 and have a high flux estimate (>5) # See if we want to remove these  
fluxes or if we want to try and recalculate these ones
```

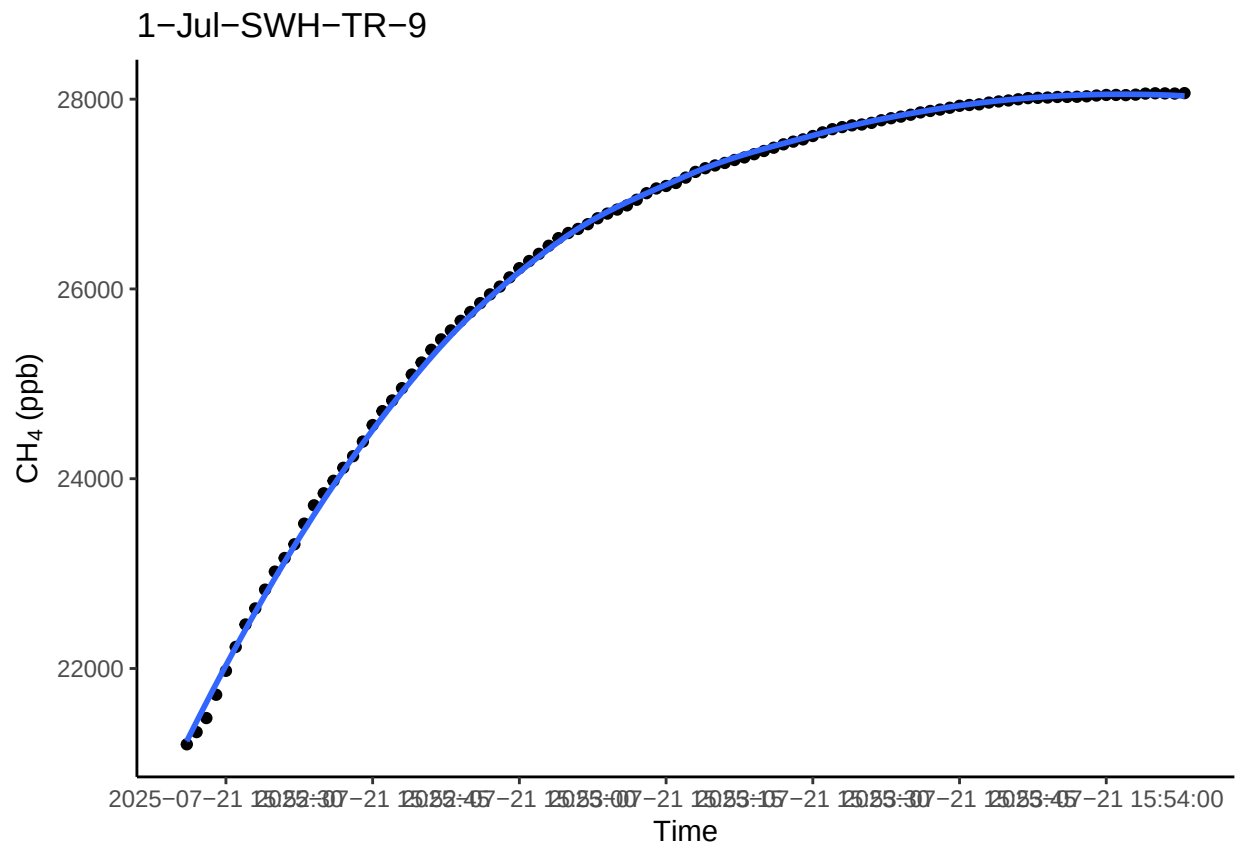
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



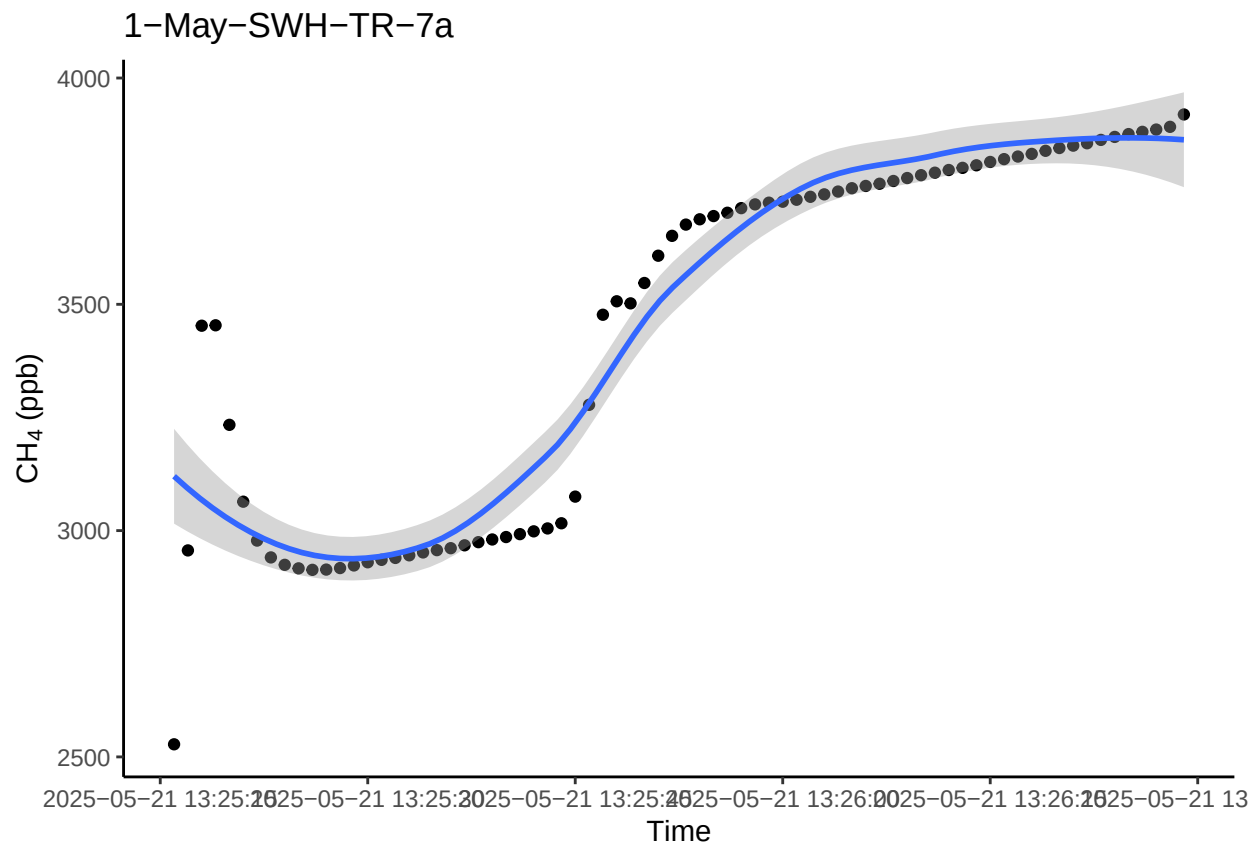
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



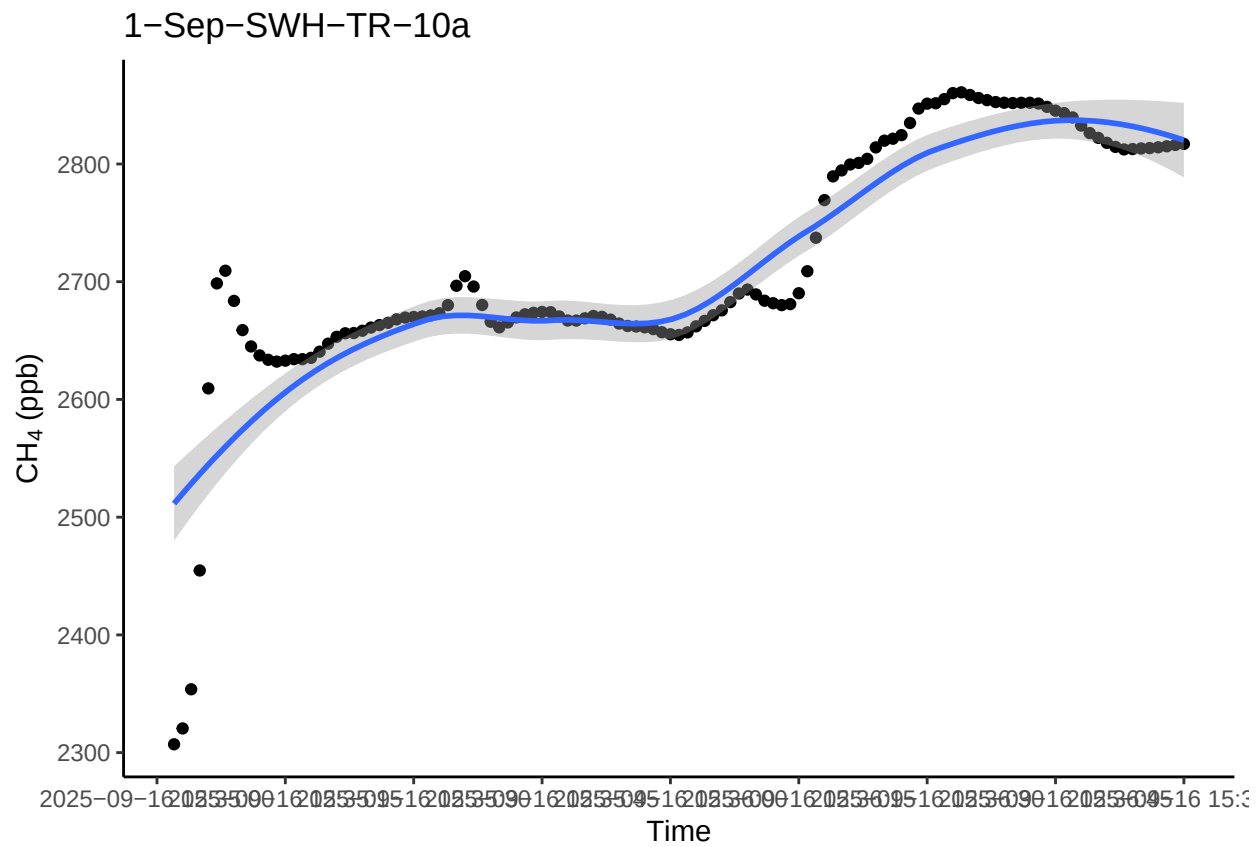
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



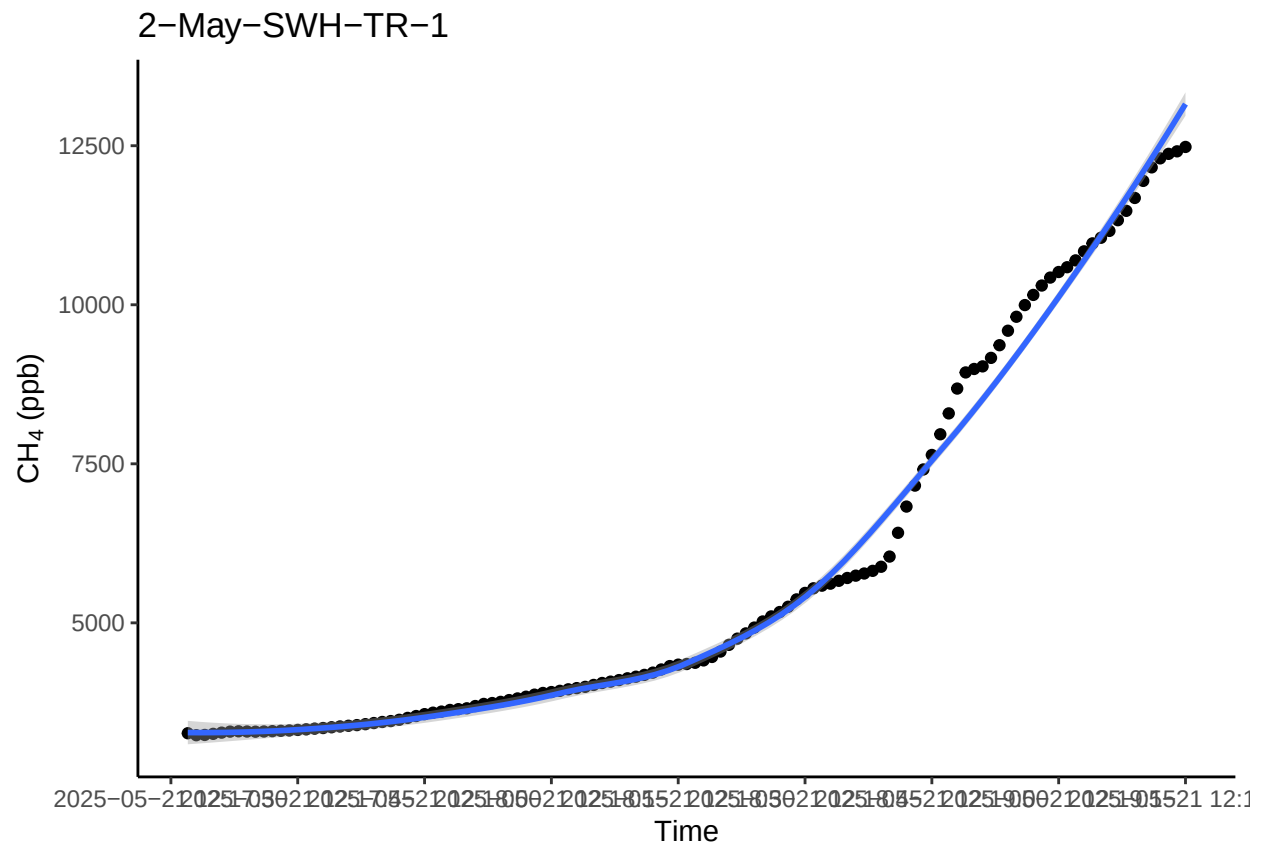
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



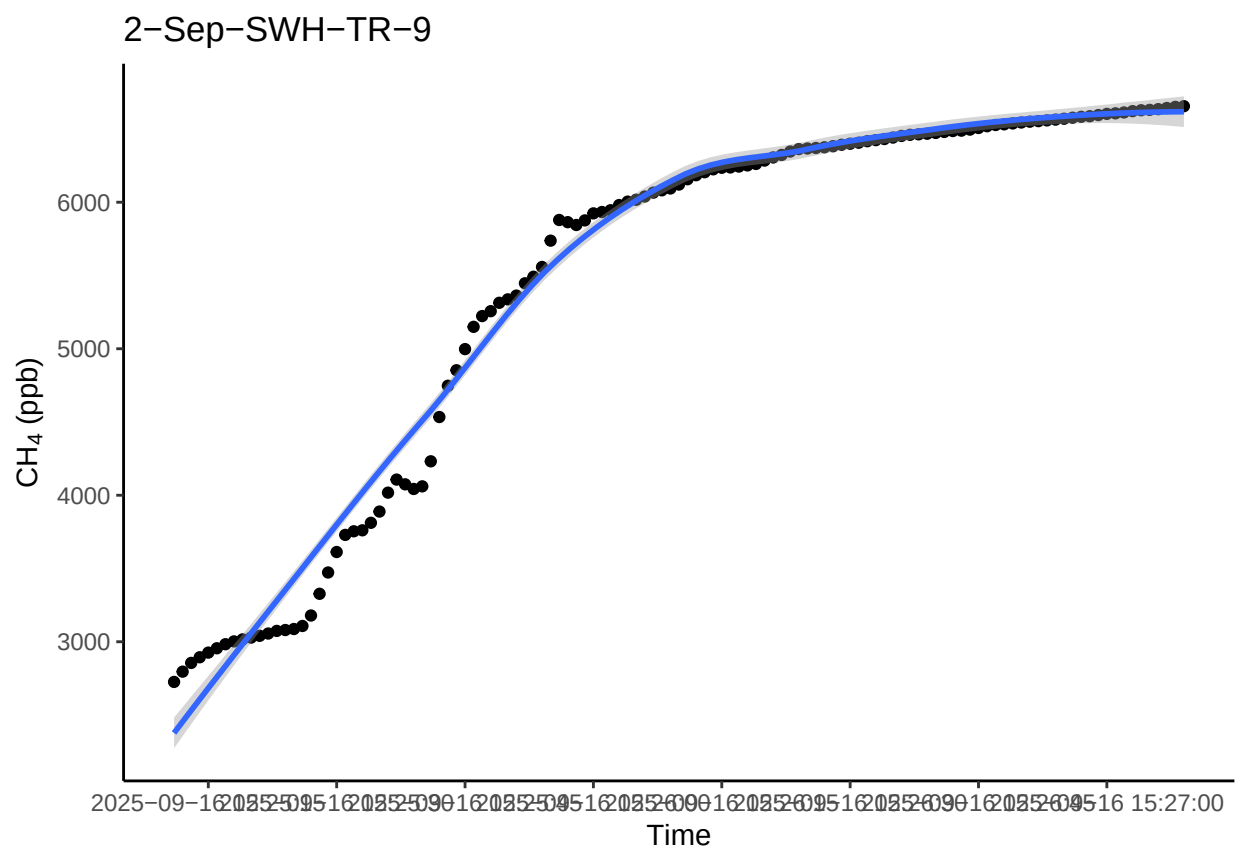
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

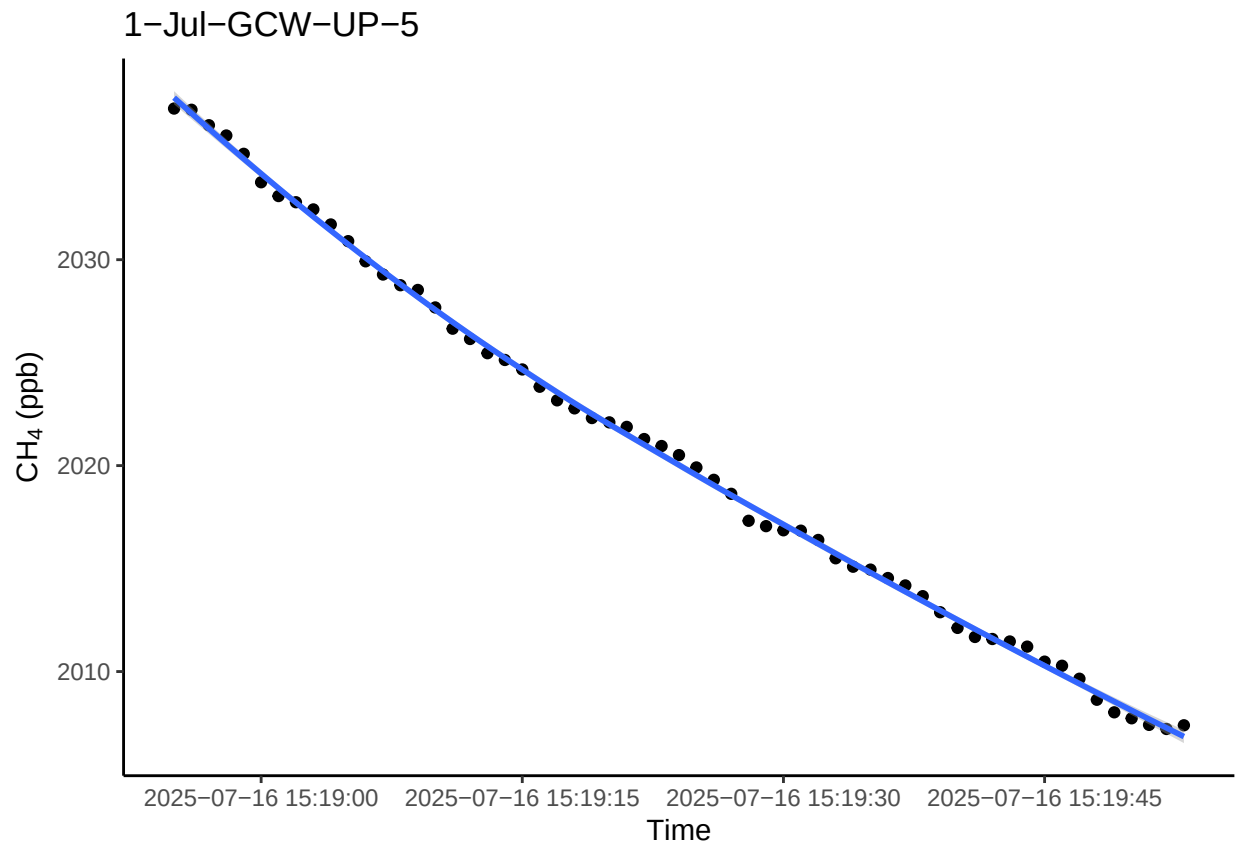



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

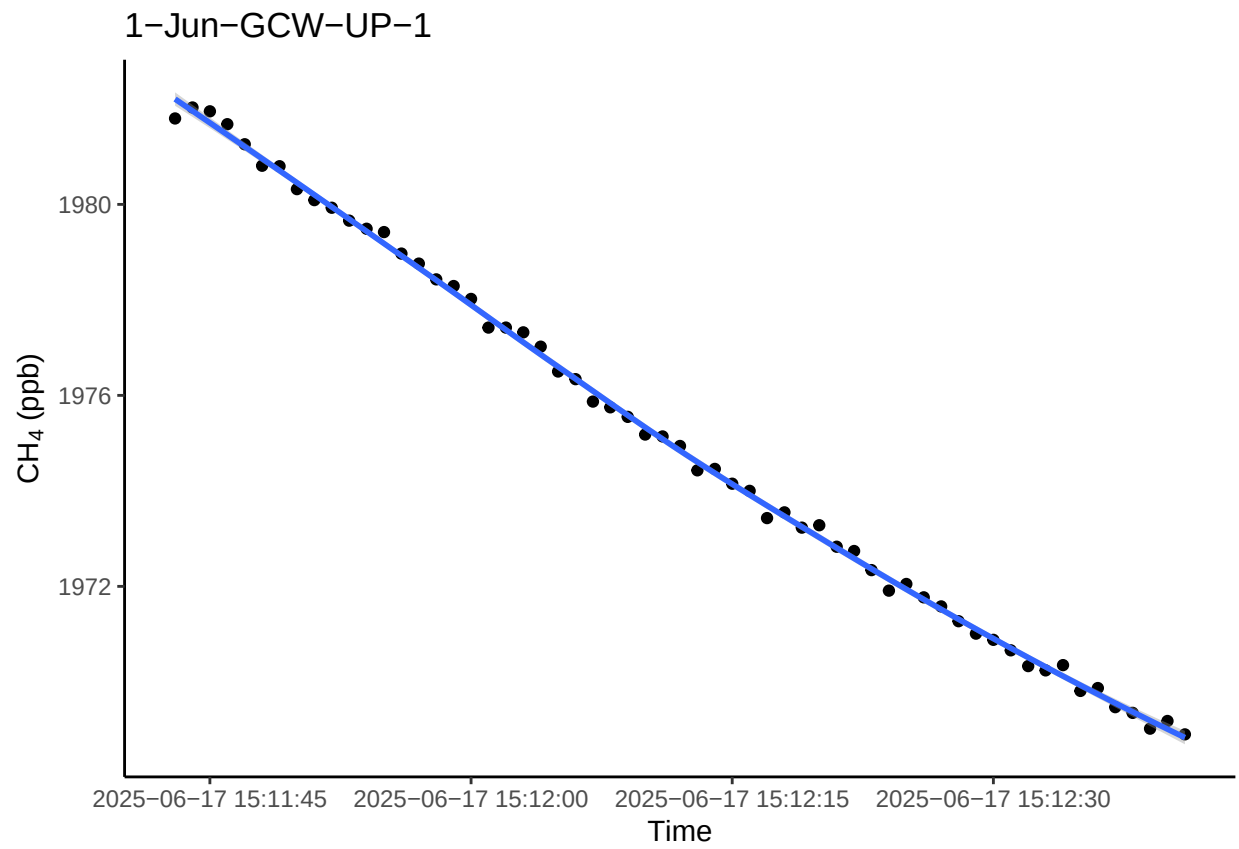


0.12 Look at the really high or really low ones to see if there is a timing issue or something

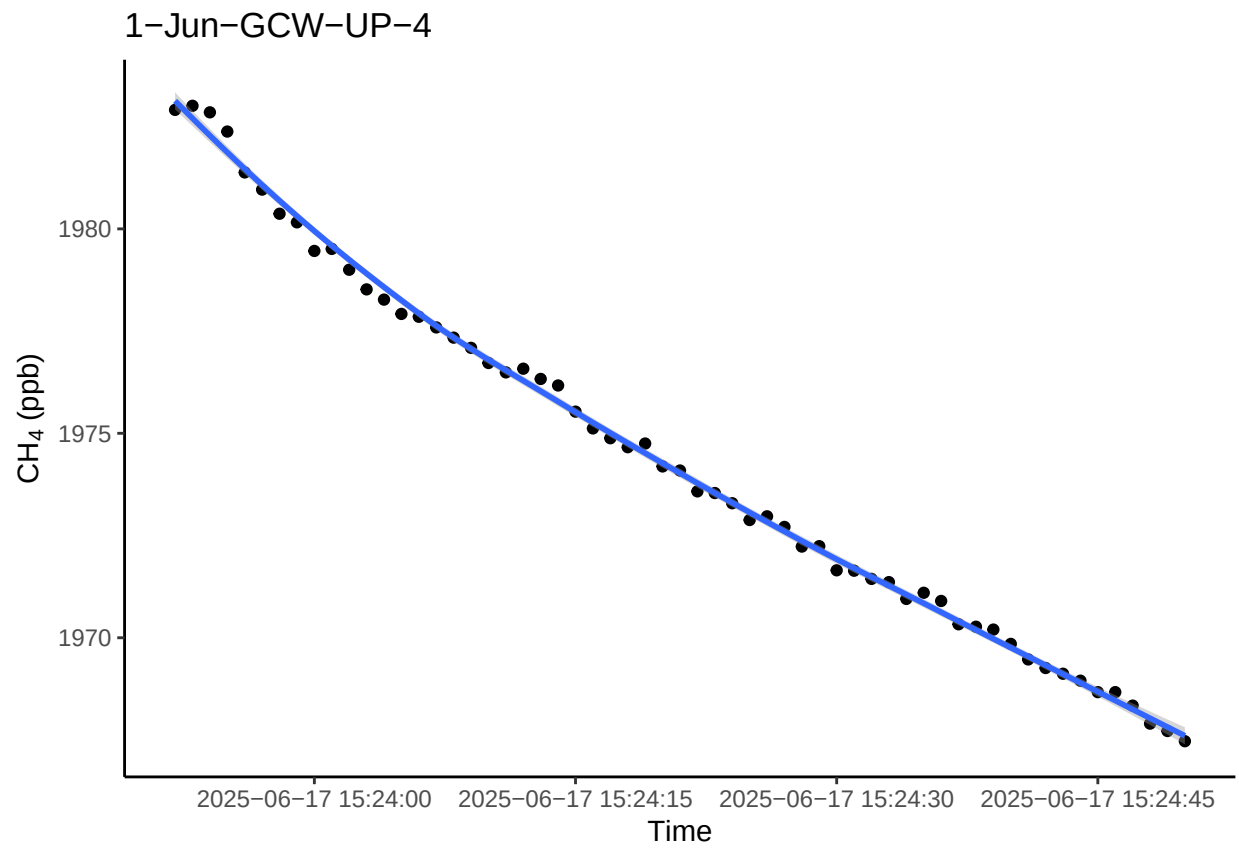
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



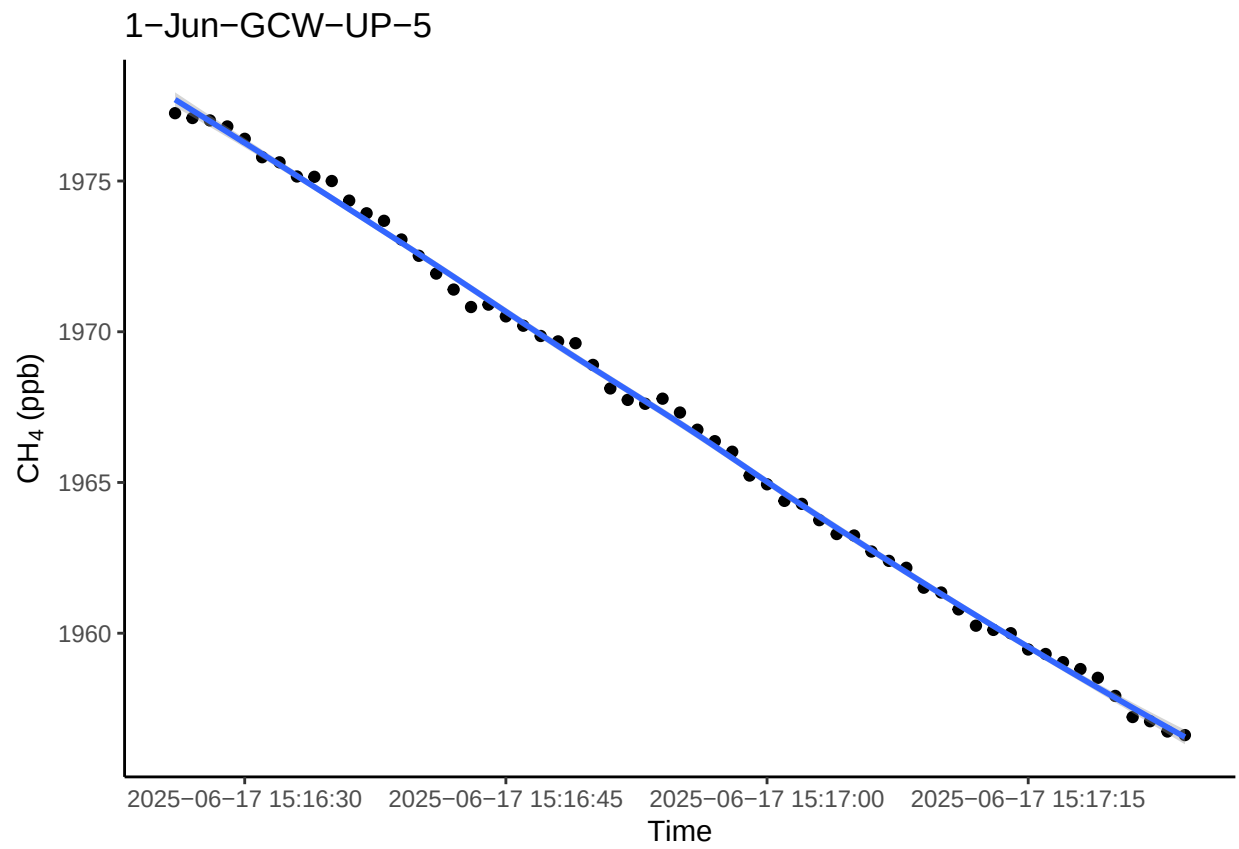
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



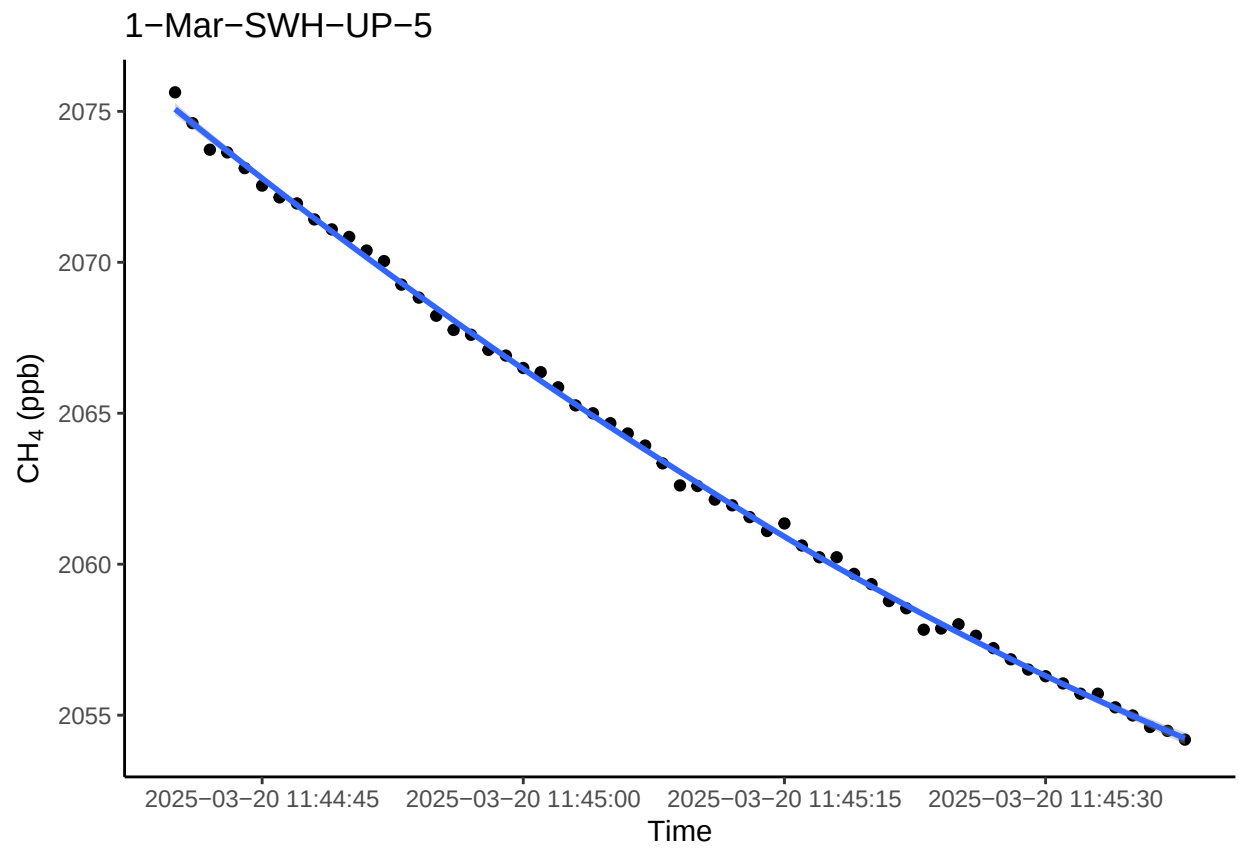
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



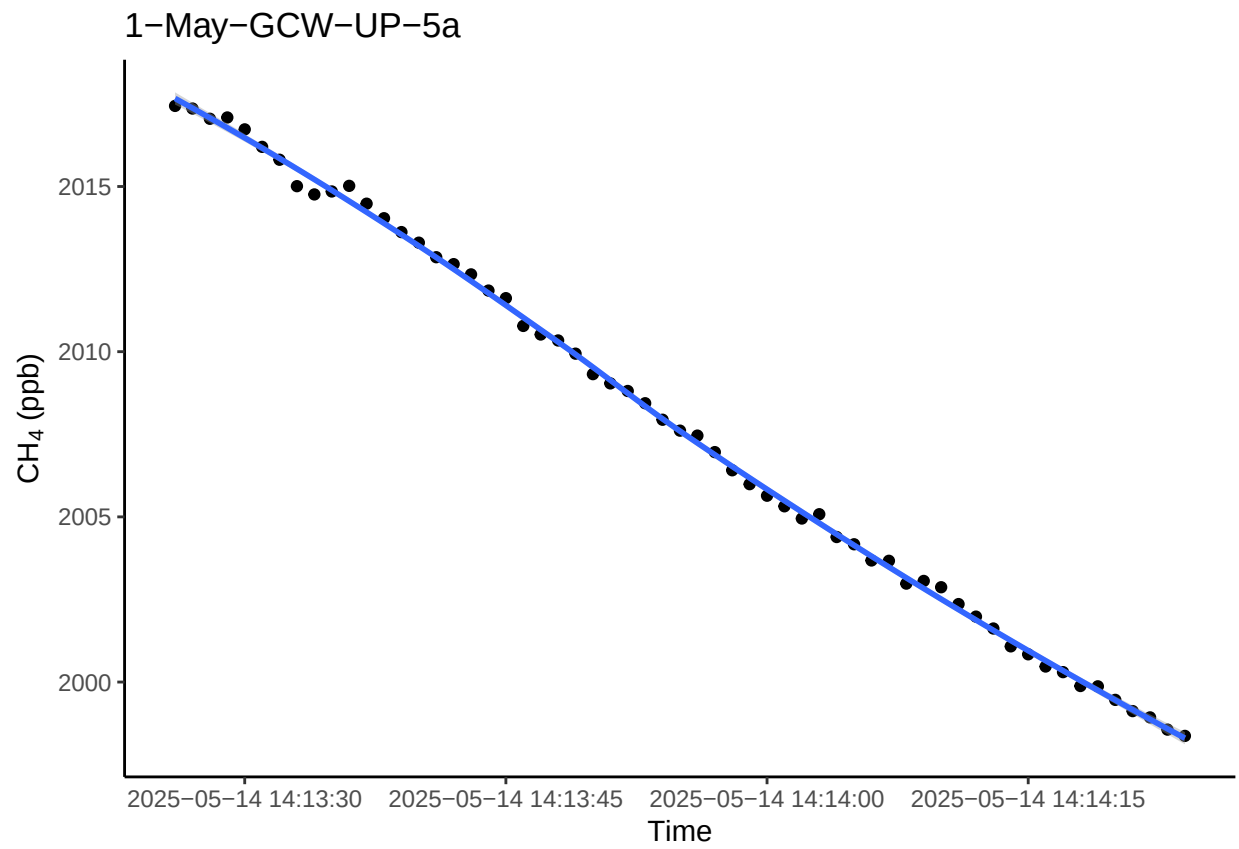
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



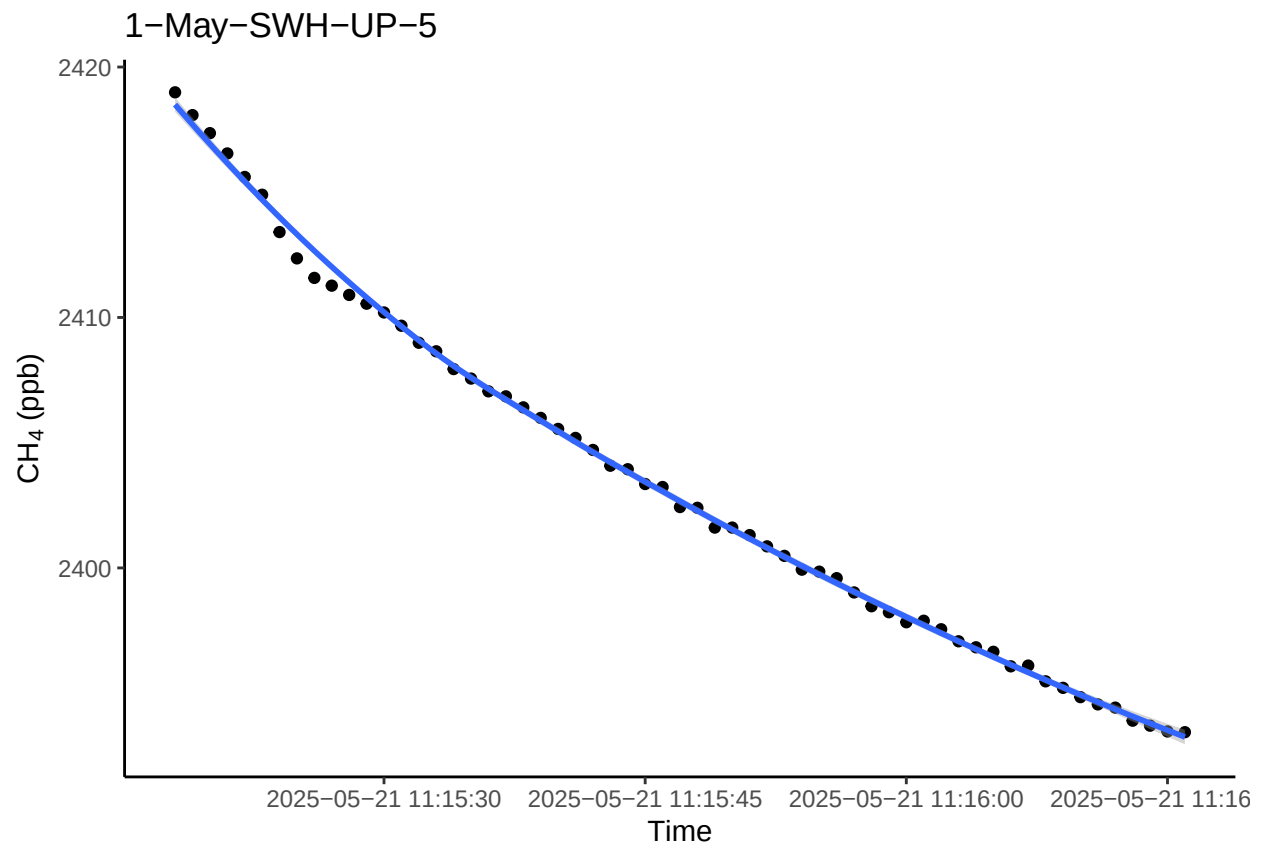
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



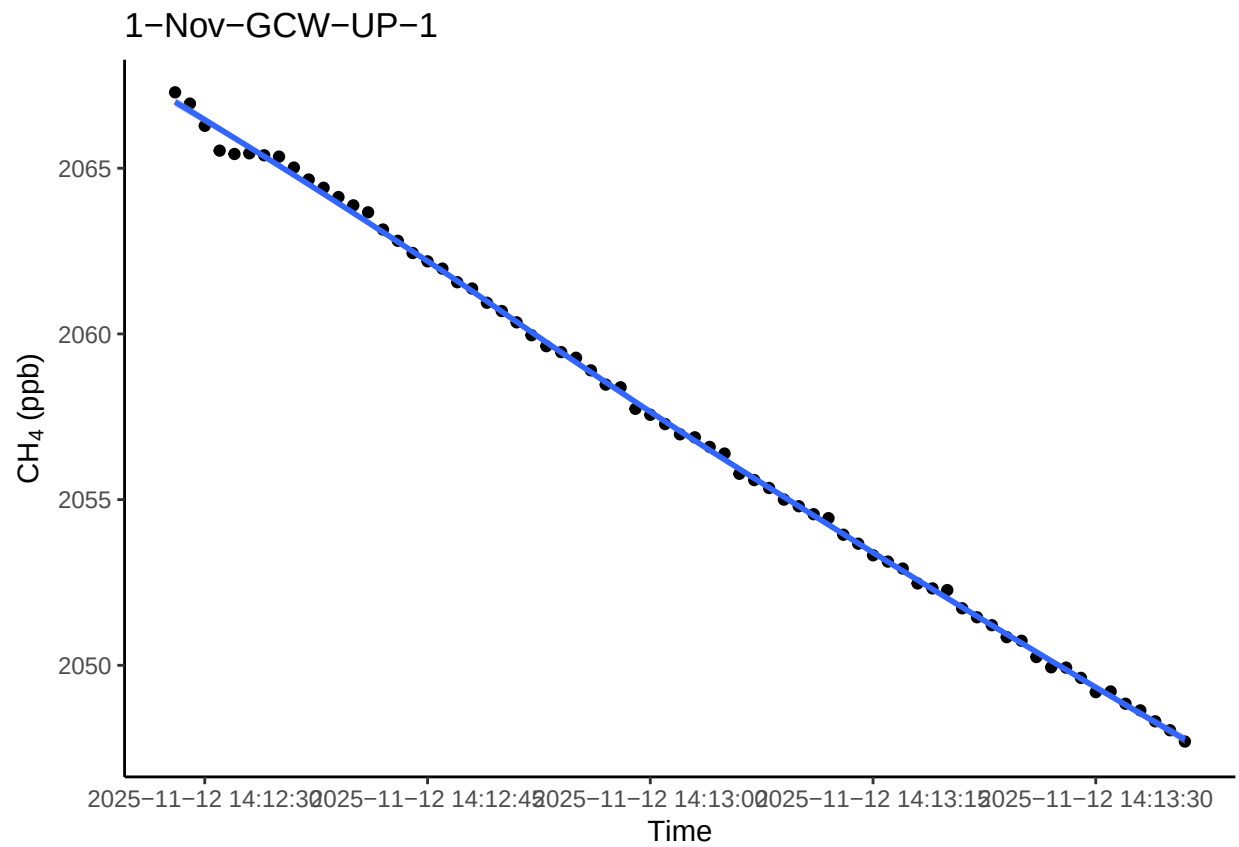
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



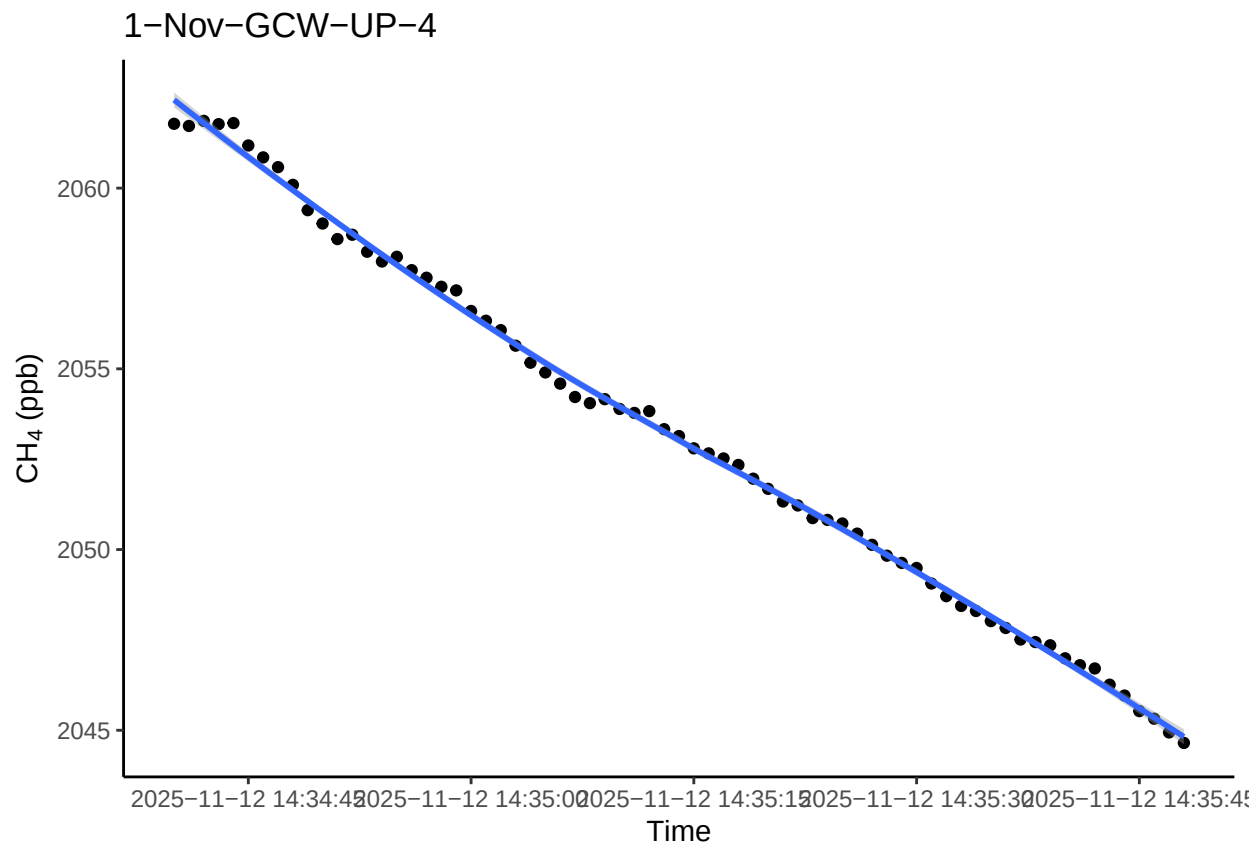
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

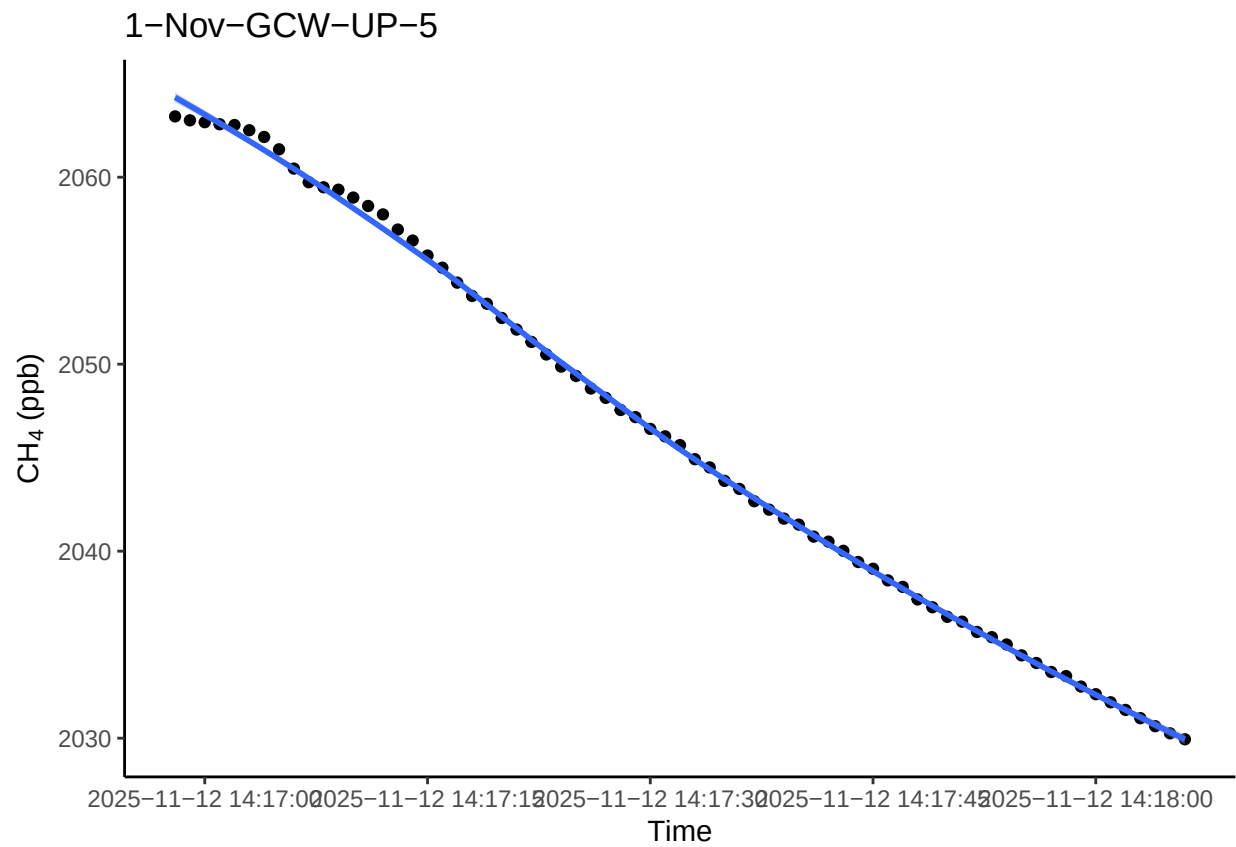
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



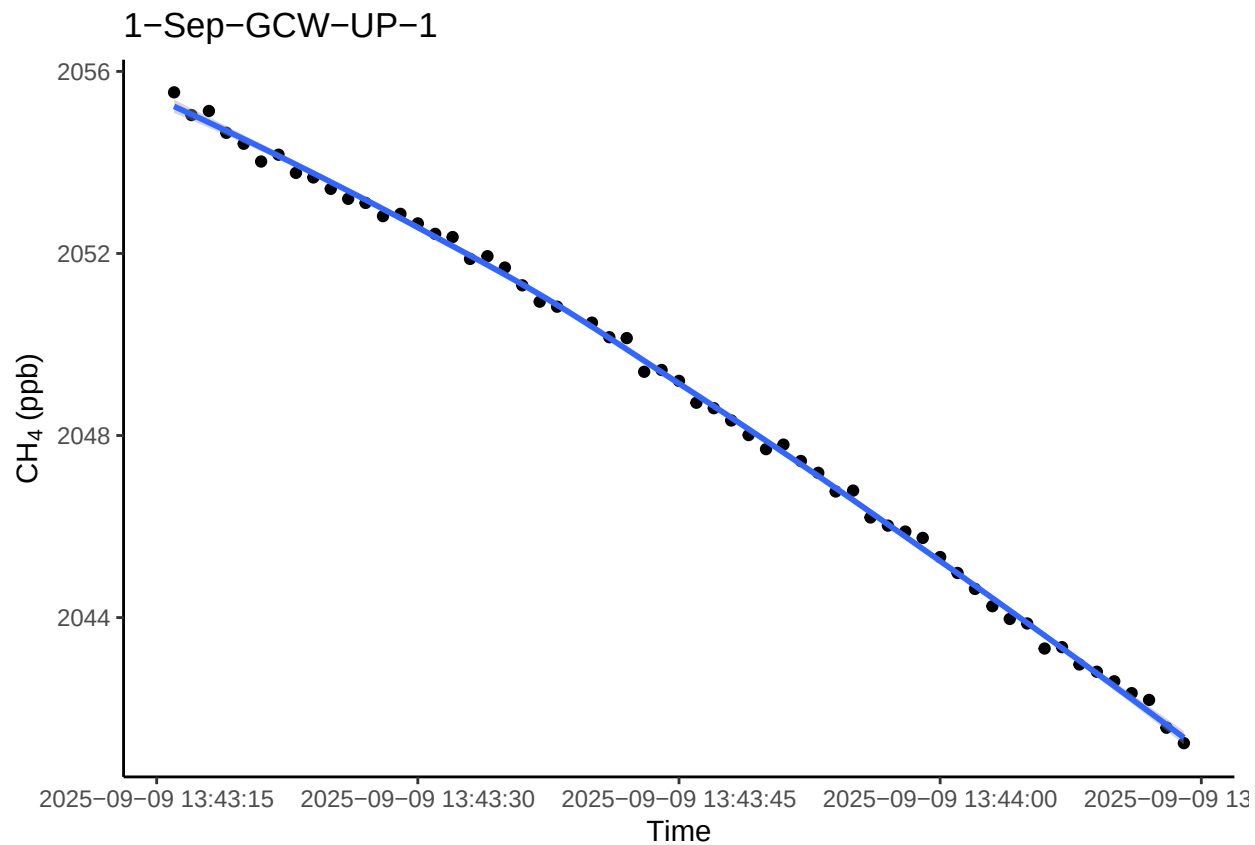
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



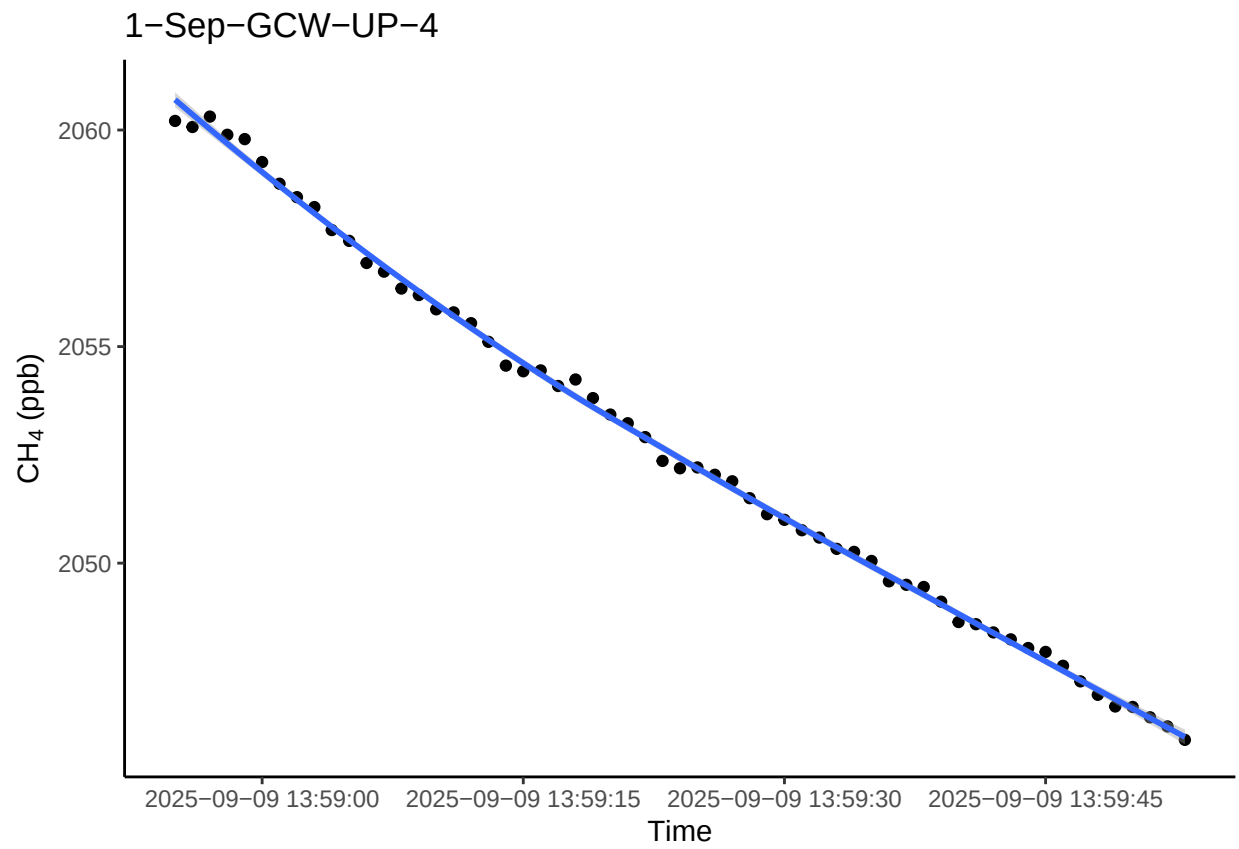
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



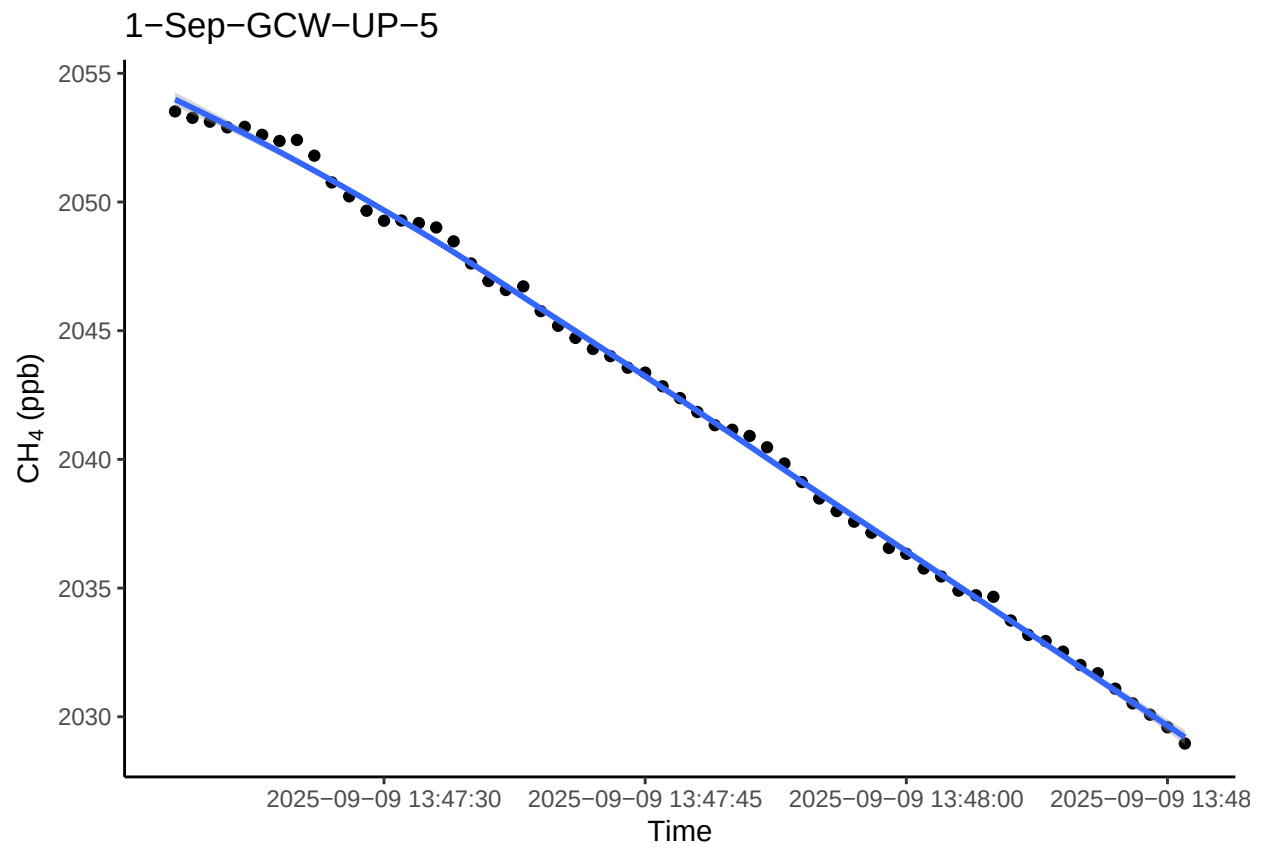
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



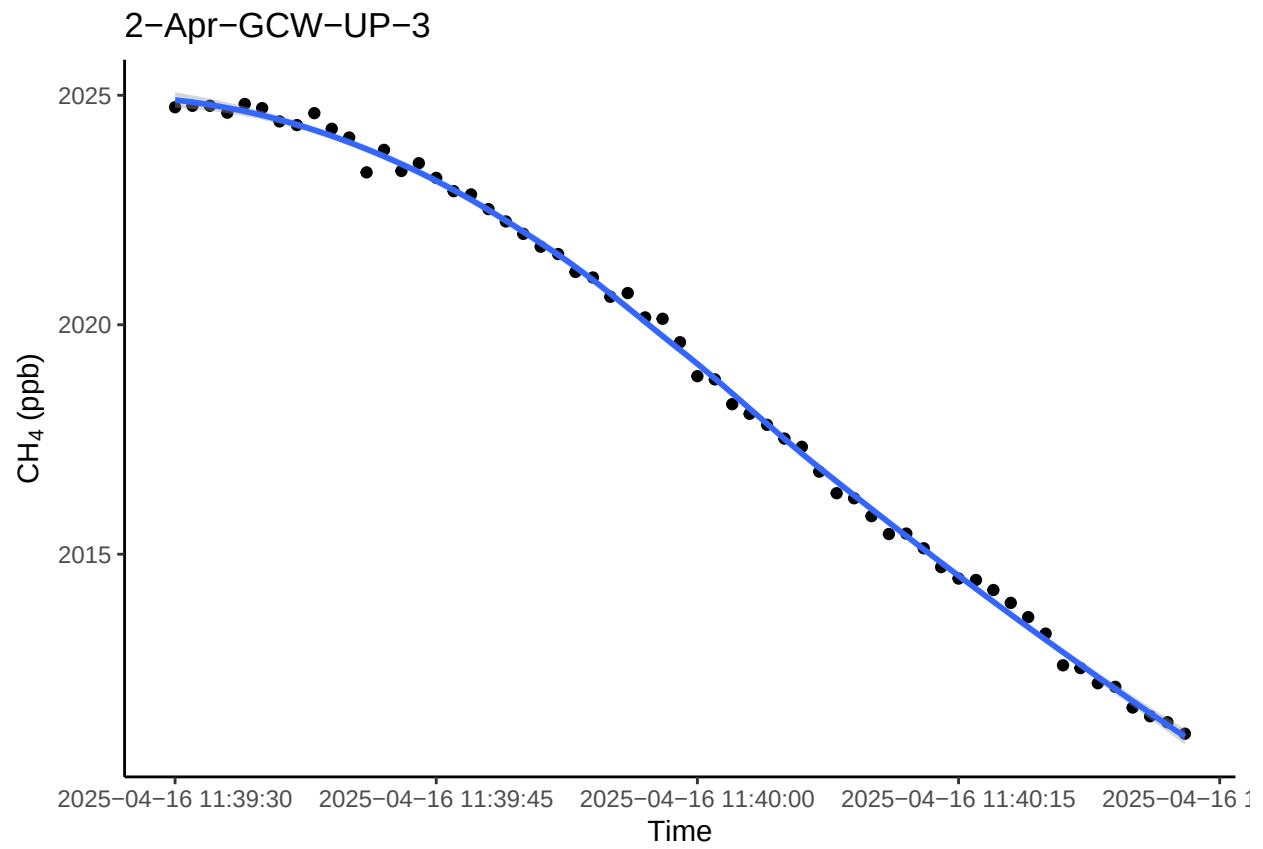
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



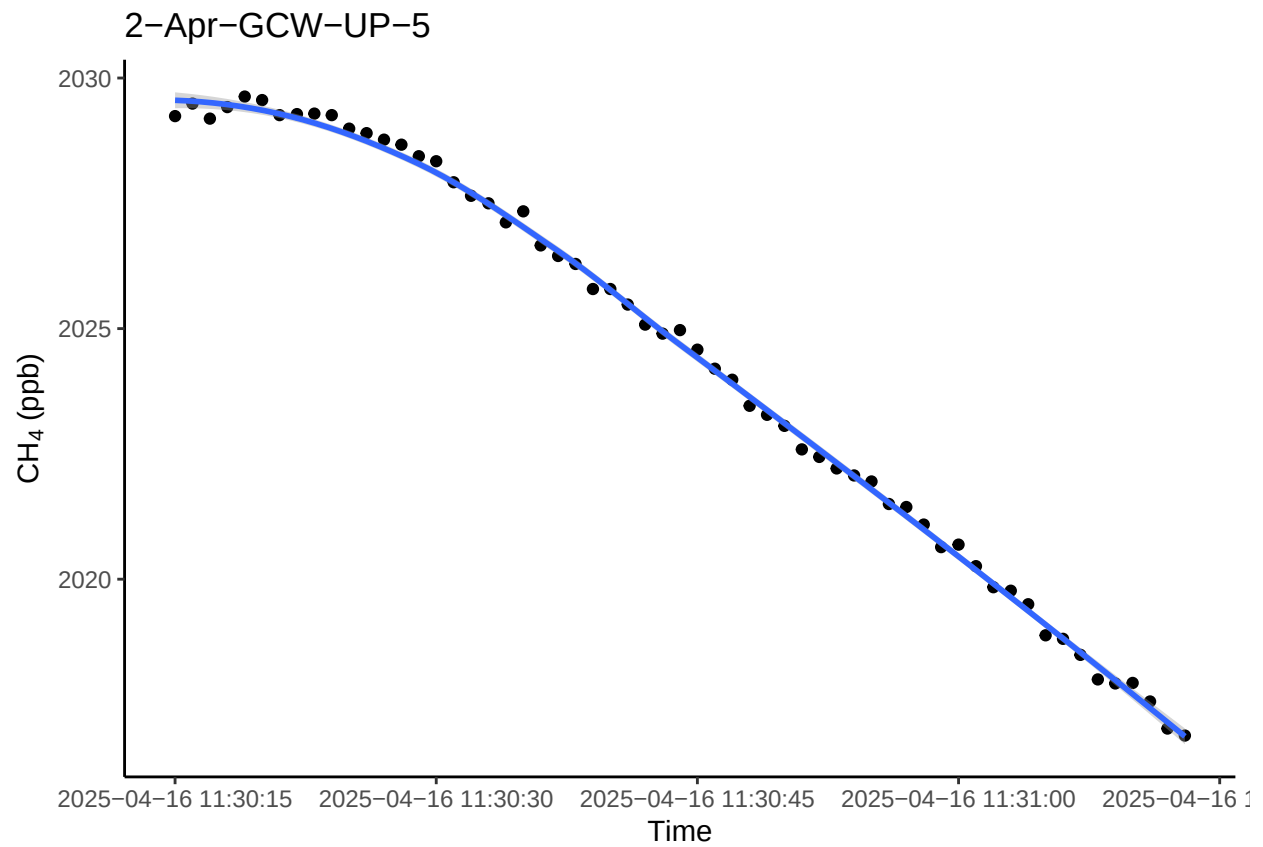
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



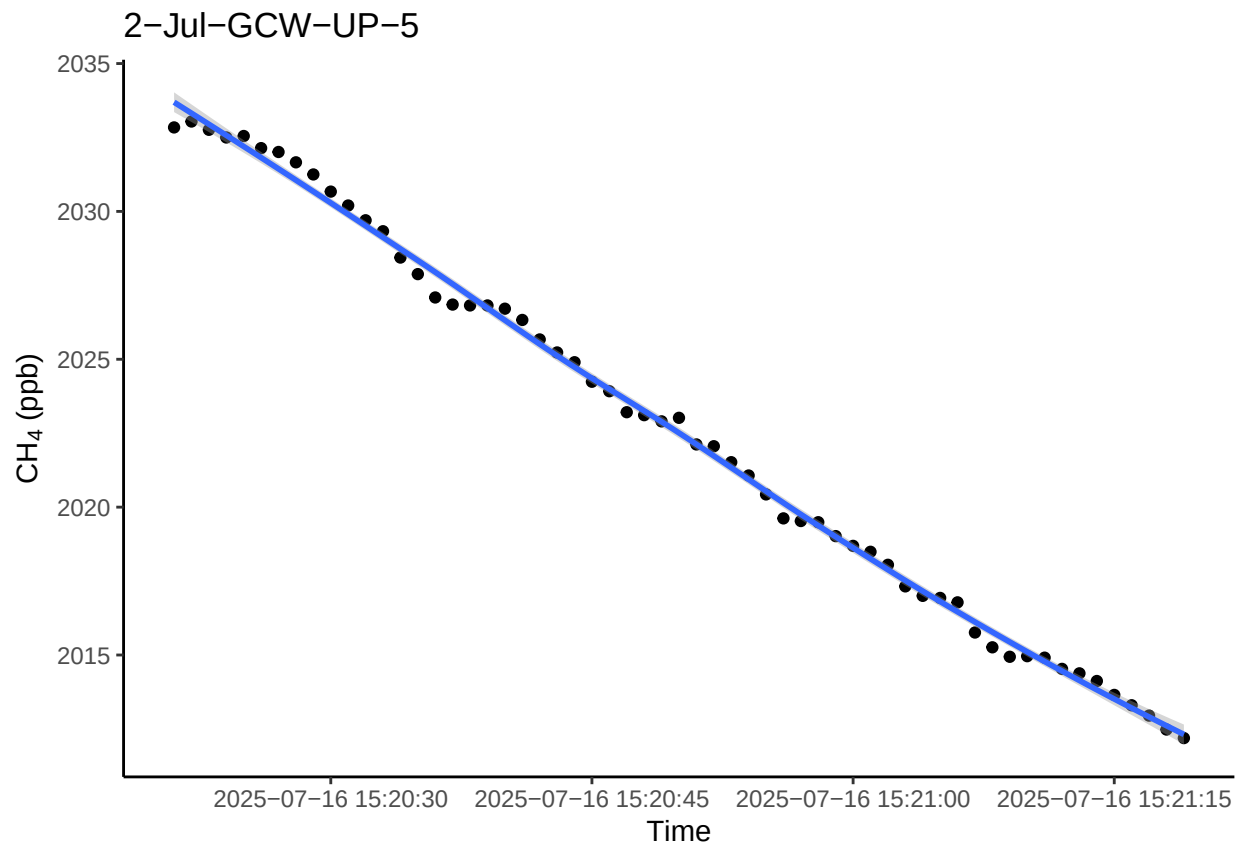
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



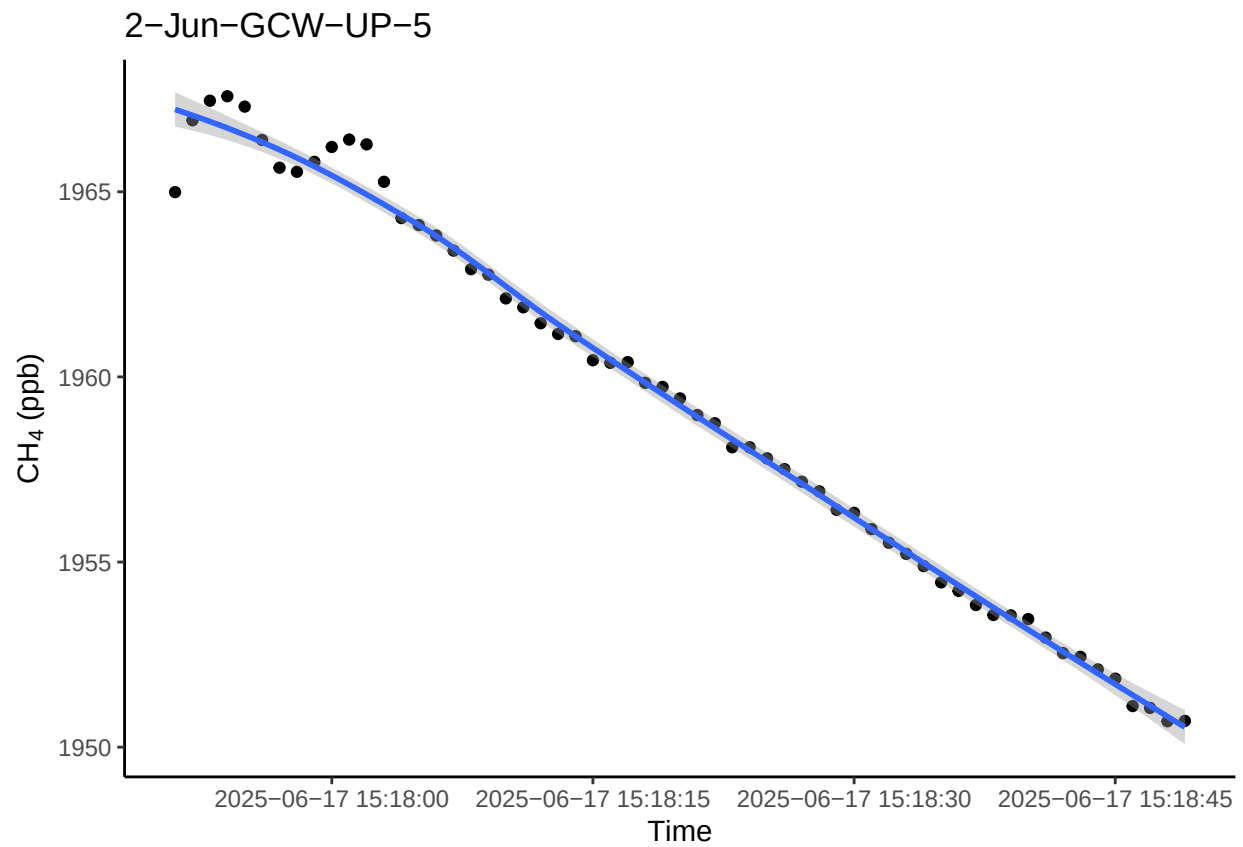
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

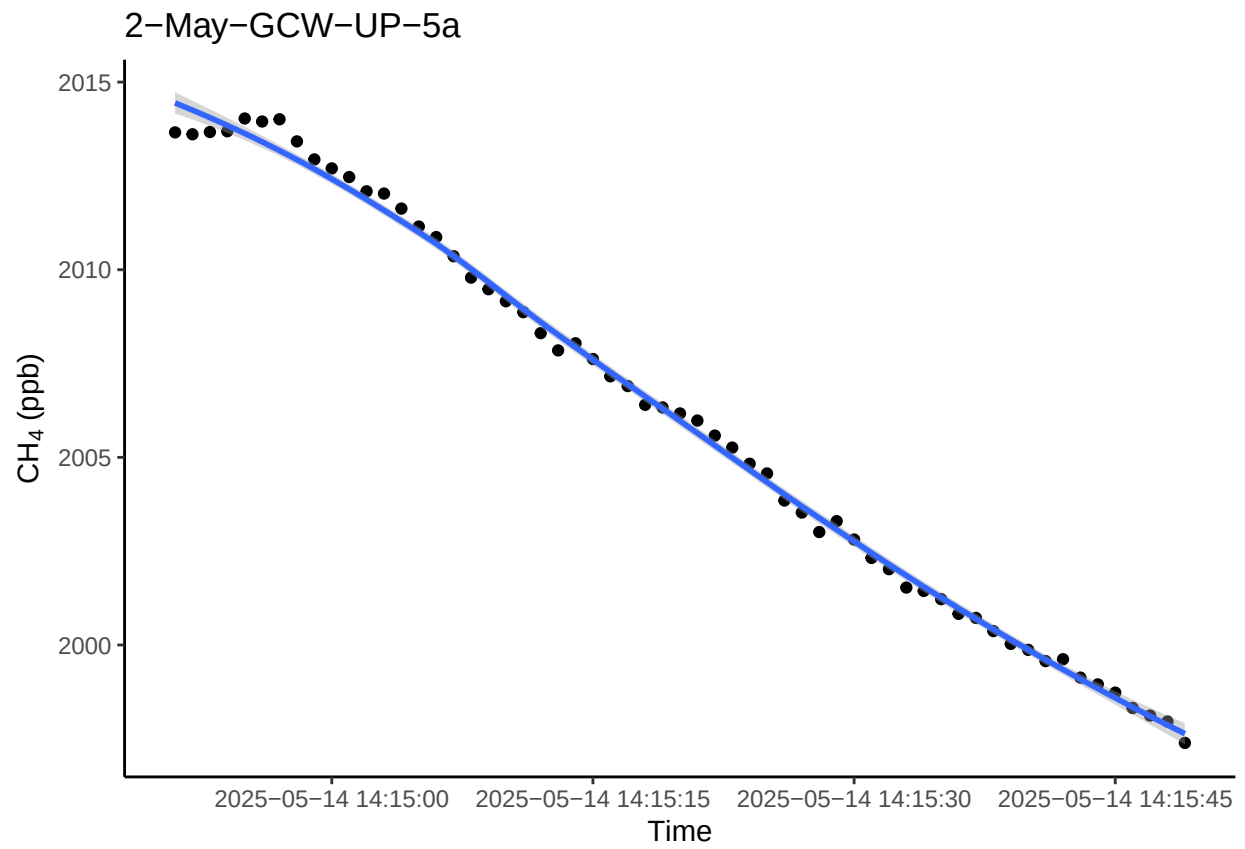
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



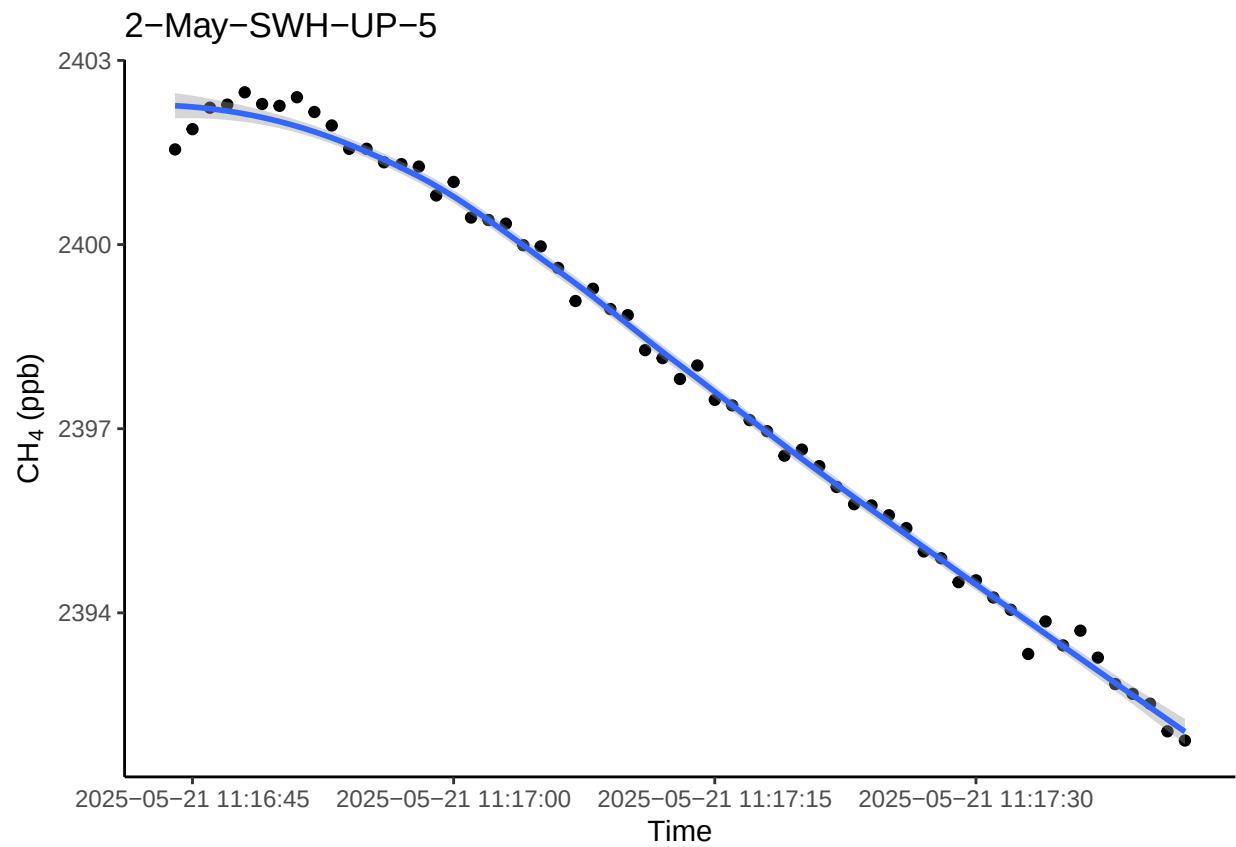
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



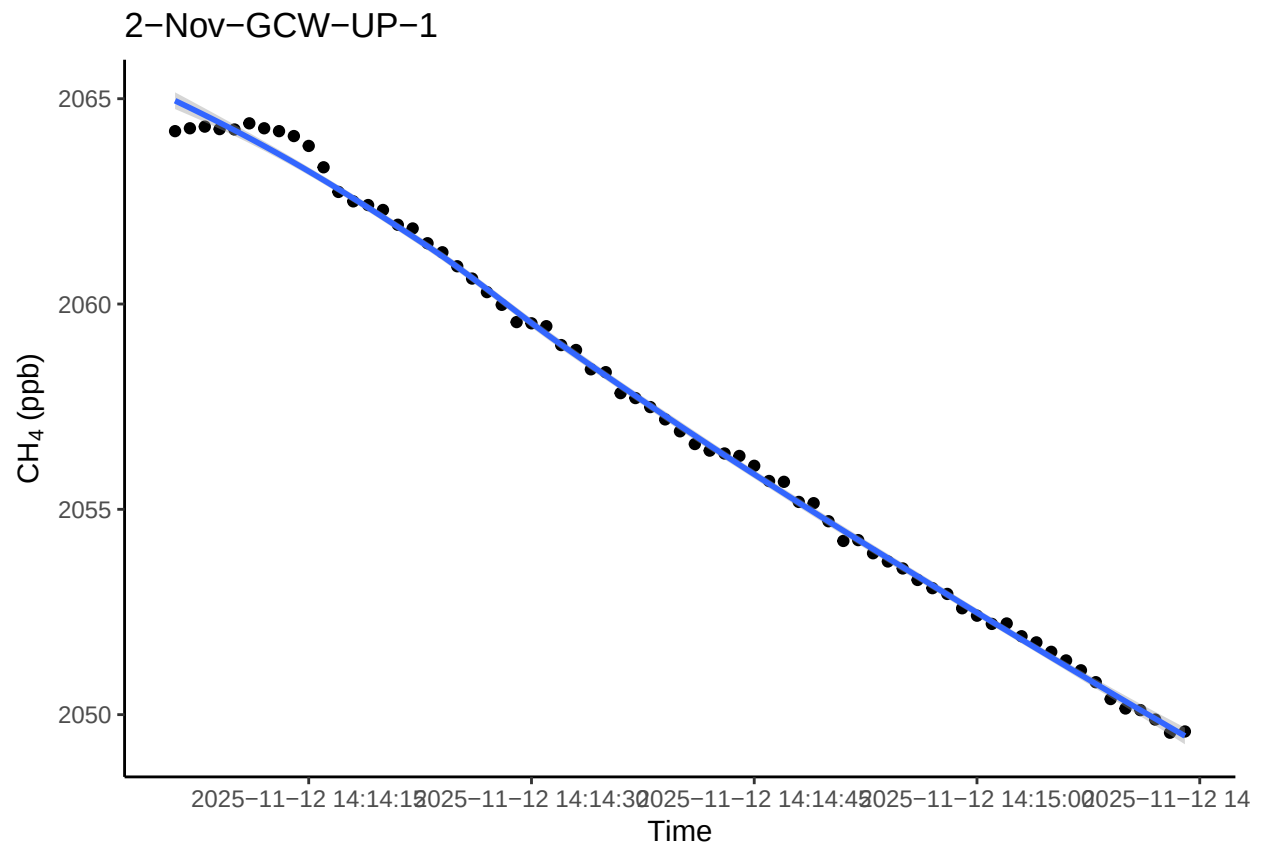
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



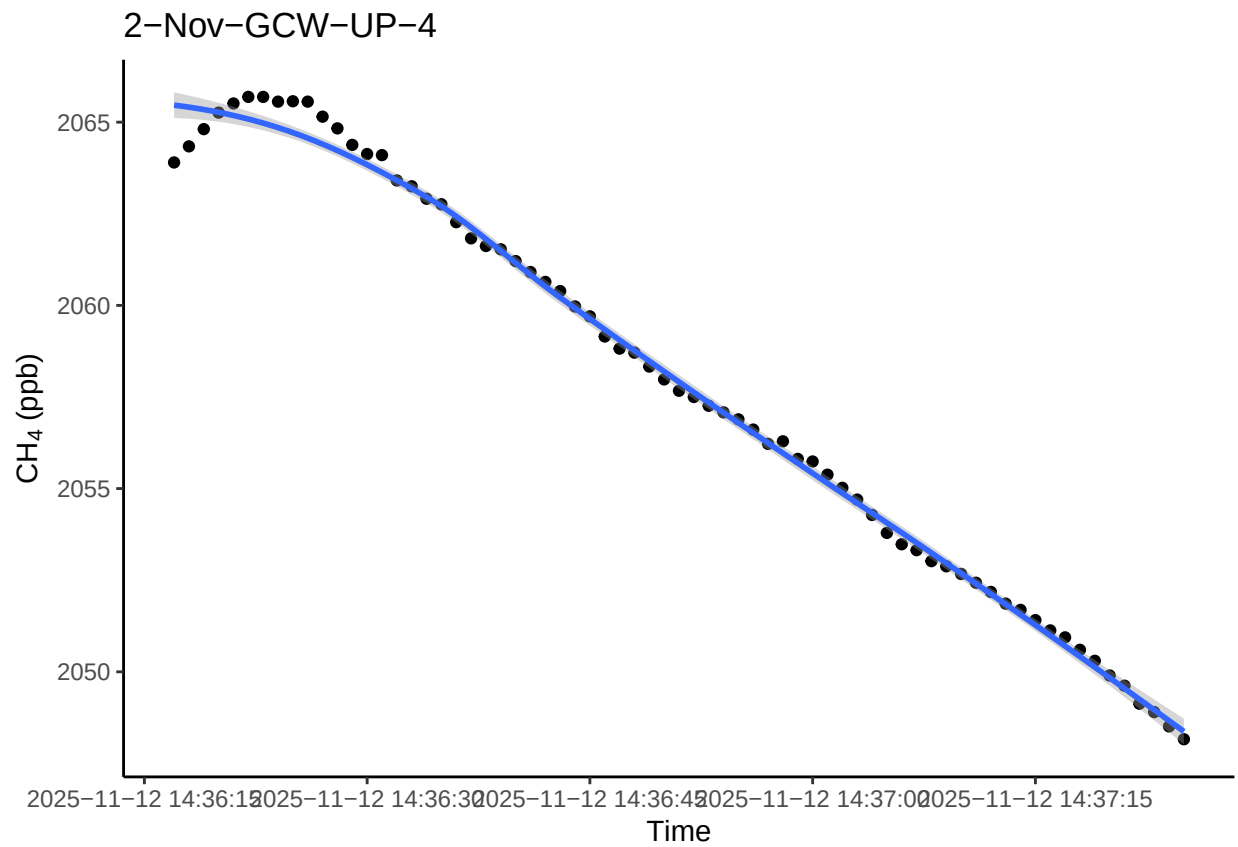
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



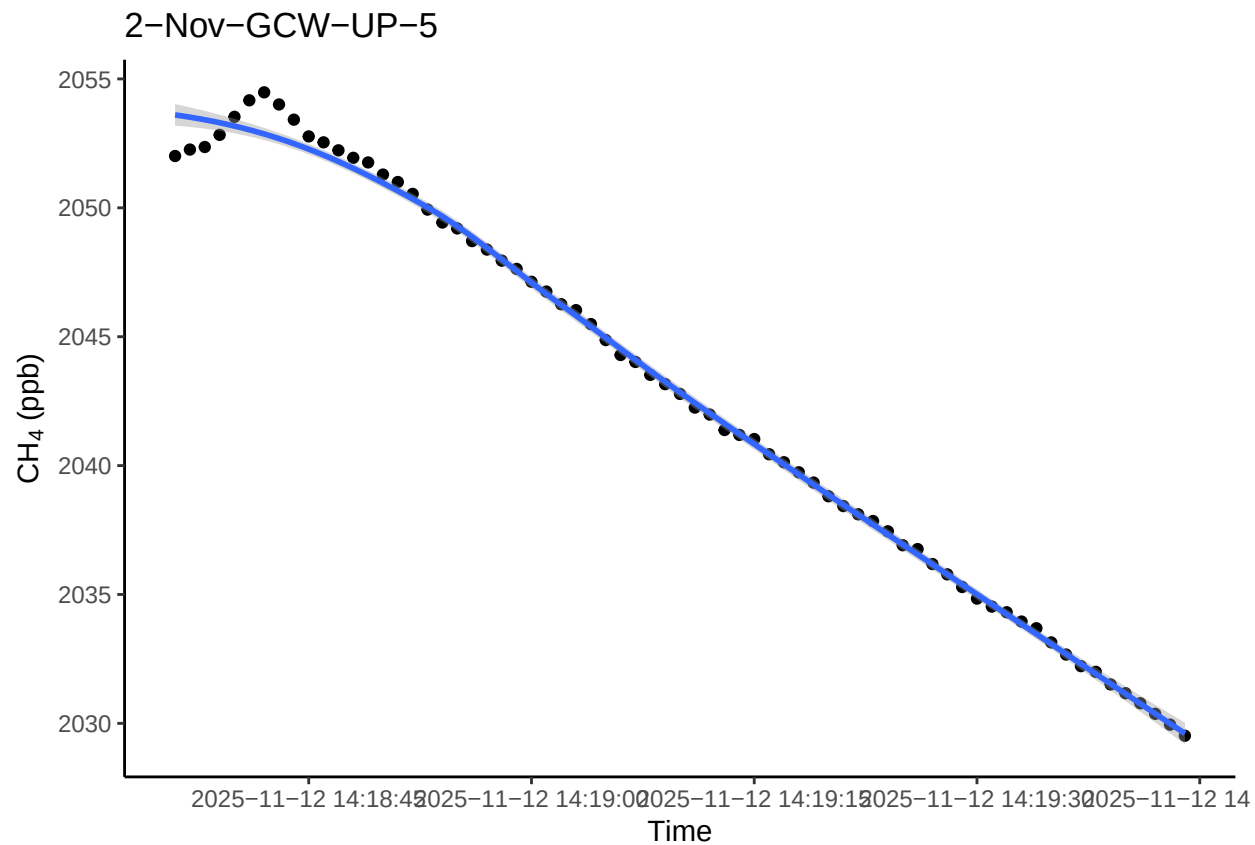
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



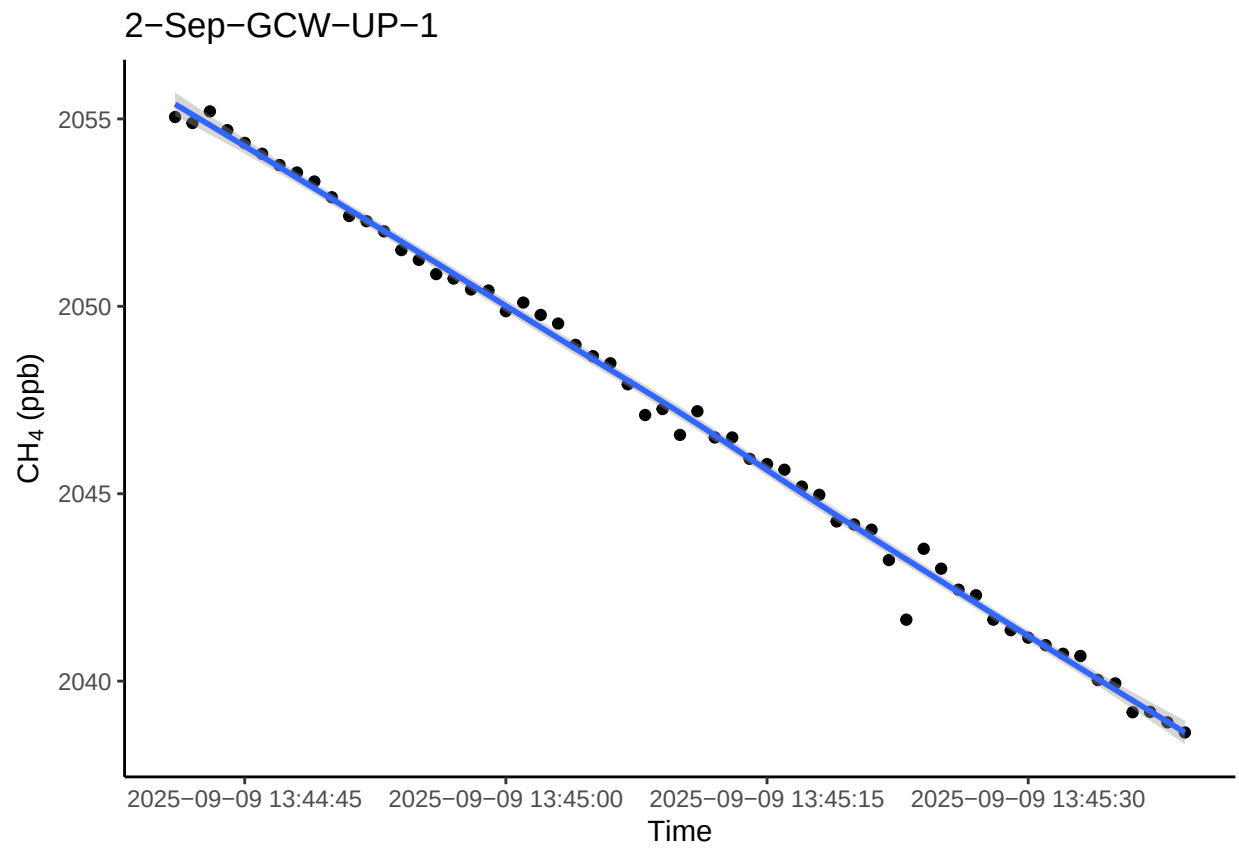
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



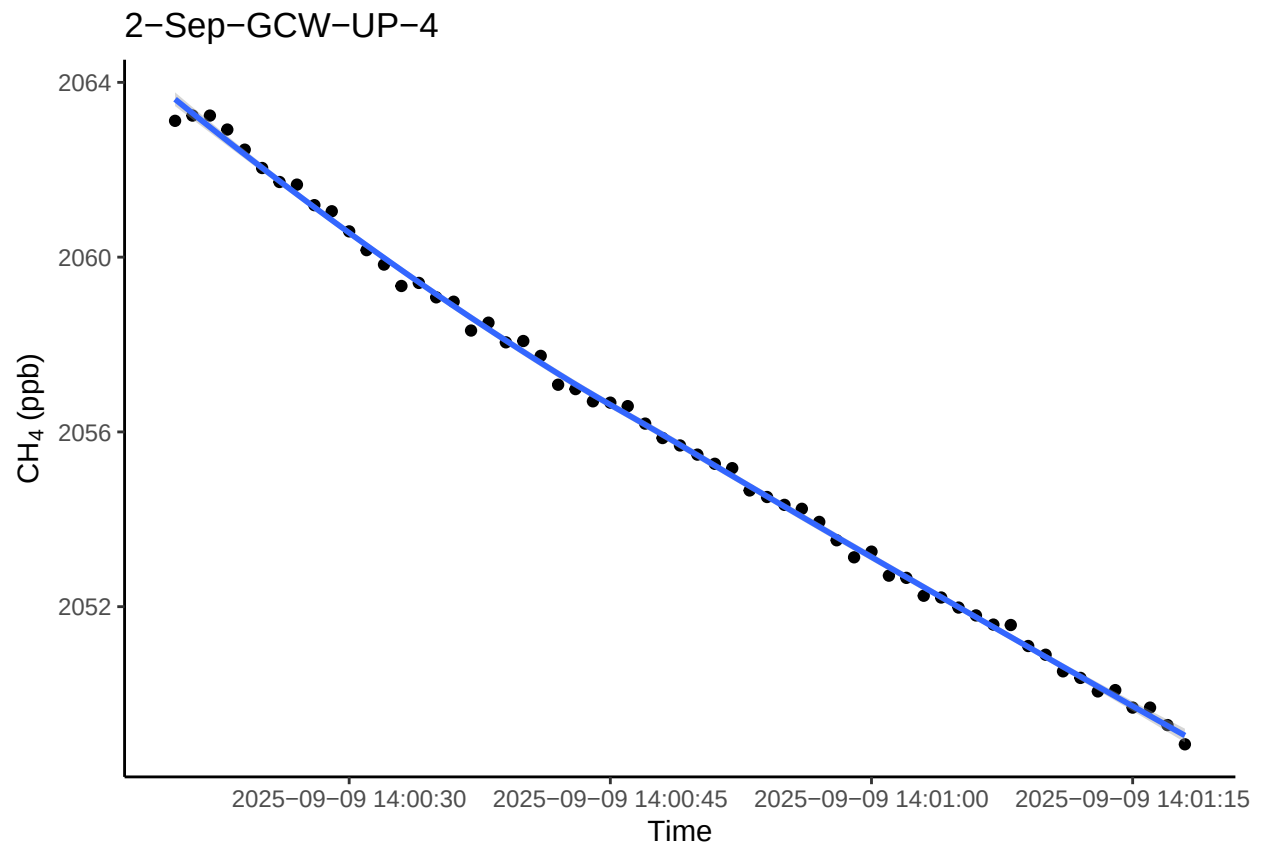
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



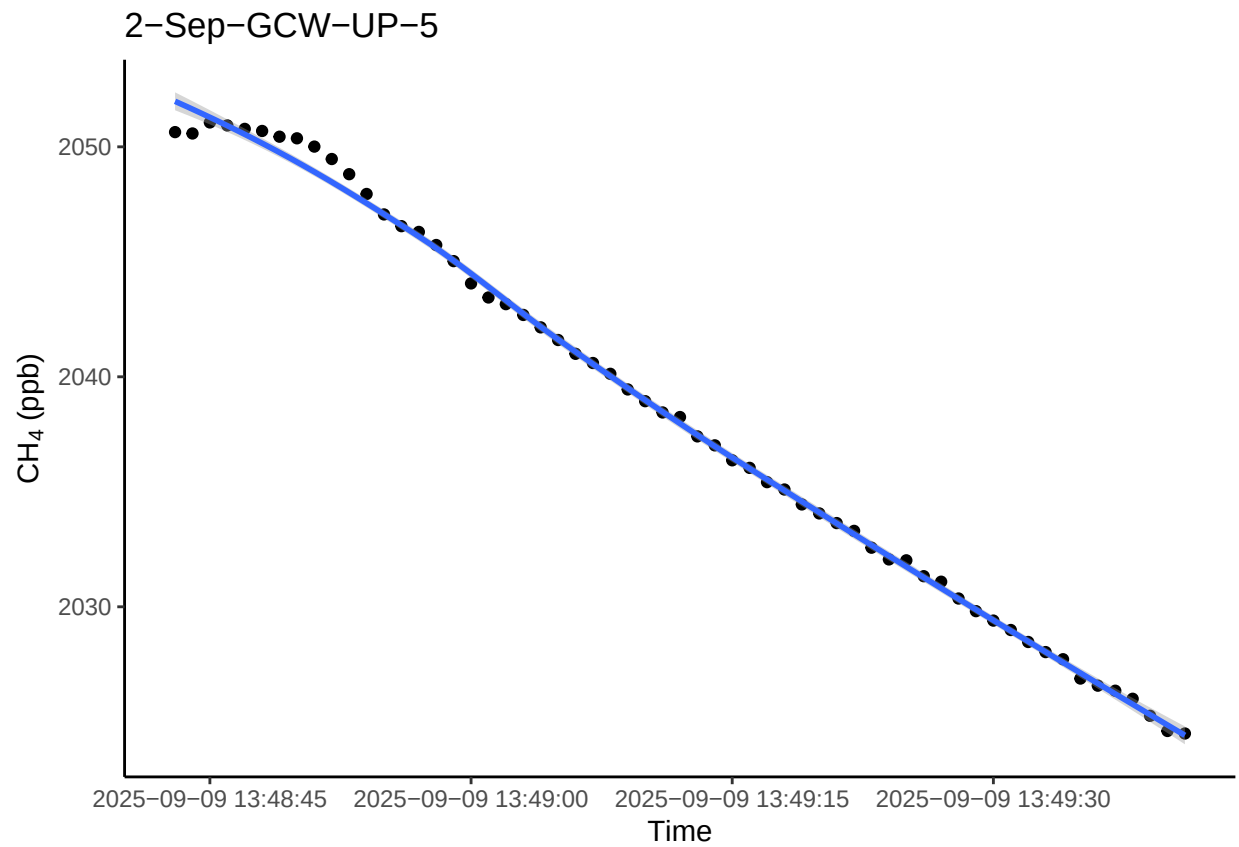
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

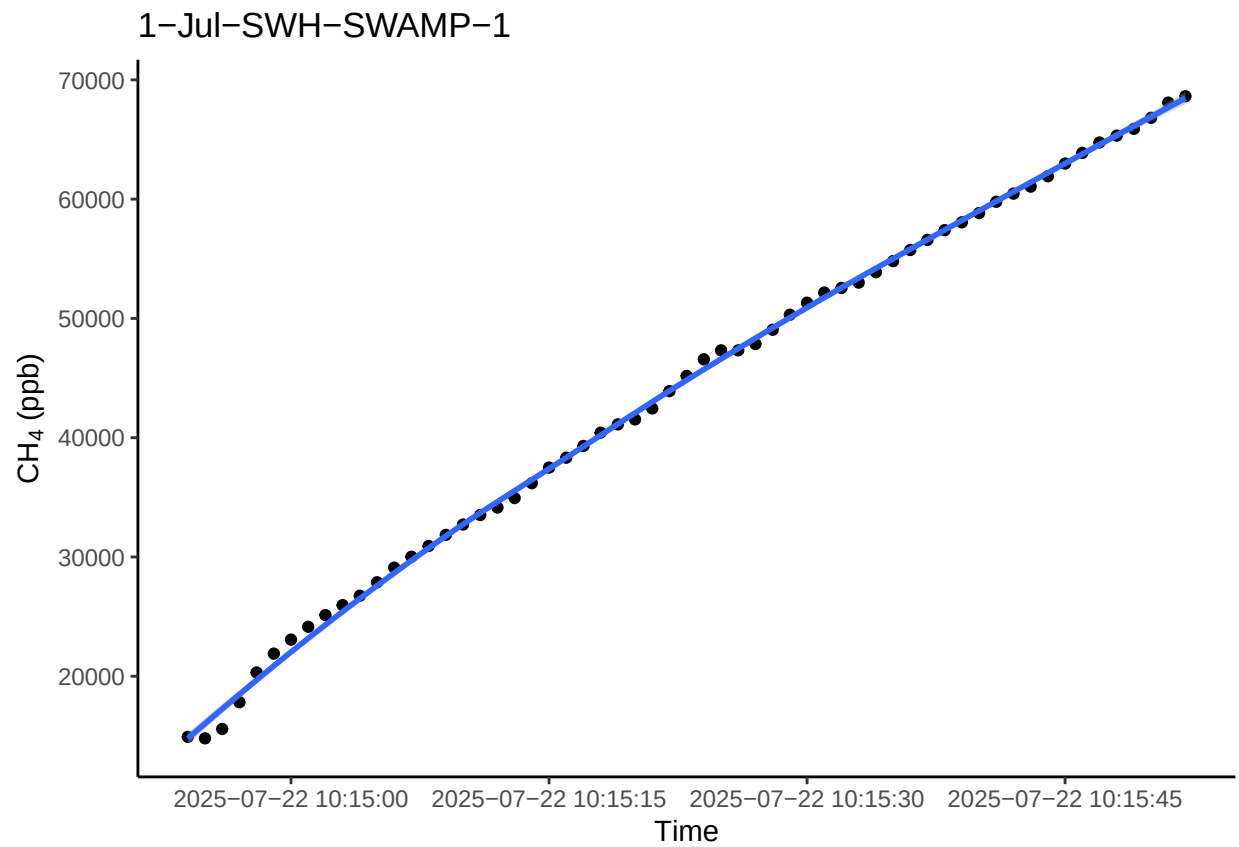
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



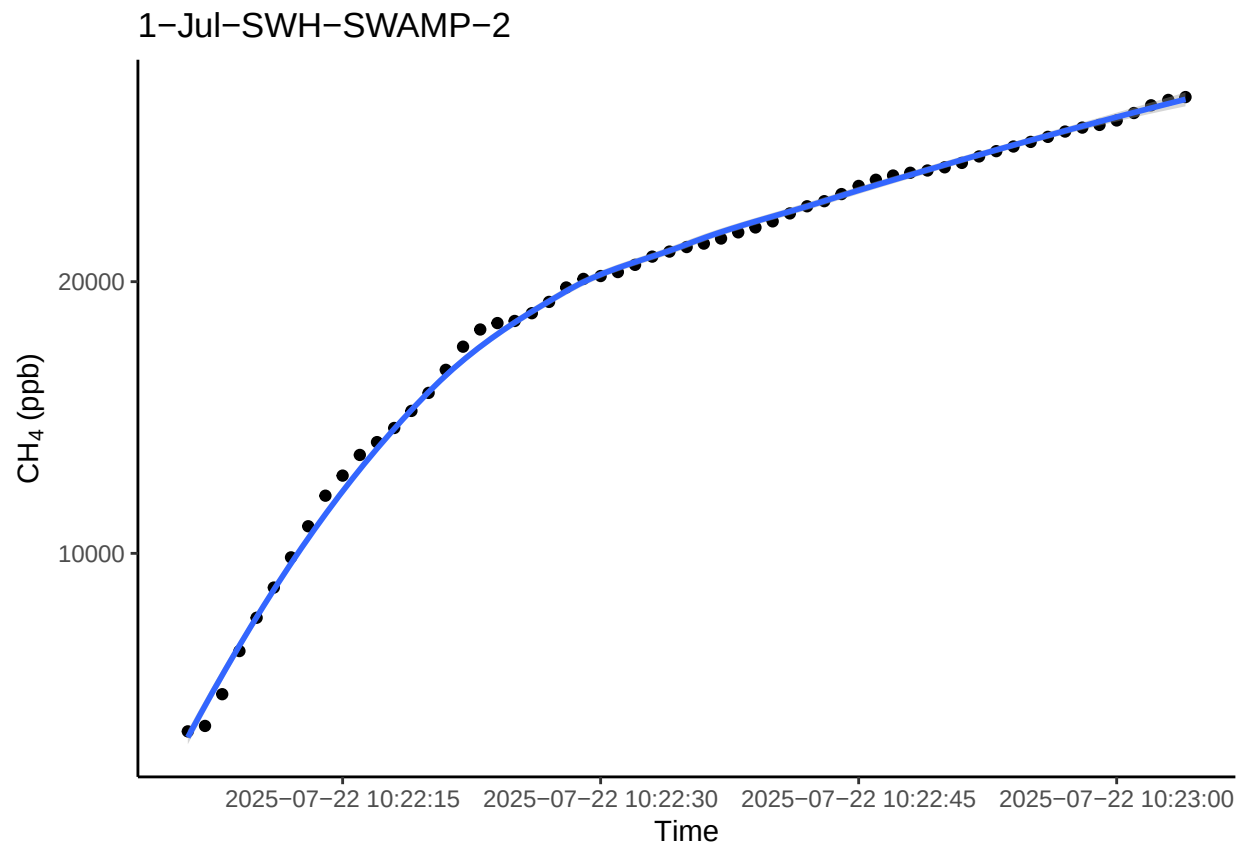
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



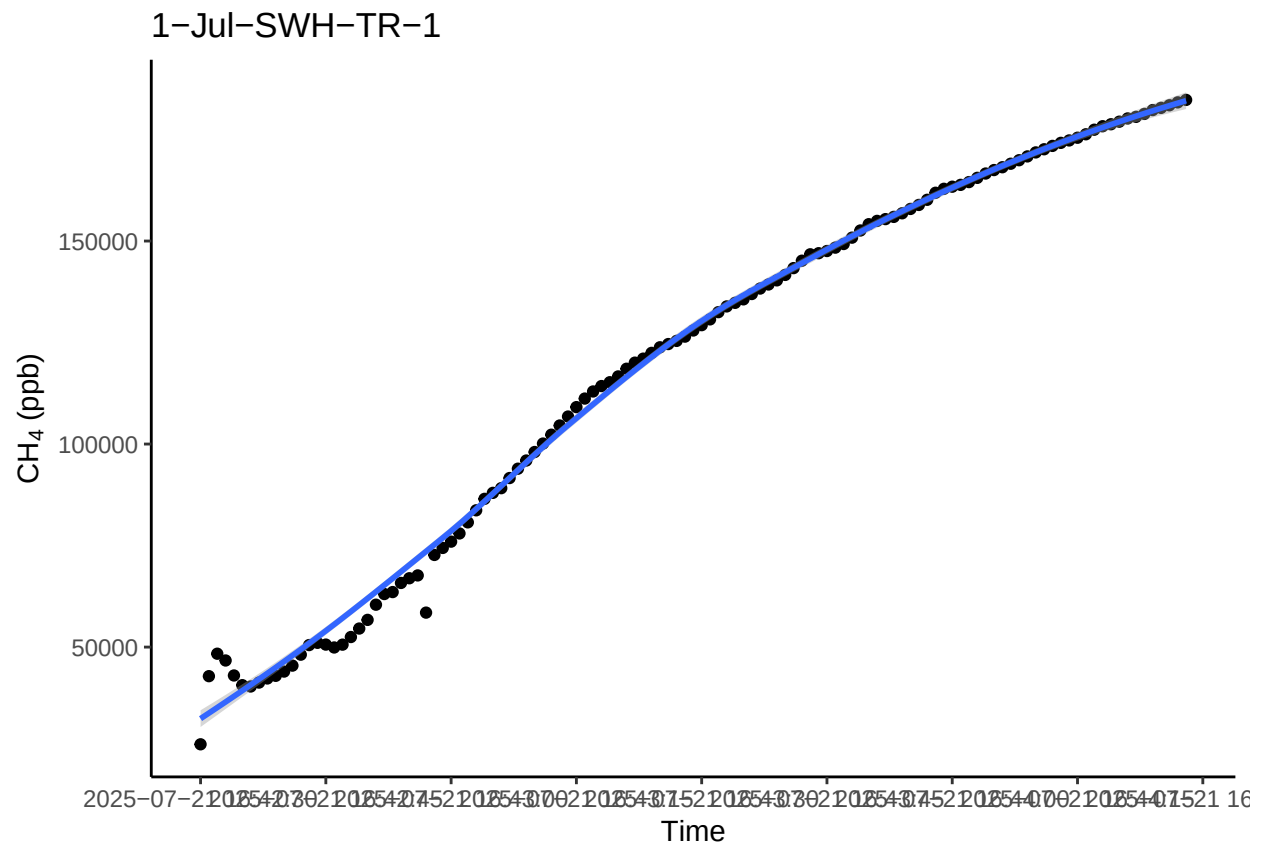
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



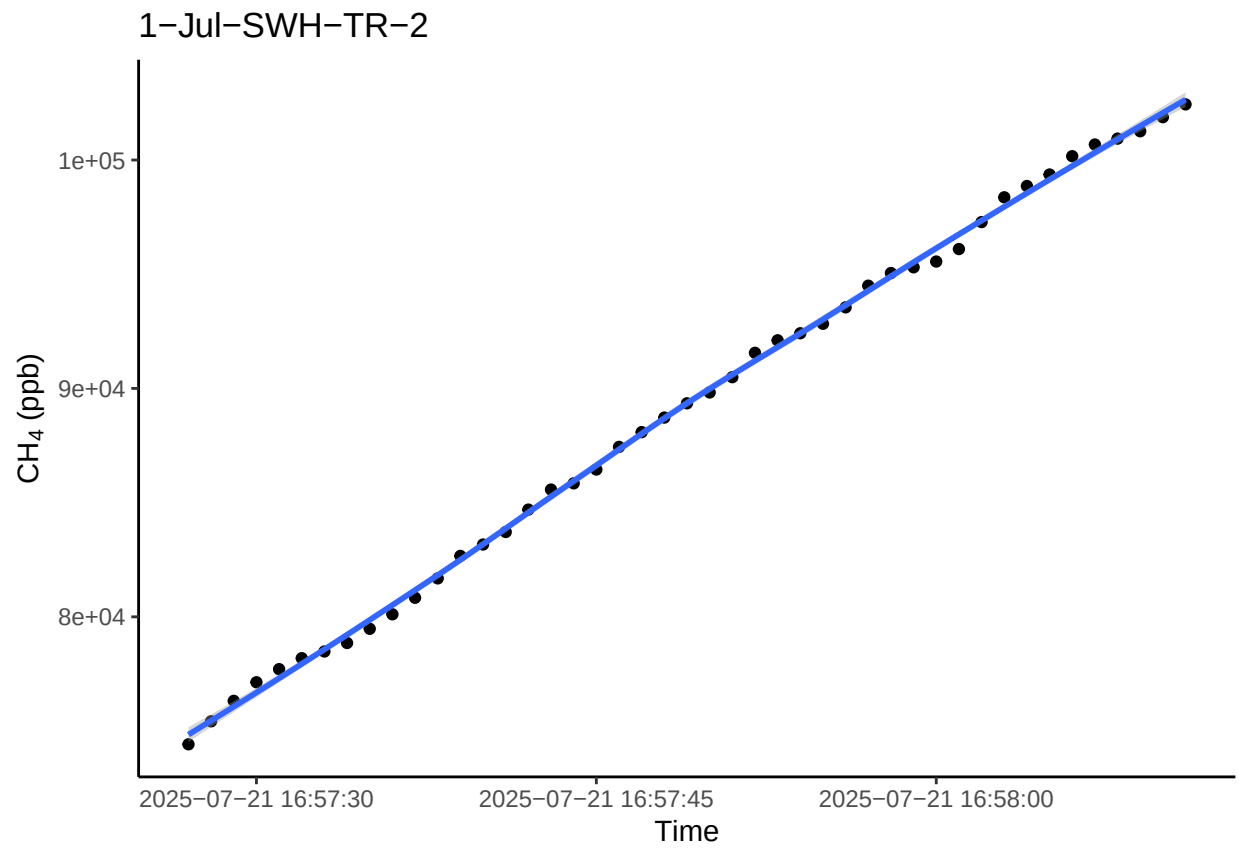
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



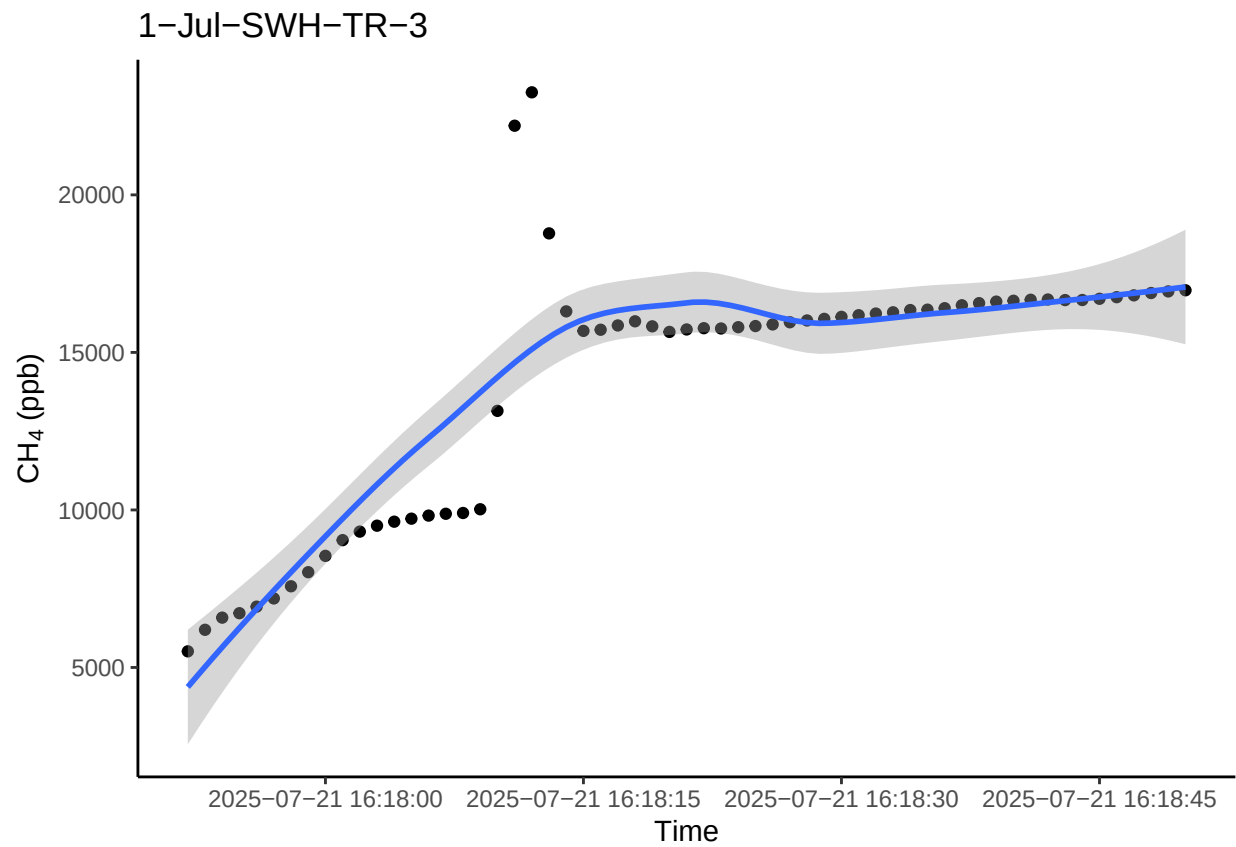
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



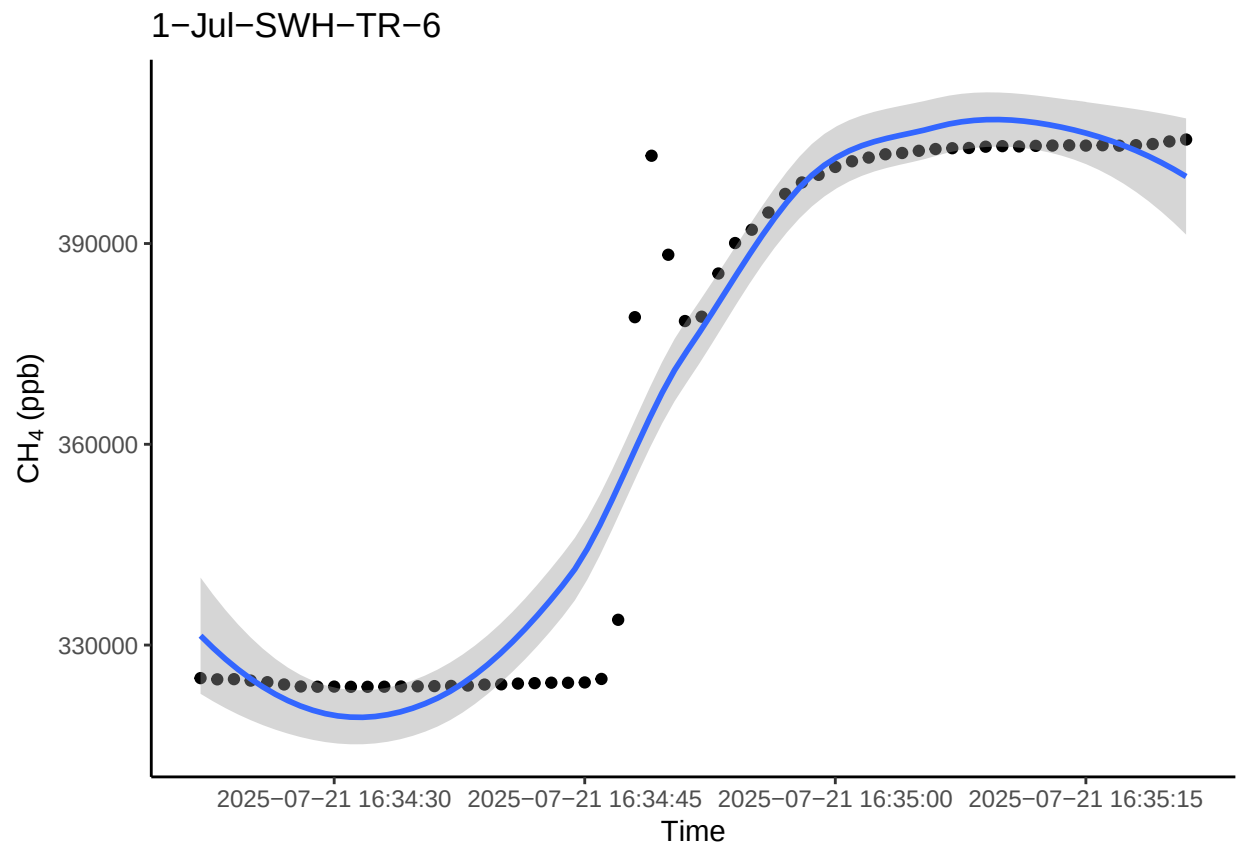
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



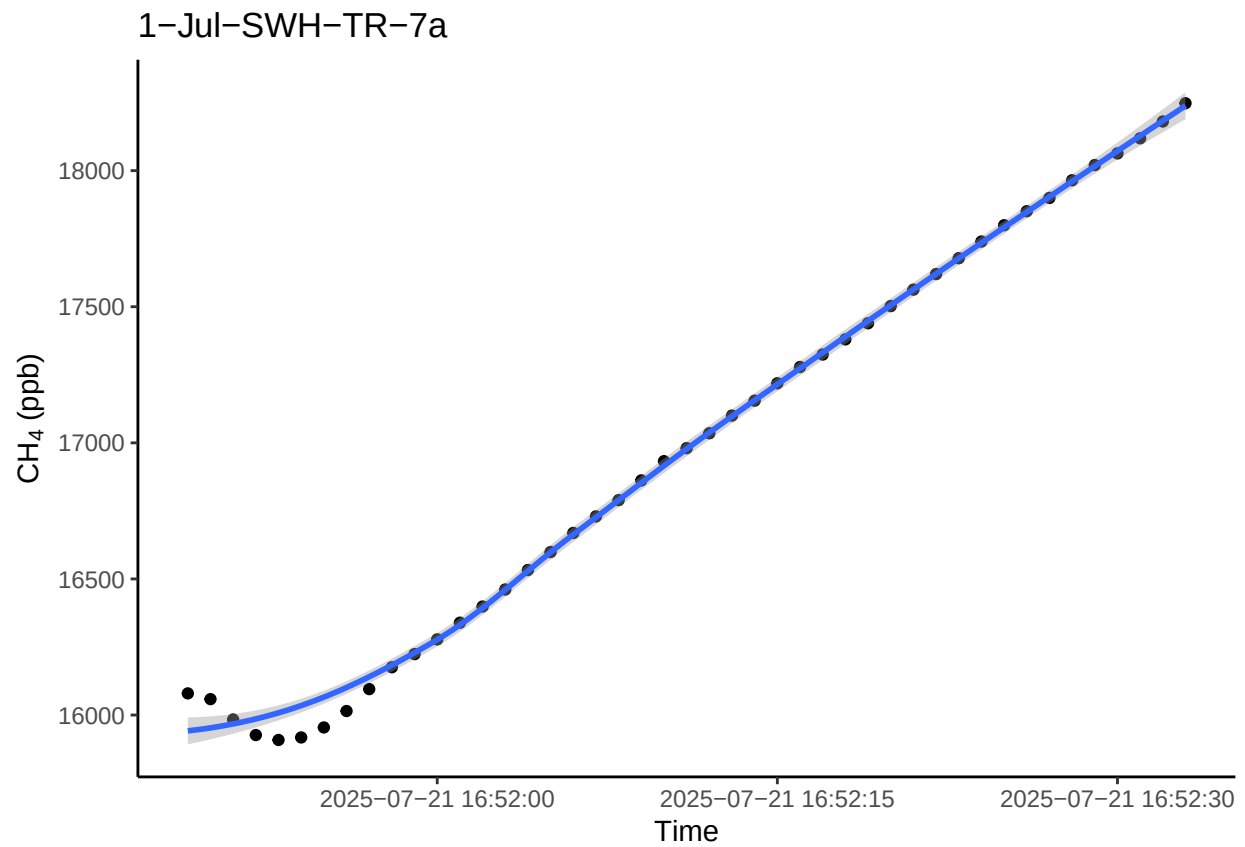
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



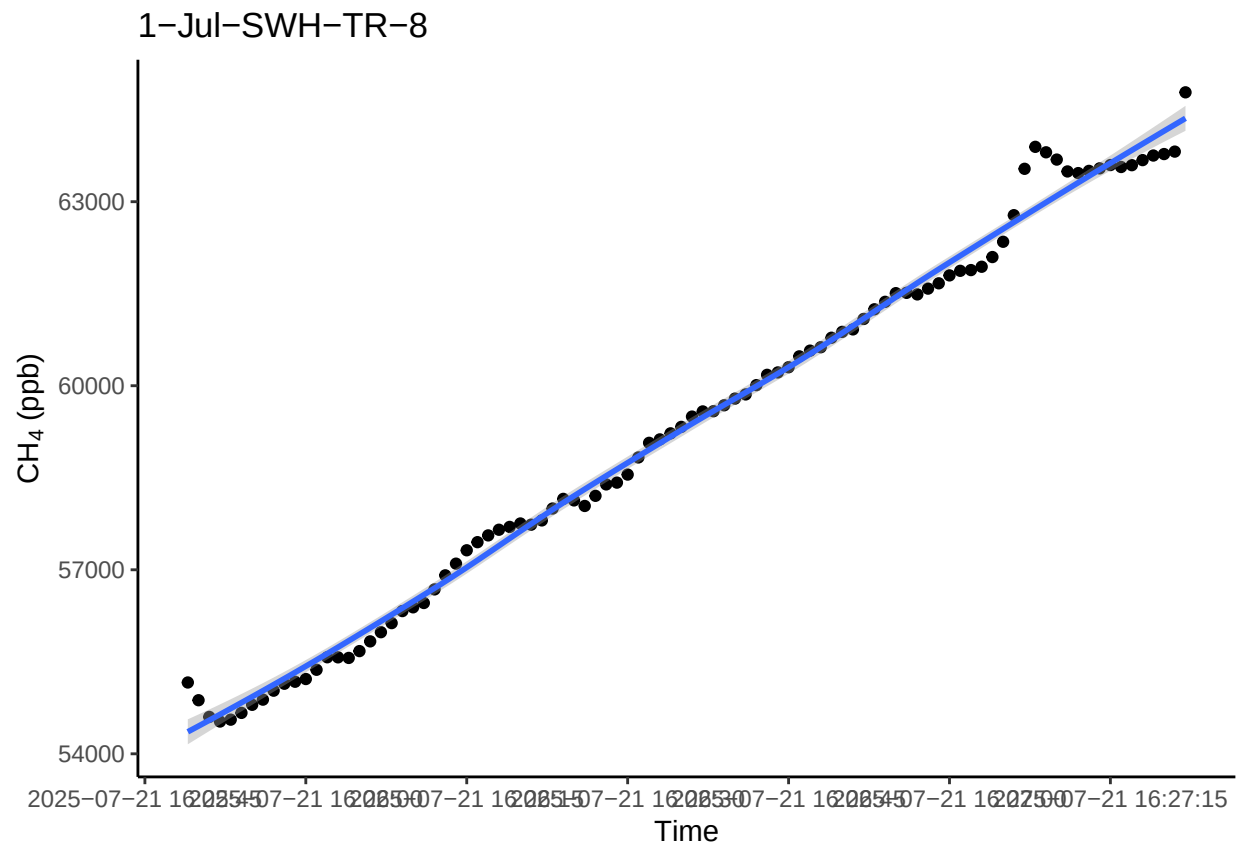
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

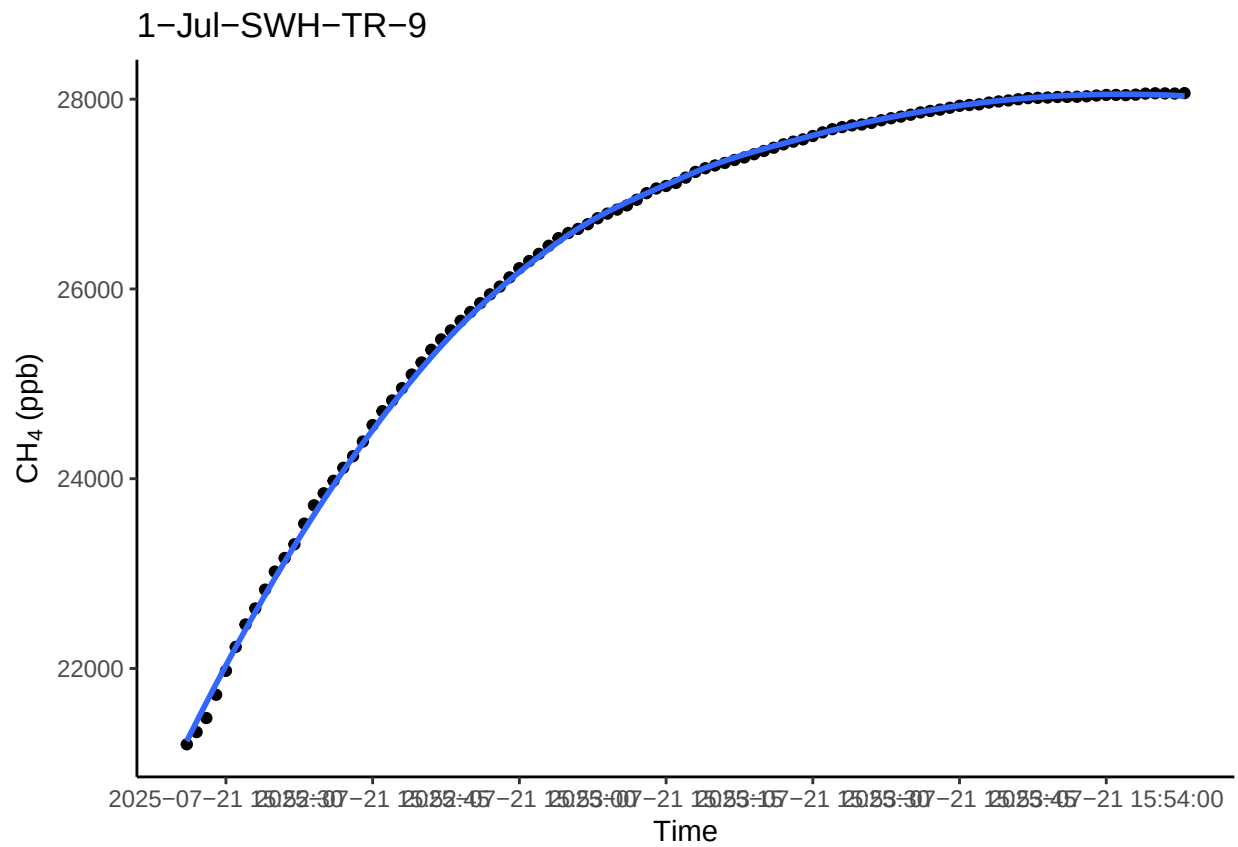
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



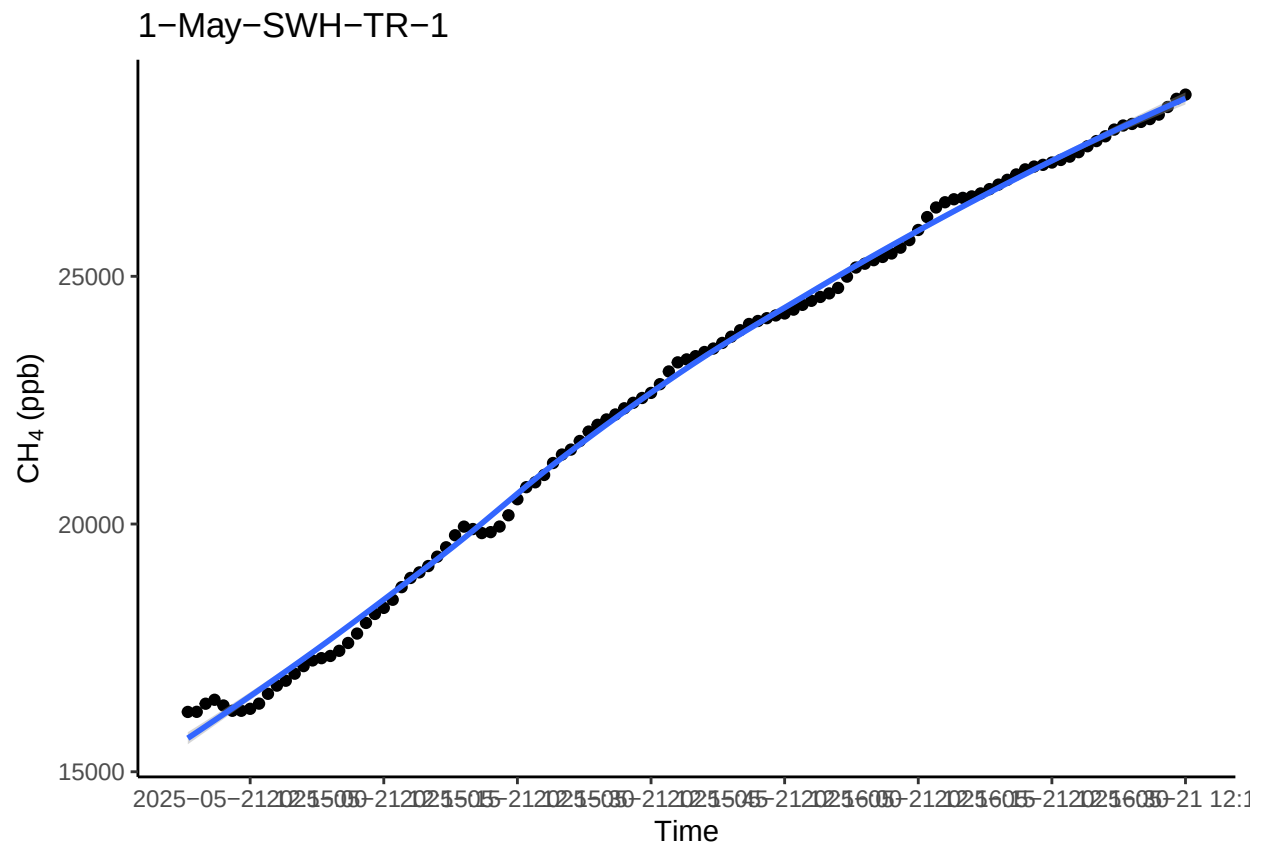
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



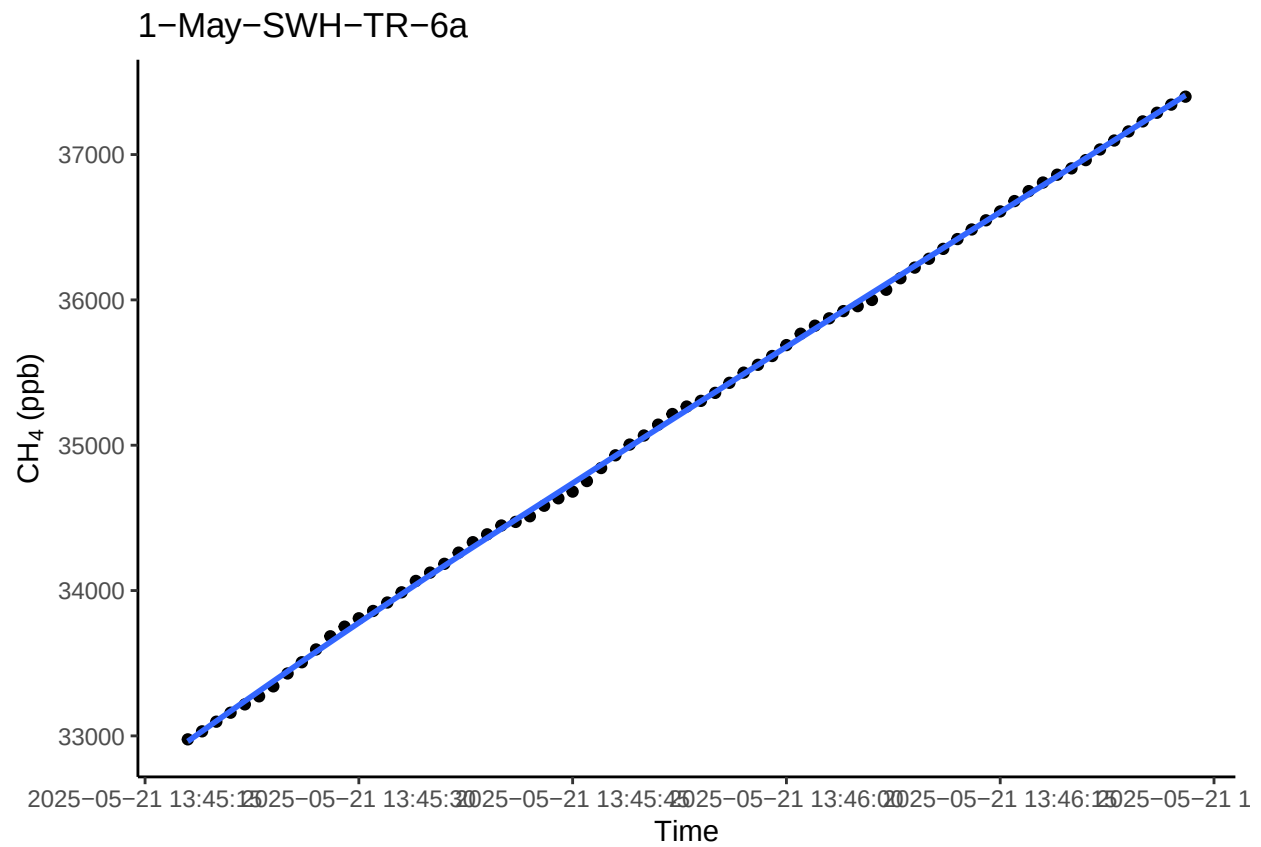
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



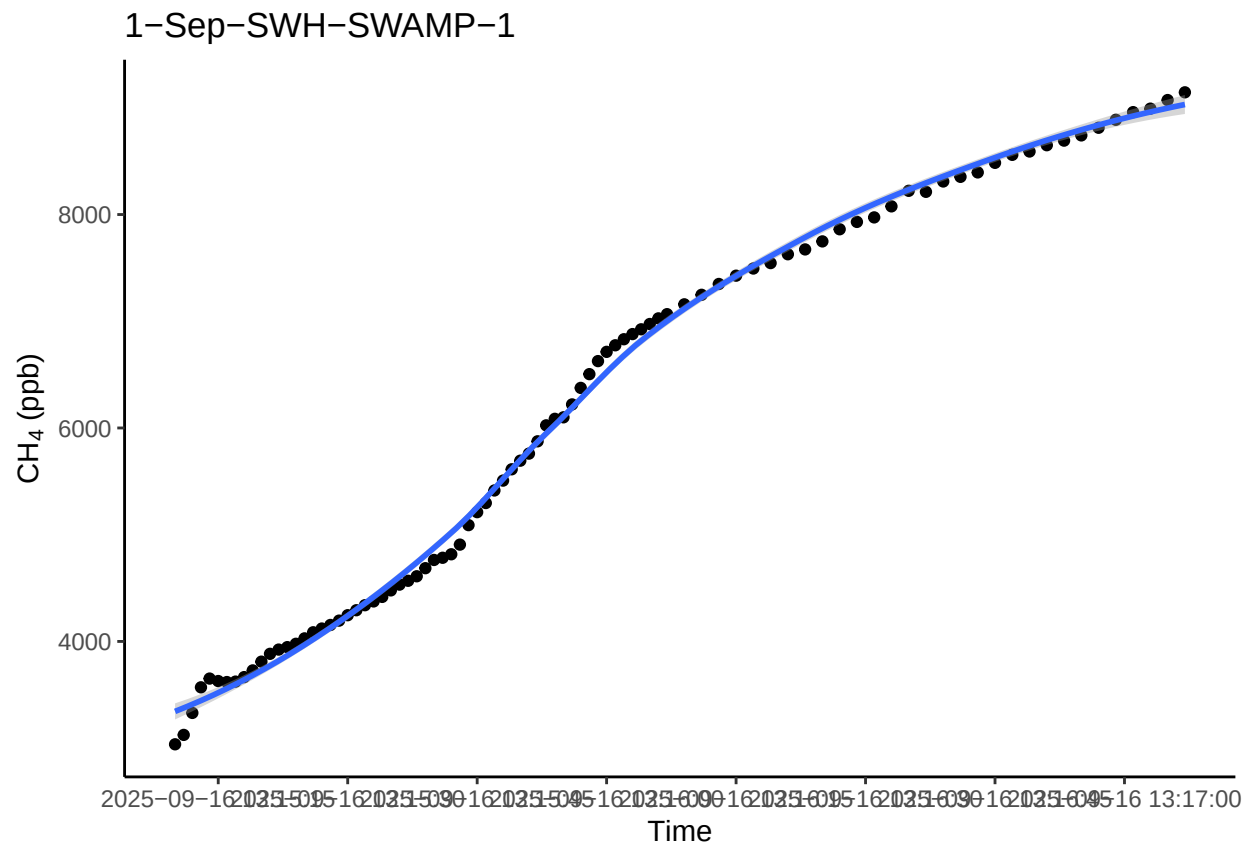
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



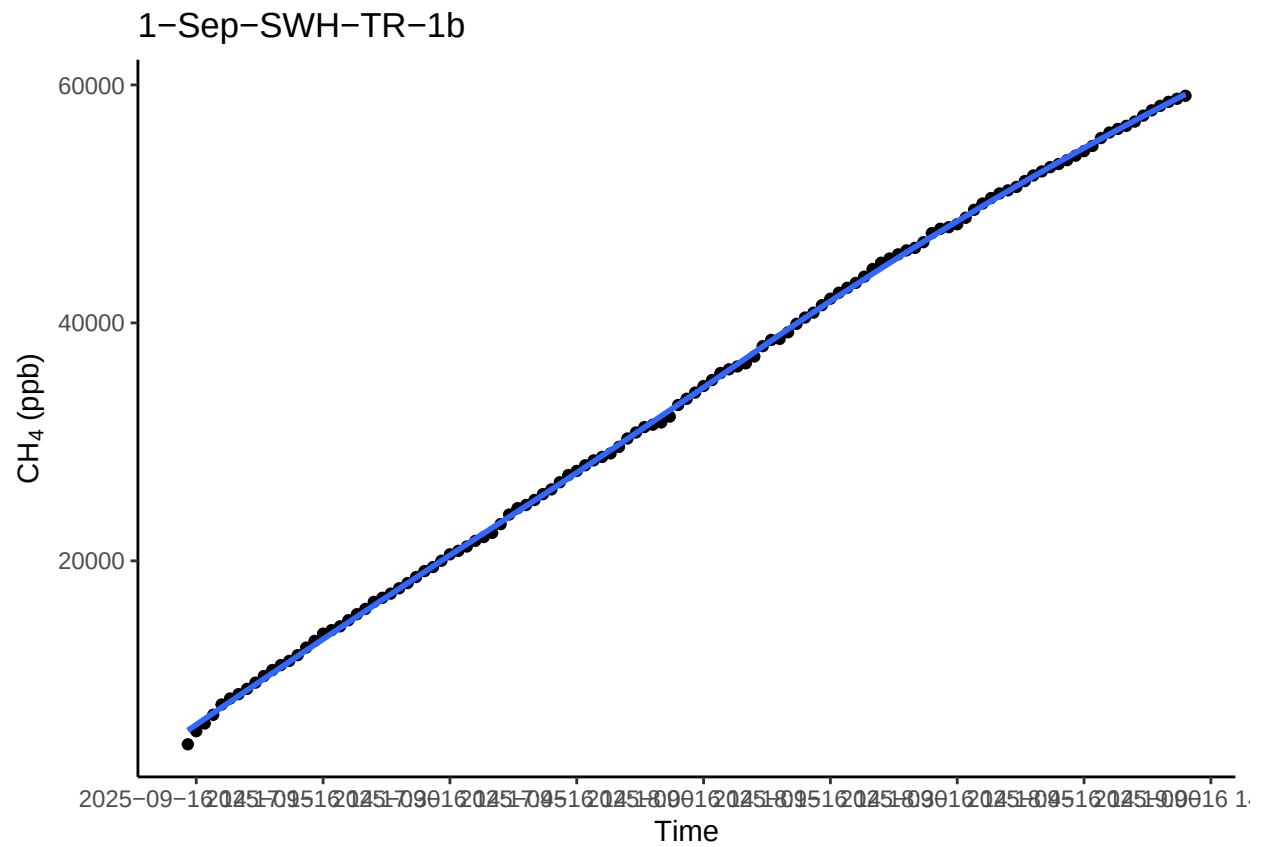
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



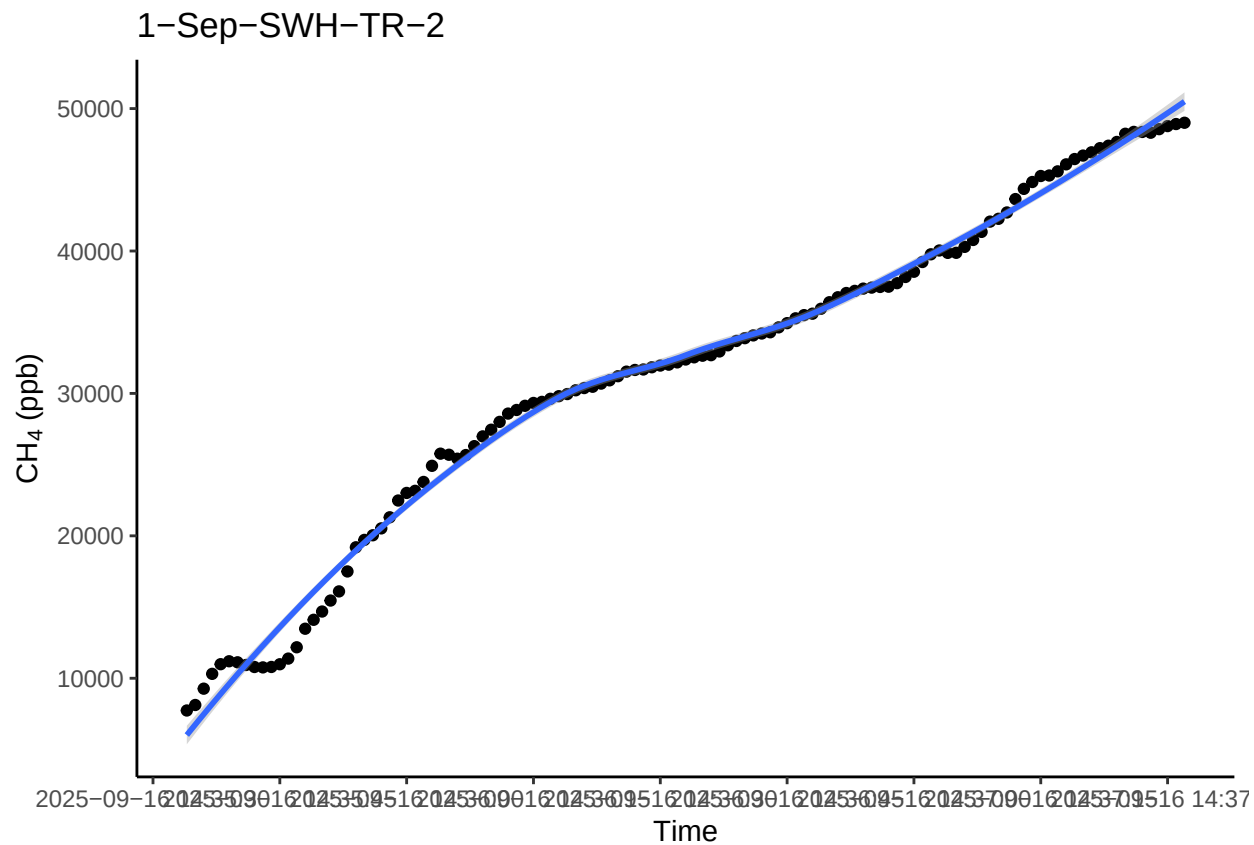
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



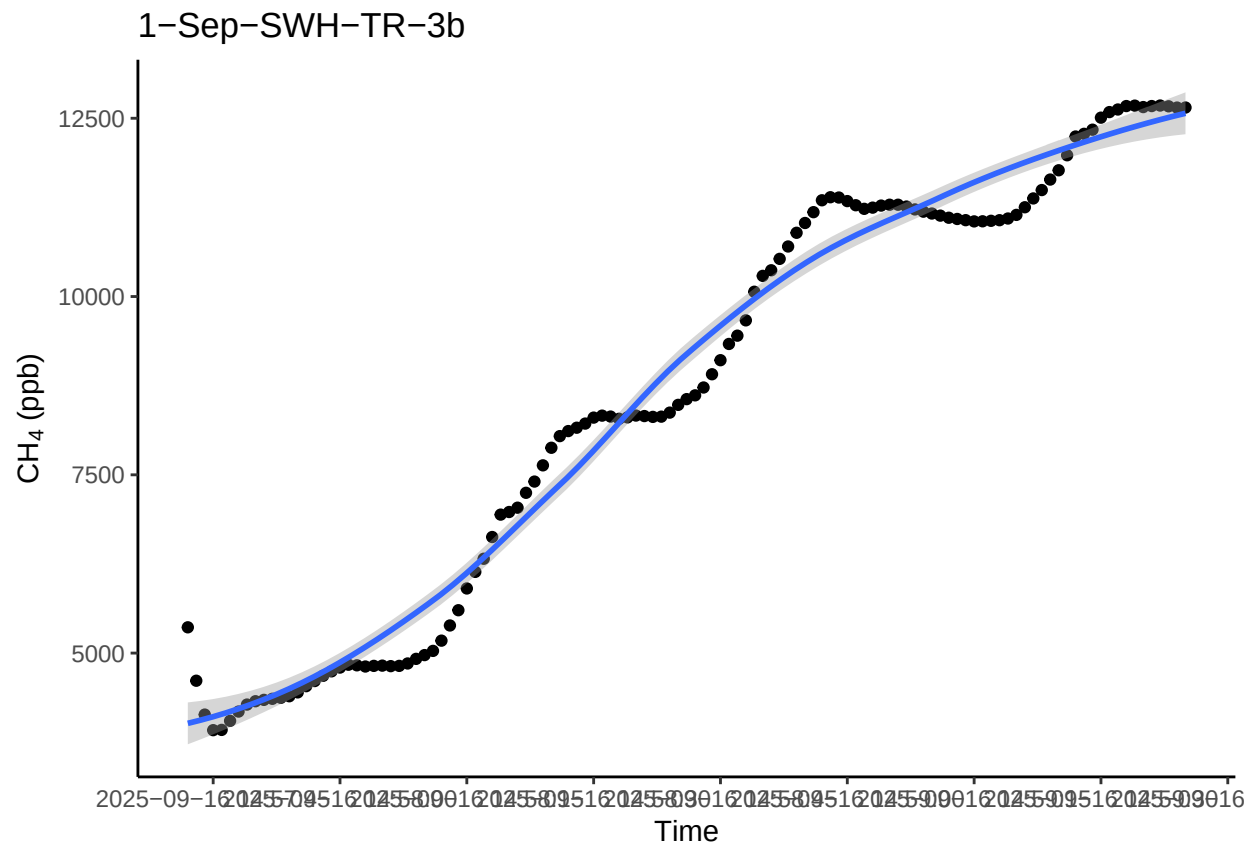
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



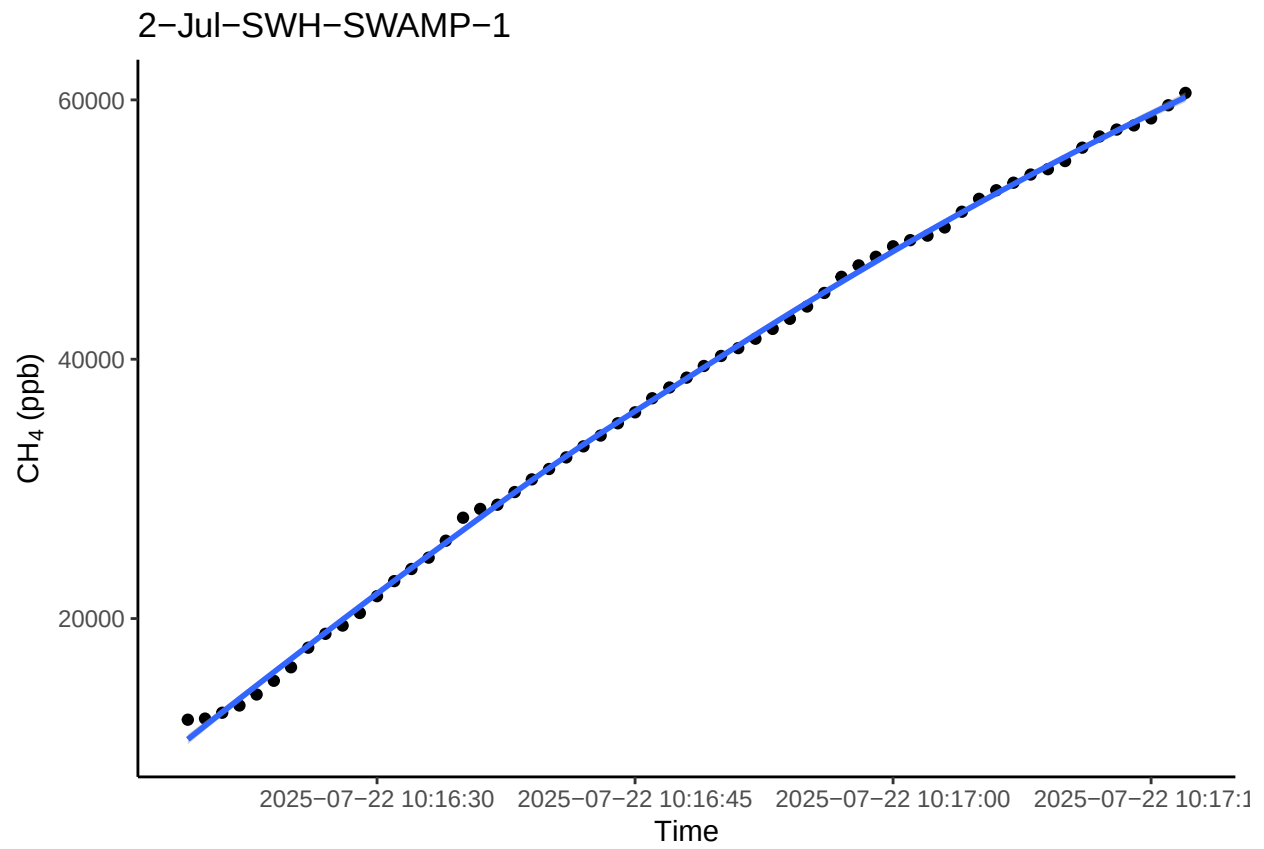
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

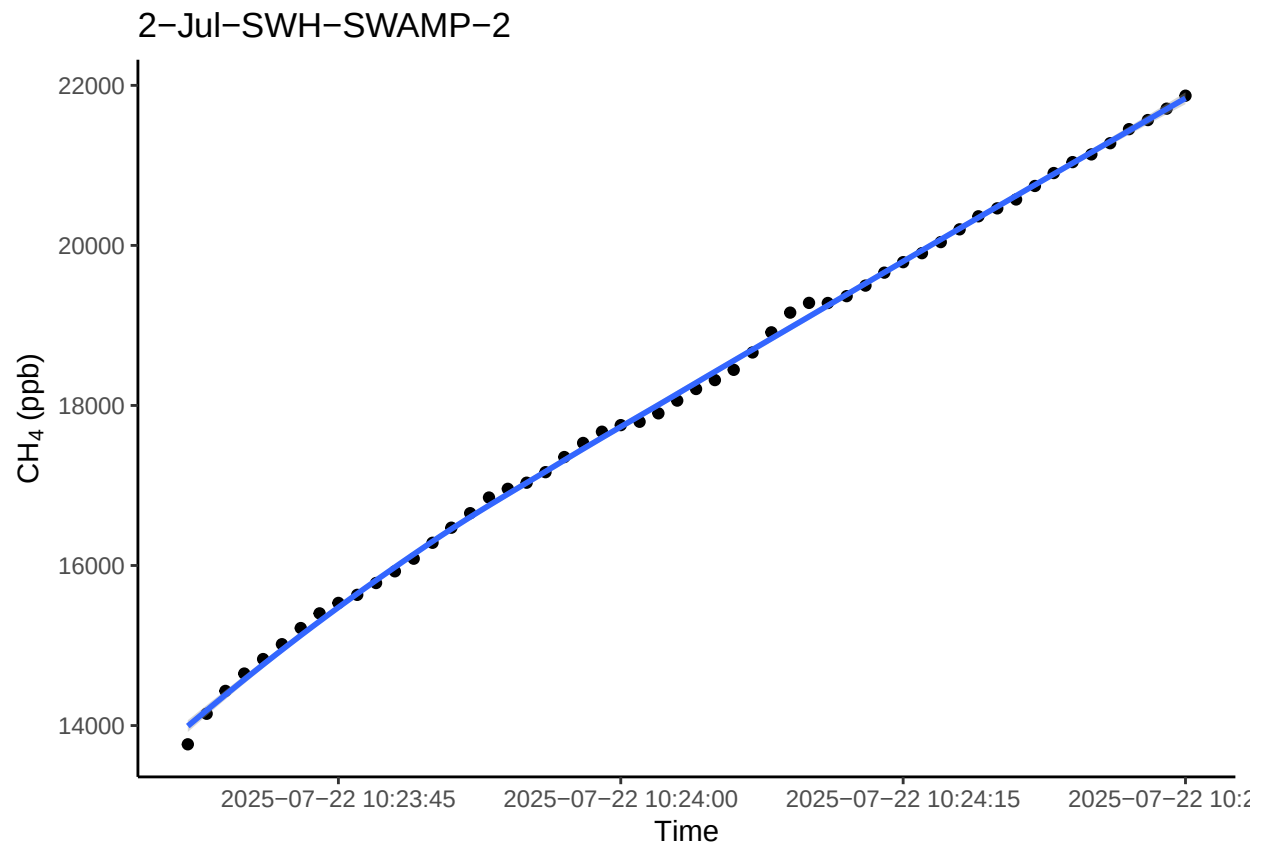
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



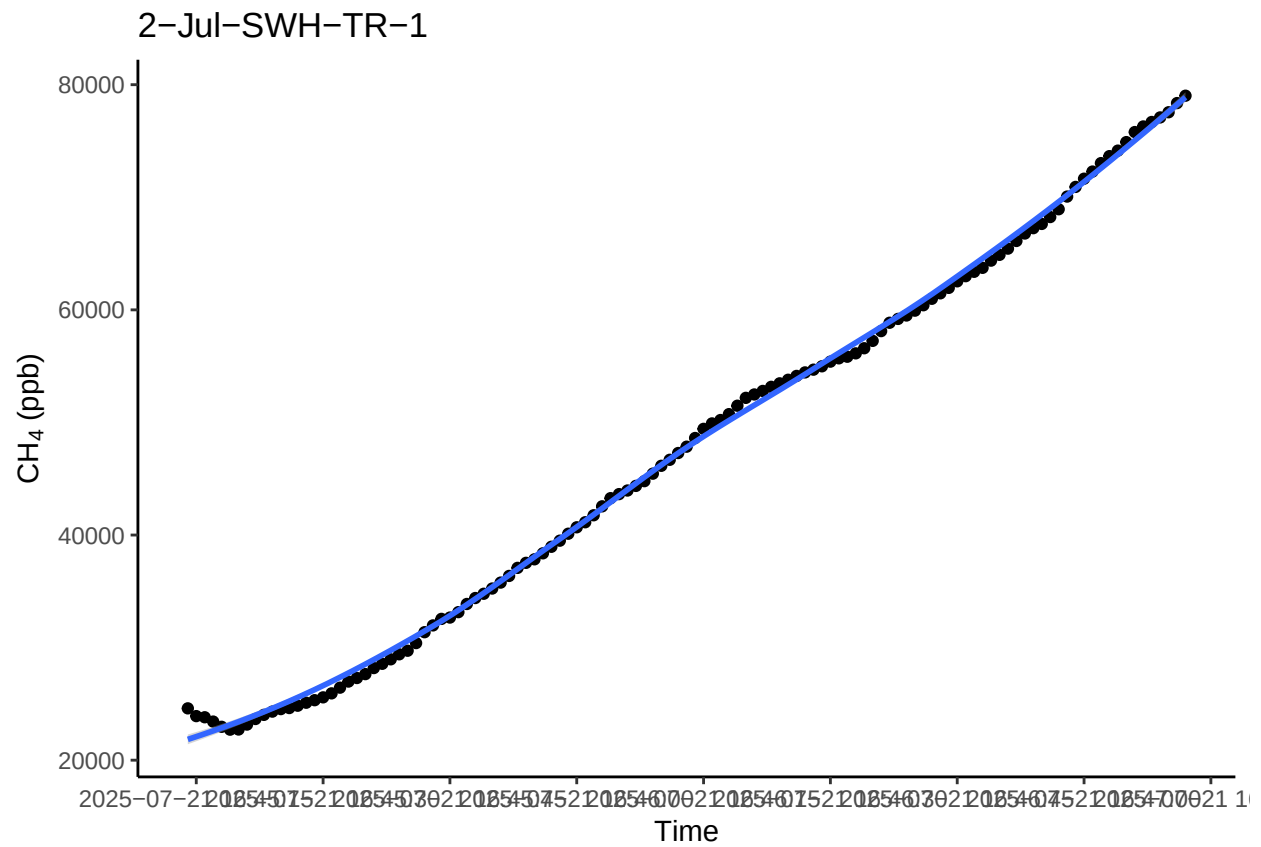
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



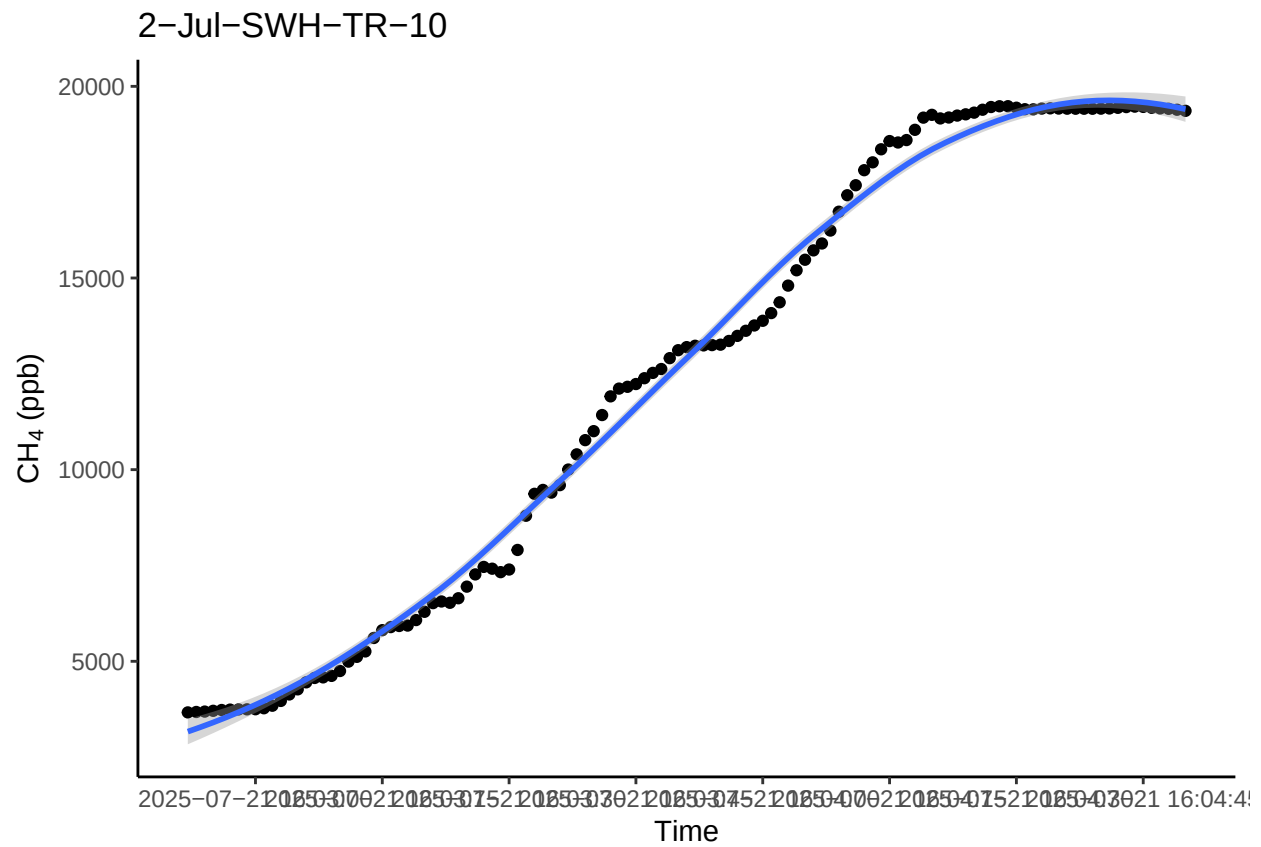
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



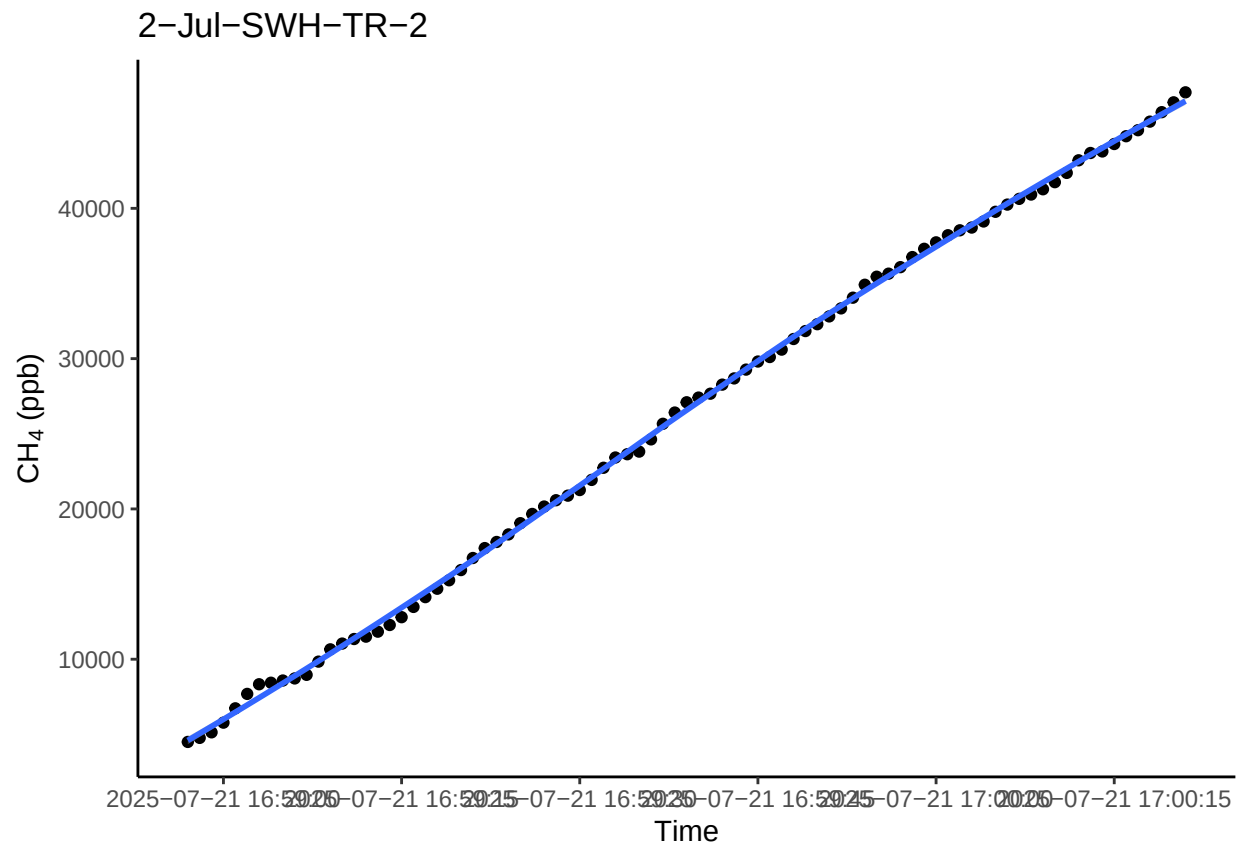
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



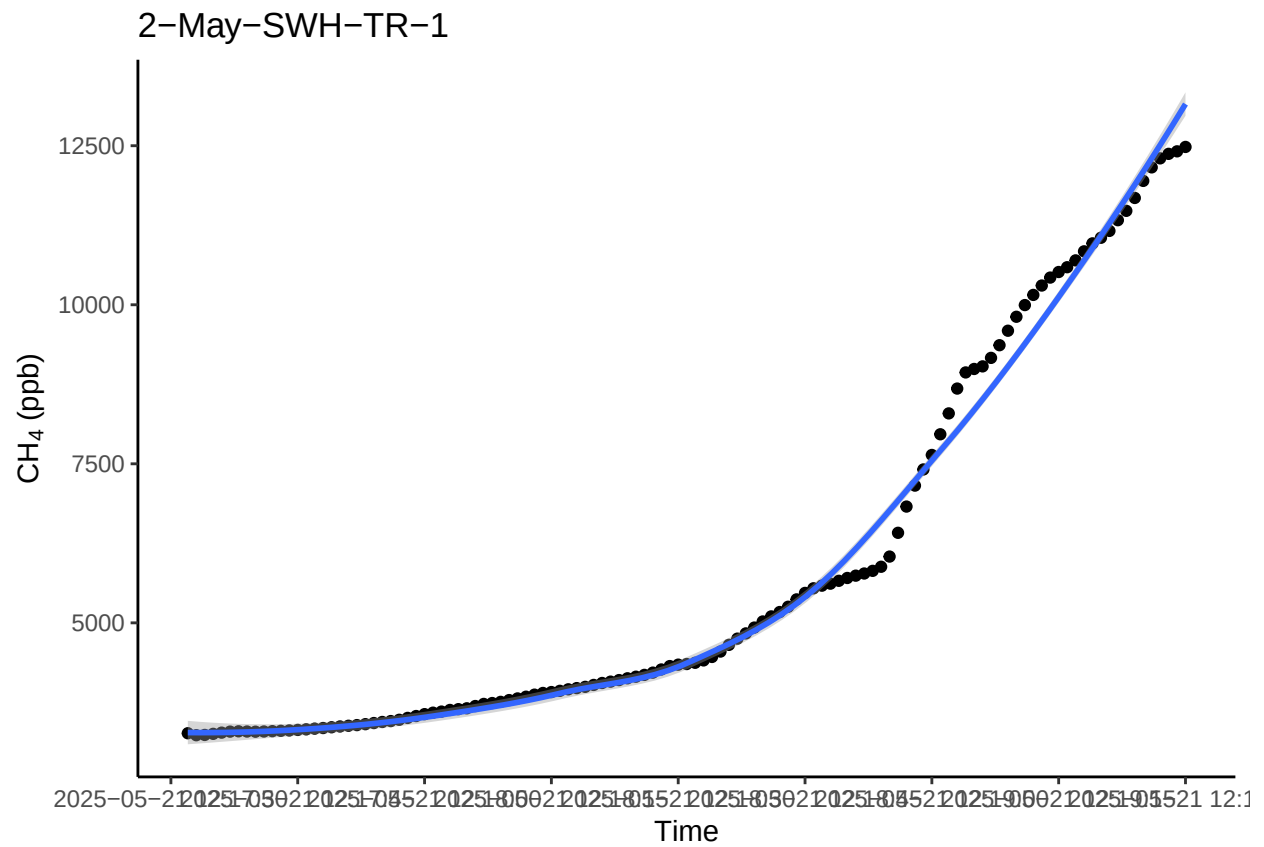
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



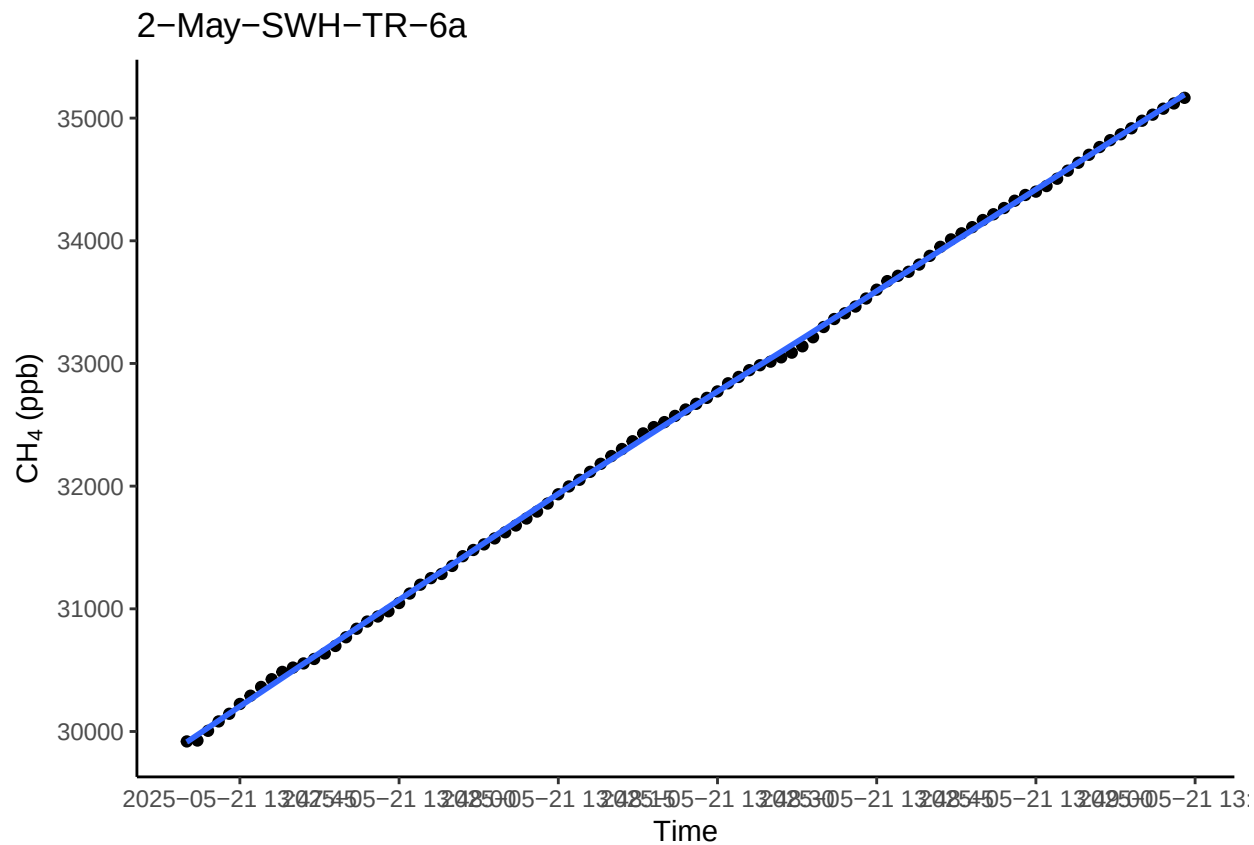
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



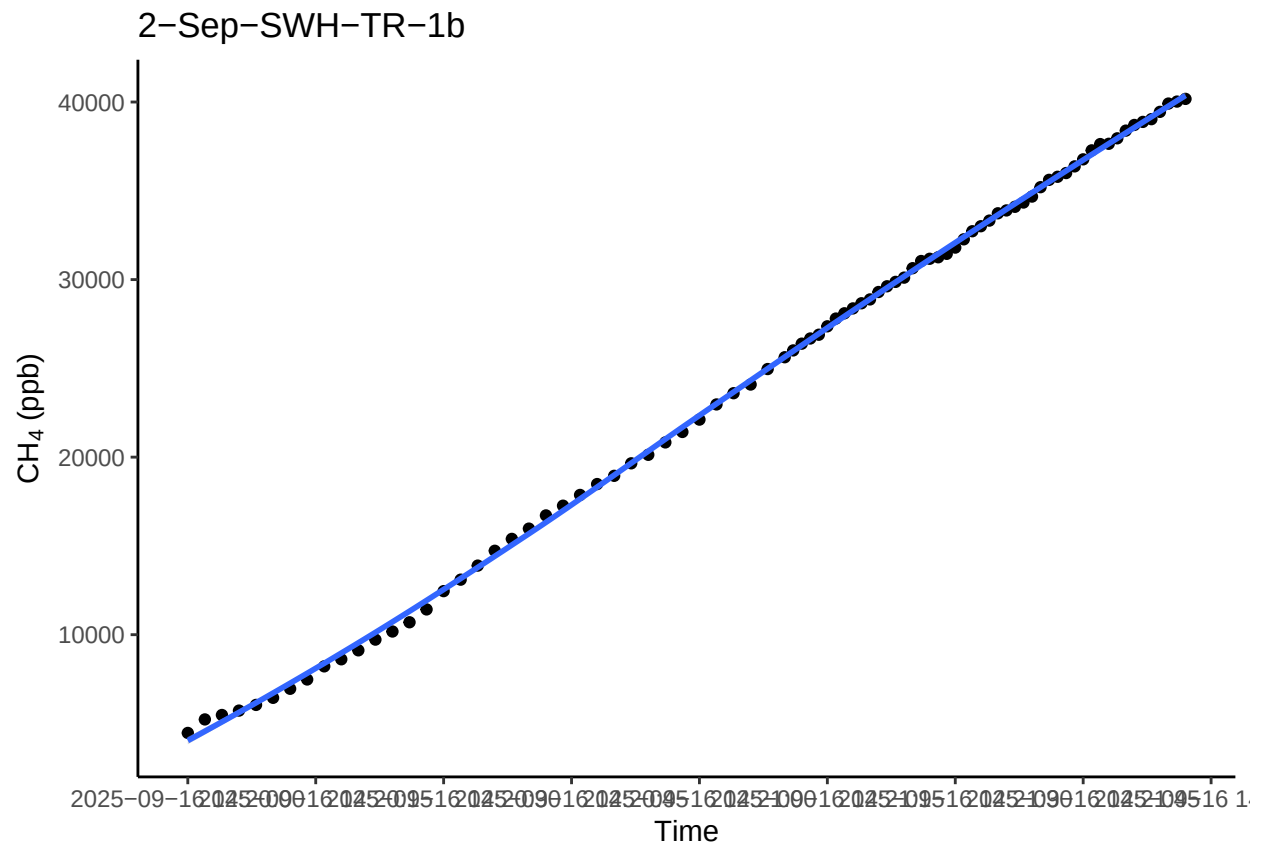
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



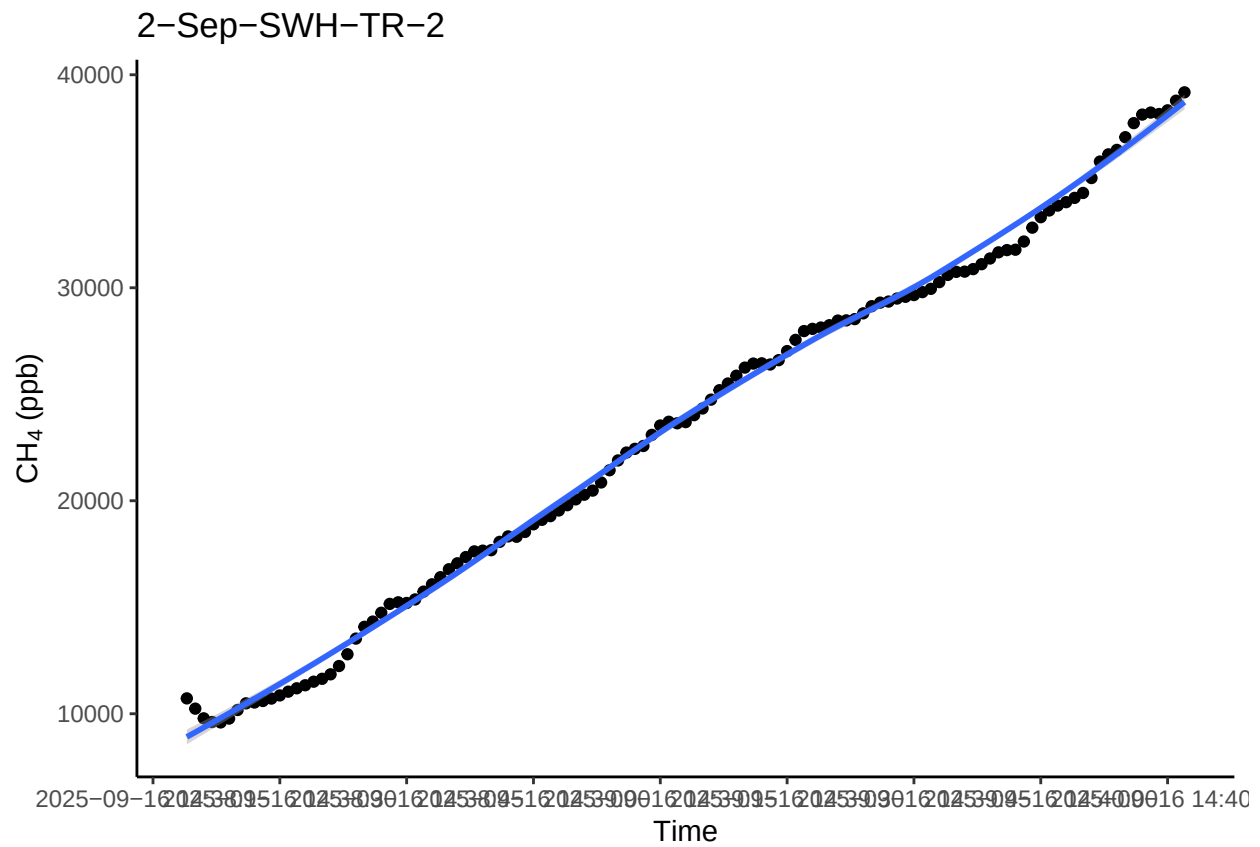
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

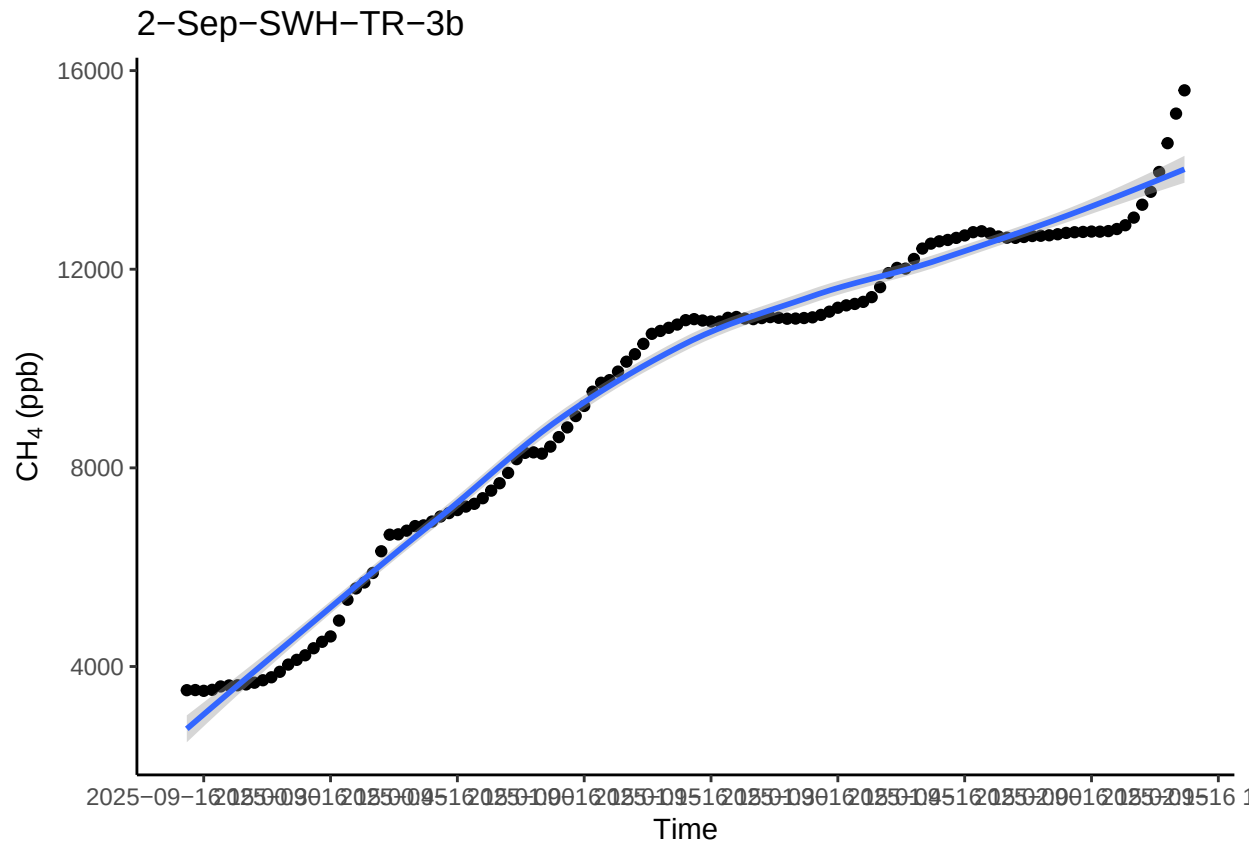
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



0.13 Check all fluxes

```
“{#r, echo=FALSE}
```

```
##Put all fluxes together fluxes_all <- as.data.frame(subset(allflux, select = Plot:TIMESTAMP_max))
all_plots <- unique(fluxes_all$Plot)
```

```
#Loop to run five random plots subset above for(P in all_plots){ r1 <- subset(alldat, Plot == P) p1 <-
ggplot(r1, aes(TIMESTAMP, ch4)) + geom_point() + theme_classic() + scale_x_datetime(breaks = “15
sec”) + ylab(expression(paste( CH [4], ” (ppb)“))) + xlab(“Time”) + labs(title = P) + geom_smooth()
print(p1) }
```

```
## Flag fluxes that do not fit criteria, low R2 etc.
```

```
““ r
```

```
##### Create a new data frame for output
```

```
# previously removed rows that were present in fluxes_weird which have a low R2 and a flux higher than
# flagged fluxes with low R squared and low flux
#allflux_filtered <- allflux[!(allflux$Plot %in% fluxes_weird$Plot), ]
allflux_filtered <- allflux
```

```
##### Poor fluxes
```

```

#List the ones that have odd/poor flux shapes - assessed visually for now
poor_flags <- c("1-Sep-SWH-SWAMP-3", "1-Jul-SWH-TR-3", "1-Jul-SWH-TR-6",
               "1-May-SWH-TR-7a", "1-Sep-SWH-TR-10a", "2-Sep-SWH-SWAMP-3",
               "2-Mar-SWH-TR-3"
               )

#add a flag for the fluxes that are v-shaped and therefore not even a flux
allflux_filtered <- allflux_filtered %>%
  mutate(qaqc_flag = ifelse(is.na(qaqc_flag) & Plot %in% poor_flags, "poor flux, need to remove", qaqc_flag))

# 1-Sep-SWH-SWAMP-3 - messy flux - remove
# 1-Jul-SWH-TR-3 - messy flux - remove
# 1-Jul-SWH-TR-6 - messy flux - remove
# 1-May-SWH-TR-7a - messy flux - remove
# 1-Sep-SWH-TR-10a - messy flux - remove
# 2-Sep-SWH-SWAMP-3 - messy flux - remove
# 2-Mar-SWH-TR-3 - poor flux from ebullition and carryover - remove

# View the result
head(allflux_filtered)

##           Plot          TIMESTAMP HM81_AIC HM81_flux.estimate HM81_p.value
## 1 1-Apr-GCW-TR-1 2025-04-16 10:58:18 181.3388          1.6327955 6.239754e-47
## 2 1-Apr-GCW-TR-2 2025-04-16 11:07:21      NA              NA              NA
## 3 1-Apr-GCW-TR-3 2025-04-16 11:11:25      NA              NA              NA
## 4 1-Apr-GCW-TR-4 2025-04-16 11:15:46 247.7671         -0.8851214 7.084537e-36
## 5 1-Apr-GCW-TR-5 2025-04-16 11:20:03 302.5536         -1.0441517 3.437872e-34
## 6 1-Apr-GCW-UP-1 2025-04-16 11:25:26      NA              NA              NA
##   HM81_r.squared HM81_RMSE  lin_AIC lin_flux.estimate lin_flux.std.error
## 1      0.9818199  1.250370 273.6901      0.5537432      0.025673190
## 2              NA          NA 167.5442     -0.5728363      0.009608188
## 3              NA          NA 161.8837     -0.6419451      0.009117570
## 4      0.9516337  2.312932 199.9005     -0.6555346      0.012964449
## 5      0.9438536  3.841250 222.9239     -1.0140967      0.016044886
## 6              NA          NA 208.2447     -1.0267931      0.014005804
##   lin_int.estimate lin_int.std.error  lin_p.value lin_r.squared lin_RMSE
## 1      18294.12          0.9022827 1.332060e-27  0.8994622 2.940393
## 2      18364.13          0.3376792 1.502279e-49  0.9855816 1.100442
## 3      18470.98          0.3204364 2.924916e-53  0.9896191 1.044250
## 4      18339.85          0.4556348 6.839649e-46  0.9800668 1.484840
## 5      18439.16          0.5638965 7.515882e-51  0.9871501 1.837647
## 6      18377.25          0.4922331 3.648665e-54  0.9904177 1.604108
##   poly_AIC poly_r.squared poly_RMSE  rob_AIC rob_converged rob_flux.estimate
## 1 148.2690      0.9908491 0.9046663 273.8418          TRUE      0.5449503
## 2 169.9530      0.9860003 1.1058212 167.7437          TRUE     -0.5753190
## 3 163.5242      0.9900629 1.0419171 162.0126          TRUE     -0.6439425
## 4 159.1735      0.9912932 1.0007781 199.9062          TRUE     -0.6562009
## 5 171.6499      0.9953830 1.1233334 222.9552          TRUE     -1.0164617
## 6 166.9547      0.9958578 1.0755441 210.7860          TRUE     -1.0458860
##   rob_flux.std.error rob_RMSE          TIMESTAMP_min          TIMESTAMP_max
## 1      0.026594310 3.360086 2025-04-16 10:57:52 2025-04-16 10:58:45
## 2      0.009243808 1.190623 2025-04-16 11:06:55 2025-04-16 11:07:48
## 3      0.010030988 0.916799 2025-04-16 11:10:59 2025-04-16 11:11:52

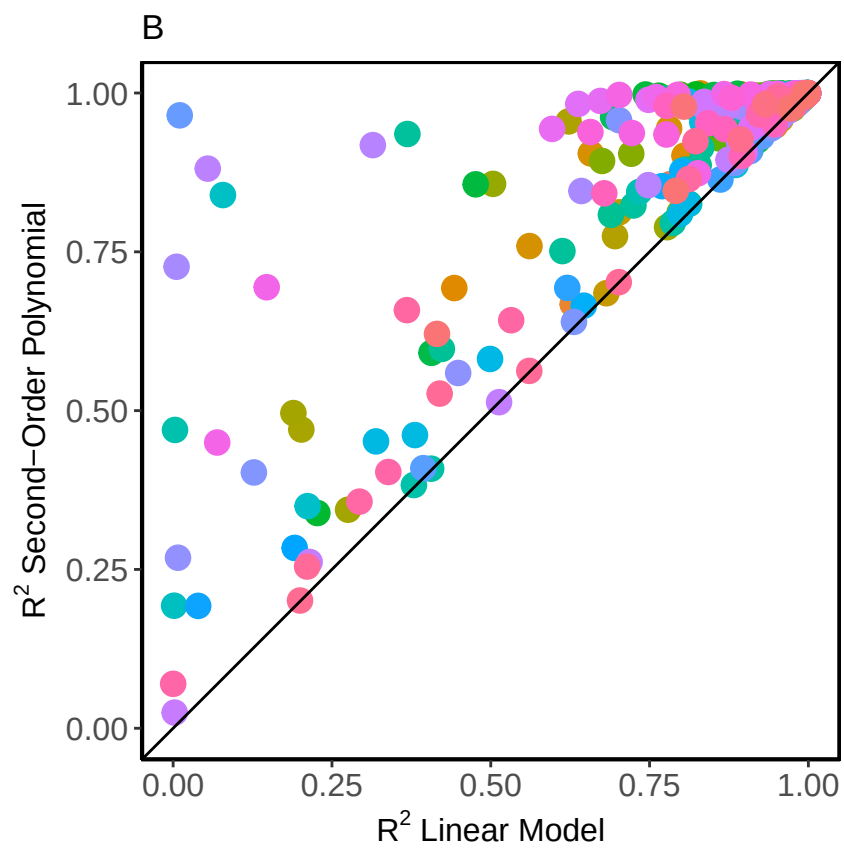
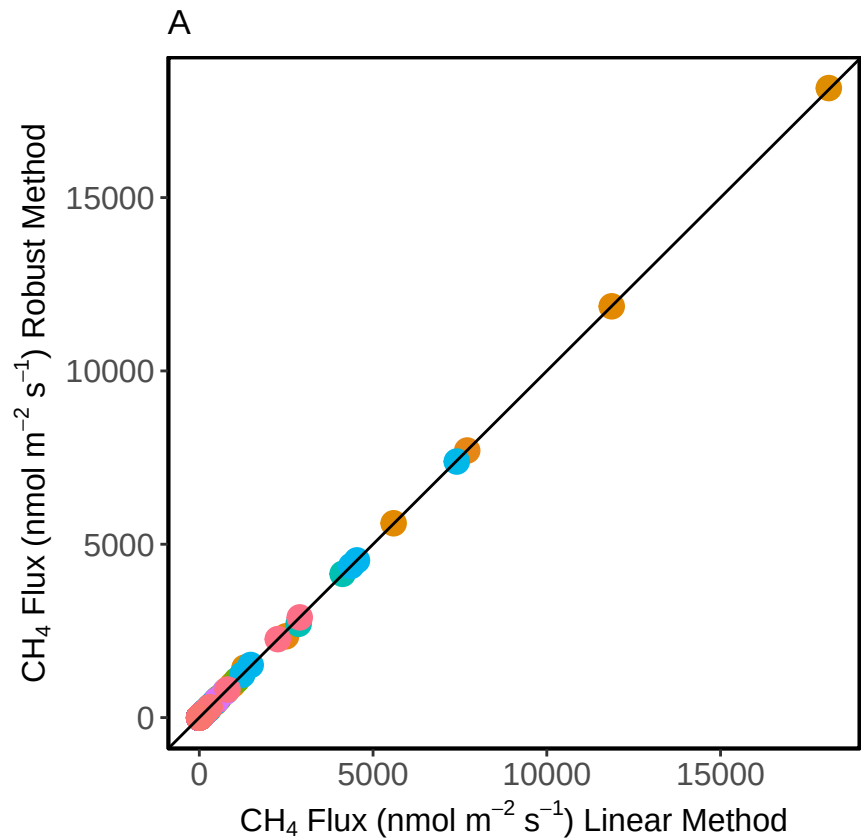
```

```

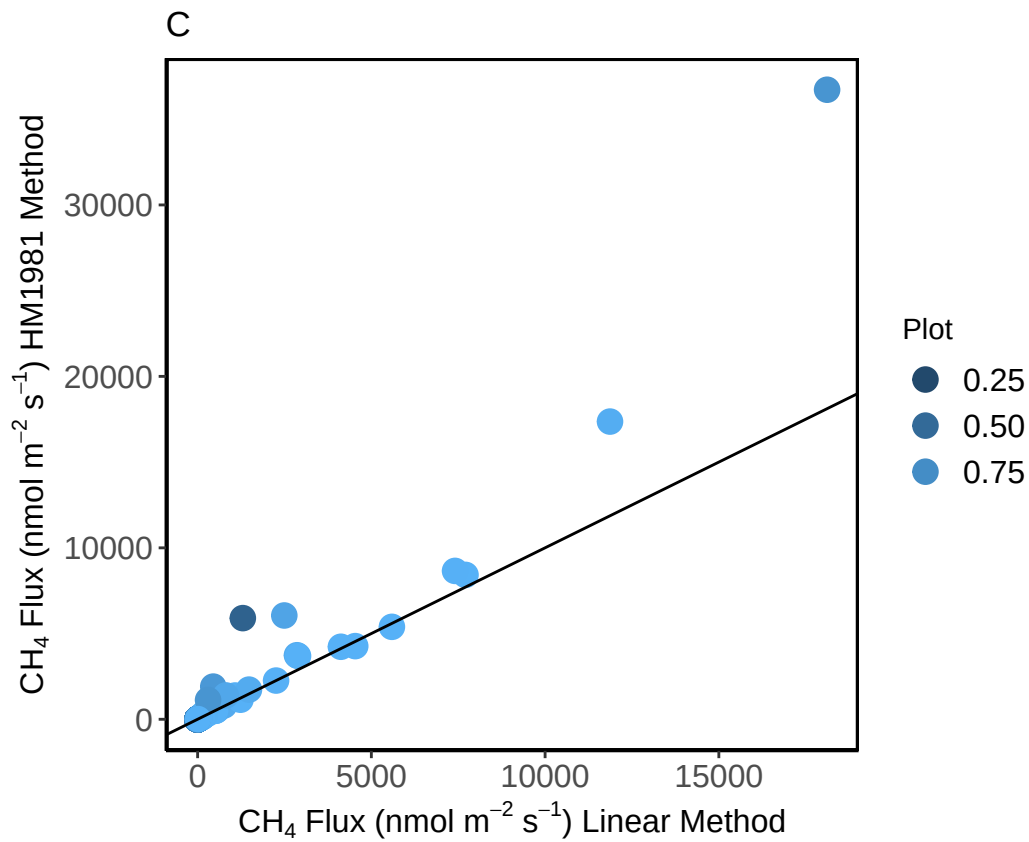
## 4      0.014016085 1.839827 2025-04-16 11:15:20 2025-04-16 11:16:13
## 5      0.016978997 2.299754 2025-04-16 11:19:37 2025-04-16 11:20:30
## 6      0.010860831 1.371665 2025-04-16 11:25:00 2025-04-16 11:25:53
##
##          qaqc_flag
## 1 low r2, but also low flux - checked
## 2
## 3
## 4
## 5
## 6

```


0.14 Do some visualization of QAQC plots to assess fluxes




```
## Warning: Removed 164 rows containing missing values or values outside the scale range
## ('geom_point()').
```



0.15 Read out final dataframe and export for use elsewhere

```
## Warning in rbind(c("1", "Apr", "GCW", "TR", "1"), c("1", "Apr", "GCW", "TR", :
## number of columns of result is not a multiple of vector length (arg 61)
```

```
##   RepNumber Month Site Zone Replicate
## 1          1   Apr  GCW   TR          1
## 2          1   Apr  GCW   TR          2
## 3          1   Apr  GCW   TR          3
## 4          1   Apr  GCW   TR          4
## 5          1   Apr  GCW   TR          5
## 6          1   Apr  GCW   UP          1
```

0.15.1 ARCHIVE